Script

Mathematics 2 for CAI - Linear Algebra

Lecture notes from K. Roegner
Based on the script of Mehrmann-Rambau-Seiler from the TU Berlin

Contents

1 Introduction

This script arose as a companion to another script “Lineare Algebra für Ingenieure” by Seiler-Mehrmann-Rambau. It was intended to supplement the rather sparse information in order to support students in their quest to master the skills of basic linear algebra.
In the following table, some measures are given that alleviate problems that arose in combination with the other script.

Problem areas	Measures
Difficulties in understanding mathematicals definitions and theorems	clear coloring schemes that show the connections and explicit explanations
Sparse details for solutions to examples	Prototypical examples with detailed solutions
Tendency to focus on formulae	When appropriate different solution strategies are discussed.
Difficulties in identifying connections due to too litte repetition of concepts	Clear relationship to previous material in new contexts

Tips:

- If you do not understand everything at once, it is normal. Some topics (such as subspaces, linear mappings, and matrix representations) are generally difficult to understand. It will require enough practice and repetition.
- All of the problems in this script can be done without a calculator. Try practicing without one.
- Use the THISuccess$^{\text{AI}}$ platform to find supplemental materials and electronic exercises.

1.1 Structure of the text

The type of box tells you about its contents.

* Definitions
* Theorems
* Algorithms

Explanations and Remarks

Examples

* Basic techniques
* Often partially computational and partially conceptual
* Tips

Exercises

* Learning checks to practice the basics from the examples
* More in-depth exercises

Questions in parentheses = food for thought

Solutions to the exercises at the end of each chapter

1.2 Organizing your work in the course

The course Mathematics 2 for CAI requires a work load of 175 hours (about 4.5 hours per week classes, 4.5 hours per week independent study + 35 hours for the exam preparation). The suggested times serve only for orientation purposes. Individual abilities also play a role in how much time needs to be invested.

Acknowledgements Thanks go to my former colleagues Ruedi Seiler, Elisabeth Ludwig, Dagmar Timmreck and Michael Heimann at the TU Berlin who gave me input and suggestions as well as the student workers in the UNITUS-Project for their efforts in making this script even better.
I wish you success in Math 2. I hope you come to enjoy this subject as much as I do!

2 The vector spaces $\mathbb{R}^n, \mathbb{C}^n, \mathbb{F}^n$

A vector $\vec{v} := \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2$ (say: v **is defined as the vector v_1, v_2 in** $\mathbb{R}$ two) can be thought of as an arrow originating at the origin $(0,0)$ (or the zero vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$) that points to the point (v_1, v_2) in the plane. (For that reason, we denote vectors $\vec{v} \in \mathbb{F}^n$ with a small arrow over the top.)

Such an arrow model can also be used in $\mathbb{R}^1$ (the real number line) or $\mathbb{R}^3$ (three-dimensional space). For $n \geq 4$ it is hard to visualize $\mathbb{R}^n$. However, it still makes sense to think about these spaces because of their useful structures that we extract (abd abstract) from or knowledge in the lower dimensional spaces $\mathbb{R}^1, \mathbb{R}^2, \mathbb{R}^3$.

We think of these objects as so-called vector spaces over a field. In general, we use the fields

- $\mathbb{R}$ (the field of **real numbers**) or
- $\mathbb{C}$ (the field of **complex numbers** $\{a + bi \mid a, b \in \mathbb{R}\}$, i.e. the set of all numbers of the form $a + bi$ with $a, b \in \mathbb{R}$ such that $i^2 = -1$.

In this section, we will consider the vector spaces $\mathbb{R}^n$ over $\mathbb{R}$ or $\mathbb{C}^n$ over $\mathbb{C}$. If we do not want to make a distinction between the two, we just use $\mathbb{F}^n$ over $\mathbb{F}$ (i.e. $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$).

The elements (or **vectors**) in $\mathbb{F}^n$ are so-called n-tuples and are denoted as **column vectors** :

$$\vec{v} := \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}, v_i \in \mathbb{F} \text{ for } \boldsymbol{i = 1, 2, \cdots, n}.$$

The v_i, $\boldsymbol{i = 1, 2, \cdots, n}$, are called **entries** or **components** of the vector.

Caution! Sometime $\boldsymbol{i}$ is considered as a complex number, sometimes as an index $\boldsymbol{i}$. The context decides the meaning. The letter i will be avoided in case we are working with complex numbers.

2.1 Vector space operations

Definition (vector addition; multiplication with scalars)

Let $\vec{u}, \vec{v} \in \mathbb{F}^n$ and $\alpha \in \mathbb{F}$. The two operations, **vector addition** and **multiplication with a scalar** α, are defined componentwise:

vector addition
$$\vec{u} + \vec{v} = \begin{bmatrix} \boldsymbol{u_1} \\ \boldsymbol{u_2} \\ \vdots \\ \boldsymbol{u_n} \end{bmatrix} + \begin{bmatrix} \boldsymbol{v_1} \\ \boldsymbol{v_2} \\ \vdots \\ \boldsymbol{v_n} \end{bmatrix} := \begin{bmatrix} \boldsymbol{u_1 + v_1} \\ \boldsymbol{u_2 + v_2} \\ \vdots \\ \boldsymbol{u_n + v_n} \end{bmatrix},$$

multiplication with a scalar
$$\alpha \begin{bmatrix} \boldsymbol{v_1} \\ \boldsymbol{v_2} \\ \vdots \\ \boldsymbol{v_n} \end{bmatrix} := \begin{bmatrix} \boldsymbol{\alpha v_1} \\ \boldsymbol{\alpha v_2} \\ \vdots \\ \boldsymbol{\alpha v_n} \end{bmatrix}.$$

Remarksn:

• We sometime just say addition instead of vector addition.

• The number $\boldsymbol{\alpha}$ is called a scalar because of the geometrical effect (e.g. in $\mathbb{R}^2$) the multiplication of a vector $\vec{v}$ with a scalar α is just a scaling of the vector $\vec{v}$. That means that $\vec{v}$ and $\alpha\vec{v}$ lie on the same line through the origin or, in the language of linear algebra, through the **zero vector** $\vec{0} := \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$.

(Make a graph including the vectors $\begin{bmatrix}1\\2\end{bmatrix}$ and $3\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}3\\6\end{bmatrix}$ to illustrate this. Try a few other scalars like -1 and $\frac{1}{2}$.)

Examples (vector space operations in $\mathbb{R}^2$)

Let $\vec{u}:=\begin{bmatrix}2\\7\end{bmatrix}, \vec{v}:=\begin{bmatrix}4\\3\end{bmatrix}, \vec{w}:=\begin{bmatrix}1\\-1\end{bmatrix}$ and $\vec{0}:=\begin{bmatrix}0\\0\end{bmatrix}\in\mathbb{R}^2$. Calculate the following vectors.

a) $\vec{u}+\vec{v}$, b) $\vec{u}-2\vec{v}$, c) $\vec{w}+\vec{0}$, d) $\vec{w}-\vec{w}$, e) $3\vec{u}-7\vec{w}$.

$\begin{bmatrix}6\\10\end{bmatrix}$ $\begin{bmatrix}-6\\1\end{bmatrix}$ $\begin{bmatrix}1\\-1\end{bmatrix}$ $\begin{bmatrix}\vec{0}\end{bmatrix}$ $\begin{bmatrix}-1\\28\end{bmatrix}$

a) $\vec{u}+\vec{v}=\begin{bmatrix}2\\7\end{bmatrix}+\begin{bmatrix}4\\3\end{bmatrix}=\begin{bmatrix}6\\10\end{bmatrix}$

b) $\vec{u}-2\vec{v}=\vec{u}+(-2)\vec{v}=\begin{bmatrix}2\\7\end{bmatrix}+\begin{bmatrix}-2\cdot4\\-2\cdot3\end{bmatrix}=\begin{bmatrix}2\\7\end{bmatrix}+\begin{bmatrix}-8\\-6\end{bmatrix}=\begin{bmatrix}2\\7\end{bmatrix}+\begin{bmatrix}-8\\-6\end{bmatrix}=\begin{bmatrix}-6\\1\end{bmatrix}$

c) $\vec{w}+\vec{0}=\begin{bmatrix}1\\-1\end{bmatrix}+\begin{bmatrix}0\\0\end{bmatrix}=\begin{bmatrix}1\\-1\end{bmatrix}=\vec{w}$

d) $\vec{w}-\vec{w}=\begin{bmatrix}1\\-1\end{bmatrix}-\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}1\\-1\end{bmatrix}+\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}=\vec{0}$

e) $3\vec{u}-7\vec{w}=3\begin{bmatrix}2\\7\end{bmatrix}-7\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}6\\21\end{bmatrix}+\begin{bmatrix}-7\\7\end{bmatrix}=\begin{bmatrix}-1\\28\end{bmatrix}$

Example (vector space operations in $\mathbb{C}^2$)

For $\vec{u}:=\begin{bmatrix}1+5i\\2-3i\end{bmatrix}, \vec{v}:=\begin{bmatrix}9\\7-8i\end{bmatrix}, \alpha=3+2i$ compute the following vectors.

a) $\vec{u}+\vec{v}$, b) $\vec{u}-\vec{v}$, c) $\alpha\vec{v}$.

$\begin{bmatrix}10+5i\\9-11i\end{bmatrix}$ $\begin{bmatrix}-8+5i\\-5+5i\end{bmatrix}$ $\begin{bmatrix}27+18i\\21+14i\\-24i+16\end{bmatrix}=\begin{bmatrix}27+18i\\37-10i\end{bmatrix}$

a) $\vec{u}+\vec{v}=\begin{bmatrix}1+5i\\2-3i\end{bmatrix}+\begin{bmatrix}9\\7-8i\end{bmatrix}=\begin{bmatrix}10+5i\\9-11i\end{bmatrix}$

b) $\vec{u}-\vec{v}=\vec{u}+(-1)\vec{v}=\begin{bmatrix}1+5i\\2-3i\end{bmatrix}+\begin{bmatrix}-9\\-7+8i\end{bmatrix}=\begin{bmatrix}-8+5i\\-5+5i\end{bmatrix}$

c) $\alpha\vec{v}=(3+2i)\begin{bmatrix}9\\7-8i\end{bmatrix}=\begin{bmatrix}(3+2i)9\\(3+2i)(7-8i)\end{bmatrix}=\begin{bmatrix}27+18i\\21-24i+14i-16i^2\end{bmatrix}\overset{(i^2=-1)}{=}\begin{bmatrix}27+18i\\37-10i\end{bmatrix}$

Exercise 1 (Learning check)

Check that $3i\begin{bmatrix}2i\\1+i\end{bmatrix}+\begin{bmatrix}2\\4i\end{bmatrix}=\begin{bmatrix}-4\\-3+7i\end{bmatrix}$.

$=\begin{bmatrix}-6\\3i-3\end{bmatrix}+\begin{bmatrix}2\\4i\end{bmatrix}=\begin{bmatrix}-4\\7i-3\end{bmatrix}$

2.2 Properties of vector spaces

Definition (Properties of the vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$)

For all $\vec{u}, \vec{v}, \vec{w} \in \mathbb{F}^n$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) and $\alpha, \beta \in \mathbb{F}$ the following properties hold.

Addition	$(\vec{u}+\vec{v})+\vec{w} = \vec{u}+(\vec{v}+\vec{w})$	Associative law
	$\vec{v}+\vec{0} = \vec{v}$	with $\vec{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ the zero vector.
	$\vec{v}+(-\vec{v}) = \vec{0}$	
	$\vec{u}+\vec{v} = \vec{v}+\vec{u}$	Commutative law
Multiplication with scalars	$\alpha(\beta\vec{v}) = (\alpha\beta)\vec{v}$	Associative law
	$\alpha(\vec{u}+\vec{v}) = \alpha\vec{u}+\alpha\vec{v}$	Distributive law
	$(\alpha+\beta)\vec{v} = \alpha\vec{v}+\beta\vec{v}$	Distributive law
	$1\vec{v} = \vec{v}$	

Remark: We can calculate with vectors almost the same way as with numbers. An important exception is that we cannot invert vectors, which means we cannot divide by vectors.

2.3 Linear combinations

Definition (Linear combination)

A **linear combination** of the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in \mathbb{F}^n$ has the form

$$\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 + \cdots + \alpha_k\vec{v}_k$$

for scalars $\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{F}$.

A linear combination of given vectors is just the sum of (constant) multiples of those vectors.

Example (linear combination)

Show that the vector $\begin{bmatrix} -3 \\ 4 \end{bmatrix}$ is a linear combination of the vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

We have to represent the vector as a sum of multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

$$\begin{aligned}
\alpha_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \alpha_2 \begin{bmatrix} 3 \\ 1 \end{bmatrix} &= \begin{bmatrix} -3 \\ 4 \end{bmatrix} \\
\begin{bmatrix} 1\alpha_1 \\ 2\alpha_1 \end{bmatrix} + \begin{bmatrix} 3\alpha_2 \\ 1\alpha_2 \end{bmatrix} &= \begin{bmatrix} -3 \\ 4 \end{bmatrix} && \text{multiplication with scalars} \\
\begin{bmatrix} 1\alpha_1 + 3\alpha_2 \\ 2\alpha_1 + 1\alpha_2 \end{bmatrix} &= \begin{bmatrix} -3 \\ 4 \end{bmatrix} && \text{vector addition} \\
1\alpha_1 + 3\alpha_2 &= -3 && \text{compare first components} \\
2\alpha_1 + 1\alpha_2 &= 4 && \text{compare second components}
\end{aligned}$$

Solve the system of equations (answer: $\alpha_1 = 3, \alpha_2 = -2$). The answer can be easily checked:

$$3\begin{bmatrix} 1 \\ 2 \end{bmatrix} - 2\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} + \begin{bmatrix} -6 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}.$$

Remark: In Chapter 4, we will learn how to find suitable scalars for larger systems of equations.

Exercise 2 (Learning check)

a) Determine scalars $\alpha_1, \alpha_2 \in \mathbb{R}$ so that the equation $\alpha_1 \begin{bmatrix}1\\0\\1\end{bmatrix} + \alpha_2 \begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}2\\-3\\2\end{bmatrix}$ is satisfied.

b) Is it possible to represent the vector $\begin{bmatrix}1\\2\\3\end{bmatrix}$ as a linear combination of the vectors $\begin{bmatrix}1\\0\\1\end{bmatrix}$ and $\begin{bmatrix}0\\1\\0\end{bmatrix}$?

2.4 Span of a set of vectors

Definition (span)

The **span** of a set of given vectors $\{\vec{v}_1, \dots, \vec{v}_k\} \subset \mathbb{F}^n$ is given by

$$\operatorname{span}\{\vec{v}_1, \dots, \vec{v}_k\} := \{\alpha_1\vec{v}_1 + \cdots + \alpha_k\vec{v}_k \mid \alpha_1, \dots, \alpha_k \in \mathbb{F}\}.$$

Remarks:

• The span of $\{\vec{v}_1, \dots, \vec{v}_k\}$ is just the set of all possible linear combinations of the vectors $\vec{v}_1, \dots, \vec{v}_k$ with scalars $\alpha_1, \dots, \alpha_k \in \mathbb{F}$.

• The scalars α_i, $i = 1, \dots, k$, are also called *coefficients* or *parameters* .

Example (span of a set)

Describe the sets $M_1 := \operatorname{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$ and $M_2 := \operatorname{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}2\\2\end{bmatrix}\right\}$ in $\mathbb{R}^2$.

Since M_1 is the set of all possible linear combinations of the single vector $\begin{bmatrix}1\\1\end{bmatrix}$, we find that

$$M_1 = \operatorname{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\} = \left\{\alpha \begin{bmatrix}1\\1\end{bmatrix} \middle| \alpha \in \mathbb{R}\right\} = \left\{\begin{bmatrix}\alpha\\\alpha\end{bmatrix} \middle| \alpha \in \mathbb{R}\right\}$$

Thus, M_1 represents the line $y = x$ in the plane $\mathbb{R}^2$.

For M_2 we note that $\begin{bmatrix}2\\2\end{bmatrix} = 2\begin{bmatrix}1\\1\end{bmatrix} \in \operatorname{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$, i.e. $\operatorname{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}2\\2\end{bmatrix}\right\} = \operatorname{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$.

Therefore, M_2 also represents the line $y = x$ in the plane.

Exercise 3 (Learning check)

Describe the set $M_3 := \operatorname{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$ in $\mathbb{R}^3$.

Exercise 4

Which of the vectors $\vec{v}_1 := \begin{bmatrix}2\\2\\2\end{bmatrix}$, $\vec{v}_2 := \begin{bmatrix}0\\0\\0\end{bmatrix}$, $\vec{v}_3 := \begin{bmatrix}3\\7\\0\end{bmatrix} \in \mathbb{R}^3$ are elements in $\operatorname{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$?

2.5 Subspaces

Definition (subspace)

Let T be a subset of $\mathbb{F}^n$ (notation: $T \subseteq \mathbb{F}^n$). T is a **subspace** of $\mathbb{F}^n$ if the following three conditions hold.

(1) T is not the *empty set* (i.e. $T \neq \emptyset$).

(2) T is *closed under addition* (i.e. for all $\vec{u}, \vec{v} \in T$ $\vec{u} + \vec{v} \in T$).

(3) T is *closed under multiplication with scalars* (i.e. for all $\vec{v} \in T, \alpha \in \mathbb{F}$ we have $\alpha\vec{v} \in T$).

Remarks:

• The symbol for a subset $\subseteq$ is often used incorrectly for the idea of subspace. There is no symbol for "T is a subspace of $\mathbb{F}^n$".

• If T is a subspace of $\mathbb{F}^n$, then T is also a vector space. That means that if a nonempty set $T \in \mathbb{F}^n$ is closed under addition and multiplication by scalars, then it satisfies all the other vector space properties.

• If T is a subspace of $\mathbb{F}^n$, then for the vectors $\vec{u}, \vec{v} \in T$ and scalars $\alpha, \beta \in \mathbb{F}$ we know, due to property (3), that $\alpha\vec{u}$ and $\beta\vec{v}$ are in T. Because of property (2) their sum $\alpha\vec{u} + \beta\vec{v}$ is also in T. You can show using an induction argument that this is true no matter how many vectors we choose so that all linear combinations of vectors in T also lies in T. In particular, this means that $\text{span}(T) = T$.

• $\mathbb{F}^n$ is a subspace of $\mathbb{F}^n$.

• $\vec{0}$ is an element of every subspace of $\mathbb{F}^n$:

Let T be a subspace of $\mathbb{F}^n$. By definition T is nonempty so we can choose an element. $\vec{v} \in T$. Every field contains the element 0 and T is per definition closed under multiplication with scalars, here with 0: $0\vec{v} = \vec{0} \in T$.

This means for a set $S \subseteq \mathbb{F}^n$ if $\vec{0} \notin M$, then S is not a subspace of $\mathbb{F}^n$.

Example (subspaces)

Which of the following sets are subspaces of $\mathbb{R}^2$?

$$M_1 := \text{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\} \qquad\qquad M_4 := \left\{\begin{bmatrix}\boldsymbol{x_1}\\\boldsymbol{x_2}\end{bmatrix} \in \mathbb{R}^2 \middle| \boldsymbol{x_2} = \boldsymbol{x_1}^2\right\}$$

Claim: $M_1 := \text{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$ is a subspace of $\mathbb{R}^2$.

Proof: We show that the three conditions in the definition of a subspace are satisfied. (M_1 is (1) not empty, (2) closed under addition and (3) closed under multiplication with scalars.)

(1) M_1 is not empty since, according to the definition of the span of a set, we get: $\begin{bmatrix}1\\1\end{bmatrix} = 1\begin{bmatrix}1\\1\end{bmatrix} \in M_1$.

(2) Let $\vec{u}, \vec{v} \in M_1$. Then $\vec{u} = \alpha_1\begin{bmatrix}1\\1\end{bmatrix}$ and $\vec{v} = \alpha_2\begin{bmatrix}1\\1\end{bmatrix}$ for suitable scalars $\alpha_1, \alpha_2 \in \mathbb{R}$. The sum

$$\vec{u} + \vec{v} = \alpha_1\begin{bmatrix}1\\1\end{bmatrix} + \alpha_2\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\alpha_1\\\alpha_1\end{bmatrix} + \begin{bmatrix}\alpha_2\\\alpha_2\end{bmatrix} = \begin{bmatrix}\alpha_1+\alpha_2\\\alpha_1+\alpha_2\end{bmatrix} = (\alpha_1+\alpha_2)\begin{bmatrix}1\\1\end{bmatrix} \in M_1$$

lies in M_1 so that M_1 is closed under addition.

(3) Let $\vec{u} \in M_1$, i.e. $\vec{u} = \alpha_1\begin{bmatrix}1\\1\end{bmatrix}$ for some $\alpha_1 \in \mathbb{R}$, und $\beta \in \mathbb{R}$. Then we have $\beta\vec{u} = (\beta \cdot \alpha_1)\begin{bmatrix}1\\1\end{bmatrix} \in M_1$. M_1 is closed under multiplication with scalars.

The three conditions are satisfied from which it follows that M_1 is a subspace of $\mathbb{R}^2$.

Claim: The set $M_4 := \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 \middle| x_2 = {x_1}^2 \right\}$ is *not* a subspace of $\mathbb{R}^2$.

Proof (with a counterexample): The vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ lies in M_4 since $1 = 1^2$ holds. Because $2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ but $\begin{bmatrix} 2 \\ 2 \end{bmatrix} \notin M_4$ (a vector in M_4 with first component 2 must have second component $2^2 = 4$, the set M_4 is not closed under multiplication with scalars. M_4 is therefore not a subspace of $\mathbb{R}^2$.

Remarks:

- When showing that a set is a **subspace** of a vector space (as with M_1), you have to verify the subspace criteria for all elements in the set and all possible scalars. That means we work with **letters**.
- When showing that a set is **not a subspace** of a vector space, (as with M_4), it is sufficient to find a concrete counterexample for which one of the subspace criteria is not satisfied.

Exercise 5 (Learning check)

Show that $M_5 := \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} \right\}$ is a subspace of $\mathbb{R}^3$.

Exercise 6

a) Is $V := \{\vec{0}\}$ a subspace of $\mathbb{F}^n$? b) Is $M_6 := \left\{ \begin{bmatrix} x \\ x+7 \end{bmatrix} \in \mathbb{R}^2 \middle| x \in \mathbb{R} \right\}$ a subspace of $\mathbb{R}^2$?

2.6 Generating set

Definition (generating set)

Let G be a subset of the subspace T, which is contained in $\mathbb{F}^n$. G is a **generating set** of T if each element in T can be represented as a linear combination of the elements in G in at least one way, i.e. $\operatorname{span} G = T$ holds. We say G generates T.

Example (generating set)

Let $T := \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \middle| z = 0 \right\}$. Which of the following subsets of T generate T?

$$G_1 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \qquad G_2 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \right\}$$

$G_1 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ generates T since $\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + (y-x) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

$G_2 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \right\}$ is *not* a generating set for T. The vector $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in T$ cannot be generated by the vectors in G_2 because the **first two components** of any linear combination must be **equal**:

$$\alpha_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 + 2\alpha_2 \\ \alpha_1 + 2\alpha_2 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{for all } \alpha_1, \alpha_2 \in \mathbb{R}.$$

Exercise 7 (Learning check)

Let $T := \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, z = 0 \right\}$. Does $G_3 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \right\} \subseteq T$ generate T?

2.7 Linear dependence and linear independence

The third vector in G_3 from the last exercise does not produce any new elements in the set span(G_1) through linear combinations with the vectors in $G_1 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ since $\begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$ already lies in the plane generated by $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

$$\begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Rearranging the equation, we get

$$\begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} - 5 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

so that the equation

$$\beta \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} - 5\beta \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 3\beta \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{for all } \beta \in \mathbb{R}.$$

also holds. The equation

$$\alpha_1 \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{1}$$

thus has infinitely many solutions, assuming our ground field is infinite. Note that the solution $\alpha_1 = \alpha_2 = \alpha_3 = 0$ is called the trivial solution.

We see that there are many solutions to equation (1) because the first vector can already be generated by the other two vectors. The generating set G_3 has more elements than the generating set G_1, leading to redundancy in that there is more than one way to express elements in T. We can check if there are redundancies in a generating set G by looking at linear combinations that produce the zero vector. If the trivial solution is the only solution, then there are no redundancies.

Definition (linear independence, linear dependence)

The vectors $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k \in \mathbb{F}^n$ are **linearly independent** if the equation

$$\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \cdots + \alpha_k \vec{v}_k = \vec{0}$$

is only satisfied by the trivial solution $\alpha_1 = \cdots = \alpha_k = 0$. If the equation has any further solution, we say that the vectors are **linearly dependent**.

The vectors $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}5\\2\\0\end{bmatrix}$ are linearly dependent since equation (1) has more than one solution, which was shown above.

The vectors $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}$ are linearly independent because the equation

$$\alpha_1 \begin{bmatrix}1\\1\\0\end{bmatrix} + \alpha_2 \begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \tag{2}$$

only possesses the trivial solution: From equation (2) it follows that $\begin{bmatrix}\alpha_1\\ \alpha_1+\alpha_2\\ 0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$ has the unique solution $\alpha_1 = \alpha_2 = 0$.

This means that for any given vector $\begin{bmatrix}x\\y\\z\end{bmatrix} \in \mathbb{R}^3$, the equation $\alpha_1 \begin{bmatrix}1\\1\\0\end{bmatrix} + \alpha_2 \begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}x\\y\\z\end{bmatrix}$ has at most one solution. (Think about why this is true.)

2.8 Basis

A basis is a set of vectors that generates a given subspace without redundancies, which means that the represetation of each vector in the subspace is unique.

Definition (basis)

Let $T \neq \{\vec{0}\}$ be a subspace of $\mathbb{F}^n$. A **basis of** T is a set $\mathcal{B} \subseteq T$ satisfying the properties

(1) $\mathcal{B}$ is a generating set,

(2) the vectors in $\mathcal{B}$ are linearly independent.

For the vector space $V = \{\vec{0}\}$, we define the basis to be $\mathcal{B}_{\text{null}} := \emptyset$.

Remarks:

• If G is a generating set of T, then every element in T can be represented as a linear combination of the vectors in G in at least one way. If G is linearly independent, then every element in T can be represented as a linear combination of the vectors in G in at most one way. If G is a basis, then each vector in T can be represented as a linear combination of the elements in G in exactly one way. This unique representation is important for most of the following chapters!

• If T is a subspace of $\mathbb{F}^n$ and $T \neq \{\vec{0}\}$, then there are many bases for T. The unique representation depends on the basis chosen.

• Bases will often be denoted by $\mathcal{B}$.

• There is a difference between the trivial vector space $\{\vec{0}\}$ and the empty set $\emptyset$, which can also be denoted by $\{\,\}$. The empty set possesses no elements. The trivial vector space contains the zero vector, which we can think of as a point in the space, and that is not nothing.

Example (bases)

Which of the following sets are bases of the vector space $T := \left\{ \begin{bmatrix}x\\y\\z\end{bmatrix} \in \mathbb{R}^3 \;\middle|\; z = 0 \right\}$?

$$G_1 := \left\{ \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix} \right\}, \quad G_3 := \left\{ \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}5\\2\\0\end{bmatrix} \right\}$$

We have already shown that $G_1 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a basis of T since G_1 is a generating set of linearly independent vectors.

$G_3 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \right\}$ is *not* a basis for T. Even though it is a generating set, the vectors are linearly dependent (see page 13).

Exercise 8

Is the set $G_4 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \right\}$ a basis of $T := \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, z = 0 \right\}$?

Exercise 9

Each element of the vector space V can be represented uniquely as a linear combination of vectors in a given basis. Determine the representation of $\begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \in T := \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, z = 0 \right\}$ with respect to the basis $\mathcal{B}_1 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ of T, respectively, with respect to the basis $\mathcal{B}_2 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \right\}$ of T.

Definition (dimension)

Every basis $\mathcal{B}$ of the vector space V contains the same number of elements $|\mathcal{B}|$. The **dimension** of V is given by $\dim V = |\mathcal{B}|$.

Example (dimension)

Determine the dimension of the following vector spaces.

a) $T := \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, z = 0 \right\}$ b) $\operatorname{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \right\}$ c) $\mathbb{R}^2$

Our previous work allows us to determine the dimension of these spaces easily.

a) $\dim T = 2$ since G_2 consists of two elements (notation: $|G_2| = 2$).

b) $\dim \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \right\} = 2$

c) $\mathcal{B}_{\text{Std}} := \left\{ \vec{e}_1 := \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{e}_2 := \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is the **standard basis** of $\mathbb{R}^2$ so that $\dim \mathbb{R}^2 = 2$.

Definition (standard basis of $\mathbb{F}^n$)
The standard basis of the vector space $\mathbb{F}^n, n \geq 1$, is given by $\mathcal{B}_{\text{std}} :=$ $\left\{ \vec{e}_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \vec{e}_2 := \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \ldots, \vec{e}_n := \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \right\}.$

Exercise 10 (Learning check)
Determine the dimension of the given vector space.
a) $\mathbb{R}^3$ b) $\text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \right\}$ c) $\{\vec{0}\}$

2.9 Suggested solutions to the exercises in Chapter 2

Exercise 1 In $\mathbb{C}^2$, $3i \begin{bmatrix} 2i \\ 1+i \end{bmatrix} + \begin{bmatrix} 2 \\ 4i \end{bmatrix} = \begin{bmatrix} 6i^2 \\ 3i+3i^2 \end{bmatrix} + \begin{bmatrix} 2 \\ 4i \end{bmatrix} = \begin{bmatrix} -6 \\ -3+3i \end{bmatrix} + \begin{bmatrix} 2 \\ 4i \end{bmatrix} = \begin{bmatrix} -4 \\ -3+7i \end{bmatrix}$, since $i^2 = -1$.

Exercise 2 a) $\alpha_1 = 2, \alpha_2 = -3$; b) No. The first and third components have to be equal. (Why is this so?)

Exercise 3 We show that the set $M_3 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} = \left\{ \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \;\middle|\; \alpha_1, \alpha_2 \in \mathbb{R} \right\}$ can be thought of as the xy-plane in $\mathbb{R}^3$. Each point (resp. the associated vector) in the xy-plane of $\mathbb{R}^3$ can be represented as a linear combination of the vectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$:

$$\begin{bmatrix} a \\ b \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 + \alpha_2 \\ 0 \end{bmatrix}.$$

Equating components yields $\alpha_1 = a$ and $\alpha_1 + \alpha_2 = b \Rightarrow \alpha_2 = b - a$.

Exercise 4 Given are the three vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in \mathbb{R}^3$. We have to decide which of the three vectors are in

$$\text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} = \left\{ \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \;\middle|\; \alpha_i \in \mathbb{R}, i = 1, 2, 3 \right\}.$$

We try to represent each of the vectors $\vec{v}_i$ $(i = 1, 2, 3)$ as a linear combination of the vectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

• $\vec{v}_1 = \begin{bmatrix}2\\2\\2\end{bmatrix}$ can be represented as a linear combination of the vectors $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ using $\alpha_1 = \alpha_2 = \alpha_3 = 1$:

$$1\begin{bmatrix}\mathbf{1}\\\mathbf{1}\\\mathbf{0}\end{bmatrix} + 1\begin{bmatrix}\mathbf{1}\\\mathbf{1}\\\mathbf{1}\end{bmatrix} + 1\begin{bmatrix}\mathbf{0}\\\mathbf{0}\\\mathbf{1}\end{bmatrix} = \begin{bmatrix}1+1+0\\1+1+0\\0+1+1\end{bmatrix} = \begin{bmatrix}2\\2\\2\end{bmatrix}.$$

(Are there other coefficients $\alpha_1, \alpha_2, \alpha_3$ that solve the equation?) Thus:

$$\begin{bmatrix}2\\2\\2\end{bmatrix} \in \text{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}.$$

• $\vec{v}_2 = \begin{bmatrix}0\\0\\0\end{bmatrix} \in \text{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ because the zero vector lies in the span of any set; here $0\begin{bmatrix}1\\1\\0\end{bmatrix} + 0\begin{bmatrix}1\\1\\1\end{bmatrix} + 0\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$.

• $\vec{v}_3 = \begin{bmatrix}3\\7\\0\end{bmatrix}$ is **not** representable as a linear combination of the vectors $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ because

$$\alpha_1\begin{bmatrix}1\\1\\0\end{bmatrix} + \alpha_2\begin{bmatrix}1\\1\\1\end{bmatrix} + \alpha_3\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}\alpha_1+\alpha_2\\\alpha_1+\alpha_2\\\alpha_2+\alpha_3\end{bmatrix} = \begin{bmatrix}3\\7\\0\end{bmatrix},$$

leads to

$$\overset{\text{component comparison}}{\rightarrow} \left[\begin{array}{ccccc}\alpha_1 & + & \alpha_2 & = & 3\\ \alpha_1 & + & \alpha_2 & = & 7\\ \alpha_2 & + & \alpha_3 & = & 0\end{array}\right] \overset{\text{II}-\text{I}}{\rightarrow} \left[\begin{array}{ccccc}\alpha_1 & + & \alpha_2 & = & 3\\ & & 0 & = & 4\\ \alpha_2 & + & \alpha_3 & = & 0\end{array}\right].$$

The second equation in the last system is a contradiction. The system of linear equations has no solution. Therefore, the vector $\begin{bmatrix}3\\7\\0\end{bmatrix}$ is not in span$\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$. $\left(\text{Notation: } \begin{bmatrix}3\\7\\0\end{bmatrix} \notin \text{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}\right)$

Only $\vec{v}_1$ and $\vec{v}_2$ are in the span of the set $\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$.

Exercise 5 $M_5 := \text{span}\left\{\begin{bmatrix}1\\5\\2\end{bmatrix}, \begin{bmatrix}-1\\0\\3\end{bmatrix}\right\}$ is a subspace of $\mathbb{R}^3$ if all three subspace criteria are satisfied:

1) M_5 is nonempty,
2) M_5 is closed under addition (for all $\vec{u}, \vec{v} \in M_5$ the sum $\vec{u} + \vec{v}$ is an element of M_5),
3) M_5 is closed under multiplication with scalars (for all $\vec{v} \in M_5, \alpha \in \mathbb{R}$ the product $\alpha\vec{v}$ is an element of M_5).

1) According to the definition of span, $\begin{bmatrix}1\\5\\2\end{bmatrix} = 1 \cdot \begin{bmatrix}1\\5\\2\end{bmatrix} + 0 \cdot \begin{bmatrix}-1\\0\\3\end{bmatrix} \in M_5$. Thus, the set M_5 is nonempty.

2) Let $\vec{u}, \vec{v} \in M_5$. Then $\vec{u} = \boldsymbol{\alpha_1} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \boldsymbol{\alpha_2} \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}$ and $\vec{v} = \boldsymbol{\beta_1} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \boldsymbol{\beta_2} \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}$ for scalars $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}$ and thus

$$\vec{u} + \vec{v} = \boldsymbol{\alpha_1} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \boldsymbol{\alpha_2} \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} + \boldsymbol{\beta_1} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \boldsymbol{\beta_2} \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} = \boldsymbol{(\alpha_1 + \beta_1)} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \boldsymbol{(\alpha_2 + \beta_2)} \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} \in M_5.$$

3) Let $\vec{v} \in M_5$ and $\alpha \in \mathbb{R}$. Then $\vec{v} = \beta_1 \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \beta_2 \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}$ with $\beta_1, \beta_2 \in \mathbb{R}$. For all $\alpha \in \mathbb{R}$

$$\alpha\vec{v} = \alpha \left(\beta_1 \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \beta_2 \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} \right) = (\alpha\beta_1) \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + (\alpha\beta_2) \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}$$

is also an element of M_5. Thus, $\alpha\vec{v} \in M_5$.

Since all three conditions are satisfied, M_5 is a subspace of $\mathbb{R}^3$.

Exercise 6 a) We will show that $V := \{\vec{0}\}$ is a subspace of $\mathbb{F}^n$.

(1) $\vec{0} \in V \Rightarrow V$ is nonempty.
(2) For all $\vec{v}, \vec{w} \in V$ we have $\vec{v} = \vec{w} = \vec{0}$. The sum $\vec{0} + \vec{0} = \vec{0}$ is an element of V.
(3) For all $\alpha \in \mathbb{F}$, $\vec{v} \in V$ the product $\alpha\vec{v} = \alpha\vec{0} = \vec{0}$ is an element of V.
The subspace criteria are satisfied, so that V is a subspace of $\mathbb{F}^n$.

b) The zero vector is not in M_6 since x and $x + 7$ cannot be zero simultaneously. The last criterion in the definition of a subspace is thus not satisfied for any vector in M_6 multiplied with the scalar 0 so that M_6 is not a subspace of $\mathbb{R}^2$.

Remark: $\vec{0} \notin M_6$ does not mean that M_6 ist empty $\left(\text{for example } \begin{bmatrix} 0 \\ 7 \end{bmatrix} \in M_6 \right)$.

Exercise 7 $G_3 := \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} \right\}$ is a generating set for T since $G_1 \subseteq G_3 \subseteq T$ and G_1 already generates T.

Exercise 8 G_4 is a basis of T:

G_4 is contained in T since the third components of the vectors in G_4 are zero. The *two* vectors in G_4 are not multiples of each other. This means that they are linearly independent. (Think about why this is so. Caution! This argument is only valid if you have two vectors. Think about why this is so.) Furthermore, G_4 is a generating set because each vector in T can be represented as a linear combination of the vectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$: $\left(\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \right) = \left(-\frac{2}{3}x + \frac{5}{3}y\right) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \left(\frac{1}{3}x - \frac{1}{3}y\right) \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$.

Exercise 9 With respect to the basis $\mathcal{B}_1$, we have

$$\begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

With respect to the basis $\mathcal{B}_2$, we have

$$\begin{bmatrix}5\\2\\0\end{bmatrix} = 0\begin{bmatrix}1\\1\\0\end{bmatrix} + 1\begin{bmatrix}5\\2\\0\end{bmatrix}.$$

The unique coefficients with respect to a given basis are called *coordinates*.

Exercise 10 a) $\mathcal{B}_{\mathrm{Std}} := \left\{ \begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix} \right\}$ is the **standard basis** of $\mathbb{R}^3$ so that $\dim \mathbb{R}^3 = 3$.

b) $\dim \operatorname{span} \left\{ \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}2\\2\\0\end{bmatrix} \right\} = 1$ since $\begin{bmatrix}1\\1\\0\end{bmatrix}$ and $\begin{bmatrix}2\\2\\0\end{bmatrix}$ are linearly independent.

c) Since the number of basis elements is zero, $\dim\{\vec{0}\} = 0$.

3 Matrices

3.1 Definition and examples of matrices

Definition (matrix)
A **matrix** is an array of numbers ordered into m rows and n columns. The **size** of the matrix is defined as $m \times n$ (m-by-n).

$$\begin{array}{ccc} & & \begin{array}{c} n = \text{number of columns} \\ \swarrow \qquad\qquad \downarrow \qquad\qquad \searrow \end{array} \\ m \;=\; \begin{array}{c}\text{number}\\ \text{of}\\ \text{rows}\end{array} & \begin{array}{c}\nearrow \\ \\ \rightarrow \\ \\ \searrow\end{array} & \begin{bmatrix} a_{1,1} & \cdots & a_{1,j} & \cdots & a_{1,n} \\ \vdots & & \vdots & & \vdots \\ a_{i,1} & \cdots & a_{i,j} & \cdots & a_{i,n} \\ \vdots & & \vdots & & \vdots \\ a_{m,1} & \cdots & a_{m,j} & \cdots & a_{m,n} \end{bmatrix} =: A \end{array}$$

$$a_{i,j} = \text{entry in the } i\text{-th row and } j\text{-th column of the matrix } A$$

Notation: $A := (a_{i,j}) := (a_{i,j})_{\substack{i=1,\dots,m \\ j=1,\dots,n}}$

Caution! $[a_{i,j}]$ denotes the 1×1 matrix with sole entry $a_{i,j}$.

Notation: $\mathbb{F}^{m,n}$ denotes the set of matrices of size $m \times n$ with entries in $\mathbb{F}$.

Remark: Vectors can be thought of as matrices with one column.

Example (matrices)
Consider the matrices

$$A = (a_{i,j}) := \begin{bmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \in \mathbb{R}^{2,3}, \quad B = (b_{k,l}) := \begin{bmatrix} i & -6 \\ 2+3i & 4 \\ 0 & -i \end{bmatrix} \in \mathbb{C}^{3,2},$$

$C = (c_{m,1}) := \begin{bmatrix} 19 \\ -7 \\ 8 \end{bmatrix} \in \mathbb{R}^{3,1} = \mathbb{R}^3$. Determine $a_{1,2}$, $a_{2,1}$, $b_{1,2}$, $b_{2,1}$, $c_{3,1}$.

The first number in the index tells you which row and the second which column. So to find $a_{1,2}$, identify the entry in the 1st row and 2nd column of A:

$$a_{1,2} = 5, \quad a_{2,1} = 7, \quad b_{1,2} = -6, \quad b_{2,1} = 2 + 3i, \quad c_{3,1} = 8.$$

3.1.1 Special types of matrices

Definition (square matrix)
A **square** matrix has the same number of rows as columns (size $n \times n$ for some $n \in \mathbb{N}$).

Here are two examples of square matrices.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Definition (special square matrices)

$\bullet A = (a_{i,j}) \in \mathbb{F}^{n,n}$ is an **upper triangular matrix** if $a_{i,j} = 0$ for all $1 \le j < i \le n$.

$\bullet D = (d_{i,j}) \in \mathbb{F}^{n,n}$ is a **diagonal matrix** if $d_{i,j} = 0$ for all $1 \le i, j \le n,\ i \ne j$.

$\bullet I_n = (e_{i,j}) \in \mathbb{F}^{n,n}$ is the $n \times n$ **identity matrix** if $e_{i,j} = 0$ for all $1 \le i, j \le n,\ i \ne j$ and $e_{i,i} = 1$ for all $1 \le i \le n$.

Examples

upper triangular matrices $\begin{bmatrix} 3 & 2 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} a_{1,1} & & * \\ & \ddots & \\ 0 & & a_{n,n} \end{bmatrix}$ ($*$ arbitrary)

diagonal matrices $\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} d_{1,1} & & 0 \\ & \ddots & \\ 0 & & d_{n,n} \end{bmatrix} =: \operatorname{diag}\{d_{1,1}, \ldots, d_{n,n}\}$

identity matrices $I_2 := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, I_n := \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$

Remarks:

• The **diagonal** of a square matrix runs from the upper left-hand corner to the bottom right-hand corner of the matrix. The components of the diagonal are the entries of the form $a_{i,i}$.

• In an uppper triangular matrix, all of the entries below the diagonal must be zeros. The entries on the diagonal and above the diagonal are arbitrary so can also be zero.

• In a diagonal matrix, all of the entries above and below the diagonal must be zeros. The entries on the diagonal are arbitrary.

• An identity matrix is a diagonal matrix with only ones on the diagonal. (Note that the i-th column of the identity matrix I_n is just the i-th standard basis vector $\vec{e}_i$ of $\mathbb{F}^n$.

3.1.2 Transpose and conjugate transpose matrices, other matrix types

Definition (transpose matrix, symmetric, antisymmetric)

Let $A := (a_{i,j}) \in \mathbb{F}^{m,n}$.

The **transpose matrix** of A is given by $A^T := (b_{i,j})$, whereby $b_{i,j} := a_{j,i}$, $1 \le i, j \le n$; i.e. $A^T \in \mathbb{F}^{n,m}$.

The matrix A is **symmetric** if $A^T = A$.

The matrix A is **antisymmetric** or **skew-symmetric** if $A^T = -A$.

To transpose A, write the i-th row of A as the i-th column of A^T for each possible i.

Example (transpose; symmetric; antisymmetric)

$A := \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{6} \end{bmatrix} \in \mathbb{R}^{2,3} \quad A^T = \begin{bmatrix} \mathbf{1} & \mathbf{4} \\ \mathbf{2} & \mathbf{5} \\ \mathbf{3} & \mathbf{6} \end{bmatrix} \in \mathbb{R}^{3,2}$

$B := \begin{bmatrix} \mathbf{1} & \mathbf{-3} \\ \mathbf{-3} & \mathbf{1} \end{bmatrix}, \quad B^T = \begin{bmatrix} \mathbf{1} & \mathbf{-3} \\ \mathbf{-3} & \mathbf{1} \end{bmatrix}$ The matrix B is **symmetric** because $B = B^T$.

$C := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad C^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ The matrix C is **antisymmetric** since $C = -C^T$.

Exercise 1 (learning check)

Given the matrices $C := \begin{bmatrix} 1+i & 1-2i & 3 \\ 4i & 5 & 6-i \end{bmatrix}$ and $D := \begin{bmatrix} 2 & 3 \\ 3 & 0 \end{bmatrix}$, determine C^T and D^T. Which of the matrices is symmetric?

Exercise 2

Determine whether each of the given statements is true or false. Justify your answer.
a) If the matrix $A \in \mathbb{F}^{m,n}$ is symmetric, then A is square.
b) If the matrix $B \in \mathbb{F}^{n,n}$ is square, then B is symmetric.
c) If the matrix $D \in \mathbb{F}^{n,n}$ is a diagonal matrix, then D is also symmetric.

Recall that a complex number has the form $z := a + bi$, whereby $i^2 = -1$. Its **conjugate** is given by

$$\overline{z} := \overline{a + bi} = a - bi.$$

Definition (conjugate transpose matrix, self-adjoint matrix, Hermitian matrix)
Let $A := (a_{j,k}) \in \mathbb{F}^{n,n}$. Its **conjugate transpose matrix** (Notation: A^*) is given by $A^* := (b_{j,k})$, wobei $b_{\boldsymbol{j},\boldsymbol{k}} := \overline{a_{\boldsymbol{k},\boldsymbol{j}}}$, $1 \leq j, k \leq n$.

The matrix A is called **self-adjoint** or **Hermitian** if $A = A^*$.

Remarks:

• The conjugate transpose matrix of A, namely A^*, is found by transposing the matrix A and then conjugating each element. This is the same as first conjugating all the elements and then transposing.

• If A is a real matrix, the conjugate transpose matrix A^* is the same as the transpose matrix A^T. (Why?)

Examples (conjugate transpose)

$C := \begin{bmatrix} \mathbf{1+i} & \mathbf{1-2i} & \mathbf{3} \\ \mathbf{4i} & \mathbf{5} & \mathbf{6-i} \end{bmatrix}$, $C^T = \begin{bmatrix} \mathbf{1+i} & \mathbf{4i} \\ \mathbf{1-2i} & \mathbf{5} \\ \mathbf{3} & \mathbf{6-i} \end{bmatrix}$, $C^* = \begin{bmatrix} \mathbf{1-i} & \mathbf{-4i} \\ \mathbf{1+2i} & \mathbf{5} \\ \mathbf{3} & \mathbf{6+i} \end{bmatrix}$

$F := \begin{bmatrix} \mathbf{1} & \boldsymbol{i} \\ \boldsymbol{-i} & \mathbf{0} \end{bmatrix}$, $F^T = \begin{bmatrix} \mathbf{1} & \boldsymbol{-i} \\ \boldsymbol{i} & \mathbf{0} \end{bmatrix}$, $F^* = \begin{bmatrix} \mathbf{1} & \boldsymbol{i} \\ \boldsymbol{-i} & \mathbf{0} \end{bmatrix}$ F is **self-adjoint** since $F = F^*$ holds.

3.2 Matrices as vector spaces

Just like $\mathbb{F}^n$, we want to consider the set $\mathbb{F}^{m,n}$ as a vector space even though we no longer can depict the vectors, which are now matrices, as arrows. Recall that we can think of a vector in $\mathbb{F}^n$ as a matrix with one column. This helps us to determine how to define the operations of addition and multiplication with a scalar to turn the set $\mathbb{F}^{m,n}$ into a vector space.

Definition (vector space operations for matrices)

• The sum of two matrices $A := (a_{i,j}), B := (b_{i,j}) \in \mathbb{F}^{m,n}$ of the same size is defined as

$$A + B := (a_{i,j} + b_{i,j}).$$

• Let $A \in \mathbb{F}^{m,n}$ and $\alpha \in \mathbb{F}$. Scalar multiplication is defined as

$$\alpha A := (\alpha a_{i,j}).$$

Just like the case with vectors in $\mathbb{F}^n$, adding matrices is done componentwise. Scalar multiplication requires each entry in the matrix to be multiplied by the scalar, just like we did with vectors in $\mathbb{F}^n$. (Compare the definitions with the ones in Chapter 2.)

Example (vector space operations for matrices)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} + \begin{bmatrix} -1 & 2 & -3 \\ 4 & -5 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 0 \\ 8 & 0 & 12 \end{bmatrix}$$

$$5 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 \cdot 1 & 5 \cdot 2 \\ 5 \cdot 3 & 5 \cdot 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix}$$

$$i \begin{bmatrix} 1+i & 2 & 3+i \\ 1-i & 3 & 3-2i \end{bmatrix} = \begin{bmatrix} i+i^2 & 2i & 3i+i^2 \\ i-i^2 & 3i & 3i-2i^2 \end{bmatrix} \overset{(i^2=-1)}{=} \begin{bmatrix} -1+i & 2i & -1+3i \\ 1+i & 3i & 2+3i \end{bmatrix}$$

3.3 Matrix multiplication

Not only can we multiply a matrix with a scalar, we can multiply a matrix with another matrix under certain conditions. The definition may seem strange at first, but it will become clear in Chapter 6 why the definition makes sense: matrix multiplication mimics the composition of so-called linear mappings, which will be defined in Section 3.5.

Definition (matrix multiplication)
Let $A := (a_{i,j}) \in \mathbb{F}^{m,n}$, $B := (b_{j,k}) \in \mathbb{F}^{n,p}$. The product of the matrices A and B is given by

$AB = (c_{i,k}) \in \mathbb{F}^{m,p}$, whereby $c_{i,k} := \sum_{j=1}^{n} a_{i,j} b_{j,k}$.

For example, $c_{1,2} = a_{1,1}b_{1,2} + a_{1,2}b_{2,2} + \cdots + a_{1,n}b_{n,2}$.

Remarks:

• We say that the rows of A get multiplied with the columns of B. This involves summing up the individual multiplications.
• The product AB is only defined when the number of columns in A equals the number of rows in B:
A matrix of size $m \times n$ times a matrix of size $n \times p$ yields a matrix of size $m \times p$.

• To get the i,k-th entry of the product, multiply the i-th row with the k-th column. (Compare with the example below.)

Example (matrix multiplication)
Given the matrices

$$A := \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \text{ and } B := \begin{bmatrix} 12 & 11 \\ 10 & 9 \\ 8 & 7 \end{bmatrix},$$

determine the products AB and BA.

$$AB = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 12 & 11 \\ 10 & 9 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 \cdot 12 + 3 \cdot 10 + 5 \cdot 8 & 1 \cdot 11 + 3 \cdot 9 + 5 \cdot 7 \\ 2 \cdot 12 + 4 \cdot 10 + 6 \cdot 8 & 2 \cdot 11 + 4 \cdot 9 + 6 \cdot 7 \end{bmatrix}$$

$$= \begin{bmatrix} 12+30+40 & 11+27+35 \\ 24+40+48 & 22+36+42 \end{bmatrix} = \begin{bmatrix} 82 & 73 \\ 112 & 100 \end{bmatrix}$$

$$BA = \begin{bmatrix} 12 & 11 \\ 10 & 9 \\ 8 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} = \begin{bmatrix} 12\cdot 1 + 11\cdot 2 & 12\cdot 3 + 11\cdot 4 & 12\cdot 5 + 11\cdot 6 \\ 10\cdot 1 + \ 9\cdot 2 & 10\cdot 3 + \ 9\cdot 4 & 10\cdot 5 + \ 9\cdot 6 \\ 8\cdot 1 + \ 7\cdot 2 & 8\cdot 3 + \ 7\cdot 4 & 8\cdot 5 + \ 7\cdot 6 \end{bmatrix} = \begin{bmatrix} 34 & 80 & 126 \\ 28 & 66 & 104 \\ 22 & 52 & 82 \end{bmatrix}$$

The example shows that, unlike the case with real numbers, matrix multiplication is non-commutative, i.e. AB does not have to equal BA. Here, the two products are not even the same size!

Another difference between real numbers and matrices is the following. In the reals, $a \cdot b = 0 \Rightarrow a = 0$ or $b = 0$. The product of two matrices can yield the zero matrix (matrix with all entries being 0) even though both matrices are not the zero matrix. This is illustrated in the exercise below.

Exercise 3 (learning check)

Using the matrices

$$A := \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, \ B := \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \ C := \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 4 & -2 \end{bmatrix},$$

calculate the products AB, BA, AC, CA if possible.

Exercise 4

Show that for any matrix $D \in \mathbb{F}^{3,3}$, multiplication with I_3 acts as the identity (i.e. $DI_3 = I_3D = D$).

Multiplying a square matrix A of size $n \times n$ with I_n is like multiplying a real number a with the identity 1 ($a \cdot 1 = 1 \cdot a = a$). We get $AI_n = I_nA = A$. This is why I_n is called the $(n \times n)$ identity matrix.

If $A \in \mathbb{F}^{m,n}$ is not square, then I_n cannot be multiplied to both sides of A. Concretely, let us consider a matrix $A \in \mathbb{R}^{2,3}$. $AI_3 = A$ is defined but I_3A is not. (Why not?) We can multiply A on the left with I_2 to get $I_2A = A$. Since I_2 and I_3 are different matrices, we cannot define a unique identity matrix for A.

These differences make it more difficult to know if matrices are invertible. In the real numbers, all numbers a not equal to 0 are invertible (i.e. we can take the reciprocal) and $a^{-1} = \frac{1}{a}$. We know that in this case $a^{-1} \cdot a = a \cdot a^{-1} = 1$. Since the identity is not defined for non-square matrices, we will restrict ourselves to the question of inverting square matrices. (The notion can be extended to left inverses and right inverses, but that goes beyond the scope of this course.)

Definition / Lemma (inverse matrix)

(Definition) Let A, B be square matrices in $\mathbb{F}^{n,n}$.
B is called the **inverse matrix** of A or simply the **inverse** of A if $AB = I_n$. (notation: $A^{-1} := B$)

(Lemma) If $A \in \mathbb{F}^{n,n}$ is **invertible** with inverse A^{-1}, then $A^{-1}A = I_n$, i.e. A is the inverse of A^{-1} so that $(A^{-1})^{-1} = A$.

Remark: Not all nonzero square matrices have an inverse matrix.

Example (inverse Matrix)

Show that the inverse matrix of $A := \begin{bmatrix} 3 & 2 \\ 8 & 5 \end{bmatrix}$ is $A^{-1} = \begin{bmatrix} -5 & 2 \\ 8 & -3 \end{bmatrix}$.

We just have to perform a matrix multiplication to verify that the equation $AA^{-1} = I_2$ is satisfied:

$$AA^{-1} = \begin{bmatrix} 3 & 2 \\ 8 & 5 \end{bmatrix} \begin{bmatrix} -5 & 2 \\ 8 & -3 \end{bmatrix} = \begin{bmatrix} 3(-5)+2\cdot 8 & 3\cdot 2+2(-3) \\ 8(-5)+5\cdot 8 & 8\cdot 2+5(-3) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

As an exercise, you can check that $A^{-1}A = I_2$ is also satisfied. How to find the inverse (if it exists) will be explained in Chapter 4.

The following matrices are especially easy to invert. How to construct such matrices is the topic of Chapter 8.

Definition (orthogonal matrix; unitary matrix)

Let $Q \in \mathbb{R}^{n,n}$. Q is called an **orthogonal matrix** if $$QQ^T = I_n = Q^TQ,$$ i.e. $Q^{-1} = Q^T$.	Let $U \in \mathbb{C}^{n,n}$. U is called a **unitary matrix** if $$UU^* = I_n = U^*U,$$ i.e. $U^{-1} = U^*$.

Exercise 5 (learning check)

Check that the matrices $Q_1 := \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$, $Q_2 := \begin{bmatrix} \frac{1}{\sqrt{5}} & 0 & -\frac{2}{\sqrt{5}} \\ 0 & 1 & 0 \\ \frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \end{bmatrix}$ are orthogonal.

Example (unitary matrix)

Show that $U = \begin{bmatrix} \mathbf{0} & \boldsymbol{i} \\ -\boldsymbol{i} & \mathbf{0} \end{bmatrix}$ is a unitary matrix.

First we compute the conjugate transpose matrix.

$$U = \begin{bmatrix} \mathbf{0} & \boldsymbol{i} \\ -\boldsymbol{i} & \mathbf{0} \end{bmatrix},\ U^T = \begin{bmatrix} \mathbf{0} & -\boldsymbol{i} \\ \boldsymbol{i} & \mathbf{0} \end{bmatrix},\ U^* = \overline{U^T} = \begin{bmatrix} \mathbf{0} & \boldsymbol{i} \\ -\boldsymbol{i} & \mathbf{0} \end{bmatrix}.$$

U satisfies the definition of a unitary matrix:

$$UU^* = \begin{bmatrix} \mathbf{0} & \boldsymbol{i} \\ -\boldsymbol{i} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{0} & \boldsymbol{i} \\ -\boldsymbol{i} & \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0}\cdot\mathbf{0} + \boldsymbol{i}\cdot(-\boldsymbol{i}) & \mathbf{0}\cdot\boldsymbol{i} + \boldsymbol{i}\cdot\mathbf{0} \\ -\boldsymbol{i}\cdot\mathbf{0} + \mathbf{0}\cdot(-\boldsymbol{i}) & -\boldsymbol{i}\cdot\boldsymbol{i} + \mathbf{0}\cdot\mathbf{0} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus, U is unitary.

3.4 Linear mappings and matrix mappings

Mappings between vector spaces are like functions. If you put in a value, you get a value as an output. This time, though, the input is not a real number but a vector. The **domain** of a mapping is the set of vectors that can be put into the mapping (inputs of the mapping) and is sometimes called the **preimage space**. The **target space** is the space of possible vectors the mapping can assume (outputs of the mapping). Some authors use the term **range** or **codomain** for the target space. If a vector arises as the output of some input (the image of the input), then the output lies in the **target space** of the mapping. The set of all such outputs is the actual **image space** of the mapping. We often just say image for image space when no confusion arises.

Definition (matrix mapping)

Let $A \in \mathbb{F}^{m,n}$. The **matrix mapping** associated with the matrix A is given by

$$A : \mathbb{F}^n \to \mathbb{F}^m; \vec{v} \mapsto A\vec{v}.$$

Remarks:

• To read the mapping aloud say, "A is the mapping from $\mathbb{F}^n$ to $\mathbb{F}^m$ that maps $\vec{v}$ (on)to $A\vec{v}$."

• In Linear Algebra, it is usually expected that the domain and a suitable target space are explicitly stated.

• Why is the matrix mapping $A \in \mathbb{F}^{m,n}$ a mapping from $\mathbb{F}^n$ to $\mathbb{F}^m$? According to the rule, $A(\vec{v}) = A\vec{v}$, i.e. the matrix multiplication has to be defined. Since the size of A is $m \times n$, the size of $\vec{v}$ has to be $n \times 1$. The product thus has the size $m \times 1$. Thus A takes a vector in $\mathbb{F}^n$ (domain) and outputs a vector in $\mathbb{F}^m$ (target space).

Example (matrix mapping)

Let A be the matrix mapping defined by the matrix $A := \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{R}^{2,3}$. Determine the images of the standard basis vectors $\vec{e}_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e}_2 := \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{e}_3 := \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, the image of an arbitrary vector $\vec{v} := \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ as well as $A(x\vec{e}_1)$, $x \in \mathbb{R}$.

The associated matrix mapping is given by $A : \mathbb{R}^3 \to \mathbb{R}^2; \vec{v} \mapsto A\vec{v}$.
The images of the standard basis vectors in $\mathbb{R}^3$ are

$$A\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} \mathbf{1} & 2 & 3 \\ \mathbf{4} & 5 & 6 \end{bmatrix} \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{1} \\ \mathbf{4} \end{bmatrix}, \quad A\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & \mathbf{2} & 3 \\ 4 & \mathbf{5} & 6 \end{bmatrix} \begin{bmatrix} 0 \\ \mathbf{1} \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{5} \end{bmatrix}, \quad A\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & \mathbf{3} \\ 4 & 5 & \mathbf{6} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \mathbf{1} \end{bmatrix} = \begin{bmatrix} \mathbf{3} \\ \mathbf{6} \end{bmatrix}.$$

(In general, $A\vec{e}_i$ is the i-th column of A.)

The image of an arbitrary vector is

$$A(\vec{v}) = A\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1x + 2y + 3z \\ 4x + 5y + 6z \end{bmatrix} = x \begin{bmatrix} 1 \\ 4 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} + z \begin{bmatrix} 3 \\ 6 \end{bmatrix}. \tag{3}$$

(The image of an arbitrary vector is a linear combination of the columns of A.)

Notice that $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Writing the equation (3) somewhat differently leads to

$$\begin{aligned} A\left(x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) &= x \begin{bmatrix} 1 \\ 4 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} + z \begin{bmatrix} 3 \\ 6 \end{bmatrix} \\ &= xA\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) + yA\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) + zA\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right). \end{aligned} \tag{4}$$

Consider $\{x\vec{e}_1 \mid x \in \mathbb{R}\} = \operatorname{span}\{\vec{e}_1\}$, the line through the origin and $\vec{e}_1$. The image $A(x\vec{e}_1) = x \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ (see equation (3)) is the line through the origin and $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$. You can check that A sends any arbitrary line running through the origin to some line through the origin.

Starting with the equation (4), we characterize general mappings that map lines onto lines, just like the matrix mapping A, at least in the real case. In this context, we consider the origin as a line of dimension 0. This type of mapping is called a linear mapping. The formulation is given for vector spaces over $\mathbb{F}^n$.

Definition (linear mapping)

Let $L : \mathbb{F}^n \to \mathbb{F}^m$ be a map. L is called **linear** if for all $\vec{u}, \vec{v} \in \mathbb{F}^n$, $\alpha \in \mathbb{F}$

$$(1)\ L(\vec{u} + \vec{v}) = L(\vec{u}) + L(\vec{v}),$$
$$(2)\ L(\alpha\vec{v}) = \alpha L(\vec{v}).$$

Remark:
A linear mapping $L : V \to W$ maps the zero vector onto the zero vector since for $\vec{v} \in V$ it follows from (2) that $L(\vec{0}) = L(0 \cdot \vec{v}) = 0 \cdot L(\vec{v}) = \vec{0}$.

Example (linear mapping)

Let L be the mapping from $\mathbb{R}^3$ to $\mathbb{R}^2$ that takes a vector $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3$ to $\begin{bmatrix} v_1 + v_2 \\ v_2 + v_3 \end{bmatrix} \in \mathbb{R}^2$. Show that L is a linear mapping.

Before we do the actual problem, let us first consider some ways to represent the linear mapping in the problem.

$$L : \mathbb{R}^3 \to \mathbb{R}^2; \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \mapsto \begin{bmatrix} v_1 + v_2 \\ v_2 + v_3 \end{bmatrix} \qquad \begin{array}{cccc} L: & \mathbb{R}^3 & \to & \mathbb{R}^2 \\ & \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} & \mapsto & \begin{bmatrix} v_1 + v_2 \\ v_2 + v_3 \end{bmatrix} \end{array}$$

These two ways can be read aloud as, "L is the mapping from $\mathbb{R}^3$ to $\mathbb{R}^2$ which maps the vector v_1, v_2, v_3 to the vector $v_1 + v_2, v_2 + v_3$."

$$L : \mathbb{R}^3 \to \mathbb{R}^2; L\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) = \begin{bmatrix} v_1 + v_2 \\ v_2 + v_3 \end{bmatrix}$$

This representation is read as, "L is the mapping from $\mathbb{R}^3$ to $\mathbb{R}^2$ with L of v_1, v_2, v_3 equal to the vector $v_1 + v_2, v_2 + v_3$."

Coming back to the example, we want to show that $L : \mathbb{R}^3 \to \mathbb{R}^2; \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \mapsto \begin{bmatrix} v_1 + v_2 \\ v_2 + v_3 \end{bmatrix}$ is a linear mapping.

To do this, we have to show that the two conditions in the definition of a linear mapping are satisfied.

$$(1)\ L(\vec{u} + \vec{v}) = L(\vec{u}) + L(\vec{v}) \qquad (2)\ L(\alpha\vec{v}) = \alpha L(\vec{v}) \qquad \text{for all } \vec{u}, \vec{v} \in \mathbb{R}^3,\ \alpha \in \mathbb{R}.$$

For (1), choose $\vec{u} := \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$, $\vec{v} := \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3$. Evaluate both sides of the equation by using the mapping rule.

$$\begin{aligned}
L(\vec{u}+\vec{v}) &= L\left(\begin{bmatrix}u_1\\u_2\\u_3\end{bmatrix}+\begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix}\right) \\
&= L\left(\begin{bmatrix}u_1+v_1\\u_2+v_2\\u_3+v_3\end{bmatrix}\right) && \text{vector addition in } \mathbb{R}^3 \\
&= \begin{bmatrix}(u_1+v_1)+(u_2+v_2)\\(u_2+v_2)+(u_3+v_3)\end{bmatrix} && \text{mapping rule} \\
&= \begin{bmatrix}(u_1+u_2)+(v_1+v_2)\\(u_2+u_3)+(v_2+v_3)\end{bmatrix} && \text{group together } u_i \text{ and } v_i \\
& && \text{(addition in } \mathbb{R} \text{ is commutative)} \\
L(\vec{u})+L(\vec{v}) &= L\left(\begin{bmatrix}u_1\\u_2\\u_3\end{bmatrix}\right)+L\left(\begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix}\right) \\
&= \begin{bmatrix}u_1+u_2\\u_2+u_3\end{bmatrix}+\begin{bmatrix}v_1+v_2\\v_2+v_3\end{bmatrix} && \text{mapping rule} \\
&= \begin{bmatrix}(u_1+u_2)+(v_1+v_2)\\(u_2+u_3)+(v_2+v_3)\end{bmatrix} && \text{vector addition in } \mathbb{R}^2 \\
&= L(\vec{u}+\vec{v})
\end{aligned}$$

For (2), let $\vec{v} := \begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix} \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$. Evaluate both sides of the equation.

$$\begin{aligned}
L(\alpha\vec{v}) &= L\left(\alpha\begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix}\right) = L\left(\begin{bmatrix}\alpha v_1\\\alpha v_2\\\alpha v_3\end{bmatrix}\right) && \text{multiplication with scalars in } \mathbb{R}^3 \\
&= \begin{bmatrix}\alpha v_1+\alpha v_2\\\alpha v_2+\alpha v_3\end{bmatrix} && \text{mapping rule} \\
&= \begin{bmatrix}\alpha(v_1+v_2)\\\alpha(v_2+v_3)\end{bmatrix} && \text{factorization} \\
\alpha L(\vec{v}) &= \alpha L\left(\begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix}\right) = \alpha\begin{bmatrix}v_1+v_2\\v_2+v_3\end{bmatrix} && \text{mapping rule} \\
&= \begin{bmatrix}\alpha(v_1+v_2)\\\alpha(v_2+v_3)\end{bmatrix} && \text{scalar multiplication in } \mathbb{R}^2 \\
&= L(\alpha(\vec{v}))
\end{aligned}$$

Since the two conditions are satisfied, L is linear.

Remarks:

• The justification for each set does not need to be given in the exam, but you should write this down for yourself until you have gained a sufficient amount of proficiency with the technique.

• With a bit more practice, you can learn to verify the conditions by starting with the left-hand side and working your way directly to the right-hand side.

Example (nonlinear mapping)

Show that the mapping $N : \mathbb{R}^3 \to \mathbb{R}^2;\ \begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix} \mapsto \begin{bmatrix}v_1\cdot v_2\\v_2\cdot v_3\end{bmatrix}$ is not linear.

To show: one of the conditions in the definition of a linear mapping is not satisfied (best done with a counterexample).

Take $\vec{v} \in \mathbb{R}^3$ to be the vector $\begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\alpha = 2$ as the scalar, for example. Calculate $N(\alpha\vec{v})$ and $\alpha N(\vec{v})$. If N is linear, both results have to be the same.

$$N(\alpha\vec{v}) = N\left(2\begin{bmatrix}1\\2\\3\end{bmatrix}\right) = N\left(\begin{bmatrix}2\\4\\6\end{bmatrix}\right) = \begin{bmatrix}2\cdot 4\\4\cdot 6\end{bmatrix} = \begin{bmatrix}8\\24\end{bmatrix}$$

$$\alpha N(\vec{v}) = 2N\left(\begin{bmatrix}1\\2\\3\end{bmatrix}\right) = 2\begin{bmatrix}1\cdot 2\\2\cdot 3\end{bmatrix} = 2\begin{bmatrix}2\\6\end{bmatrix} = \begin{bmatrix}4\\12\end{bmatrix}$$

Since $N(\alpha\vec{v}) \neq \alpha N(\vec{v})$ for $\alpha = 2$ and $\vec{v} = \begin{bmatrix}1\\2\\3\end{bmatrix}$, the mapping N is nicht linear.

Exercise 6 (learning check)

Check if the following mappings are linear:

$$L_1 : \mathbb{R}^3 \to \mathbb{R}^2; \vec{x} \mapsto \vec{0} \qquad L_2 : \mathbb{R}^3 \to \mathbb{R}^2; \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} \mapsto \begin{bmatrix}1\\1\end{bmatrix}$$

Definition (kernel and image of a linear mapping)

Let $L : \mathbb{F}^n \to \mathbb{F}^m$ be a linear mapping. The kernel and image of L are defined by

$$\mathbf{ker}(L) := \{\vec{v} \in \mathbb{F}^n \mid L(\vec{v}) = \vec{0}\} \subseteq \mathbb{F}^n, \qquad \mathbf{im}(L) := \{L(\vec{v}) \mid \vec{v} \in \mathbb{F}^n\} \subseteq \mathbb{F}^m.$$

Remarks:

• The kernel of a linear mapping $L : V \to W$ is the set of all vectors in the domain (V) that get mapped under L to the zero vector.

• The image of the linear mapping $L : V \to W$ is the set of all vectors in the target space (W) that arise under L from some vector in the domain (V). The image of L does not need to be the entire target space. (For example, the image of the linear mapping $L : \mathbb{R}^2 \to \mathbb{R}^3; \begin{bmatrix}x\\y\end{bmatrix} \mapsto \begin{bmatrix}x\\y\\0\end{bmatrix}$ is geometrically speaking the xy-plane in $\mathbb{R}^3$.)

Example (kernel and image)

Determine the kernel and the image of the linear mapping $L : \mathbb{R}^3 \to \mathbb{R}^2; \begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix} \mapsto \begin{bmatrix}v_1+v_2\\v_2+v_3\end{bmatrix}$.

According to the definition of the kernel

$$\ker(L) = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; L\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}.$$

The mapping rule for L is

$$L\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) = \begin{bmatrix} v_1 + v_2 \\ v_2 + v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Comparing the components leads to

$$\begin{array}{ccccccc} v_1 & + & v_2 & = & 0 & \qquad \boldsymbol{v_1} & \boldsymbol{= -v_2} \\ v_2 & + & v_3 & = & 0 & \qquad \boldsymbol{v_3} & \boldsymbol{= -v_2}. \end{array}$$

An element $\begin{bmatrix} \boldsymbol{v_1} \\ \boldsymbol{v_2} \\ \boldsymbol{v_3} \end{bmatrix} \in \ker(L)$ has the form $\begin{bmatrix} \boldsymbol{-v_2} \\ \boldsymbol{v_2} \\ \boldsymbol{-v_2} \end{bmatrix}$ or (with the free parameter $\alpha \in \mathbb{R}$) $\begin{bmatrix} -\alpha \\ \alpha \\ -\alpha \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ so that

$$\ker(L) = \left\{ \alpha \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \;\middle|\; \alpha \in \mathbb{R} \right\} = \operatorname{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

Since $L\left(\begin{bmatrix} a \\ 0 \\ b \end{bmatrix}\right) = \begin{bmatrix} a \\ b \end{bmatrix}$ for arbitrary $a, b \in \mathbb{R}$, $\operatorname{im}(L)$ spans the entire target space $\mathbb{R}^2$.

$$\operatorname{im}(L) = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \;\middle|\; a, b \in \mathbb{R} \right\} = \mathbb{R}^2.$$

Exercise 7 (learning check)
Determine the kernel and image of the linear mapping $L_1 : \mathbb{R}^3 \to \mathbb{R}^2; \vec{x} \mapsto \vec{0}$.

Exercise 8
Let $L : \mathbb{R}^n \to \mathbb{R}^m$ be a linear mapping. Prove the following statements.

a) $\ker(L)$ is a subspace of $\mathbb{R}^n$.

b) $\operatorname{im}(L)$ is a subspace of $\mathbb{R}^m$.

Theorem (linear mapping; matrix mapping)

(1) Matrix mapping are always linear.

(2) Every linear mapping $\boldsymbol{L} : \mathbb{F}^n \to \mathbb{F}^m$ can be represented as a matrix A mapping (called the associated matrix). The i-th column of A is the image of the i-th standard basis vector $\vec{e}_i$ of $\mathbb{F}^n$.

(3) If $\boldsymbol{L} : \mathbb{F}^n \to \mathbb{F}^m$ and $\boldsymbol{M} : \mathbb{F}^p \to \mathbb{F}^n$ are linear mappings with **associated matrices** A resp. B, then the **composition**

$$\boldsymbol{L} \circ \boldsymbol{M} : \mathbb{F}^p \to \mathbb{F}^m$$

is a linear mapping with associated matrix AB.

Remarks:

• (With respect to (2)) If A is the associated matrix of the linear mapping L, then the i-th column of A is just $A\vec{e}_i$. That means that the images of the standard basis vectors in $\mathbb{F}^n$ under the matrix mapping $A : \mathbb{F}^n \to \mathbb{F}^m; \vec{x} \mapsto A\vec{x}$ and the linear mapping L are the same. Due to the linearity, we can show that $L(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{F}^n$. Furthermore, $\operatorname{im}(A) = \operatorname{span}\{\text{columns of } A\}$.

• (With respect to (3)) When composing matrices, as is the convention with functions, we write from right to left. In the theorem, M is first applied to a vector in $\mathbb{F}^p$ to get a vector in $\mathbb{F}^n$. The result can be put into L and is mapped to an element in $\mathbb{F}^m$.

Example (associated matrix; composition of linear mappings)

Let

$$L_1 : \mathbb{R}^2 \to \mathbb{R}^3; \begin{bmatrix} \boldsymbol{v_1} \\ \boldsymbol{v_2} \end{bmatrix} \mapsto \begin{bmatrix} 2\boldsymbol{v_1} - 3\boldsymbol{v_2} \\ 5\boldsymbol{v_2} - 6\boldsymbol{v_1} \\ 4\boldsymbol{v_1} + 7\boldsymbol{v_2} \end{bmatrix} \qquad L_2 : \mathbb{R}^3 \to \mathbb{R}^2; \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \mapsto \begin{bmatrix} -v_1 \\ v_2 + v_3 \end{bmatrix}$$

be linear mappings.

a) Determine the associated matrices A_1 and A_2 for L_1 and L_2.

b) Compute the composition $L_3 := L_1 \circ L_2$.

c) Find the associated matrix for L_3 as well as the product $A_1 A_2$.

a) We compute the images of the standard basis vectors in the respective domains.

$$A_1 := \left[L_1\left(\begin{bmatrix} \mathbf{1} \\ \mathbf{0} \end{bmatrix}\right) \quad L_1\left(\begin{bmatrix} \mathbf{0} \\ \mathbf{1} \end{bmatrix}\right) \right] = \begin{bmatrix} 2 & -3 \\ -6 & 5 \\ 4 & 7 \end{bmatrix}$$

$$A_2 := \left[L_2\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) \quad L_2\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) \quad L_2\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) \right] = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

b) $L_3 = L_1 \circ L_2$ is a mapping from $\mathbb{R}^3$ to $\mathbb{R}^3$. In the upper portion of the diagram it is shown how the maps transform the vector spaces. In the lower portion we track the effects on the first standard basis vector of $\mathbb{R}^3$.

In order to find the image of an arbitrary vector in $\mathbb{R}^3$, consider the composition.

$$\begin{aligned} L_3\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) = L_1\left(L_2\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right)\right) &= L_1\left(\begin{bmatrix} \boldsymbol{-v_1} \\ \boldsymbol{v_2 + v_3} \end{bmatrix}\right) \\ &= \begin{bmatrix} 2(\boldsymbol{-v_1}) - 3(\boldsymbol{v_2 + v_3}) \\ 5(\boldsymbol{v_2 + v_3}) - 6(\boldsymbol{-v_1}) \\ 4(\boldsymbol{-v_1}) + 7(\boldsymbol{v_2 + v_3}) \end{bmatrix} \\ &= \begin{bmatrix} -2v_1 - 3v_2 - 3v_3 \\ 6v_1 + 5v_2 + 5v_3 \\ -4v_1 + 7v_2 + 7v_3 \end{bmatrix}. \end{aligned}$$

c) The columns of the associated matrix for L_3 are the images of the standard basis vectors.

$$[L_3(\vec{e}_1)\ L_3(\vec{e}_2)\ L_3(\vec{e}_3)] = \begin{bmatrix} -2 & -3 & -3 \\ 6 & 5 & 5 \\ -4 & 7 & 7 \end{bmatrix}$$

$$A_1 A_2 = \begin{bmatrix} 2 & -3 \\ -6 & 5 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -3 & -3 \\ 6 & 5 & 5 \\ -4 & 7 & 7 \end{bmatrix}.$$

$L_3 = L_1 \circ L_2 : \mathbb{R}^3 \to \mathbb{R}^3$ is thus the mapping with associated matrix $A_1 A_2$.

Exercise 9 (learning check)

Use the linear mappings L_1, L_2 from the previous example on page 31.

a) Determine the composition $L_4 := L_2 \circ L_1$.
b) Find the matrix associated to L_4 as well as the product $A_2 A_1$.
c) Let $L_5 : \mathbb{R}^2 \to \mathbb{R}^4$ be a linear mapping. Which of the following compositions are defined?

$$L_1 \circ L_5,\ L_5 \circ L_1,\ L_2 \circ L_5,\ L_5 \circ L_2$$

3.5 Suggested solutions to the exercises in Chapter 3

Exercise 1 $C^T := \begin{bmatrix} 1+i & 4i \\ 1-2i & 5 \\ 3 & 6-i \end{bmatrix} \neq C \Rightarrow C$ nicht symmetrisch

$D^T = \begin{bmatrix} 2 & 3 \\ 3 & 0 \end{bmatrix} = D \Rightarrow D$ symmetrisch

Exercise 2 a) The statement is correct. If $A \in \mathbb{F}^{m,n}$ mit $A = A^T$, then the number of rows must equal the number of columns. This is because when transposing, the rows become columns.
b) The statement is false. The statement says that when a matrix is square, then it must be symmetric. Naturally, we can find matrices that are simultaneously square and symmetiric. The question is if all square matrices are symmetric. As a counterexmaple, consider $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. B has size 2×2 so it is a square matrix. $B^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \neq B \Rightarrow B$ is however not symmetric. There exists a square matrix that is not symmetric. The statement is therefore false.
c) The statement is true. (In this situation, it is not sufficient to give an example. We have to argue that all diagonal matrices are symmetric.) Let $D := (d_{i,j}) \in \mathbb{F}^{n,n}$ be a diagonal matrix and $D^T := (c_{i,j})$ be its transpose with $c_{i,j} = d_{j,i}$. For $1 \leq i, j \leq n$ with $i \neq j$, we know that $d_{i,j} = 0 = d_{j,i} = c_{i,j}$. For $i = j$ we have $d_{i,i} = c_{i,i}$. It follows that $d_{i,j} = c_{i,j}$ for all $1 \leq i, j \leq n$. Since $D = D^T$ we can conclude that D is symmetric.

Exercise 3 $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, AC is not defined. The number of columns in A is two, which is different from the number of rows in C (three). $CA = \begin{bmatrix} 0 & 5 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

Exercise 4 Set $D := \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Then

$$DI_3 = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} a \cdot 1 + b \cdot 0 + c \cdot 0 & a \cdot 0 + b \cdot 1 + c \cdot 0 & a \cdot 0 + b \cdot 0 + c \cdot 1 \\ d \cdot 1 + e \cdot 0 + f \cdot 0 & d \cdot 0 + e \cdot 1 + f \cdot 0 & d \cdot 0 + e \cdot 0 + f \cdot 1 \\ g \cdot 1 + h \cdot 0 + i \cdot 0 & g \cdot 0 + h \cdot 1 + i \cdot 0 & g \cdot 0 + h \cdot 0 + i \cdot 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = D$$

A similar computation yields $I_3 D = D$. (Are you able to give a theoretical argument?)

Exercise 5

$$Q_1Q_1^T = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{3}{5}\cdot\frac{3}{5}+\frac{4}{5}\cdot(\frac{4}{5}) & \frac{3}{5}\cdot(-\frac{4}{5})+\frac{4}{5}\cdot\frac{3}{5} \\ -\frac{4}{5}\cdot\frac{3}{5}+\frac{3}{5}\cdot(\frac{4}{5}) & -\frac{4}{5}\cdot(-\frac{4}{5})+\frac{3}{5}\cdot\frac{3}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

From $Q_1Q_1^T = I_2$ it follows that $Q_1^TQ_1 = I_2$ (see the lemma on page 24).

$$Q_2Q_2^T = \begin{bmatrix} \frac{1}{\sqrt{5}} & 0 & -\frac{2}{\sqrt{5}} \\ 0 & 1 & 0 \\ \frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} \\ 0 & 1 & 0 \\ -\frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{1}{5}+\frac{4}{5} & 0 & \frac{2}{5}-\frac{2}{5} \\ 0 & 1 & 0 \\ \frac{2}{5}-\frac{2}{5} & 0 & \frac{4}{5}+\frac{1}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

Exercise 6 The linearity of the mappings L_1, L_2 need to be considered. It needs to be checked if for all $\vec{u}, \vec{v} \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$, $i = 1, 2$ the properties (1) and (2) hold with

$$\text{(1) } L_i(\vec{u}+\vec{v}) = L_i(\vec{u}) + L_i(\vec{v}),$$
$$\text{(2) } L_i(\alpha\vec{v}) = \alpha L_i(\vec{v}).$$

Claim: The mapping $L_1 : \mathbb{R}^3 \to \mathbb{R}^2; \vec{v} \mapsto \vec{0}$ is linear.
Proof: Check the conditions.
For (1). Let $\vec{u} := \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \vec{v} := \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3$.

$$\begin{aligned}
L_1(\vec{u}+\vec{v}) &= L_1\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) & \\
&= L_1\left(\begin{bmatrix} u_1+v_1 \\ u_2+v_2 \\ u_3+v_3 \end{bmatrix}\right) & \text{vector addition} \\
&= \begin{bmatrix} 0 \\ 0 \end{bmatrix} & \text{mapping rule} \\
L_1(\vec{u}) + L_1(\vec{v}) &= L_1\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) + L_1\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) & \\
&= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} & \text{mapping rule} \\
&= \begin{bmatrix} 0 \\ 0 \end{bmatrix} & \text{vector addition} \\
&= L_1(\vec{u}+\vec{v}) &
\end{aligned}$$

For (2). Let $\vec{v} := \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3$, $\alpha \in \mathbb{R}$.

$$\begin{aligned}
L_1(\alpha\vec{v}) &= L_1\left(\alpha\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) \\
&= L_1\left(\begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \alpha v_3 \end{bmatrix}\right) && \text{scalar multiplication} \\
&= \begin{bmatrix} 0 \\ 0 \end{bmatrix} && \text{mapping rule}
\end{aligned}$$

$$\begin{aligned}
\alpha L_1(\vec{v}) &= \alpha L_1\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) \\
&= \alpha\begin{bmatrix} 0 \\ 0 \end{bmatrix} && \text{mapping rule} \\
&= \begin{bmatrix} 0 \\ 0 \end{bmatrix} && \text{scalar multiplication} \\
&= L_1(\alpha\vec{v})
\end{aligned}$$

Since both conditions are satisfied, L_1 is linear.

Claim: $L_2 : \mathbb{R}^3 \to \mathbb{R}^2; \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is not a linear mapping.

Proof: Using $\vec{v} := \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\alpha = 2$, we will show that (2) does not hold.

$$L_2(\alpha\vec{v}) = L_2\left(2\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = L_2\left(\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\alpha L_2(\vec{v}) = 2L_2\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Since $L_2(\alpha\vec{v}) \neq \alpha L_2(\vec{v})$ for $\alpha = 2$ and $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, L_2 is not linear.

Remark: Sometimes it is difficult to find a counterexample.

Exercise 7 $\ker(L_1) = \mathbb{R}^3$ and $\operatorname{im}(L_1) = \left\{\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}$ since every vector in $\mathbb{R}^3$ gets mapped to the zero vector.

Exercise 8 **a)** Given is a linear mapping L from $\mathbb{R}^n$ to $\mathbb{R}^m$. To show: $\ker(L)$ is a subspace of $\mathbb{R}^n$.

$\ker(L) := \{\vec{v} \in \mathbb{R}^n \mid L(\vec{v}) = \vec{0}\} \subseteq \mathbb{R}^n$ is a subspace of $\mathbb{R}^n$ if all of the following conditions are satisfied.

1) $\ker(L) \neq \emptyset$

2) For all $\vec{v}_1, \vec{v}_2 \in \ker(L)$, the sum $\vec{v}_1 + \vec{v}_2$ is in $\ker(L)$,

3) For all $\vec{v} \in \ker(L), \alpha \in \mathbb{R}$, the product $\alpha\vec{v}$ is in $\ker(L)$.

For 1): Let $\vec{v} \in \mathbb{R}^n$. Then $L(\vec{0}) = L(0\vec{v}) \overset{\text{linearity}}{=} 0 \cdot L(\vec{v}) = \vec{0} \Rightarrow \vec{0} \in \ker(L) \Rightarrow \ker(L) \neq \emptyset$

Remark: The computation here is independent of the linear mapping. A linear mapping always maps the zero vector to the zero vector.

For 2): Let $\vec{v}_1, \vec{v}_2 \in \ker(L)$. It follows that $\boldsymbol{L(\vec{v}_1) = \vec{0}}, \boldsymbol{L(\vec{v}_2) = \vec{0}}$. To show: $\vec{v}_1 + \vec{v}_2 \in \ker(L)$, i.e. $L(\vec{v}_1 + \vec{v}_2) = \vec{0}$.

$$L(\vec{v}_1 + \vec{v}_2) \overset{\text{linearity}}{=} \boldsymbol{L(\vec{v}_1)} + \boldsymbol{L(\vec{v}_2)} = \vec{\boldsymbol{0}} + \vec{\boldsymbol{0}} = \vec{0}.$$

For 3): Let $\vec{v} \in \ker(L)$ (i.e. $\boldsymbol{L(\vec{v}) = \vec{0}}$) and $\alpha \in \mathbb{R}$. Then

$$L(\boldsymbol{\alpha}\vec{v}) \overset{\text{linearity}}{=} \boldsymbol{\alpha L(\vec{v})} = \boldsymbol{\alpha\vec{0}} = \vec{0}$$

Thus $\alpha\vec{v}$ is also an element in $\ker(L)$.

The three subspace criteria are satisfied so that $\ker(L)$ is a subspace of $\mathbb{R}^n$.

b) For the linear mapping L we wish to show that $\mathrm{im}(L) := \{L(\vec{v}) \mid \vec{v} \in \mathbb{R}^n\} \subset \mathbb{R}^m$ is a subspace of $\mathbb{R}^m$. As in part a), we will check the subspace criteria.

For 1): $L(\vec{0}) = \vec{0} \Rightarrow \vec{0} \in \mathrm{im}(L) \Rightarrow \mathrm{im}(L) \neq \emptyset$

For 2): Let $\vec{w}_1, \vec{w}_2 \in \mathrm{im}(L)$, i.e. there exist vectors $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$ with $L(\vec{v}_1) = \vec{w}_1, L(\vec{v}_2) = \vec{w}_2$.
To show: $\vec{w}_1 + \vec{w}_2 \in \mathrm{im}(L)$.

$$\vec{w}_1 + \vec{w}_2 = L(\vec{v}_1) + L(\vec{v}_2) \overset{\text{linearity}}{=} L(\vec{v}_1 + \vec{v}_2)$$

and, since $\mathbb{R}^n$ is a vector space, $\vec{v}_1 + \vec{v}_2$ is also in $\mathbb{R}^n$. Thus $\vec{w}_1 + \vec{w}_2$ lies in $\mathrm{im}(L)$.

For 3): Let $\vec{w} \in \mathrm{im}(L)$ (i.e. there exists a vector $\vec{v} \in \mathbb{R}^n$ with $L(\vec{v}) = \vec{w}$). Choose any $\alpha \in \mathbb{R}$. It follows that

$$\alpha\vec{w} = \alpha L(\vec{v}) \overset{\text{linearity}}{=} L(\alpha\vec{v})$$

with $\alpha\vec{v} \in \mathbb{R}^n$, so that $\alpha\vec{w} \in \mathrm{im}(L)$ holds.

All of the subspace criteria are satisfied so that $\mathrm{im}(L)$ is a subspace of $\mathbb{R}^m$.

Exercise 9 The linear mappings L_1 and L_2 are defined on page 31.

a) The composition is given by

$$L_4 = L_2 \circ L_1 : \mathbb{R}^2 \to \mathbb{R}^2; \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} -2x_1 + 3x_2 \\ -2x_1 + 12x_2 \end{bmatrix}.$$

b) The matrix associated to L_4 is $\begin{bmatrix} -2 & 3 \\ -2 & 12 \end{bmatrix}$.

The product of the matrices

$$C := A_2 A_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -6 & 5 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ -2 & 12 \end{bmatrix}$$

is the same as the matrix in part b).
c) The image of L_5 (a subspace of $\mathbb{R}^4$) is not contained in the domain of L_1 ($\mathbb{R}^2$) nor in the domain of L_2 ($\mathbb{R}^3$) so that $L_1 \circ L_5$ and $L_2 \circ L_5$ are not defined.

The image of L_1 (a subspace of $\mathbb{R}^3$) is not contained in the domain of L_5 ($\mathbb{R}^2$) so that $L_5 \circ L_1$ is not defined.

The image of L_2 (a subspace of $\mathbb{R}^2$) is contained in the domain of L_5 ($\mathbb{R}^2$). Thus $L_5 \circ L_2$ is defined.

4 Gaussian elimination, Gauss-Jordan elimination

A procedure to systematically solve **systems of linear equations** is Gaussian elimination. The more general form is Gauss-Jordan elimination, but for readability we just use the term Gauss elimination.

4.1 Systems of linear equations (SLE) and matrices

Definition (system of linear equations; coefficient matrix; augmented matrix; (in)homogenity)

A system of the form
$$\begin{array}{ccccccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \cdots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \cdots & + & a_{2,n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \cdots & + & a_{m,n}x_n & = & b_m \end{array}$$
is called a **system of linear equations** (abbreviation: SLE) with m equations in n *unknowns* (or *variables*) $x_1, \ldots, x_n$.

The **coefficient matrix** (abbreviation: CM) is defined as $A := \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$. The matrix

$[A|\vec{b}] := \left[\begin{array}{cccc|c} a_{1,1} & a_{1,2} & \cdots & a_{1,n} & b_1 \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} & b_m \end{array}\right]$ is called the **augmented matrix** (abbreviation: AM).

The SLE is said to be **homogeneous** if $\vec{b} := \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \vec{0}$, otherwise **inhomogeneous**. The vector $\vec{b}$ is the **homogenity** if $\vec{b} = \vec{0}$, otherwise the **inhomogenity**.

Example (SLE in 3 variables (or unknowns))

Determine the coefficient matrix and the augmented matrix for the given system of linear equations. Identify the inhomogenity.

$$\begin{array}{ccccccc} x_3 & - & \frac{1}{2}x_2 & & & = & -\frac{3}{2} \\ x_1 & + & x_2 & - & 2x_3 & = & 5 \\ 3x_1 & - & 2x_2 & + & 4x_3 & = & 0 \end{array}$$

The entries in the i-th column of A correspond to the coefficients of the variable x_i. After rearranging the system $\begin{array}{ccccccc} 0x_1 & - & \frac{1}{2}x_2 & + & x_3 & = & -\frac{3}{2} \\ x_1 & + & x_2 & - & 2x_3 & = & 5 \\ 3x_1 & - & 2x_2 & + & 4x_3 & = & 0 \end{array}$, we can consider it in the form $A\vec{x} = \vec{b}$ for $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$:

$$\boxed{\text{coefficient matrix}} \to A := \begin{bmatrix} 0 & -\frac{1}{2} & 1 \\ 1 & 1 & -2 \\ 3 & -2 & 4 \end{bmatrix}, \quad \vec{b} := \begin{bmatrix} -\frac{3}{2} \\ 5 \\ 0 \end{bmatrix}. \leftarrow \boxed{\begin{array}{c}\text{inhomogeneity} \\ \text{(or right-hand side)}\end{array}}$$

(Multiply A with the vector $\vec{x}$. What do you get?) The AK is
$$[A|\vec{b}] = \left[\begin{array}{ccc|c} 0 & -\frac{1}{2} & 1 & -\frac{3}{2} \\ 1 & 1 & -2 & 5 \\ 3 & -2 & 4 & 0 \end{array}\right]. \longleftarrow \boxed{\begin{array}{c}\text{augmented} \\ \text{coefficient} \\ \text{matrix}\end{array}}$$

4.2 The three elementary row operations

Recall that to find all solutions of an SLE (the **solution set**), you can operate on the equations according to specific rules that do not change the solution set. These are as follows.

1. The order of the equations can be changed.
2. An equation can be multiplied with any nonzero number.
3. A multiple of one equation can be added to another equation.

Using the matrix representation of an SLE, it is easy to solve the system it represents. The variables are ignored until the end. This usually results in fewer computational errors. The arrangement of the coefficients in the matrix contains all the information needed so that these do not need to be carried along until the end.

The rules for operating on systems of equations can be modified to operate on the rows of the matrix representation of the system.

Definition (elementary row operations)

- exchanging two rows
- multiplying a row with a nonzero number
- adding a multiple of one row to another

Remark: The elementary row operations have no effect on the solution set!

As an example, we apply the three elementary row operations to the augmented matrix from the preceding example.

$$[A|\vec{b}] = \left[\begin{array}{ccc|c} 0 & -\frac{1}{2} & 1 & -\frac{3}{2} \\ 1 & 1 & -2 & 5 \\ 3 & -2 & 4 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} \mathbf{0} & -\mathbf{\frac{1}{2}} & \mathbf{1} & -\mathbf{\frac{3}{2}} \\ \mathbf{1} & \mathbf{1} & -\mathbf{2} & \mathbf{5} \\ 3 & -2 & 4 & 0 \end{array}\right] \quad \underset{\longrightarrow}{\substack{\text{(2 exchange rows)} \\ \mathbf{I \leftrightarrow II}}} \quad \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{1} & -\mathbf{2} & \mathbf{5} \\ \mathbf{0} & -\mathbf{\frac{1}{2}} & \mathbf{1} & -\mathbf{\frac{3}{2}} \\ 3 & -2 & 4 & 0 \end{array}\right]$$

$$\underset{\longrightarrow}{\substack{\text{(mult. a row with} \\ \text{a number } (\neq 0)) \\ \mathbf{2 \cdot II}}} \quad \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{1} & -\mathbf{2} & \mathbf{5} \\ \mathbf{0} & -\mathbf{1} & \mathbf{2} & -\mathbf{3} \\ 3 & -2 & 4 & 0 \end{array}\right]$$

$$\underset{\longrightarrow}{\substack{\text{(add multiple of a} \\ \text{row to another row)} \\ \mathrm{III}+\mathbf{(-3)\cdot I}}} \quad \left[\begin{array}{ccc|c} \mathbf{1} & \mathbf{1} & -\mathbf{2} & \mathbf{5} \\ \mathbf{0} & -\mathbf{1} & \mathbf{2} & -\mathbf{3} \\ 0 & -5 & 10 & -15 \end{array}\right]$$

4.3 (Reduced) row echelon form - (r)ref

Definition (leading coefficient of a row; row echelon form (ref); reduced row echelon form (rref))

The **leading coefficent** of a row is the first nonzero entry (from left to right) in the row.

Conditions for row echelon form (ref):

1. The lower the row is in the matrix, the further to the right the leading coefficient is.
2. There are no leading coefficients below a row of zeros.

Condition for reduced row echelon form (rref):

1. The matrix is in ref.
2. Each **leading coefficient** is a 1.
3. All **entries above a leading coefficient** are 0.

Remarks:

• A row of zeros has no leading coefficient.

• The rref is unique, but the ref is, in general, not.

Gaussian elimination is a *systematic* procedure that can be applied to a matrix to bring it into rref. The procedure will be illustrated using the example in this section. Try to describe the procedure after working through the details.

Example (ref; rref)

Bring the augmented matrix (AK) $\left[\begin{array}{rrr|r} 1 & 1 & -2 & 5 \\ 0 & -1 & 2 & -3 \\ 0 & -5 & 10 & -15 \end{array}\right]$ into rref.

$$\left[\begin{array}{rrr|r} \mathbf{1} & \mathbf{1} & -2 & 5 \\ 0 & \mathbf{-1} & 2 & -3 \\ 0 & \mathbf{-5} & 10 & -15 \end{array}\right]$$

The AK is **not** in ref. The **leading coefficient** in the 3rd row ($\mathbf{-5}$) does not lie to the right of the **leading coefficient** in the 2nd row ($\mathbf{-1}$).

$$\downarrow \text{III} -5\,\text{II}$$

$$\left[\begin{array}{rrr|r} 1 & \mathbf{1} & -2 & 5 \\ 0 & \mathbf{-1} & 2 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The matrix is in ref but **not** in rref. The **leading coefficient** in the 2nd row is $\mathbf{-1}$ and not 1. Furthermore the entry $\mathbf{1}$ above the leading coefficient in the 2nd row is not 0.

$$\downarrow (-1)\cdot \text{II}$$

$$\left[\begin{array}{rrr|r} 1 & \mathbf{1} & -2 & 5 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The matrix is still **not** in rref. The entry $\mathbf{1}$ above the leading coefficient in the 2nd row is $\neq 0$.

$$\downarrow \text{I} - \text{II}$$

$$\left[\begin{array}{rrr|r} 1 & 0 & 0 & 2 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The matrix is in **rref**.

Definition (leading coefficient variable, nonleading coefficient variable)

Let $A\vec{x} = \vec{b}$ be an SLE in the variables $x_1, x_2, \ldots, x_n$. Bring the matrix A into some ref $\tilde{A}$. If the i-th column in $\tilde{A}$ contains a leading coefficient, then x_i is called a **leading coefficient variable** (abbreviation: lcv). Otherwise x_i is a nonleading coefficient variable (abbreviation: nlcv).

In the previous example, x_1 and x_2 are lcvs and x_3 is an nlcv.

Exercise 1 (Learning check)

Determine whether the following matrices are in **ref** resp. **rref**. If a matrix is not in rref, apply Gaussian elimination until it is.

$$\text{a)} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix} \quad \text{b)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 3 & 0 \end{bmatrix} \quad \text{c)} \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 0 & 1 & 2 \\ 4 & 0 & -3 & 22 \end{bmatrix}$$

4.4 Rank and the number of solutions

Definition (rank of a matrix)

rank A = maximal number of linearly independent rows in A

= maximal number of linearly independet columns in A

= number of **leading coefficients** in a (r)ref of A

Example (rank of a matrix)
Determine the rank of A and the rank of $[A|\vec{b}]$ for

$$A = \begin{bmatrix} 0 & -\frac{1}{2} & 1 \\ 1 & 1 & -2 \\ 3 & -2 & 4 \end{bmatrix} \text{ and } [A|\vec{b}] = \left[\begin{array}{ccc|c} 0 & -\frac{1}{2} & 1 & -\frac{3}{2} \\ 1 & 1 & -2 & 5 \\ 3 & -2 & 4 & 0 \end{array}\right].$$

$$\text{rank}(A) = \text{rank} \begin{bmatrix} 0 & -\frac{1}{2} & 1 \\ 1 & 1 & -2 \\ 3 & -2 & 4 \end{bmatrix} = \text{rank} \begin{bmatrix} \mathbf{1} & 1 & -2 \\ 0 & \mathbf{-1} & 2 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{2}$$

$$\text{rank}[A|\vec{b}] = \text{rank} \left[\begin{array}{ccc|c} 0 & -\frac{1}{2} & 1 & -\frac{3}{2} \\ 1 & 1 & -2 & 5 \\ 3 & -2 & 4 & 0 \end{array}\right] = \text{rank} \left[\begin{array}{ccc|c} \mathbf{1} & 0 & 0 & 2 \\ 0 & \mathbf{1} & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right] = \mathbf{2}$$

The rank can be used to determine whether an SLE $A\vec{x} = \vec{b}$ has no soultions, one solution or infinitely many solutions.

Theorem (number of solutions)

1. $\text{rank}[A|\vec{b}] = \text{rank}(A)$ $\Rightarrow$ the SLE has at least one solution

1a) $\text{rank}(A) < \#$ of columns in A $\Rightarrow$ the SLE has infinitely many solutions (see the example above)

1b) $\text{rank}(A) = \#$ of columns in A $\Rightarrow$ the SLE has a unique solution

2. $\text{rank}[A|\vec{b}] \neq \text{rank}(A)$ $\Rightarrow$ the SLE has no solutions

Example (solution set)

Determine the solution set $\mathcal{S}$ of the real SLE $A\vec{x} = \vec{b}$ for the matrix $A := \begin{bmatrix} 0 & -\frac{1}{2} & 1 \\ 1 & 1 & -2 \\ 3 & -2 & 4 \end{bmatrix}$ and the vector $\vec{b} = \begin{bmatrix} -\frac{3}{2} \\ 5 \\ 0 \end{bmatrix}$.

$$\left[\begin{array}{ccc|c} 0 & -\frac{1}{2} & 1 & -\frac{3}{2} \\ 1 & 1 & -2 & 5 \\ 3 & -2 & 4 & 0 \end{array}\right] \xrightarrow{\text{rref - see the example on page 38}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right] =: [\tilde{A}|\tilde{\vec{b}}]$$

$\text{rank}(A) = \text{rank}[A|\vec{b}] = 2$ means that the SLE has at least one solution hat.
$\text{rank}[A|\vec{b}] = \text{rank}(A) = 2 < 3 =$ number of columns in A; i.e. the solution set $\mathcal{S}$ contains infinitely many elements.

Solve for the **leading coefficient variables** of A (columns in $\tilde{A}$ with a leading coefficient): x_1, x_2 in terms of the **nonleading coefficient variable** of A (columns in $\tilde{A}$ without a leading coefficient): x_3

$$x_1 = 2, x_2 = 3 + 2x_3 \longrightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 + 2x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} \mathbf{2} + \mathbf{0}x_3 \\ \mathbf{3} + \mathbf{2}x_3 \\ \mathbf{0} + \mathbf{1}x_3 \end{bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{3} \\ \mathbf{0} \end{bmatrix} + \boldsymbol{x_3} \begin{bmatrix} \mathbf{0} \\ \mathbf{2} \\ \mathbf{1} \end{bmatrix}$$

Choose a parameter $s \in \mathbb{R}$ (since it is a *real* SLE) for x_3:

$$\mathcal{S} := \left\{ \underbrace{\begin{bmatrix} \mathbf{2} \\ \mathbf{3} \\ \mathbf{0} \end{bmatrix}}_{\boxed{\text{particular solution}}\uparrow} + \underbrace{\boldsymbol{s} \begin{bmatrix} \mathbf{0} \\ \mathbf{2} \\ \mathbf{1} \end{bmatrix}}_{\uparrow\boxed{\text{solutions of the homogeneous system } A\vec{x} = \vec{0}}} \;\middle|\; \boldsymbol{s} \in \mathbb{R} \right\}. \quad \textbf{parametrization}$$

The solution set in this example has to be real due to the way the problem was stated. The SLE was said to be real, which means that the parameter is real ($s \in \mathbb{R}$).

The solution set of an arbitrary SLE $A\vec{x} = \vec{b}$ can be expressed as
$\mathcal{S} = \left\{ \vec{x}_p + \vec{x}_h \,|\, \vec{x}_p \textbf{ is a particular solution of the SLE } A\vec{x} = \vec{b},\ \vec{x}_h \in \textbf{ker}(A) \right\}$.

The vector $\vec{x}_p$ is a particular solution, i.e. $A\vec{x}_p = \vec{b}$. The vectors $\vec{x}_h$ are in $\ker(A)$, i.e. they are solutions of the homogeneous SLE $A\vec{x} = \vec{0}$.

Going back to the example, we check our solution set. Due to linearity, we get

$$\begin{aligned} A\left(\begin{bmatrix} \mathbf{2} \\ \mathbf{3} \\ \mathbf{0} \end{bmatrix} + \boldsymbol{s}\begin{bmatrix} \mathbf{0} \\ \mathbf{2} \\ \mathbf{1} \end{bmatrix}\right) &= A\left(\begin{bmatrix} \mathbf{2} \\ \mathbf{3} \\ \mathbf{0} \end{bmatrix}\right) + A\left(\boldsymbol{s}\begin{bmatrix} \mathbf{0} \\ \mathbf{2} \\ \mathbf{1} \end{bmatrix}\right) \\ &= A\left(\begin{bmatrix} \mathbf{2} \\ \mathbf{3} \\ \mathbf{0} \end{bmatrix}\right) + \boldsymbol{s}A\left(\begin{bmatrix} \mathbf{0} \\ \mathbf{2} \\ \mathbf{1} \end{bmatrix}\right) \\ &= \begin{bmatrix} -\frac{3}{2} \\ 5 \\ 0 \end{bmatrix} + \boldsymbol{s}\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{3}{2} \\ 5 \\ 0 \end{bmatrix}. \end{aligned}$$

Exercise 2 (Learning check)

Find the solution set $\mathcal{S}$ of the real SLE $A\vec{x} = \vec{b}$ for the given matrix A and vector $\vec{b}$.

a) $A := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $\vec{b} := \begin{bmatrix} 1 \\ 1 \\ 6 \end{bmatrix}$ b) $A := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 3 & 0 \end{bmatrix}$, $\vec{b} := \begin{bmatrix} -5 \\ 4 \\ -12 \end{bmatrix}$

c) $A := \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$, $\vec{b} := \begin{bmatrix} 6 \\ 0 \\ 3 \end{bmatrix}$

Algorithm (basis for the kernel)

Input: $A \in \mathbb{F}^{m,n}$
Output: A basis for the kernel of A.

1. Bring A into rref ($\tilde{A}$). (Alternatively, bring $[A|\vec{0}]$ into rref ($[\tilde{A}|\vec{0}]$).)
2. Solve for the lcv of $\tilde{A}$ in terms of the nlcv of $\tilde{A}$.
3. Let k be the number of nlcv's. If $k = 0$, then a basis for the kernel of A is $\{\,\}$. Break.
4. ($k \neq 0$) For $i = 1, \dots, k$ set the i-th nlcv to 1 and all other nlcv's to 0 in the equations from Step 2 to get the i-th basis vector in a basis for the kernel of A.

Example (algorithm - basis for the kernel resp. kernel of a matrix)

Consider the matrix $A := \begin{bmatrix} 1 & 2 & 4 & 3 & 5 \\ 1 & 2 & 2 & 3 & 3 \\ 2 & 4 & 6 & 6 & 8 \end{bmatrix} \in \mathbb{F}^{3,5}$. Find a basis for the kernel of the homogeneous system $A\vec{x} = \vec{0}$ as well as all solutions.

The solution set of the homogeneous system $A\vec{x} = \vec{0}$ is just the kernel of the matrix A:

$$\ker(A) = \{\vec{x} \in \mathbb{F}^5 | A\vec{x} = \vec{0}\}.$$

Apply the above algorithm to find a basis $\mathcal{B}$ of the kernel. Take the span of these vectors (span($\mathcal{B}$) = the set of all linear combinations of the basis vectors) to get all of the solutions to the homogeneous system.

1.
$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 3 & 5 & 0 \\ 1 & 2 & 2 & 3 & 3 & 0 \\ 2 & 4 & 6 & 6 & 8 & 0 \end{array}\right] \xrightarrow{\mathrm{II}-\mathrm{I},\,\mathrm{III}-2\mathrm{I}} \left[\begin{array}{ccccc|c} 1 & 2 & 4 & 3 & 5 & 0 \\ 0 & 0 & -2 & 0 & -2 & 0 \\ 0 & 0 & -2 & 0 & -2 & 0 \end{array}\right] \xrightarrow{\mathrm{III}-\mathrm{II}} \left[\begin{array}{ccccc|c} 1 & 2 & 4 & 3 & 5 & 0 \\ 0 & 0 & -2 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]_{\text{ref}}$$

$$\xrightarrow{-\frac{1}{2}\mathrm{II}} \left[\begin{array}{ccccc|c} 1 & 2 & 4 & 3 & 5 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]_{\text{ref}} \xrightarrow{\mathrm{I}-4\mathrm{II}} \left[\begin{array}{ccccc|c} 1 & 2 & 0 & 3 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]_{\text{rref}}$$

2. lcv: x_1, x_3 nlcv: x_2, x_4, x_5
$x_1 + 2x_2 + 3x_4 + x_5 = 0, x_3 + x_5 = 0 \Rightarrow x_1 = -2x_2 - 3x_4 - x_5, x_3 = -x_5$

3. $k = 3$

4. $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_2 - 3x_4 - x_5 \\ x_2 \\ -x_5 \\ x_4 \\ x_5 \end{bmatrix}$ Set the nclv's successively to 1: $\boldsymbol{x_2 := 1, x_4 := 0, x_5 := 0}$
$\boldsymbol{x_2 := 0, x_4 := 1, x_5 := 0}$
$\boldsymbol{x_2 := 0, x_4 := 0, x_5 := 1}$

$$\text{A basis for } \ker(A) \text{ is: } \left\{ \begin{bmatrix} -2\\1\\0\\0\\0 \end{bmatrix}, \begin{bmatrix} -3\\0\\0\\1\\0 \end{bmatrix}, \begin{bmatrix} -1\\0\\-1\\0\\1 \end{bmatrix} \right\}.$$

$$\ker(A) = \left\{ r\begin{bmatrix} -2\\1\\0\\0\\0 \end{bmatrix} + s\begin{bmatrix} -3\\0\\0\\1\\0 \end{bmatrix} + t\begin{bmatrix} -1\\0\\-1\\0\\1 \end{bmatrix} \middle| \, r,s,t \in \mathbb{F} \right\} = \text{span}\left\{ \begin{bmatrix} -2\\1\\0\\0\\0 \end{bmatrix}, \begin{bmatrix} -3\\0\\0\\1\\0 \end{bmatrix}, \begin{bmatrix} -1\\0\\-1\\0\\1 \end{bmatrix} \right\}$$

Remarks:

• The number of nlcv's is the dimension of the kernel of A.

• The elementary row operations have no effect on the homogeneity. That is why it is possible to just consider A instead of $[A|\vec{0}]$ when looking for the rref.

Algorithm (basis for the image)

1. Put the matrix A into ref or rref ($\tilde{A}$).
2. Identify the lcv's of $\tilde{A}$.
3. Set $\mathcal{B} = \{\}$. For all suitable i, if x_i is an lcv, add the i-th column of A to $\mathcal{B}$ and call the reult $\mathcal{B}$.

Example (algorithm - basis of the image of a matrix)

Determine a basis of the image space of the matrix $A := \begin{bmatrix} 1&2&3&4&5\\1&2&3&6&7\\1&2&3&7&8 \end{bmatrix} \in \mathbb{F}^{3,5}$.

1. $A := \begin{bmatrix} 1&2&3&4&5\\1&2&3&6&7\\1&2&3&7&8 \end{bmatrix} \to \overset{*}{\cdots} \to \begin{bmatrix} \mathbf{1}&2&3&0&1\\0&0&0&\mathbf{1}&1\\0&0&0&0&0 \end{bmatrix}$ rref

2. lcv: x_1, x_4

3. A basis for the image of A (**1**. and **4**. Spalte von A): $\left\{ \begin{bmatrix} 1\\1\\1 \end{bmatrix}, \begin{bmatrix} 4\\6\\7 \end{bmatrix} \right\}$

$*$ In the examination, please show your steps. Here, you should verify the results whenever you see $\to \cdots \to$.

Remarks: The 2nd and 3rd columns of A are each a multiple of the 1st column (so linearly dependent on the 1st column). Thus x_2 and x_3 are nlcv's. The 4th column is linearly independent from the first column (why?) so that x_4 is an lcv. The 5th column is a linear combination of the 1st and 4th columns. Thus x_5 is a nlcv. The algorithm identifies the 1st and 4th columns as linearly independent, which are the columns we take from A. Note that A must be put into *row echelon form* $\tilde{A}$. We identify the lcv in $\tilde{A}$ and take those columns from the original matrix A to get a basis for the image of A.

IMPORTANT !

The column vectors of the matrix form a generating set for the image of A.

dim(im(A)) = rank(A) = number of lcv's in $\tilde{A}$

n = number of columns of A
  = number of lcv's + number of nlcv's of $\tilde{A}$
  = dim(im(A)) + dim(ker(A))

Exercise 3 (Learning check)

Determine a basis of the kernel and a basis of the image of the matrix $A := \begin{bmatrix} 1&2&4&3&5\\1&2&2&3&3\\2&4&6&6&8 \end{bmatrix} \in \mathbb{F}^{3,5}$.

algorithmus (inversion)

To invert a square matrix use the following strategy.

$$[A|I] \overset{\text{rref}}{\to} [I|A^{-1}].$$

If the rref of A is not the identity, then the matrix A is not invertible.

To explain why the algorithm works, assume that A is an invertible matrix in $\mathbb{R}^{2,2}$. Since $AA^{-1} = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ by definition, we search for a matrix $A^{-1} = [\vec{c}_1\ \vec{c}_2]$ such that $A\vec{c}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \vec{e}_1$ and $A\vec{c}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \vec{e}_2$. That means we have to solve the system of linear equations $[A|\vec{e}_1]$ and $[A|\vec{e}_2]$. Applying Gaussian elimination to both systems in the same way leads to $\vec{c}_1$ and $\vec{c}_2$. To save time and effort, we augment the coefficient matrix with both $\vec{e}_1$ and $\vec{e}_2$: $[A|\vec{e}_1\ \vec{e}_2] = [A|I]$.

Remark: This principle can be extended to more general systems of linear equations having the same coefficient matrix.

Example (matrix inversion)

Given the matrix $A := \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, determine the inverse matrix A^{-1}.

The inverse matrix of A can be calculated as follows.

$$[A|I_2] = \left[\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array}\right] \overset{\text{I}-\text{II}}{\to} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 2 & 1 & 0 & 1 \end{array}\right] \overset{\text{II}-2\text{I}}{\to} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -2 & 3 \end{array}\right] = [I_2|A^{-1}]$$

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

To check the answer, simply multiply to get the identity matrix: $AA^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$.

Example (a noninvertible matrix)

Check whether $B := \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ is invertible.

$$[B|I_2] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array}\right] \overset{\text{II}-2\text{I}}{\to} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & -1 & 1 \end{array}\right]$$

The **row echelon form of B** contains a **zero row**. The rows of B are linearly dependent so that the rref of B cannot be I_2. Thus, B is **not invertible**.

Exercise 4 (Learning check)

Invert the matrix $B := \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix}$.

Exercise 5 (Learning check)

Is the matrix $A := \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$ invertible?

Algorithm (coordinates of a vector with respect to a given basis)

Given the basis $\mathcal{B} := \left\{\vec{b}_1, \dots, \vec{b}_n\right\}$ of $\mathbb{F}^n$, $\vec{v} \in \mathbb{F}^n$, the coordinates of $\vec{v}$ with respect to (abbreviation: wrt) $\mathcal{B}$ are the coefficients needed to linearly combine the basis vectors to produce $\vec{v}$.

1. Set $B := \begin{bmatrix} \vec{b}_1 & \cdots & \vec{b}_n \end{bmatrix}$.
2. Bring the augmented matrix into rref: $[B|\vec{v}] \xrightarrow{\text{rref}} [I_n|\vec{\alpha}]$
3. The coordinaten are $\alpha_1, \dots, \alpha_n \quad (\vec{v} = \alpha_1\vec{b}_1 + \cdots + \alpha_n\vec{b}_n)$.

Example (algorithm - coordinates of a vector wrt a given basis)

Let

$$\mathcal{B} := \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} \right\}$$

be a basis of $\mathbb{R}^3$. Find the representation of the vector $\vec{v}_1 := \begin{bmatrix} 4 \\ 8 \\ 0 \end{bmatrix}$ respectively $\vec{v}_2 := \begin{bmatrix} 9 \\ -2 \\ 14 \end{bmatrix}$ as a linear combination of the basis vectors in $\mathcal{B}$. Form the coordinaten vector $(\vec{v}_i)_{\mathcal{B}}$, $i = 1, 2$.

We are searching for scalars $\boldsymbol{\alpha_1}, \boldsymbol{\alpha_2}, \boldsymbol{\alpha_3} \in \mathbb{R}$ with

$$\boldsymbol{\alpha_1} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \boldsymbol{\alpha_2} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \boldsymbol{\alpha_3} \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} = v_i, \quad i = 1, 2.$$

Comparing components, we get the following system of linear equations.

$$\begin{array}{ccccccccc} 1\alpha_1 & + & 1\alpha_2 & + & 2\alpha_3 & = & 4 & \text{bzw.} & 9 \\ 0\alpha_1 & + & 2\alpha_2 & + & 0\alpha_3 & = & 8 & \text{bzw.} & -2 \\ 1\alpha_1 & + & 0\alpha_2 & + & 3\alpha_3 & = & 0 & \text{bzw.} & 14 \end{array}$$

Form the AM $[B|\vec{v}_1\,\vec{v}_2]$ (see the remark on page 43), bring it into rref, and solve for $\vec{v}_1, \vec{v}_2$.

$$\left[\begin{array}{ccc|cc} 1 & 1 & 2 & 4 & 9 \\ 0 & 2 & 0 & 8 & -2 \\ 1 & 0 & 3 & 0 & 14 \end{array}\right] \to \cdots \to \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 4 & -1 \\ 0 & 0 & 1 & 0 & 4 \end{array}\right]$$

$$\vec{v}_1 := \begin{bmatrix} 4 \\ 8 \\ 0 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

$$\vec{v}_2 := \begin{bmatrix} 9 \\ -2 \\ 14 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

The coordinate vectors of $\vec{v}_1$ resp. $\vec{v}_2$ wrt the basis $\mathcal{B}$ are $(\vec{v}_1)_{\mathcal{B}} = \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}$ bzw. $(\vec{v}_2)_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.

Remarks:

- The coordinate vector $(\vec{v}_i)_{\mathcal{B}}$, $i = 1, 2$, is a solution of the SLE $B\vec{\alpha} = \vec{v}_i$ for $\vec{\alpha} := \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$.

• The notation $(\vec{v}_i)_{\mathcal{B}}$ for $\vec{\alpha}$ emphasizes the relationship between the vector $\vec{v}$, the basis $\mathcal{B}$ and the solution.
• The matrix B is always invertible since $\mathcal{B}$ is a basis. The equation $B\vec{\alpha} = \vec{v}_i$ has the solution set

$$B^{-1}B\vec{\alpha} = B^{-1}\vec{v}_i,$$

i.e. $I\vec{\alpha} = \vec{\alpha} = B^{-1}\vec{v}_i$.

Show that the inverse matrix of $B = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$ is the matrix $B^{-1} = \begin{bmatrix} 3 & -\frac{3}{2} & -2 \\ 0 & \frac{1}{2} & 0 \\ -1 & \frac{1}{2} & 1 \end{bmatrix}$. Calculate the products $B^{-1}\vec{v}_i$, $i = 1, 2$.

Exercise 6 (Learning check)

Let $\mathcal{B}$ be defined as $\mathcal{B} := \left\{ \vec{b}_1 := \begin{bmatrix} -2 \\ -4 \\ -6 \end{bmatrix}, \vec{b}_2 := \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \vec{b}_3 := \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} \right\}$.

a) Show that $\mathcal{B}$ is a basis of $\mathbb{R}^3$.
b) Determine the coordinate vectors $(\vec{v}_1)_{\mathcal{B}}$ and $(\vec{v}_2)_{\mathcal{B}}$ for the vectors $\vec{v}_1 := \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ resp. $\vec{v}_2 := \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$.

4.5 Suggested solutions to the exercises in Chapter 4

Exercise 1

1. $\begin{bmatrix} \mathbf{1} & -1 & 0 \\ 0 & \mathbf{1} & 2 \\ 0 & 0 & \mathbf{4} \end{bmatrix}$ is in ref but not in rref (not all leading coefficients are 1s, there are nonzero entries over the leading coefficients in the 2nd and 3rd rows).

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 2 \\ \mathbf{0} & \mathbf{0} & \mathbf{4} \end{bmatrix}_{\text{ref}} \overset{\frac{1}{4}\text{III}}{\to} \begin{bmatrix} 1 & -1 & 0 \\ \mathbf{0} & \mathbf{1} & \mathbf{2} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}_{\text{ref}} \overset{\text{II}-2\text{III}}{\to} \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ 0 & 0 & 1 \end{bmatrix}_{\text{ref}} \overset{\text{I}+\text{II}}{\to} \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{\text{rref}}$$

Comment: A matrix can be put in different refs. A given matrix has only one rref.

2. $\begin{bmatrix} \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{2} \\ 0 & \mathbf{3} & 0 \end{bmatrix}$ is not in ref (the leading coefficient in the 2nd row lies to the right of the leading coefficient in the 3rd row).

$$\begin{bmatrix} 1 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{2} \\ \mathbf{0} & \mathbf{3} & \mathbf{0} \end{bmatrix} \overset{\text{II}\leftrightarrow\text{III}}{\to} \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{0} & \mathbf{3} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{2} \end{bmatrix}_{\text{ref}} \overset{\frac{1}{3}\text{II},\,\frac{1}{2}\text{III}}{\to} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{\text{rref}}$$

3. $\begin{bmatrix} \mathbf{1} & 0 & 0 & 7 \\ 0 & 0 & \mathbf{1} & 2 \\ \mathbf{4} & 0 & -3 & 22 \end{bmatrix}$ is not in ref (only zeros can lie below a leading coefficient).

$$\begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 0 & 1 & 2 \\ \mathbf{4} & \mathbf{0} & \mathbf{-3} & \mathbf{22} \end{bmatrix} \overset{\text{III}-4\text{I}}{\to} \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 0 & 1 & 2 \\ \mathbf{0} & \mathbf{0} & \mathbf{-3} & \mathbf{-6} \end{bmatrix} \overset{\text{III}+3\text{II}}{\to} \begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 0 & 1 & 2 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}_{\text{rref}}$$

Comment: The entries in the 4th column can be nonzero because there is no leading coefficient in that column.

Exercise 2

1. $A := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $\vec{b} := \begin{bmatrix} 1 \\ 1 \\ 6 \end{bmatrix}$

$$[A|\vec{b}] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 6 \end{array}\right] \overset{\text{III}-\text{I}}{\to} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 5 \end{array}\right] \overset{\text{III}-\text{II}}{\to} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 4 \end{array}\right]$$

The **(r)ref of A** contains a **zero row** but (r)ref of $[A|\vec{b}]$ does not so that $\boldsymbol{\mathcal{L} = \emptyset}$ holds.
(The SLE has **no solution** because $\operatorname{rank}(A)$ (= number of rows in a ref of A) $= 2 < 3 = \operatorname{rank}([A|\vec{b}])$.)

2. $A := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 3 & 0 \end{bmatrix}$, $\vec{b} := \begin{bmatrix} -5 \\ 4 \\ -12 \end{bmatrix}$

$$[A|\vec{b}] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & -5 \\ 0 & 0 & 2 & 4 \\ 0 & 3 & 0 & -12 \end{array}\right] \overset{\text{II}\leftrightarrow\text{III}}{\rightarrow} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -5 \\ 0 & 3 & 0 & -12 \\ 0 & 0 & 2 & 4 \end{array}\right] \overset{\frac{1}{3}\text{II}, \frac{1}{2}\text{III}}{\rightarrow} \left[\begin{array}{ccc|c} \overbrace{1 \quad 0 \quad 0}^{\textbf{identity matrix}} & & & \mathbf{-5} \\ 0 & 1 & 0 & \mathbf{-4} \\ 0 & 0 & 1 & \mathbf{2} \end{array}\right]$$

$$\mathcal{S} = \left\{ \begin{bmatrix} \mathbf{-5} \\ \mathbf{-4} \\ \mathbf{2} \end{bmatrix} \right\}$$

(The SLE has **exactly one solution** because $\text{rank}(A) = 3 = \text{rank}([A|\vec{b}])$ = number of columns in A.)

3. $A := \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$, $\vec{b} := \begin{bmatrix} 6 \\ 0 \\ 3 \end{bmatrix}$

$$[A|\vec{b}] = \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 3 \end{array}\right] \overset{\text{II}\leftrightarrow\text{III}}{\rightarrow} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 6 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \overset{\frac{1}{3}\text{II}}{\rightarrow} \begin{array}{c} \begin{array}{cccc} \boldsymbol{x_1} & \boldsymbol{x_2} & \boldsymbol{x_3} & \boldsymbol{x_4} \end{array} \\ \left[\begin{array}{cccc|c} \mathbf{1} & 0 & \mathbf{2} & 0 & \mathbf{6} \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{1} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]_{\text{rref}} \end{array}$$

Leading coefficient variables: $\boldsymbol{x_1}, \boldsymbol{x_4}$; Nonleading coefficient variables (parameters): $\boldsymbol{x_2}$, $\boldsymbol{x_3}$

Solve in terms of the leading coefficient variables (LCV): $\boldsymbol{x_1} = \mathbf{6}-\mathbf{2}\boldsymbol{x_3}$, $\boldsymbol{x_4} = \mathbf{1}$.

$$\mathcal{L} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \middle| \; x_1 = 6 - 2x_3,\ x_4 = 1, \boldsymbol{x_2}, \boldsymbol{x_3} \in \mathbb{R} \right\}$$

Alternatively,
a vector $\vec{x}$ is the solution set has the form

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 6 - 2x_3 \\ x_2 \\ x_3 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 + 0x_2 - 2x_3 \\ 0 + 1x_2 + 0x_3 \\ 0 + 0x_2 + 1x_3 \\ 1 + 0x_2 + 0x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

Choose parameters $s, t \in \mathbb{R}$ (one for each nonleading coefficient variable (nlcv)) and write down the solution set:

$$\mathcal{S} = \left\{ \overset{\substack{\text{particular solution} \\ \downarrow}}{\begin{bmatrix} 6 \\ 0 \\ 0 \\ 1 \end{bmatrix}} + \overbrace{s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}}^{\in \ker(A)} \middle| \; s, t \in \mathbb{R} \right\}.$$

(The SLE has infinitely many solutions: $\text{rank}(A = 2) = \text{rank}([A|\vec{b}]) <$ number of columns of A.)
To check the solution, multiply the coefficient matrix A with an element from $\mathcal{S}$. What should the result be? What happens if you multiply A with an element in $\ker(A)$?

Exercise 3 1. $A := \begin{bmatrix} 1 & 2 & 4 & 3 & 5 \\ 1 & 2 & 2 & 3 & 3 \\ 2 & 4 & 6 & 6 & 8 \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{\text{rref}} =: \tilde{A}$

2. lcv: x_1, x_3

3. A basis of the image of A is the (**1**st and **3**rd columns of A): $\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix} \right\}$.

Remark: The number of lcv's is the same as the dimension of the image of A.

To determine a basis of the kernel of A, put A in rref. This matrix was already computed as $\tilde{A}$. We get two equations in the lcv's x_1, x_3 and nlcv's x_2, x_4, x_5: $x_1 + 2x_2 + 3x_4 + x_5 = 0$ and $x_3 + x_5 = 0$. An element in $\ker(A)$ has the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_2 - 3x_4 - x_5 \\ x_2 \\ -x_5 \\ x_4 \\ x_5 \end{bmatrix}.$$

Set $x_2 = 1, x_4 = 0, x_5 = 0$ for a first basis vector, $x_2 = 0, x_4 = 1, x_5 = 0$ for a second basis vector, and $x_2 = 0, x_4 = 0, x_5 = 1$ for a third basis vector. A basis of the kernel of A iss

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Exercise 4

The invertible matrix $B := \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix}$ can be inverted as follows: $[B|I_3] \xrightarrow{\text{rref}} [I_3|B^{-1}]$.

$$[B|I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\text{II}-2\text{I},\text{III}-3\text{I}} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 2 & -3 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{2\text{II}-\text{III}} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 2 & -1 \end{array}\right] \xrightarrow[\text{II}+\text{III}]{\text{I}-\text{III}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \mathbf{2} & \mathbf{-2} & \mathbf{1} \\ 0 & 1 & 0 & \mathbf{-3} & \mathbf{3} & \mathbf{-1} \\ 0 & 0 & 1 & \mathbf{-1} & \mathbf{2} & \mathbf{-1} \end{array}\right]$$

$$B^{-1} = \begin{bmatrix} \mathbf{2} & \mathbf{-2} & \mathbf{1} \\ \mathbf{-3} & \mathbf{3} & \mathbf{-1} \\ \mathbf{-1} & \mathbf{2} & \mathbf{-1} \end{bmatrix}$$

Check answer: $BB^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 \\ -3 & 3 & -1 \\ -1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$

Exercise 5

$$[A|I_3] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \overset{\text{II}-2\text{I},\ \text{III}-2\text{I}}{\longrightarrow} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 2 & -2 & -2 & 0 & 1 \end{array}\right]$$

$$\overset{\text{III}-2\text{II}}{\longrightarrow} \left[\begin{array}{c|c} \underbrace{\begin{matrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{matrix}}_{\tilde{A}} & \begin{matrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & -2 & 1 \end{matrix} \end{array}\right]$$

Because $\tilde{A}$ contains a zero row, alternatively $\tilde{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \neq I_3$, the rows resp. the columns of A are linearly dependent. A is not invertible.

Exercise 6

a) To show that $\mathcal{B}$ is a basis of $\mathbb{R}^3$, we have to show that $\mathcal{B}$ is a linearly independent generating set. Since every basis of $\mathbb{R}^3$ contains exactly three linearly independent vectors and $\mathcal{B}$ contains exactly three vectors, it remains to show that the set is linearly independent (alternatively, that $\mathcal{B}$ is a generating set). The rank of $B := \begin{bmatrix} \mathbf{-2} & \mathbf{0} & \mathbf{2} \\ \mathbf{-4} & \mathbf{1} & \mathbf{1} \\ \mathbf{-6} & \mathbf{-1} & \mathbf{4} \end{bmatrix}$ is three, so that the rows of B are linearly independent. Since B is square and the row rank is equal to the column rank, the column vectors of B are also linearly independent.

Put B in ref to determine the rank:

$$\begin{bmatrix} -2 & 0 & 2 \\ -4 & 1 & 1 \\ -6 & -1 & 4 \end{bmatrix} \overset{\text{II}-2\text{I},\ \text{III}-3\text{I}}{\longrightarrow} \begin{bmatrix} -2 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & -1 & -2 \end{bmatrix} \overset{\text{II}+\text{III}}{\longrightarrow} \begin{bmatrix} -2 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & -5 \end{bmatrix}_{\text{ref}}.$$

The rank of B is three (a ref of B contains three leading coefficients -2, 1, -5). Thus $\mathcal{B}$ is a basis of $\mathbb{R}^3$.

b) If $\vec{v}_i = \boldsymbol{\alpha_1}\vec{\boldsymbol{b}}_1 + \alpha_2\vec{\boldsymbol{b}}_2 + \alpha_3\vec{\boldsymbol{b}}_3$, the coordinate vector of $\vec{v}_i$ is

$$(\vec{v}_i)_{\mathcal{B}} = \begin{bmatrix} \boldsymbol{\alpha_1} \\ \alpha_2 \\ \alpha_3 \end{bmatrix}.$$

We are looking for scalars $\boldsymbol{\alpha_1}, \alpha_2, \alpha_3 \in \mathbb{R}$ with

$$\boldsymbol{\alpha_1} \begin{bmatrix} \mathbf{-2} \\ \mathbf{-4} \\ \mathbf{-6} \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \alpha_3 \begin{bmatrix} \mathbf{2} \\ \mathbf{1} \\ \mathbf{4} \end{bmatrix} = \vec{v}_i, \quad i = 1, 2.$$

Coefficient comparison leads to the following SLE:

$$\begin{array}{rcrcrcll} \mathbf{-2\alpha_1} & + & 0\alpha_2 & + & \mathbf{2\alpha_3} & = & \mathbf{1} & \text{bzw.} \quad -2 \\ \mathbf{-4\alpha_1} & + & 1\alpha_2 & + & \mathbf{1\alpha_3} & = & \mathbf{2} & \text{resp.} \quad 1 \\ \mathbf{-6\alpha_1} & + & (-1)\alpha_2 & + & \mathbf{4\alpha_3} & = & \mathbf{3} & \text{bzw.} \quad 4. \end{array}$$

Form the AM and put in rref:

$$\left[\begin{array}{ccc|cc} \mathbf{-2} & \mathbf{0} & \mathbf{2} & \mathbf{1} & \mathbf{-2} \\ \mathbf{-4} & \mathbf{1} & \mathbf{1} & \mathbf{2} & \mathbf{1} \\ \mathbf{-6} & \mathbf{-1} & \mathbf{4} & \mathbf{3} & \mathbf{4} \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{ccc|cc} 1 & 0 & 0 & -\frac{1}{2} & -2 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & -3 \end{array}\right].$$

Thus the desired coordinate vectors are $(\vec{v}_1)_{\mathcal{B}} = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$, $(\vec{v}_2)_{\mathcal{B}} = \begin{bmatrix} -2 \\ -4 \\ -3 \end{bmatrix}$.

First check: $-\frac{1}{2}\begin{bmatrix} -2 \\ -4 \\ -6 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + 0\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \vec{v}_1$

(That's right. We could have seen that.)

Second check: $$-2\begin{bmatrix} -2 \\ -4 \\ -6 \end{bmatrix} - 4\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} - 3\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} = \vec{v}_2$$

5 Generalized vector spaces

In this chapter we extend the definition of a vector space to other sets of objects. We introduced $\mathbb{F}^n$ and its subspaces in Chapter 2. In Chapter 3, we saw how to define addition and multiplication with scalars for matrices in $\mathbb{F}^{m,n}$ so that $\mathbb{F}^{m,n}$ has the structure of a vector space. This idea can also be applied to the polynomials $\mathbb{F}[x] := \{$polynomials in the variable x with coefficients from $\mathbb{F}\}$. In doing so, we regard a polynomial p as a linear combination of powers of x:

$$p(x) := p_n x^n + \cdots + p_1 x + p_0, \quad p_n, \ldots, p_1, p_0 \in \mathbb{F}.$$

The addition $p+q$ of the polynomials p and q of the form

$$p(x) := p_n x^n + \cdots + p_1 x + p_0, \quad q(x) := q_n x^n + \cdots + q_1 x + q_0 \in \mathbb{F}[x]$$

is defined by adding the coefficients with the same power of x:

$$\begin{aligned} p+q: \quad (p+q)(x) &:= p(x) + q(x) = (p_n x^n + \cdots + p_1 x + p_0) + (q_n x^n + \cdots + q_1 x + q_0) \\ &:= (p_n + q_n)x^n + \cdots + (p_1 + q_1)x + (p_0 + q_0). \end{aligned}$$

In order to multiply a polynomial with a scalar $\alpha \in \mathbb{F}$, we multiply every coefficient with α:

$$\alpha p: \quad (\alpha p)(x) := \alpha(p(x)) = \alpha(p_n x^n + \cdots + p_1 x + p_0) := (\alpha p_n)x^n + \cdots + (\alpha p_1)x + (\alpha p_0).$$

Under these operations, the coefficients of polynomials behave similarly to the components of vectors in $\mathbb{F}^n$. Note that there is no finite basis for $\mathbb{F}[x]$. There is also no standard basis. Often we just use $\{x^0 = 1, x^1, x^2, \ldots\}$.

The notation for polynomials is sometimes confusing. We denote a polynomial with p but then it appears as $p(x)$. What is the difference? The notation p just refers to an element in the vector space $\mathbb{F}[x]$ with the name p. As soon as we are evaluating the polynomial at "x", we use $p(x)$. The difference may seem a bit contrived, but this distinction is very important in some areas of mathematics. You will, however, not be penalized in this course for using the wrong notation for p or $p(x)$.

5.1 Vector spaces over the field $\mathbb{F}$

Definition (vector space over $\mathbb{F}$)

The quadruple $\{V, \mathbb{F}, \text{Plus}, \text{Mult}\}$ is called a *vector space* over $\mathbb{F}$ if the following properties hold.

- V is a set and $\mathbb{F}$ a field ($\mathbb{F}$ is often $\mathbb{R}$ or $\mathbb{C}$).
- Plus $(+)$ is a binary mapping: $V \times V \to V$. The set is **closed under Plus**.
- Mult $(\,\cdot\,)$ is a binary mapping $\mathbb{F} \times V \to V$. The set is **closed under Mult**.
- There is a special element $\vec{0}$ in V, the so-called **zero vector**, such that for every element $\vec{v} \in V$ the equation $\vec{v} + \vec{0} = \vec{v}$ holds. The zero vector is also called the **null vector**.
  (**existence of the additive identity**)
- For each element $\vec{v} \in V$, there is an element $-\vec{v} \in V$ with $\vec{v} + (-\vec{v}) = \vec{0}$.
  (**existence of the additive inverse**)
- For all $\vec{u}, \vec{v}, \vec{w} \in V$, $\alpha, \beta \in \mathbb{F}$ the following rules hold:

$$\begin{array}{rcll} (\vec{u}+\vec{v})+\vec{w} &=& \vec{u}+(\vec{v}+\vec{w}) & \text{associativity of +} \\ \vec{u}+\vec{v} &=& \vec{v}+\vec{u} & \text{commutativity of +} \\ \alpha\cdot(\beta\cdot\vec{v}) &=& (\alpha\beta)\cdot\vec{v} & \text{associativity of } \cdot \\ \alpha\cdot(\vec{u}+\vec{v}) &=& \alpha\cdot\vec{u}+\alpha\cdot\vec{v} & \text{first distributive law} \\ (\alpha+\beta)\cdot\vec{v} &=& \alpha\cdot\vec{v}+\beta\cdot\vec{v} & \text{second distributive law} \\ 1\cdot\vec{v} &=& \vec{v} & \text{multiplicative identity} \end{array}$$

To define a vector space, we need a set V, a ground field $\mathbb{F}$, and two operations that describe how elements in V are added and how an element of V gets multiplied by an element of the ground field $\mathbb{F}$. These four objects completely describe the vector space. Often we just refer to the vector space V with no special reference to the other objects if the ground field is clear and the operations are the "usual" operations of Plus and Mult.

The operations in the definition look perhaps a bit strange. "Plus" is a mapping of the form $V \times V \to V$. That means that we take two vectors from V, "add" them together and get a vector in V as an answer. "Mult" takes a number from the ground field $\mathbb{F}$ and a vector from V and turns it into a vector in V. We denote Plus using the usual notation "$+$" and Mult with "$\cdot$" or just by juxtapositioning the scalar and the vector in V. Note that the definition of multiplication requires the scalar to multiply from the left and not the right.

The concepts of generating sets, linear independence, basis and dimension can be defined for generalized vector spaces in a manner analogous to Chapter 2. The following table contains the most important vector spaces for this course.

Theorem (important vector spaces over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$)

notation	a basis	dimension
$\mathbb{F}^n$ (see Chapter 2)	$\left\{ \vec{e}_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \vec{e}_2 := \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \cdots, \vec{e}_{\boldsymbol{n}} := \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \right\}$	$\dim \mathbb{F}^{\boldsymbol{n}} = \boldsymbol{n}$
$\mathbb{F}[x] :=$ {polynomials with coefficients in $\mathbb{F}$}	$\{1 = x^0, x = x^1, x^2, x^3, \dots\}$	$\dim \mathbb{F}[x]$ is infinite
$\mathbb{F}_{\leq n}[x] :=$ {polynomials of degree $\leq n$ with coefficients in $\mathbb{F}$}	$\{1 = x^{\mathbf{0}}, x = x^{\mathbf{1}}, \dots, x^{\boldsymbol{n}}\}$	$\dim \mathbb{F}_{\leq \boldsymbol{n}}[x] = \boldsymbol{n} + \mathbf{1}$
$\mathbb{F}^{m,n} :=$ {$m \times n$ matrices with entries in $\mathbb{F}$} (see Chapter 3)	$\{E_{ij} \mid 1 \leq i \leq \boldsymbol{m}, 1 \leq j \leq \boldsymbol{n}\}$ E_{ij} has a 1 in the (i,j)-position otherwise 0	$\dim \mathbb{F}^{\boldsymbol{m},\boldsymbol{n}} = \boldsymbol{mn}$

5.2 Subspaces

Definition (subspace)

A subset T of a vector space V over the field $\mathbb{F}$ (notation: $T \subseteq V$) is called a **subspace** if the following properties hold.

(1) T is non-empty ($T \neq \emptyset$).
(2) T is closed under Plus (for all $\vec{u}, \vec{v} \in T$, $\vec{u} + \vec{v} \in T$).
(3) T is closed under Mult (for all $\vec{v} \in T$ and $\alpha \in \mathbb{F}$, $\alpha\vec{v} \in T$).

Comments:

- $\{\vec{0}\}$ and V are *always* subspaces of V.
- There is no symbol for "is a subspace of".
- A set T that satisfies the definition of a subspace is in itself a vector space because it inherits all the properties of a vector space from the bigger vector space V.

Why was $\mathbb{F}_{\leq n}[x]$ inserted into the table in the preceding section?

At the beginning of this chapter, we claimed that the set of all polynomials with coefficients from the field $\mathbb{F}$ equipped with the usual addition of polynomials and multiplication with scalars is a vector space. (We leave it to the interested reader to check that all properties are indeed satisfied.) This vector space is infinite dimensional, but we are primarily interested in finite dimensional vector spaces in this course. We are thus seeking a finite dimensional subspace of $\mathbb{F}[x]$.

Let $P_n, n \geq 1$ a natural number, be the set of all polynomials of degree n with respect to the usual addition and multiplication with scalars. P_n is *not* a vector space with the given operation of addition. For example, the sum of the two polynomials $p(x) := \mathbf{2x^n + 1}$ and $q(x) = \mathbf{-2x^n}$ is not in P_n because

$$(p+q)(x) = \mathbf{2x^n + 1 - 2x^n} = 1$$

is a constant polynomial of degree 0 so that $p + q \notin P_n$.

Clearly, the sum of two polynomials in P_n can be smaller than n but never larger than n. For this reason we consider $\mathbb{F}_{\leq n}[x]$, the set of all polynomials in $\mathbb{F}[x]$ of degree less than or equal to n. The zero polynomial p_{zerol} is in $\mathbb{F}_{\leq n}[x]$ so that $\mathbb{F}_{\leq n}[x]$ is non-empty. The sum of two arbitrary polynomials in $\mathbb{F}_{\leq n}[x]$ is again a polynomial of degree less than or equal to n. Multiplication with a nonzero scalar does not change the degree of the polynomial, and certainly multiplying by zero yields the zero polynomial, which is also in $\mathbb{F}_{\leq n}[x]$. Thus, $\mathbb{F}_{\leq n}[x]$ is closed with respect to addition and multiplication with scalars. We conclude that $\mathbb{F}_{\leq n}[x]$ is a subspace of $\mathbb{F}[x]$.

Example (subspace with polynomials)

Check whether the set

$$P := \{p \in \mathbb{R}_{\leq 2}[x] \,|\, p(0) = 0\}$$

is a subspace of $\mathbb{R}_{\leq 2}[x]$.

We check the three subspace conditions:

(1) (P is not empty) The zero polynomial (i.e. $p_{\text{null}}(x) = 0x^2 + 0x + 0$) has the property $p_{\text{null}}(0) = 0$ so that $p_{\text{null}} \in P$ holds.

(2) (P is closed under addition) Let $p, q \in P$; i.e. $p(0) = 0 = q(0)$. For $p(x) := p_2x^2 + p_1x + p_0$ and $q(x) := q_2x^2 + q_1x + q_0$ we must therefore have

$$p(0) = p_2(0^2) + p_1(0) + p_0 = p_0 = 0 \qquad q(0) = q_2(0^2) + q_1(0) + q_0 = q_0 = 0.$$

Therefore p and q are of the form:

$$\begin{array}{rcccc} p(x) & = & p_2x^2 & + & p_1x \\ q(x) & = & q_2x^2 & + & q_1x. \end{array}$$

Next we check whether $p + q$ lies in P:

$$(p+q)(x) = (p_2+q_2)x^2 + (p_1+q_1)x$$

$$(p+q)(0) = (p_2+q_2)0^2 + (p_1+q_1)0 = 0 \quad \Rightarrow p+q \in P.$$

(3) (P is closed under multiplication by scalars) Let $p \in P$, $\alpha \in \mathbb{R}$. Then

$$(\alpha p)(x) = \alpha p_2x^2 + \alpha p_1x$$

$$(\alpha p)(0) = \alpha p_2(0^2) + \alpha p_1(0) = 0 \Rightarrow \alpha p \in P.$$

Because the three subspace criteria are satisfied, P is a subspace of $\mathbb{R}_{\leq 2}[x]$.

Example (subspace with matrices)

Show that $D := \left\{ \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \,\middle|\, a_{11}, a_{22} \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^{2,2}$.

We check the three subspace conditions:

(1) (D not empty) The zero matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ has an entry of zero in the positions $(1,2)$ and $(2,1)$ and is thus in D.

(2) (D closed under addition) Let $A, B \in D$; i.e. $a_{12} = a_{21} = 0$ and $b_{12} = b_{21} = 0$. The sum is given by

$$A + B = \begin{bmatrix} a_{11} + b_{11} & 0 + 0 \\ 0 + 0 & a_{22} + b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} + b_{11} & 0 \\ 0 & a_{22} + b_{22} \end{bmatrix}.$$

$A + B$ has zero entries in the $(1, 2)$ and $(2, 1)$ positions so that $A + B \in D$ holds.

(3) (D closed under multiplication with scalars) Let $A \in D$, $\alpha \in \mathbb{R}$. Because

$$\alpha A = \begin{bmatrix} \alpha \cdot a_{11} & \alpha \cdot 0 \\ \alpha \cdot 0 & \alpha \cdot a_{22} \end{bmatrix} = \begin{bmatrix} \alpha \cdot a_{11} & 0 \\ 0 & \alpha \cdot a_{22} \end{bmatrix},$$

it follows that αA has entries of zero in the $(1, 2)$ and $(2, 1)$ positions, i.e. $\alpha A \in D$.

Because the three subspace criteria are satisfied, D a subspace of $\mathbb{R}^{2,2}$.

The next two examples serve to illustrate that all criteria of a subspace must be shown. Just because a non-empty set is closed under addition does not mean that the set is necessarily closed under multiplication with scalars and vice versa.

Example (closed under addition but not multiplication)

Show that the set

$$\mathbb{Z}^2 := \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 \,|\, x_1, x_2 \in \mathbb{Z} \right\} \subset \mathbb{R}^2$$

is *not* a subspace of $\mathbb{R}^2$.

$\mathbb{Z}^2$ is non-empty: $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{Z}^2$, because 0 is an integer. Furthermore, $\mathbb{Z}^2$ is closed under addition (the sum of two integers is again an integer). The set is not closed under multiplication with scalars (from the field $\mathbb{R}$). For example,

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathbb{Z}^2, \frac{1}{2} \in \mathbb{R} \text{ but } \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \notin \mathbb{Z}^2,$$

since $\frac{1}{2}$ is not an integer. Therefore $\mathbb{Z}^2$ is not a subspace of $\mathbb{R}^2$.

Exercise 1 (learning check)

Determine whether

$$\mathbb{Q}^2 := \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 \,|\, x_1, x_2 \in \mathbb{Q} \right\}$$

is a subspace of $\mathbb{R}^2$, whereby $\mathbb{Q} := \left\{ \frac{p}{q} \,|\, p, q \in \mathbb{Z}, q \neq 0 \right\}$ denotes the field of rational numbers.

Example (closed under multiplication but not under addition)

Let

$$X := \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} \,|\, x \in \mathbb{R} \right\} = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \subset \mathbb{R}^2 \qquad Y := \left\{ \begin{bmatrix} 0 \\ y \end{bmatrix} \,|\, y \in \mathbb{R} \right\} = \operatorname{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^2$$

(geometrically speaking, X is the x-axis and Y the y-axis). Define the union of the sets X, Y through

$$M := X \cup Y = \left\{ \vec{m} \in \mathbb{R}^2 \,|\, \vec{m} \in X \text{ oder } \vec{m} \in Y \right\}.$$

Is M a subspace of $\mathbb{R}^2$?

Obviously M is the union of two non-empty sets and as such is also non-empty either. If $\vec{m} \in M$, then $\vec{m}$ is in X or Y. For all $\alpha \in \mathbb{R}$ we have

$$\alpha\vec{m} \in X = \operatorname{span}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\} \text{ or } \alpha\vec{m} \in Y = \operatorname{span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}$$

and thus a subset of $X \cup Y = M$ so that M is closed under multiplication with scalars.

The sum of the vectors $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix} \in M$ is however not an element of M because $\begin{bmatrix}1\\1\end{bmatrix}$ lies neither on the x-axis nor on the y-axis. M is not closed under addition and is thus not a subspace of $\mathbb{R}^2$.

Exercise 2

Determine which of the following subsets of $\mathbb{R}_{\leq 2}[x]$ resp. $\mathbb{R}^{2,2}$ are subspaces of $\mathbb{R}_{\leq 2}[x]$ resp. $\mathbb{R}^{2,2}$.

a) $M_1 := \{p_2x^2 + p_1x + p_0 \in \mathbb{R}_{\leq 2}[x] \mid \sum_{i=0}^{2} p_i = 0\}$

b) $M_2 := \{p_2x^2 + p_1x + p_0 \in \mathbb{R}_{\leq 2}[x] \mid p_2p_1p_0 = 0\}$

c) $M_3 := \{A \in \mathbb{R}^{2,2} \mid A \text{ is orthogonal}\}$

d) $M_4 := \{A \in \mathbb{R}^{2,2} \mid AB = BA \text{ for a particular matrix } B \in \mathbb{R}^{2,3}\}$

e) $M_5 := \{A \in \mathbb{R}^{2,2} \mid AC = CA \text{ for a particular } C \in \mathbb{R}^{2,2}\}$

Definition (span; generating set; linear (in)dependence; basis; dimension)

Let V be a vector space with scalars from the field $\mathbb{F}$ and $\vec{v}_1, \ldots, \vec{v}_k \in V$.

The **span** of the set $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is the set of all possible linear combinations of the vectors in the set with coefficients from $\mathbb{F}$:

$$\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\} := \{\alpha_1\vec{v}_1 + \cdots + \alpha_k\vec{v}_k \mid \alpha_1, \ldots, \alpha_k \in \mathbb{F}\}.$$

The set of vectors $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is **linearly independent**, if the **only** solution of the SLE

$$\alpha_1\vec{v}_1 + \cdots + \alpha_k\vec{v}_k = \vec{0}$$

is the trivial solution (i.e. $\alpha_1 = \cdots = \alpha_k = 0$ is the only solution). Otherwise the set is **linearly dependent**.[a]

[a] We also speak of the vectors being linearly independent or dependent.

The set $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is called a **generating set** of the vector space V if

$$\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\} = V.$$

If the set $\mathcal{B} := \{\vec{v}_1, \ldots, \vec{v}_k\}$ is a linearly dependent generating set, we call $\mathcal{B}$ a **basis** of V. The **dimension** of V is the number of elements in a basis.

Example (span; generating set; linear independence; basis)

The set $P := \{p \in \mathbb{R}_{\leq 2}[x] \mid p(0) = 0\}$ is a subspace of $\mathbb{R}_{\leq 2}[x]$. Check whether $M := \{\boldsymbol{x^2}, \boldsymbol{x}, \boldsymbol{x(x+1)}\} \subseteq P$ is a basis of P.

We first check if M is a generating set:

It has already been shown (see page 53), that $p \in P$ is of the form $p(x) = p_2x^2 + p_1x$. The polynomial p can be written as a linear combination of vectors in M:

$$p(x) = p_2x^2 + p_1x = p_2 \cdot \boldsymbol{x^2} + p_1 \cdot \boldsymbol{x} + 0 \cdot \boldsymbol{x(x+1)},$$

so that M is a generating set for P ($P = \text{span } M$).

On the other hand we have:

$$p(x) = p_2x^2 + p_1x = \mathbf{0} \cdot \boldsymbol{x^2} + (\boldsymbol{p_1} - \boldsymbol{p_2}) \cdot \boldsymbol{x} + \boldsymbol{p_2} \cdot \boldsymbol{x(x+1)}.$$

The representation is not unique since for $p_2 \neq 0$ we have:

$$(\boldsymbol{p_2} - \mathbf{0}) \cdot \boldsymbol{x^2} + (\boldsymbol{p_1} - (\boldsymbol{p_1} - \boldsymbol{p_2})) \cdot \boldsymbol{x} + (\mathbf{0} - \boldsymbol{p_2}) \cdot \boldsymbol{x(x+1)}$$

$$= p_2x^2 + p_2x - p_2x^2 - p_2x = p_{\text{zero}}.$$

So the zero polynomial can be represented by a non-trivial linear combination of the polynomials in M (i.e. the equation $\alpha_1x^2 + \alpha_2x + \alpha_3x(x+1) = 0$ has a nontrivial solution different from $\alpha_1 = \alpha_2 = \alpha_3 = 0$). It follows that the vectors in M are linearly dependent so that M is not a basis for P.

Exercise 3

Show that $\mathcal{B} := \{x^2, x(x+1)\}$ is a basis of $P := \{p \in \mathbb{R}_{\leq 2}[x] \mid p(0) = 0\}$. What is the dimension of P?

5.3 Coordinate mappings

Actually, we want to discuss coordinate vectors. However, we can already introduce this related topic a bit in advance.

Definition (coordinate vector; coordinate mapping)

Given the vector space V of dimension n over the field $\mathbb{F}$ and a basis $\mathcal{B} := \{\vec{b}_1, \ldots, \boldsymbol{\vec{b}_n}\}$ write the vector $\vec{v}$ as a linear combination of the basis elements:

$$\vec{v} = \boldsymbol{\alpha_1}\vec{b}_1 + \cdots + \boldsymbol{\alpha_n}\boldsymbol{\vec{b}_n}.$$

This is the unique representation of $\vec{v}$ with respect to the basis $\mathcal{B}$.

The **coordinate vector** of $\vec{v}$ with respect to the basis $\mathcal{B}$ is given by

$$\vec{v}_{\mathcal{B}} := \begin{bmatrix} \boldsymbol{\alpha_1} \\ \vdots \\ \boldsymbol{\alpha_n} \end{bmatrix}.$$

The **coordinate mapping** of V with respect to $\mathcal{B}$ is the mapping

$$K_{\mathcal{B}} : V \to \mathbb{F}^n; \vec{v} \mapsto \begin{bmatrix} \boldsymbol{\alpha_1} \\ \vdots \\ \boldsymbol{\alpha_n} \end{bmatrix},$$

which maps each vector in V onto its coordinate vector with respect to the given basis $\mathcal{B}$.

What is the difference between coordinate vectors and coordinate mappings? A coordinate vector is an element in the vector space $\mathbb{R}^n$. For a coordinate mapping we need the domain and image space as well as a rule to map from V to $\mathbb{R}^n$.

Example (coordinate mapping with polynomials)
Determine the coordinate mapping of

$$P := \{p \in \mathbb{R}_{\leq 2}[x] \mid p(0) = 0\}$$

with respect to the basis $\mathcal{B} := \{x^2, \boldsymbol{x(x+1)}\}$.

A general element of P is of the form $ax^2 + bx$. Because dim $P = 2$, the coordinate map has the form

$$\begin{array}{rccc} K_{\mathcal{B}}: & P & \to & \mathbb{R}^2 \\ & \boldsymbol{ax^2+bx} & \mapsto & \begin{bmatrix} \boldsymbol{\alpha_1} \\ \boldsymbol{\alpha_2} \end{bmatrix}, \end{array}$$

whereby $\alpha_1 x^2 + \boldsymbol{\alpha_2 x(x+1)} = \boldsymbol{ax^2 + bx}$; i.e.

$$\alpha_1 x^2 + \alpha_2 x^2 + \alpha_2 x = (\alpha_1 + \alpha_2)x^2 + \alpha_2 x = ax^2 + bx.$$

Coefficient comparison leads to:

$$\begin{array}{ccccccc} x^2 & : & \alpha_1 & + & \alpha_2 & = & a \\ x & : & & & \boldsymbol{\alpha_2} & = & \boldsymbol{b} \\ & & & \Rightarrow & \boldsymbol{\alpha_1} & = & \boldsymbol{a-b}. \end{array}$$

Hence the coordinate mapping of P with respect to $\mathcal{B}$ is

$$\begin{array}{rccc} K_{\mathcal{B}}: & P & \to & \mathbb{R}^2 \\ & ax^2+bx & \mapsto & \begin{bmatrix} \boldsymbol{a-b} \\ \boldsymbol{b} \end{bmatrix}. \end{array} \tag{5}$$

Comment: $K_{\mathcal{B}}(ax^2+bx) = \begin{bmatrix} a-b \\ b \end{bmatrix}$ is a vector and not a mapping! If we are looking for a coordinate mapping, then the answer needs to be in the form of a mapping (5).

Exercise 4 (bases and coordinate mapping with matrices)

a) Show that the set

$$\mathcal{B} := \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

is a basis of $\mathbb{R}^{3,2}$.

b) Determine the coordinate mapping of $V := \mathbb{R}^{3,2}$ with respect to the basis $\mathcal{B}$.

Exercise 5 (bases and coordinate mapping with polynomials)

Consider the following bases of $\mathbb{R}_{\leq 2}[x]$:

$$\mathcal{B}_1 := \{1, \boldsymbol{x+1}, \boldsymbol{x^2+x+1}\}, \; \mathcal{B}_2 := \{x+1, \boldsymbol{2}, \boldsymbol{x^2+x+1}\}.$$

a) Determine the coordinate vectors of

$$x^2, \; x+5, \; 3x+3$$

with respect to the basis $\mathcal{B}_1$ and with respect to the basis $\mathcal{B}_2$.

b) Find the coordinate mapping of $\mathbb{R}_{\leq 2}[x]$ with respect to the basis $\mathcal{B}_1$ and with respect to $\mathcal{B}_2$.

c) Set the basis vectors in $\mathcal{B}_i$ into $K_{\mathcal{B}_i}$ for $i = 1, 2$. What do you notice?

5.4 Suggested solutions to the exercises in Chapter 5

Exercise 1 Choose for example the vector $\begin{bmatrix}1\\0\end{bmatrix} \in \mathbb{Q}^2$ and the scalar $\sqrt{2} \in \mathbb{R}$. Since $\sqrt{2} \notin \mathbb{Q}$, it follows that $\sqrt{2}\begin{bmatrix}1\\0\end{bmatrix} \notin \mathbb{Q}^2$, so that $\mathbb{Q}^2$ is not closed with respect to multiplication by scalars.

Exercise 2 For each set M_i, all subspace criteria

(1) $M_i \neq \emptyset$,
(2) M_i is closed under addition,
(3) M_i is closed under multiplication with scalars,

must be examined (in the case of a subspace of $\mathbb{R}_{\leq 2}[x]$ or $\mathbb{R}^{2,2}$) or a counterexample (i.e. not a subspace of $\mathbb{R}_{\leq 2}[x]$ or $\mathbb{R}^{2,2}$) has to be produced.

a) We check if all three subspace criteria hold for M_1.

(1) The zero polynomial $p_{\text{zero}}(x) = 0x^2 + 0x + 0$ is in M_1 because the sum of the coefficients is zero $(0 + 0 + 0 = 0)$; i.e. M_1 is non-empty.

(2) Choose $p_2x^2 + p_1x + p_0, q_2x^2 + q_1x + q_0 \in M_1$; i.e. the conditions $p_0 + p_1 + p_2 = 0$ und $q_0 + q_1 + q_2 = 0$ are satisfied. The sum

$$(p_2x^2 + p_1x + p_0) + (q_2x^2 + q_1x + q_0) = (p_2 + q_2)x^2 + (p_1 + q_1)x + (p_0 + q_0)$$

lies in M_1 because he sum of the coefficients of this polynomial also gives zero:

$$(p_0 + q_0) + (p_1 + q_1) + (p_2 + q_2) = (p_0 + p_1 + p_2) + (q_0 + q_1 + q_2) = 0 + 0 = 0.$$

It follows that M_1 is closed under addition.

(3) Let $p_2x^2 + p_1x + p_0 \in M_1$ (i.e. $p_0 + p_1 + p_2 = 0$) and $\alpha \in \mathbb{R}$. M_1 is closed under multiplication with scalars if $\alpha(p_2x^2 + p_1x + p_0) = (\alpha p_2)x^2 + (\alpha p_1)x + (\alpha p_0)$ lies in M_1. This is the case since $\alpha p_0 + \alpha p_1 + \alpha p_2 = \alpha(p_0 + p_1 + p_2) = \alpha \cdot 0 = 0$. It thus follows that M_1 is closed with respect to multiplication with scalars.

The subspace criteria are satisfied. M_1 is a subspace of $\mathbb{R}_{\leq 2}[x]$.

b) M_2 is not a subspace of $\mathbb{R}_{\leq 2}[x]$ since M_2 is not closed with respect to addition:

$$3x^2 + 0x + 4,\ 5x^2 + 7x + 0 \in M_2\ (3 \cdot 0 \cdot 4 = 0, 5 \cdot 7 \cdot 0 = 0),$$

$$(3x^2 + 0x + 4) + (5x^2 + 7x + 0) = 8x^2 + 7x + 4 \notin M_2\ (8 \cdot 7 \cdot 4 \neq 0).$$

c) The set M_3 consists of matrices $A \in \mathbb{R}^{2,2}$ for which $AA^T = I_2$. M_3 is non-empty:

$$\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} \Rightarrow I_2 \in M_2.$$

The set M_3 is, however, not closed with regard to scalar multiplication: We choose $I_2 \in M_3, 0 \in \mathbb{R}$ and consider $0 \cdot I_2 = \begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}$:

$$\begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}\begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}^T = \begin{bmatrix}0 & 0\\0 & 0\end{bmatrix} \neq \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}.$$

Thus M_3 is not a subspace of $\mathbb{R}^{2,2}$.

d) There is no matrix $A \in \mathbb{R}^{2,2}$ for which matrix multiplication BA is defined because the number of columns in B is not equal to the number of lines in A, i.e. M_4 is empty. So M_4 is not a subspace of $\mathbb{R}^{2,2}$.

e) M_5 is different from M_4 because here the multiplication of A by $C \in \mathbb{R}^{2,2}$ is to the right (i.e. AC) and is also defined from the left (i.e. CA).
We check whether the three subspace criteria hold for M_5:
(1) Since

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = C \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

we have $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in M_5$, so that M_5 is non-empty.

(2) Choose $A_1, A_2 \in M_5$. The rule defining M_5 requires that

$$\boldsymbol{A_1 C} = \boldsymbol{C A_1} \quad \text{and} \quad \boldsymbol{A_2 C} = \boldsymbol{C A_2}. \qquad (*)$$

We check whether $A_1 + A_2 \in M_5$ holds:

$$\boldsymbol{(A_1 + A_2) C} = A_1 C + A_2 C \overset{\text{due to } (*)}{=} C A_1 + C A_2 = \boldsymbol{C(A_1 + A_2)}.$$

Thus, $A_1 + A_2 \in M_5$.
(3) Let $A \in M_5$ (d.h. $\boldsymbol{AC} = \boldsymbol{CA}$), $\alpha \in \mathbb{R}$. We check whether $\alpha A \in M_5$ holds:

$$\boldsymbol{(\alpha A) C} = \alpha(AC) = \alpha(CA) = \boldsymbol{C(\alpha A)}.$$

The subspace criteria are satisfied. M_5 is a subspace of $\mathbb{R}^{2,2}$.

Exercise 3 $\mathcal{B}$ is a generating set for the vector space P since every polynomial in P has the form $ax^2 + bx$. Furthermore $ax^2 + bx$ can be written as a linear combination of the polynomials in $\mathcal{B}$:

$$(a - b)x^2 + b(x(x + 1)) = ax^2 - bx^2 + bx^2 + bx = ax^2 + bx.$$

The two polynomials are linearly independent:

$$\alpha_1 x^2 + \alpha_2 x(x + 1) = 0$$

or

$$(\alpha_1 + \alpha_2)x^2 + \alpha_2 x = 0x^2 + 0x.$$

Coefficient comparison for x^2 yields $\alpha_1 + \alpha_2 = 0$. Coefficient comparison for x yields $\alpha_2 = 0$, so that also $\alpha_1 = 0$.

The dimension of P is 2 since a basis (nämlich $\mathcal{B}$) consists of two elements.

Exercise 4

a) It has to be shown that $\mathcal{B}$ is a linearly independent generating set is. This is the case when every arbitrary vector $\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \in \mathbb{R}^{3,2}$ can be written as a linear combination of the vectors in $\mathcal{B}$ in a *unique* way, i.e., if the following LGS has *exactly one* solution:

$$\alpha_1 \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} + \alpha_4 \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} + \alpha_5 \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} + \alpha_6 \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}.$$

Component comparison leads to the following SLE:

$$\begin{array}{rrrrrrcl}
1\alpha_1 & +1\alpha_2 & +1\alpha_3 & +1\alpha_4 & +1\alpha_5 & & = & a \\
1\alpha_1 & & & & & & = & b \\
& 1\alpha_2 & & & & & = & c \\
& & 1\alpha_3 & & & & = & d \\
& & & 1\alpha_4 & & & = & e \\
& & & & 1\alpha_5 & +1\alpha_6 & = & f.
\end{array}$$

Bring the AM $[A|\vec{b}]$ in rref using the Gauss algorithm:

$$\left[\begin{array}{cccccc|c} 1&1&1&1&1&0&a\\ 1&0&0&0&0&0&b\\ 0&1&0&0&0&0&c\\ 0&0&1&0&0&0&d\\ 0&0&0&1&0&0&e\\ 0&0&0&0&1&1&f \end{array}\right] . \to \dots(\text{Gauss})\dots \to \left[\begin{array}{cccccc|c} 1&0&0&0&0&0&b\\ 0&1&0&0&0&0&c\\ 0&0&1&0&0&0&d\\ 0&0&0&1&0&0&e\\ 0&0&0&0&1&0&a-b-c-d-e\\ 0&0&0&0&0&1&-a+b+c+d+e+f. \end{array}\right]$$

Since $\text{rank}(A) = \text{rank}([A|\vec{b}])$ = number of columns in A . the SLE has exactly one solution. Thus $\mathcal{B}$ is a basis o $\mathbb{R}^{3,2}$.

b) The coordinate vector $\vec{v}_{\mathcal{B}}$ of $\vec{v} \in \mathbb{R}^{3,2}$ with respect to the basis $\mathcal{B}$ contains the coefficients needed to form the linear combination of the basis vectors resulting in $\vec{v}$. We can read these off from the rref in part a) :

$$\vec{v} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} = b \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} + d \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} + e \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$+(a-b-c-d-e) \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} + (-a+b+c+d+e+f) \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

The coordinate mapping from $\mathbb{R}^{3,2}$ with respect to $\mathcal{B}$ maps a vector in $\mathbb{R}^{3,2}$ to its coordinate vector with respect to that basis:

$$\begin{array}{rccc} K_{\mathcal{B}}: & \mathbb{R}^{3,2} & \to & \mathbb{R}^6 \\ & \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} & \mapsto & \begin{bmatrix} b \\ c \\ d \\ e \\ a-b-c-d-e \\ -a+b+c+d+e+f \end{bmatrix}. \end{array}$$

Exercise 5 **a)** The coordinate vector $\vec{v}_{\mathcal{B}_i} := \begin{bmatrix} \alpha_{i,1} \\ \alpha_{i,2} \\ \alpha_{i,3} \end{bmatrix}$ of $\vec{v} \in \mathbb{R}_{\leq 2}[x]$ with respect to the basis $\mathcal{B}_i := \{\vec{b}_{i,1}, \vec{b}_{i,2}, \vec{b}_{i,3}\}, i = 1, 2,$ contains the coefficients needed to combine the basis vectors linearly to produce the given vector $\vec{v}$:

$$\alpha_{i,1}\vec{b}_{i,1} + \alpha_{i,2}\vec{b}_{i,2} + \alpha_{i,3}\vec{b}_{i,3} = \vec{v}.$$

The results are as follows.

$x^2_{\mathcal{B}_1} = \begin{bmatrix}0\\-1\\1\end{bmatrix}$ since $x^2 = 0(1) + (-1)(x+1) + \mathbf{1(x^2+x+1)}$	$x^2_{\mathcal{B}_2} = \begin{bmatrix}-1\\0\\1\end{bmatrix}$ since $x^2 = (-1)(x+1) + 0(2) + \mathbf{1(x^2+x+1)}$
$(x+5)_{\mathcal{B}_1} = \begin{bmatrix}4\\1\\0\end{bmatrix}$ since $x+5 = 4(1) + \mathbf{1(x+1)} + \mathbf{0(x^2+x+1)}$ since	$(x+5)_{\mathcal{B}_2} = \begin{bmatrix}1\\2\\0\end{bmatrix}$ since $x+5 = 1(x+1) + 2(2) + \mathbf{0(x^2+x+1)}$
$(3x+3)_{\mathcal{B}_1} = \begin{bmatrix}0\\3\\0\end{bmatrix}$ since $3x+3 = 0(1) + \mathbf{3(x+1)} + \mathbf{0(x^2+x+1)}$	$(3x+3)_{\mathcal{B}_2} = \begin{bmatrix}3\\0\\0\end{bmatrix}$ since $3x+3 = 3(x+1) + 0(1) + \mathbf{0(x^2+x+1)}$

b) The coordinate mapping of $\mathbb{R}_{\leq 2}[x]$ with respect to $\mathcal{B}_i, i = 1, 2$, maps a vector in $\mathbb{R}_{\leq 2}[x]$ (a polynomial of degree 2) onto the associated coordinate vector with respect to the basis $\mathcal{B}_i$. We show the details here for $K_{\mathcal{B}_1}$. For a general polynomial $ax^2 + bx + c \in \mathbb{R}_{\leq 2}[x]$, we search for scalars $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ with:

$$\alpha_1(1) + \alpha_2(x+1) + \alpha_3(x^2+x+1) = ax^2 + bx + c$$

or

$$\alpha_3(x^2) + (\alpha_2 + \alpha_3)(x) + (\alpha_1 + \alpha_2 + \alpha_3)(1) = ax^2 + bx + c.$$

Comparison of coefficients leads to the SLE
$$\begin{array}{lccccccc} x^0 = 1: & \alpha_1 & + & \alpha_2 & + & \alpha_3 & = & c \\ x^1: & & & \alpha_2 & + & \alpha_3 & = & b \\ x^2: & & & & & \alpha_3 & = & a. \end{array}$$

Build the AM and bring it in rref: $\left[\begin{array}{ccc|c} 1 & 1 & 1 & c \\ 0 & 1 & 1 & b \\ 0 & 0 & 1 & a \end{array}\right] \overset{\text{I}-\text{II},\text{II}-\text{III}}{\longrightarrow} \left[\begin{array}{ccc|c} 1 & 0 & 0 & c-b \\ 0 & 1 & 0 & b-a \\ 0 & 0 & 1 & a \end{array}\right]$.

The coordinate mapping is thus $K_{\mathcal{B}_1}: \mathbb{R}_{\leq 2}[x] \to \mathbb{R}^3; ax^2 + bx + c \mapsto \begin{bmatrix} c-b \\ b-a \\ a \end{bmatrix}$.

(Note: Set x^2, $x+5$ and $3x+3$ into $K_{\mathcal{B}_1}$. What do you notice? For example $K_{\mathcal{B}_1}(x^2) = \begin{bmatrix}0\\-1\\1\end{bmatrix}$ is the coordinate vector of x^2 with respect to the basis $\mathcal{B}$ of $\mathbb{R}_{\leq 2}[x]$.)

A similar calculation yields

$$K_{\mathcal{B}_2}: \mathbb{R}_{\leq 2}[x] \to \mathbb{R}^3; \quad ax^2 + bx + c \mapsto \begin{bmatrix} b-a \\ (c-b)/2 \\ a \end{bmatrix}.$$

c) In each case, we get the standard basis of $\mathbb{R}^3$. For example,

$$x^2 + x + 1 = 0 \cdot 1 + 0(x+1) + 1(x^2+x+1)$$

so that $K_{\mathcal{B}_1}(x^2+x+1) = \begin{bmatrix}0\\0\\1\end{bmatrix}$ holds. Setting x^2+x+1 into $K_{\mathcal{B}_1}$ leads to $\begin{bmatrix}0\\0\\1\end{bmatrix}$.

6 Linear mappings

6.1 Definitions and examples

Definition (linear mapping, homomorphism)
Let V, W be vector spaces over the field $\mathbb{F}$. The mapping $L : \boldsymbol{V} \to \boldsymbol{W}$ is called **linear** if the following properties hold:

$$\text{for all } \vec{u}, \vec{v} \in V,\ \alpha \in \mathbb{F}$$

$$\begin{array}{lrcl} (1) & L(\vec{u} + \vec{v}) & = & L(\vec{u}) + L(\vec{v}) \\ (2) & L(\alpha \cdot \vec{v}) & = & \alpha \cdot L(\vec{v}). \end{array}$$

Linear mappings are also called **homomorphisms**.

Recall that the vector space V is the **domain** of the mapping and the vector space W is called the **image** or **target space** of the mapping. The **image** of the mapping is the set $\{L(\vec{v}) | \vec{v} \in V\}$. The **preimage** of any element $\vec{w} \in \text{image}(L) \subseteq W$, is given by the set of all elements in V that map onto $\vec{w}$ $(L^{-1}(\vec{w}) = \{\vec{v} \in V | L(\vec{v}) = \vec{w}\})$.

Tip: $L(\vec{\mathbf{0}}_{\boldsymbol{V}}) = \vec{\mathbf{0}}_{\boldsymbol{W}}$ (Set $\alpha := 0$ in (2).)

With a linear mapping $L : V \to W$ it does not matter whether the vector space operations are performed in V and then the result mapped or if first the elements are mapped and then the vector space operations are performed in W. The results are the same.

Example (linear mapping)

Show that the following mapping is linear.

$$\begin{array}{rccc} L : & \mathbb{R}_{\leq 2}[\boldsymbol{x}] & \to & \mathbb{R}^{\mathbf{2,2}} \\ & \boldsymbol{a}x^2 + \boldsymbol{b}x + \boldsymbol{c} & \mapsto & \begin{bmatrix} \boldsymbol{a} & \boldsymbol{b} \\ \boldsymbol{c} & \boldsymbol{a} \end{bmatrix} \end{array}$$

The mapping L is linear if for all

$$p(x) := p_2 x^2 + p_1 x + p_0, \quad q(x) := q_2 x^2 + q_1 x + q_0 \in \mathbb{R}_{\leq 2}[x], \quad \alpha \in \mathbb{R}$$

the two properties $\left\{ \begin{array}{lrcl} 1) & L(p + q) & = & L(p) + L(q) \\ 2) & L(\alpha \cdot p) & = & \alpha \cdot L(p) \end{array} \right\}$ hold.

1) $L(p + q) = L((p_2 x^2 + p_1 x + p_0) + (q_2 x^2 + q_1 x + q_0))$

$$= L((\underbrace{\boldsymbol{p_2} + \boldsymbol{q_2}}_{\boldsymbol{a}})x^2 + (\underbrace{\boldsymbol{p_1} + \boldsymbol{q_1}}_{\boldsymbol{b}})x + (\underbrace{\boldsymbol{p_0} + \boldsymbol{q_0}}_{\boldsymbol{c}}))$$

$$= \begin{bmatrix} \boldsymbol{p_2} + \boldsymbol{q_2} & \boldsymbol{p_1} + \boldsymbol{q_1} \\ \boldsymbol{p_0} + \boldsymbol{q_0} & \boldsymbol{p_2} + \boldsymbol{q_2} \end{bmatrix} = \begin{bmatrix} \boldsymbol{p_2} & \boldsymbol{p_1} \\ \boldsymbol{p_0} & \boldsymbol{p_2} \end{bmatrix} + \begin{bmatrix} \boldsymbol{q_2} & \boldsymbol{q_1} \\ \boldsymbol{q_0} & \boldsymbol{q_2} \end{bmatrix}$$

$$= L(\boldsymbol{p_2} x^2 + \boldsymbol{p_1} x + \boldsymbol{p_0}) + L(\boldsymbol{q_2} x^2 + \boldsymbol{q_1} x + \boldsymbol{q_0}) = L(p) + L(q) \qquad \checkmark$$

2) $$L(\alpha \cdot p) = L(\alpha(p_2 x^2 + p_1 x + p_0)) = L((\underbrace{\boldsymbol{\alpha p_2}}_{\boldsymbol{a}})x^2 + (\underbrace{\boldsymbol{\alpha p_1}}_{\boldsymbol{b}})x + (\underbrace{\boldsymbol{\alpha p_0}}_{\boldsymbol{c}}))$$

$$= \begin{bmatrix} \boldsymbol{\alpha p_2} & \boldsymbol{\alpha p_1} \\ \boldsymbol{\alpha p_0} & \boldsymbol{\alpha p_2} \end{bmatrix} = \alpha \begin{bmatrix} \boldsymbol{p_2} & \boldsymbol{p_1} \\ \boldsymbol{p_0} & \boldsymbol{p_2} \end{bmatrix} = \alpha \cdot L(\boldsymbol{p_2} x^2 + \boldsymbol{p_1} x + \boldsymbol{p_0}) = \alpha \cdot L(p) \qquad \checkmark$$

The defining properties are satisfied so the mapping is linear.

Exercise 1 (learning check)

Which of the following mappings are linear?

$$L_1 : \mathbb{R}^{2,3} \to \mathbb{R}^{3,2}; A \mapsto A^T \qquad L_2 : \mathbb{R}_{\leq 2}[x] \to \mathbb{R}_{\leq 2}[x]; ax^2 + bx + c \mapsto cx^2 + x + a$$

$$L_3 : \mathbb{R}_{\leq 2}[x] \to \mathbb{R}^{2,2}; ax^2 + bx + c \mapsto \begin{bmatrix} a & b \\ a & bc \end{bmatrix}$$

6.2 The vector space of linear mappings

Theorem
Let V, W be vector spaces over $\mathbb{F}$. The set of all linear mappings (or homomorphisms) from V to W

$$\hom(V, W) := \{\text{all linear mappings from } V \text{ to } W\}$$

is a vector space over $\mathbb{F}$.

The addition and multiplication with scalars for two elements $L, M \in \hom(V, W)$ and $\alpha \in \mathbb{F}$ is defined as:

$$\begin{aligned} L + M : \quad V &\to W; \quad \vec{v} \mapsto L(\vec{v}) + M(\vec{v}) \\ \alpha L : \quad V &\to W; \quad \vec{v} \mapsto \alpha L(\vec{v}). \end{aligned}$$

Example (linear combination of linear mappings)
Let $L, M : \mathbb{R}^2 \to \mathbb{R}_{\leq 1}[x]$ be the linear mappings given by

$$\boldsymbol{L}\left(\begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{bmatrix}\right) = \boldsymbol{ax - b}, \ \boldsymbol{M}\left(\begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{bmatrix}\right) = \boldsymbol{-ax + 2a}.$$

Determine $3L - 5M$.

$3L - 5M$ is calculated as follows.

$$\begin{aligned} (3L - 5M)\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) &= 3\boldsymbol{L}\left(\begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{bmatrix}\right) - 5\boldsymbol{M}\left(\begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \end{bmatrix}\right) \\ &= 3(\boldsymbol{ax - b}) - 5(-ax + 2a) \\ &= 8ax + (-10a - 3b) \end{aligned}$$

Example
Determine, if possible, the sum of the linear mappings

$$\begin{aligned} L : \quad \mathbb{R}^2 &\to \mathbb{R}_{\leq 1}[x], & N : \quad \mathbb{R}_{\leq 1}[x] &\to \mathbb{R}_{\leq 1}[x] \\ \begin{bmatrix} a \\ b \end{bmatrix} &\mapsto ax - b & ax + b &\mapsto 3bx - a. \end{aligned}$$

The sum is *not* defined because the **domains** are different.

6.3 The composition of linear mappings

Definition (composition)

Consider the vector spaces V, W, X over $\mathbb{F}$ and the linear mappings

$$\boldsymbol{L} : \boldsymbol{V} \to \boldsymbol{W}, \ \boldsymbol{M} : \boldsymbol{W} \to \boldsymbol{X}.$$

The **composition** (or **product**) $\boldsymbol{M} \circ \boldsymbol{L} : \boldsymbol{V} \to \boldsymbol{X}$ is given by

$$(M \circ L)(\vec{v}) \to M(L(\vec{v})).$$

Example (composition of linear mappings)

Define the vector spaces $V := \mathbb{R}^2, W := \mathbb{R}_{\leq 1}[x]$ and the linear mappings $L : V \to W, \ M : W \to V$ through

$$L\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = ax - b, \ M(ax+b) = \begin{bmatrix} 3a \\ a - 2b \end{bmatrix}.$$

Determine the compositions $M \circ L$ resp. $L \circ M$.

• The composition $M \circ L$ is a mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$, whereby

$$(M \circ L)\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = M\left(L\left(\begin{bmatrix} c \\ d \end{bmatrix}\right)\right) = M(cx - d).$$

We set c for a and $-d$ for b into the mapping M and get

$$(M \circ L)\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = \begin{bmatrix} 3c \\ c + 2d \end{bmatrix}.$$

• The composition $L \circ M$ is a mapping from $\mathbb{R}_{\leq 1}[x]$ to $\mathbb{R}_{\leq 1}[x]$, whereby

$$(L \circ M)(cx + d) = L\left(M\left(cx + d\right)\right) = L\left(\begin{bmatrix} 3c \\ c - 2d \end{bmatrix}\right).$$

We set $3c$ for a and $c - 2d$ for b into the mapping L and get

$$L\left(\begin{bmatrix} 3c \\ c - 2d \end{bmatrix}\right) = 3cx - (c - 2d).$$

Definition (invertibility of linear mappings)

Let V, W be vector spaces over the field $\mathbb{F}$ and $L : V \to W$ a linear mapping.

L is **invertible** if there exists a linear mapping $M : W \to V$ such that $(M \circ L)(\vec{v}) = \vec{v}$ for all vectors $\vec{v} \in V$.

Comments

Let $\boldsymbol{L} : V \to W$ be invertible with inverse $\boldsymbol{M}$.

• $M \circ L = I_V$ is the identity mapping: $I_V(\vec{v}) = \vec{v}$ for all $\vec{v} \in V$.

• Notation: $L^{-1} := \boldsymbol{M}$

• If dim(V) is finite, then $L \circ \boldsymbol{M} = I_W$.

Example (inverse mappings)

Are the mappings

$$L : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x]; ax + b \mapsto (2a - 3b)x + (7b - 5a) \text{ und}$$
$$M : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x]; \boldsymbol{a}x + \boldsymbol{b} \mapsto -(7\boldsymbol{a} + 3\boldsymbol{b})x - (5\boldsymbol{a} + 2\boldsymbol{b})$$

inverse mappings?

L and M are inverse mappings since $\mathbb{R}_{\leq 1}[x]$ is finite dimensional and $M \circ L = I_{\mathbb{R}_{\leq 1}[x]}$:

$$\begin{aligned}
(M \circ L)(\boldsymbol{cx + d}) &= M(L(cx + d)) \\
&= M(\underbrace{(\mathbf{2c - 3d})x + (\mathbf{7d - 5c})}_{L(cx+d)}) \\
&= -(7\underbrace{(\mathbf{2c - 3d})}_{\boldsymbol{a}} + 3\underbrace{(\mathbf{7d - 5c})}_{\boldsymbol{b}})x - (5\underbrace{(\mathbf{2c - 3d})}_{\boldsymbol{a}} + 2\underbrace{(\mathbf{7d - 5c})}_{\boldsymbol{b}}) \\
&= -(14c - 21d + 21d - 15c)x - (10c - 15d + 14d - 10c) \\
&= \boldsymbol{cx + d}.
\end{aligned}$$

6.4 Kernel, image, solution space of a system of linear equations

Theorem (image of a subspace is a subspace)

A linear mapping $L : V \to W$ retains the structure of a vector space, i.e. if U is a subspace of V, then $L(U)$ (the image of U) is a subspace of W.

Proof: Let U be a subspace of V. We need to show that the set $L(U)$ is not empty and that $L(U)$ is closed with respect to addition and multiplication with scalars.

1. Since $\vec{0}_V \in U$ and L is a linear mapping, it follows that $L(\vec{0}_V) = \vec{0}_W \in L(U)$, so that $L(U) \neq \emptyset$.
2. Let $\vec{w}_1, \vec{w}_2 \in L(U)$, i.e. there exist $\vec{u}_1, \vec{u}_2 \in U$ with $L(\vec{u}_1) = \vec{w}_1, L(\vec{u}_2) = \vec{w}_2$. To show: $\vec{w}_1 + \vec{w}_2 \in L(U)$.
$$\vec{w}_1 + \vec{w}_2 = L(\vec{u}_1) + L(\vec{u}_2) \overset{\text{linearity}}{=} L(\vec{u}_1 + \vec{u}_2)$$
Since U is a subspace of V, $\vec{u}_1 + \vec{u}_2 \in U$, so that
$$\vec{w}_1 + \vec{w}_2 = L(\vec{u}_1 + \vec{u}_2) \in L(U). \quad \checkmark$$
3. Let $\vec{w} \in L(U), \alpha \in \mathbb{F}$, i.e. there exists a $\vec{u} \in U$ with $L(\vec{u}) = \vec{w}$. To show: $\alpha\vec{w} \in L(U)$.
$$\alpha\vec{w} = \alpha L(\vec{u}) \overset{\text{Linearität}}{=} L(\alpha\vec{u})$$
Since U is a subspace of V, it follows that $\alpha\vec{u} \in U$ so that $\alpha\vec{w} = L(\alpha\vec{u}) \in L(U)$ holds. $\checkmark$

$L(U)$ is a subspace of W.

Due to this fact, it is possible to apply theorem of linear algebra to the image of a subspace because it is itself a vector space.

Definition (injectivity)

Let $L : V \to W$ be a linear mapping. L is called **injective** if for all $\vec{v}_1, \vec{v}_2 \in V$,

$$L(\vec{v}_1) = L(\vec{v}_2) \Rightarrow \vec{v}_1 = \vec{v}_2.$$

The statement "L is **injective**" is equivalent to the following statements:

$$\mathbf{ker}(\boldsymbol{L}) = \{\vec{0}\} \text{ and } \mathbf{dim}(\mathbf{ker}(\boldsymbol{L})) = \mathbf{0}$$

We verify the first equivalence.

$(\Rightarrow)$: Assume the linear mapping L is injective. Let $\vec{v}_1$ and $\vec{v}_2 := \vec{0}$ be elements in the kernel of L. Due to the definition of the kernel, it follows that $L(\vec{v}_1) = \vec{0} = L(\vec{0})$. Now, since L is injective, it follows that $\vec{v}_1 = \vec{0}$. Thus $\ker(L) = \{\vec{0}\}$.

$(\Leftarrow)$: Assume that $\ker(L) = \{\vec{0}\}$. If there exist $\vec{v}_1, \vec{v}_2 \in V$ with

$$L(\vec{v}_1) = L(\vec{v}_2),$$

we can rewrite the equation as

$$L(\vec{v}_1) - L(\vec{v}_2) = \vec{0}.$$

Because of the linearity of the mapping, this is the same as

$$L(\vec{v}_1 - \vec{v}_2) = \vec{0},$$

i.e. $\vec{v}_1 - \vec{v}_2 \in \ker(L)$. By assumption, the kernel of L only contains the zero vector so that $\vec{v}_1 - \vec{v}_2 = \vec{0}$ or equivalently $\vec{v}_1 = \vec{v}_2$. The definition of injectivity is thus satisfied.

Comment: Yet another formulation of the definition says that L is injective if every element in the image L has at most one element in its preimage.

Definition (surjectivity)

Let $L : V \to \boldsymbol{W}$ be a linear mapping. L is called **surjective** if

$$\{L(\vec{v}) \,|\, \vec{v} \in V\} = \boldsymbol{W}.$$

If V is finite dimensional, then L is **surjective**, exactly when

$$\mathbf{im}(\boldsymbol{L}) = \boldsymbol{W}, \text{ d.h. } \mathbf{dim}(\mathbf{im}(\boldsymbol{L})) = \mathbf{dim}(\boldsymbol{W}).$$

Comments:

• Another wording of the definition: L is surjective if every element in the image space W has at least one element in its preimage.

• This definition demonstrates that there is potentially a difference between the target space and the actual image of the mapping. Consider $L : \mathbb{R}^3 \to \mathbb{R}^2; \vec{v} \mapsto \vec{0}$. The target space is $\mathbb{R}^2$, but the image of L is $\{\vec{0}\}$. This mapping is definitely not surjective.

Definition (bijectivity)

Let $L : V \to W$ be a linear mapping. L is called **bijective** if L is **injective and surjective**.

Comments:

• An alternative formulation uses the notion of preimage. L is bijective if every vector in W has exactly one element in its preimage.

• The linear mapping $L : V \to V$ is bijective if and only if L is invertible.

Example (injectivity, surjectivity, bijectivity)

For each of the following linear mappings, decide which of the properties injective/surjective/bijective hold.

$L_1 : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x];\ ax + b \mapsto bx + a$
$L_2 : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x];\ ax + b \mapsto bx + b$
$L_3 : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 2}[x];\ ax + b \mapsto ax^2 + (a + b)x + b$
$L_4 : \mathbb{R}_{\leq 2}[x] \to \mathbb{R}_{\leq 1}[x];\ ax^2 + bx + c \mapsto cx + (a + b)$

• L_1 is **injective** because

$$L_1(ax + b) = bx + a = 0 = 0x + 0(i.e. a = b = 0)$$

$$\text{implies } \ker(L_1) = \{\vec{0}\}.$$

L_1 is **surjective** because the preimage of $ax + b \in \mathbb{R}_{\leq 1}[x]$ contains exactly one element, namely, the polynomial $bx + a$.
L_1 is **bijective** because it is injective and surjective.

• L_2 is **neither injective**

$$(\text{from } L(2x) = L(2x + 0) = 0x + 0 \text{ it follows that } 2x \in \ker(L_2),\ \text{i.e. } \ker(L_2) \neq \{\vec{0}\})$$

nor surjective (for example, the preimage of $3x + 1$ is empty because $3x + 1 \neq b(x + 1)$ for all $b \in \mathbb{R}$).
L_2 is **not bijective** because it is not injective (alternatively it is not surjective).

• L_3 is **injective**: For $ax + b \in \ker(L_3)$ we get

$$L_3(ax + b) = ax^2 + (a + b)x + b = 0x^2 + 0x + 0.$$

Comparing the coefficients of x^2 and x^0 we obtain $a = 0$ resp. $b = 0$, i.e. $\ker(L_3) = \{\vec{0}\}$.

L_3 is, however, **not surjective**. For example, the preimage of $x^2 + 1$ is the empty set:
$L_3(ax + b) = ax^2 + (a + b)x + b = x^2 + 1$ leads to the SLE $a = 1,\ a + b = 0,\ b = 1$, which has no solution.

L_3 is **not bijective** because it is not simultaneously injective and surjective.

• L_4 is **not injective** because $L_4(x^2 - x) = \vec{0}$ implies $x^2 - x \in \ker(L_4)$, i.e. $\ker(L_4) \neq \{\vec{0}\}$.

L_4 is **surjective**. The preimage of $ax + b \in \mathbb{R}_{\leq 1}[x]$ contains $bx + a$ ($L_4(bx + a) = ax + b$).

L_4 is **not bijective** because it is not simultaneously injective and surjective.

Theorem (dimension theorem)

Let V, W be vector spaces over the field $\mathbb{F}$ and $L : V \to W$ a linear mapping. The following equation holds.:

$$\dim(V) = \dim(\ker(L)) + \dim(\text{im}(L)).$$

In the previous example, L_4 is a lineare mapping from a 3-dimensional vector space $V := \mathbb{R}_{\leq 2}[x]$ into a 2-dimensional vector space $W := \mathbb{R}_{\leq 1}[x]$. Thus $\dim(\text{im}(L_4))$ can be at most 2. From the dimension theorem, it follows that $\dim(\ker(L_4))$ has to be at least one so that L_4 cannot be injective. (This observation is true for

every linear mapping from $\mathbb{R}_{\leq 2}[x]$ to $\mathbb{R}_{\leq 1}[x]$.)

The dim(im(L_4)) is also at least 2 because the images $L(1) = x$ and $L(x^2) = 1$ are linearly independent. Thus the dimension of im(L_4) is simultaneously at most two and at least two, so exactly two. The dimension theorem tells us that dim(ker(L_4)) $= 1$.

Exercise 2

Under which conditions can a linear mapping $L : \mathbb{R}^{3,2} \to \mathbb{R}^{2,4}$ be injective / surjective / bijective?

Theorem (solution set of an abstract SLE)
Let V, W be vector spaces over the field $\mathbb{F}$, $L : V \to W$ a linear mapping, and $\vec{b} \in W$.
The solution set of the SLE $L(\vec{x}) = \vec{b}$ is given by

$$\mathcal{S} := \{\vec{x} \in V \mid L(\vec{x}) = \vec{b}\} = \{\vec{x}_P + \vec{x}_H \mid \vec{x}_H \in \text{ker}(L)\}.$$

$\uparrow$ particular (or special) solution

Recall that $\vec{x}_p$ is any solution of the SLE, i.e. $L(\vec{x}_p) = \vec{b}$, and $\vec{x}_h$ is any solution of the homogeneous SLE, i.e. $L(\vec{x}_h) = \vec{0}$. The sum is also a solution of the original SLE with right-hand side $\vec{b}$:

$$L(\vec{x}_p + \vec{x}_h) \overset{\text{linearity}}{=} L(\vec{x}_p) + L(\vec{x}_h) = \vec{b} + \vec{0} = \vec{b}.$$

Thus $\vec{x}_p + \vec{x}_h$ also solves the SLE.

Example
Determine the solution set of the SLE $L_4(q) = 5x + 3$ whereby

$$L_4 : \mathbb{R}_{\leq 2}[x] \to \mathbb{R}_{\leq 1}[x];\ ax^2 + bx + c \mapsto cx + (a + b).$$

From previous work, we know that dim(ker(L_4)) $= 1$ and $x^2 - x \in$ ker(L_4). Thus $\{x^2 - x\}$ must be a basis of the kernel of L_4. Furthermore,

$$q_p = 3x^2 + 5$$

is a particular solution of the equation

$$L_4(q) = 5x + 3.$$

The solution set is thus

$$\mathcal{S} = \left\{3x^2 + 5 + s(x^2 - x) \mid s \in \mathbb{R}\right\}.$$

Exercise 3

A linear mapping $L : \mathbb{R}^{2,2} \to \mathbb{R}^2$ is given by:

$$L\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad L\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ -7 \end{bmatrix},$$

$$L\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad L\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Determine the following:

(a) $L\left(\begin{bmatrix} 3 & -2 \\ 2 & 17 \end{bmatrix}\right)$, $= L\left(3\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + (-2)\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 2\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 17\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right)$

$= 3L\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) - 2L\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) + 2L\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) + 17L\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right)$

(b) the solution set of the SLE $L(\vec{x}) = \begin{bmatrix} 1 \\ 16 \end{bmatrix}$, $= 3\begin{bmatrix} 2 \\ 3 \end{bmatrix} - 2\begin{bmatrix} 5 \\ -7 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \vec{0}$

(c) a basis for $\ker(L)$,

(d) a basis for $\operatorname{im}(L)$, $= \begin{bmatrix} -2 \\ 25 \end{bmatrix}.$

(e) whether the mapping L is injective/surjective/bijective.

6.5 Suggested solutions to the exercises in Chapter 6

Exercise 1 $L_i : V_i \to W_i$ $(i = 1, 2, 3)$ is a linear mapping if for all $\vec{u}, \vec{v} \in V_i$, $\alpha \in \mathbb{R}$ the following conditions are satisfied:

$$L_i(\vec{u} + \vec{v}) = L_i(\vec{u}) + L_i(\vec{v}), \quad (+ \text{ addition in } V_i \ + \text{ addition in } W_i),$$
$$L_i(\alpha \cdot \vec{v}) = \alpha \cdot L_i(\vec{v}), \quad (\cdot \text{ scalar multiplication in } V_i \ \cdot \text{ scalar multiplication in } W_i).$$

If L is a linear mapping, then with $\alpha = 0$: $L(0 \cdot \vec{0}) = 0L(\vec{0}) = \vec{0}$. This means that if $L_i(\vec{0}) \neq \vec{0}$, then L_i cannot be a linear mapping.

- $L_1(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$. L_1 might be linear. We check the two conditions. For all $A := [a_{i,j}]$, $B := [b_{i,j}] \in \mathbb{R}^{2,3}$, and $\alpha \in \mathbb{R}$:

  1. $$\boldsymbol{L_1(A+B)} = L_1([a_{i,j}] + [b_{i,j}]) = L_1([a_{i,j} + b_{i,j}])$$
  $$= [a_{j,i} + b_{j,i}] = [a_{j,i}] + [b_{j,i}] = A^T + B^T = \boldsymbol{L_1(A)+L_1(B)},$$

  2. $$\boldsymbol{L_1(\alpha \cdot A)} = L_1(\alpha \cdot [a_{i,j}]) = L_1([\alpha \cdot a_{i,j}])$$
  $$= [\alpha \cdot a_{j,i}] = \alpha \cdot [a_{j,i}] = \alpha \cdot A^T = \boldsymbol{\alpha \cdot L_1(A)}.$$

  The conditions are satisfied so that L_1 is a linear mapping.

- L_2 is not a linear mapping since $L_2(0x^2 + 0x + 0) = 0x^2 + \mathbf{1}x + 0 \neq 0x^2 + 0x + 0$.

- $L_3(0x + 0) = 0$, which means that L_3 has a shot at being linear. Note though that $x, 1 \in \mathbb{R}_{\leq 2}[x]$ is on the one hand

$$L_3(x) + L_3(1) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & \mathbf{0} \end{bmatrix}.$$

On the other hand, we get
$$L_3(x+1) = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \neq L_3(x) + L_3(1).$$
So L_3 cannot be linear.

Alternatively (in case you did not find a counterexample): Let's check the first condition of a linear mapping. Choose $ax^2+bx+c,\ dx^2+ex+f \in \mathbb{R}_{\leq 2}[x]$.

1. On the one hand,
$$\begin{aligned} L_3((ax^2+bx+c)+(dx^2+ex+f)) &= L_3((a+d)x^2+(b+e)x+(c+f)) \\ = \begin{bmatrix} a+d & b+e \\ a+d & (b+e)(c+f) \end{bmatrix} &= \begin{bmatrix} a+d & b+e \\ a+d & \boldsymbol{bc}+\boldsymbol{bf}+\boldsymbol{ec}+\boldsymbol{ef} \end{bmatrix}. \end{aligned}$$
On the other hand,
$$\begin{aligned} L_3(ax^2+bx+c)+L_3(dx^2+ex+f) &= \begin{bmatrix} a & b \\ a & bc \end{bmatrix} + \begin{bmatrix} d & e \\ d & ef \end{bmatrix} \\ &= \begin{bmatrix} a+d & b+e \\ a+d & \boldsymbol{bc}+\boldsymbol{ef} \end{bmatrix} \end{aligned}$$
$L_3((ax^2+bx+c)+(dx^2+ex+f)) \neq L_3(ax^2+bx+c)+L_3(dx^2+ex+f)$ for $\boldsymbol{bf}+\boldsymbol{ec} \neq 0$ (Choose, for example, $b=f=1, c=e=0$). Thus L_3 is not a linear mapping.

Exercise 2 According to the dimension theorem
$$\dim(\mathbb{R}^{3,2}) = 6 = \dim(\ker(L)) + \dim(\operatorname{im}(L)).$$
This means that $\dim(\operatorname{im}(L)) \leq 6 < 8 = \dim(\mathbb{R}^{2,4})$, so that L cannot be surjective. Consequently, L cannot be bijective.

L is injective if $\dim(\ker(L)) = 0$ (i.e. $\dim(\operatorname{im}(L)) = 6$) gilt.

Exercise 3 (a)
$$\begin{aligned} L\left(\begin{bmatrix} 3 & -2 \\ 2 & 17 \end{bmatrix}\right) &= L\left(3\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + (-2)\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 2\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 17\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) \\ &\overset{\text{linearity}}{=} L\left(3\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) + L\left(-2\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) + L\left(2\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) + L\left(17\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) \\ &\overset{\text{linearity}}{=} 3L\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) - 2L\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) + 2\boldsymbol{L}\left(\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{1} & \mathbf{0} \end{bmatrix}\right) + 17\boldsymbol{L}\left(\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{bmatrix}\right) \\ &= 3\begin{bmatrix} 2 \\ 3 \end{bmatrix} - 2\begin{bmatrix} 5 \\ -7 \end{bmatrix} + 2\begin{bmatrix} \mathbf{1} \\ \mathbf{1} \end{bmatrix} + 17\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ 9 \end{bmatrix} + \begin{bmatrix} -10 \\ 14 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 25 \end{bmatrix} \end{aligned}$$

(b) We are looking for all matrices $\begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} \in \mathbb{R}^{2,2}$ mit $L\left(\begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 16 \end{bmatrix}$:
$$\begin{aligned} \begin{bmatrix} 1 \\ 16 \end{bmatrix} &= L\left(\begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}\right) \\ &= L\left(x_1\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + x_2\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + x_3\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + x_4\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) \\ &= x_1L\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) + x_2L\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) + x_3L\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) + x_4L\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) \\ &= x_1\begin{bmatrix} 2 \\ 3 \end{bmatrix} + x_2\begin{bmatrix} 5 \\ -7 \end{bmatrix} + x_3\begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_4\begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned}$$

Coefficient comparison leads to the SLE

$$\left\{\begin{array}{rcrcrcrcr} 2x_1 & + & 5x_2 & + & 1x_3 & + & 0x_4 & = & 1 \\ 3x_1 & - & 7x_2 & + & 1x_3 & + & 0x_4 & = & 16. \end{array}\right.$$

Bring AM into rref:

$$\stackrel{\text{AM}}{\to} \left[\begin{array}{rrrr|r} 2 & 5 & 1 & 0 & 1 \\ 3 & -7 & 1 & 0 & 16 \end{array}\right] \stackrel{2\text{II}-3\text{I}}{\to} \left[\begin{array}{rrrr|r} 2 & 5 & 1 & 0 & 1 \\ 0 & -29 & -1 & 0 & 29 \end{array}\right] \stackrel{\frac{1}{2}\text{I},-\frac{1}{29}\text{II}}{\to} \left[\begin{array}{rrrr|r} 1 & \frac{5}{2} & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{29} & 0 & -1 \end{array}\right]$$

$$\stackrel{\text{I}-\frac{5}{2}\text{II}}{\to} \left[\begin{array}{rrrr|r} 1 & 0 & \frac{12}{29} & 0 & 3 \\ 0 & 1 & \frac{1}{29} & 0 & -1 \end{array}\right].$$

This leads to the equations

$$\begin{array}{rcrcrcr} x_1 & + & & & \frac{12}{29}x_3 & = & 3 \\ & & x_2 & + & \frac{1}{29}x_3 & = & -1. \end{array}$$

leading coefficient variables: x_1, x_2
nonleading coefficient variables: x_3, x_4 (free parameters: s for x_3, t for x_4).

Solve equations for the lcv's:

$$x_1 = 3 - \frac{12}{29}s \qquad x_2 = -1 - \frac{1}{29}s$$

The solution set is thus

$$\mathcal{S} = \left\{ \left[\begin{array}{rr} 3 - \frac{12}{29}s & -1 - \frac{1}{29}s \\ s & t \end{array}\right] \mid s, t \in \mathbb{R} \right\}.$$

(c) Another representation (particular solution + ker(L))

$$\mathcal{S} = \left\{ \left[\begin{array}{rr} 3 & -1 \\ 0 & 0 \end{array}\right] + s \left[\begin{array}{rr} -\frac{12}{29} & -\frac{1}{29} \\ 1 & 0 \end{array}\right] + t \left[\begin{array}{rr} 0 & 0 \\ 0 & 1 \end{array}\right] \mid s, t \in \mathbb{R} \right\}$$

lets us read off the kernel of L:

$$\ker(L) = \left\{ s \left[\begin{array}{rr} -\frac{12}{29} & -\frac{1}{29} \\ 1 & 0 \end{array}\right] + t \left[\begin{array}{rr} 0 & 0 \\ 0 & 1 \end{array}\right] \mid s, t \in \mathbb{R} \right\}.$$

A basis for ker(L) can be read off:

$$\left\{ \left[\begin{array}{rr} -\frac{12}{29} & -\frac{1}{29} \\ 1 & 0 \end{array}\right], \left[\begin{array}{rr} 0 & 0 \\ 0 & 1 \end{array}\right] \right\}.$$

(d) im(L) consists of all linear combinations of the image vectors:

$$\text{Bild}(L) = \text{span}\left\{ \left[\begin{array}{r} 2 \\ 3 \end{array}\right], \left[\begin{array}{r} 5 \\ -7 \end{array}\right], \left[\begin{array}{r} 1 \\ 1 \end{array}\right], \left[\begin{array}{r} 0 \\ 0 \end{array}\right] \right\} = \text{span}\left\{ \left[\begin{array}{r} 2 \\ 3 \end{array}\right], \left[\begin{array}{r} 5 \\ -7 \end{array}\right] \right\}$$

because

$$\begin{array}{rcrcr} \left[\begin{array}{r} 0 \\ 0 \end{array}\right] & = & 0\left[\begin{array}{r} 2 \\ 3 \end{array}\right] & + & 0\left[\begin{array}{r} 5 \\ -7 \end{array}\right] \\ \left[\begin{array}{r} 1 \\ 1 \end{array}\right] & = & \frac{12}{29}\left[\begin{array}{r} 2 \\ 3 \end{array}\right] & + & \frac{1}{29}\left[\begin{array}{r} 5 \\ -7 \end{array}\right]. \end{array}$$

We can argue that the vectors $\begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ -7 \end{bmatrix}$ are linearly independent.

$$\alpha_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 5 \\ -7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ leads to the SLE } \begin{array}{rcrcr} 2\alpha_1 & + & 5\alpha_2 & = & 0 \\ 3\alpha_1 & - & 7\alpha_2 & = & 0. \end{array}$$

Bring the AM in rref:

$$\begin{bmatrix} 2 & 5 \\ 3 & -7 \end{bmatrix} \overset{\mathrm{I}-\frac{2}{3}\mathrm{II}}{\rightarrow} \begin{bmatrix} 2 & 5 \\ 0 & \frac{29}{3} \end{bmatrix}.$$

The rank of the AM is 2 (2 leading coefficients in ref, i.e. 2 linear independent rows resp. columnsin AM). The set $\Rightarrow \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ -7 \end{bmatrix} \right\}$ is also a basis of the image of L.

(e) L is not injective because $\ker(L) \neq \{\vec{0}\}$ (for example, $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in \ker(L)$).

L is surjective because $\dim(\mathrm{im}(L)) = 2 = \dim(\mathbb{R}^2) =$ dimension of the image space.

L is not bijective since L is not simultaneously injective and surjective.

7 Coordinate mappings and matrix representatives

Recall that there is a difference between a generating set and a basis. Each basis is a generating set, but not every generating set is a basis. The set $G := \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is a **generating set** of $\mathbb{R}^2$ because G is clearly a subset of $\mathbb{R}^2$ and every vector $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$ can be represented as a linear combination of the three vectors

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad + \quad b \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad + \quad 0 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad = \quad \begin{bmatrix} a \\ b \end{bmatrix} \qquad (1)$$

or also

$$(a-b) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad + \quad 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad + \quad b \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad = \quad \begin{bmatrix} a \\ b \end{bmatrix}. \qquad (2)$$

The *scalars* or *coefficients* before the vectors in this generating set can obviously not be defined uniquely if $b \neq 0$.

The vectors are linearly **dependent** because from $(1) - (2)$ it follows that $b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} - b \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, i.e. the equation

$$\alpha_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

has a nontrivial solution.

For a **basis**, i.e. a linearly **independent** generating set, this cannot happen. For a given finite-dimensional vector space $V \neq \{\vec{0}\}$ with a given basis $\mathcal{B} := \{\vec{b}_1, \ldots, \vec{b}_n\}$ of V, every vector in V has at least one representation as a linear combination of the basis vectors since $\mathcal{B}$ is a generating set. The fact that $\mathcal{B}$ is linearly independent means that there is at most one such representation as a linear combination of the basis vector in $\mathcal{B}$. If there were at least two such representations of the vector $\vec{v} \in V$, we would have

$$\alpha_1 \vec{b}_1 + \cdots + \alpha_n \vec{b}_n = \vec{v}, \qquad (3)$$

$$\beta_1 \vec{b}_1 + \cdots + \beta_n \vec{b}_n = \vec{v}, \qquad (4)$$

so that we obtain through (3) $-$ (4) the equation

$$(\alpha_1 - \beta_1)\vec{b}_1 + \cdots + (\alpha_n - \beta_n)\vec{b}_n = \vec{0}.$$

The linear independence of the vectors means that $\alpha_i - \beta_i = 0$ for $i = 1, \ldots, n$. Each vector $\vec{v} \in V$ thus has to have exactly one representation as a linear combination of the given basis vectors. The coefficients of this unique linear combination defines the coordinate vector $\vec{v}_{\mathcal{B}}$ of $\vec{v}$ with respect to the basis $\mathcal{B}$. The coordinate mapping takes each vector $\vec{v} \in V$ and sends it to the associated coordinate vector $\vec{v}_{\mathcal{B}}$.

You might be wondering why this is of interest. Surely you noticed that the composition of mappings of $\mathbb{R}^n \to \mathbb{R}^n$ is very simple when calculated with the associated matrices. Only one matrix multiplication has to be carried out here, instead of inserting one mapping into the other. The more zeros in the matrix, the easier. It is particularly easy to do matrix multiplication when one of the matrices is diagonal.

Furthermore a finite-dimensional abstract vector space can be identified with $\mathbb{F}^n$ using coordinates and coordinate mappings. Dealing with vectors in $\mathbb{F}^n$ is usually easier.

7.1 Coordinate vectors and coordinate mappings

Definition (coordinate vector; coordinate mapping)

Given a vector space V of dimension n over the field $\mathbb{F}$, let $\mathcal{B} := \{\vec{b}_1, \ldots, \vec{b}_n\}$ be a basis of V. Assume $\vec{v} = \alpha_1\vec{b}_1 + \cdots + \alpha_n\vec{b}_n$ is the unique linear combination of the basis vectors in $\mathcal{B}$ yielding $\vec{v} \in V$.

The **coordinate vector** of $\vec{v}$ with respect to the basis $\mathcal{B}$ is given by $\vec{v}_{\mathcal{B}} := \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$.

The **coordinate mapping** of V with respect to $\mathcal{B}$ is given by

$$K_{\mathcal{B}} : V \to \mathbb{F}^n; \vec{v} \mapsto \vec{v}_{\mathcal{B}} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}.$$

This mapping sends every vector in V onto its associated coordinate vector with respect to the basis $\mathcal{B}$.

Remarks

- If the basis changes, the coordinates of the vector generally change as well.
- The order of the basis vectors must be taken into account. We consider a basis as an ordered set.
- Coordinate mappings are linear.

Example (coordinate mapping)

Let

$$\mathcal{B}_1 := \{3x + 7, 2x + 5\}$$

be a basis of the vector space $\mathbb{R}_{\leq 1}[x]$. Find the coordinate mapping of $\mathbb{R}_{\leq 1}[x]$ with respect to the basis $\mathcal{B}_1$. Check your answer.

The coordinate mapping of $\mathbb{R}_{\leq 1}[x]$ with respect to $\mathcal{B}_1$ sends each polynomial $p \in \mathbb{R}_{\leq 1}[x], p(x) := ax + b$ onto the associated coordinate vector with respect to the basis $\mathcal{B}_1 := \{3x + 7, 2x + 5\}$, i.e.

$$\begin{array}{rrcl} K_{\mathcal{B}_1} : & \mathbb{R}_{\leq 1}[x] & \to & \mathbb{R}^2 \\ & ax + b & \mapsto & \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \end{array} \quad \text{for} \quad \alpha_1(3x + 7) + \alpha_2(2x + 5) = ax + b.$$

Comparing coefficients leads to the following SLE:

$$\begin{array}{rcrcrcl} x^1 : & & 3\alpha_1 & + & 2\alpha_2 & = & a \\ x^0 = 1 : & & 7\alpha_1 & + & 5\alpha_2 & = & b. \end{array}$$

The augmented matrix of the SLE is put in rref using Gaussian elimination.

$$\left[\begin{array}{cc|c} 3 & 2 & a \\ 7 & 5 & b \end{array}\right] \overset{\text{II}-2\text{I}}{\to} \left[\begin{array}{cc|c} 3 & 2 & a \\ 1 & 1 & -2a + b \end{array}\right]$$

We try to get a 1 in the first column without dividing. (This is not always possible.)

$$\overset{\text{I}\leftrightarrow\text{II}}{\to} \left[\begin{array}{cc|c} 1 & 1 & -2a + b \\ 3 & 2 & a \end{array}\right]$$

$$\overset{\text{II}-3\text{I}}{\to} \left[\begin{array}{cc|c} 1 & 1 & -2a + b \\ 0 & -1 & 7a - 3b \end{array}\right] \overset{\text{I+II},-\text{II}}{\to} \left[\begin{array}{cc|c} 1 & 0 & 5a - 2b \\ 0 & 1 & -7a + 3b \end{array}\right].$$

The coordinate mapping is therefore

$$\begin{array}{rrcl} K_{\mathcal{B}_1} : & \mathbb{R}_{\leq 1}[x] & \to & \mathbb{R}^2 \\ & ax + b & \mapsto & \begin{bmatrix} 5a - 2b \\ -7a + 3b \end{bmatrix}. \end{array} \tag{6}$$

The coordinate mapping can be checked by plugging in the basis vectors in $\mathcal{B}_1$. The resulting coordinate vector for the ith basis vector is just the ith standard basis vector because

$$\mathbf{3x+7} = 1(3x+7) + \mathbf{0}(\mathbf{2x+5}) \qquad \mathbf{2x+5} = 0(3x+7) + \mathbf{1}(\mathbf{2x+5}).$$

Now we calculate using the coordinate mapping to check our work.

$$K_{\mathcal{B}_1}(\mathbf{3x+7}) = \begin{bmatrix} 5(3)-2(7) \\ -7(3)+3(7) \end{bmatrix} = \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix} \qquad K_{\mathcal{B}_1}(\mathbf{2x+5}) = \begin{bmatrix} 5(2)-2(5) \\ -7(2)+3(5) \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{1} \end{bmatrix}.$$

Because of the linearity of $K_{\mathcal{B}_1}$, we can now demonstrate that if the coordinate mapping produces the standard basis vectors for the basis vectors in $K_{\mathcal{B}_1}$, then the mapping is correct for any arbitrary vector in V. Let $\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$ be the coordinate vector of the polynomial $ax+b \in \mathbb{R}_{\leq 1}[x]$ with respect to the basis $\mathcal{B}_1$, i.e. $ax+b = \alpha_1(3x+7) + \alpha_2(2x+5)$. We wish to check that $K_{\mathcal{B}_1}$ really takes $ax+b$ to its coordinate vector with respect to the basis $\mathcal{B}_1$:

$$K_{\mathcal{B}_1}(ax+b) = K_{\mathcal{B}_1}(\alpha_1(3x+7)+\alpha_2(2x+5)) = \alpha_1 K_{\mathcal{B}_1}(3x+7)+\alpha_2 K_{\mathcal{B}_1}(2x+5) = \alpha_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}.$$

This argument is true in general.

Exercise 1 (learning check)

Given the basis $\mathcal{B}_2 := \{x+2, x+3\}$ of the vector space $\mathbb{R}_{\leq 1}[x]$, determine the coordinate mapping of $\mathbb{R}_{\leq 1}[x]$ with respect to $\mathcal{B}_2$. Check your answer.

Remark: Coordinate mappings are not only linear, they are also invertible. These maps are very easy to invert.

Example (inverse of a coordinate mapping)

Let $\mathcal{B}_1 := \{3x+7, \mathbf{2x+5}\}$ be the basis for $\mathbb{R}_{\leq 1}[x]$ from the previous example. Find the inverse: $K_{\mathcal{B}_1}^{-1}$.

$K_{\mathcal{B}_1}$ is the coordinate mapping from $\mathbb{R}_{\leq 1}[x]$ to $\mathbb{R}^2$ with respect to the basis $\mathcal{B}_1$. It follows that $K_{\mathcal{B}_1}^{-1}$ is a mapping from $\mathbb{R}^2$ to $\mathbb{R}_{\leq 1}[x]$:

$$K_{\mathcal{B}_1}^{-1} : \mathbb{R}^2 \to \mathbb{R}_{\leq 1}[x]; \quad \begin{bmatrix} c \\ d \end{bmatrix} \mapsto fx+g.$$

What is $fx+g$? This is the (unique!) polynomial in $\mathbb{R}_{\leq 1}[x]$ with coordinate vector $\begin{bmatrix} c \\ \boldsymbol{d} \end{bmatrix}$ with respect to the given basis (otherwise $K_{\mathcal{B}_1}$ would not map the polynomial $fx+g$ to $\begin{bmatrix} c \\ \boldsymbol{d} \end{bmatrix}$). Now we just need to calculate an appropriate linear combination of the basis polynomials: $fx+g = c(3x+7) + \boldsymbol{d}(\mathbf{2x+5}) = (3c+2d)x + (7c+5d)$. The inverse of the coordinate mapping $K_{\mathcal{B}_1}$ is hence

$$K_{\mathcal{B}_1}^{-1} : \mathbb{R}^2 \to \mathbb{R}_{\leq 1}[x]; \quad \begin{bmatrix} c \\ d \end{bmatrix} \mapsto (3c+2d)x + (7c+5d).$$

Exercise 2 (learning check)

Let $\mathcal{B}_2 := \{x+2, x+3\}$ be the basis of $\mathbb{R}_{\leq 1}[x]$ from the previous exercise. Find $K_{\mathcal{B}_2}^{-1}$.

7.2 Transition matrix or change of basis matrix

In this section we consider an n-dimensional vector space V over the field $\mathbb{F}$ with two bases $\mathcal{B}_1$, $\mathcal{B}_2$. How do we get from the coordinate vector of a given vector $\vec{v} \in V$ with respect to (wrt) $\mathcal{B}_1$ ($\vec{v}_{\mathcal{B}_1}$) to the coordinate vector of $\vec{v}$ wrt $\mathcal{B}_2$ ($\vec{v}_{\mathcal{B}_2}$)? We can do this with a matrix mapping because it is a linear mapping from $\mathbb{F}^n$ to $\mathbb{F}^n$. This matrix map S is the so-called transition matrix or change of basis matrix going from $\mathcal{B}_1$ to $\mathcal{B}_2$.

Definition (transition or change of basis matrix)

Let V be a vector space over $\mathbb{F}$. The transition matrix $T \in \mathbb{F}^{n,n}$ when changing from the basis $\mathcal{B}_1$ to the basis $\mathcal{B}_2$ of V is the matrix mapping associated with the mapping $K_{\mathcal{B}_2} \circ K_{\mathcal{B}_1}^{-1}$. We write $T = K_{\mathcal{B}_2} \circ K_{\mathcal{B}_1}^{-1}$, even though the two mappings on the right are generally not matrix mappings.

To illustrate the situation, consider the following *commutative diagram*.

$$\begin{tikzcd}
& V \arrow[dl, "K_{\mathcal{B}_2}"'] \arrow[dr, "K_{\mathcal{B}_1}"] & \\
\mathbb{F}^n & & \mathbb{F}^n \arrow[ll, "T"]
\end{tikzcd}$$

The diagram is called commutative because it does not matter whether we directly map $\vec{v} \in V$ to the copy of $\mathbb{F}^n$ on the left-hand side using $K_{\mathcal{B}_2}$ or if we first apply $K_{\mathcal{B}_1}$ and then use T. The same result is obtained.

$$K_{\mathcal{B}_2}(\vec{v}) = T(K_{\mathcal{B}_1}(\vec{v})) \text{ for all } \vec{v} \in V, \text{ i.e. } K_{\mathcal{B}_2} = T \circ K_{\mathcal{B}_1}.$$

(As explained in Section 3.4, we write the composition from right to left.)

The diagram can be read like a map. The arrows show in which direction the "streets" run: $K_{\mathcal{B}_1}, K_{\mathcal{B}_2}$ (between V and a copy of $\mathbb{F}^n$) and T (between the two copies of $\mathbb{F}^n$). Since coordinate mappings are invertible, it is possible to "drive" in the opposite direction, which explains what we mean by $K_{\mathcal{B}_1}^{-1}$. It also means that T is invertible. In fact, $T^{-1} = K_{\mathcal{B}_1} \circ K_{\mathcal{B}_2}^{-1}$.

Comments

- Change of basis matrices are also called transition matrices because they tell you how to transition from the coordinates with respect to one basis to those of the other basis.
- We sometimes denote $T = K_{\mathcal{B}_2} \circ K_{\mathcal{B}_1}^{-1}$ by $T_{\mathcal{B}_1 \to \mathcal{B}_2}$.

Example (transition matrix, change of basis)

Consider the bases $\mathcal{B}_1 := \{3x+7, 2x+5\}, \mathcal{B}_2 := \{x+2, x+3\}$, of the vector space $\mathbb{R}_{\leq 1}[x]$. Determine the transition matrix T under a change of basis from $\mathcal{B}_1$ to $\mathcal{B}_2$.

The formula $T = K_{\mathcal{B}_2} \circ K_{\mathcal{B}_1}^{-1}$ needs to be calculated. We already know $K_{\mathcal{B}_1}^{-1}$ (see the example on page 74), and the mapping $K_{\mathcal{B}_2}$ was left as an exercise (see page 75). Here are the results.

$$K_{\mathcal{B}_1}^{-1} : \mathbb{R}^2 \to \mathbb{R}_{\leq 1}[x]; \begin{bmatrix} c \\ d \end{bmatrix} \mapsto (3c+2d)x + (7c+5d) \text{ and } K_{\mathcal{B}_2} : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}^2; \; \boldsymbol{a}x + \boldsymbol{b} \mapsto \begin{bmatrix} 3\boldsymbol{a} - \boldsymbol{b} \\ -2\boldsymbol{a} + \boldsymbol{b} \end{bmatrix}$$

Now we need to calculate T.

$$T\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = (K_{\mathcal{B}_2} \circ K_{\mathcal{B}_1}^{-1})\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = K_{\mathcal{B}_2}\left(K_{\mathcal{B}_1}^{-1}\left(\begin{bmatrix} c \\ d \end{bmatrix}\right)\right) = K_{\mathcal{B}_2}((\mathbf{3c+2d})x + (\mathbf{7c+5d})),$$

wobei

$$K_{\mathcal{B}_2}((\mathbf{3c+2d})x + (\mathbf{7c+5d})) = \begin{bmatrix} 3(\mathbf{3c+2d}) - (\mathbf{7c+5d}) \\ -2(\mathbf{3c+2d}) + (\mathbf{7c+5d}) \end{bmatrix} = \begin{bmatrix} 2c+1d \\ 1c+1d \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}$$

This defines the matrix mapping $T : \mathbb{R}^2 \to \mathbb{R}^2; \begin{bmatrix} c \\ d \end{bmatrix} \mapsto \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}$. The transition matrix T under a change of basis from $\mathcal{B}_1$ to $\mathcal{B}_2$ is therefore

$$T = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Alternatively:

We can construct the columns of the matrix T. The i-th column of T is $T\vec{e}_i$, whereby $\vec{e}_i$ is the i-th standard basis vector of $\mathbb{R}^2$. We know that $K_{\mathcal{B}_1}^{-1}(\vec{e}_i)$ is the i-th basis vector in $\mathcal{B}_1$. So all we have to do is put the i-th basis vector from $\mathcal{B}_1$ into $K_{\mathcal{B}_2}$. For $i = 1, 2$ we get the columns:

$$K_{\mathcal{B}_2}(3x+7) = \begin{bmatrix} 3(3) - 7 \\ -2(3) + 7 \end{bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{1} \end{bmatrix} \text{ and } K_{\mathcal{B}_2}(2x+5) = \begin{bmatrix} 3(2) - 5 \\ -2(2) + 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and thus } T = \begin{bmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}.$$

Schematically we get the first column like this.

$$\begin{tikzcd}
 & & 3x+7 \arrow[dddll, mapsto] & & \\
 & & V \arrow[dl, "K_{\mathcal{B}_2}"'] \arrow[dr, "K_{\mathcal{B}_1}"] & & \\
 & \mathbb{F}^n & & \mathbb{F}^n \arrow[ll, "T"] & \\
\begin{bmatrix} 2 \\ 1 \end{bmatrix} & & & & \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \arrow[llll, mapsto] \arrow[uuull, mapsto]
\end{tikzcd}$$

> **Exercise 3** (learning check)
>
> Consider the bases $\mathcal{B}_1 := \{3x+7, 2x+5\}, \mathcal{B}_2 := \{x+2, x+3\}$ of the vector space $\mathbb{R}_{\leq 1}[x]$. Determine the transition matrix S under a change of basis from $\mathcal{B}_2$ to $\mathcal{B}_1$ using the method in the last example. What is the product of the matrix multiplication ST? Why?

7.3 Matrix representations of linear transformations

Let V, W be finite-dimensional vector spaces. The linear mappings in $\hom(V, W)$ can be represented by matrices after applying the respective coordinate mappings. We will restrict ourselves to the case $L \in \hom(V, V)$ in this course. Thus the domain and image space are identical.

> **Definition** (matrix representative, representing matrix)
>
> Let V be an n-dimensional vector space over the field $\mathbb{F}$ with a given basis $\mathcal{B}$. The **matrix representative** or **representing matrix** $L_{\mathcal{B}}$ of L wrt $\mathcal{B}$ is the associated matrix mapping $L_{\mathcal{B}} : \mathbb{F}^n \to \mathbb{F}^n; \vec{v} \mapsto (K_{\mathcal{B}} \circ L \circ K_{\mathcal{B}}^{-1})(\vec{v})$. Notation: $L_{\mathcal{B}} = K_{\mathcal{B}} \circ L \circ K_{\mathcal{B}}^{-1}$.

Using a diagram, we can read off the formula

$$L_{\mathcal{B}} = K_{\mathcal{B}} \circ L \circ K_{\mathcal{B}}^{-1}$$

from the definition.

$$\begin{tikzcd}
V \arrow[r, "L"] \arrow[d, "K_{\mathcal{B}}"'] & V \arrow[d, "K_{\mathcal{B}}"] \\
\mathbb{F}^n \arrow[r, "L_{\mathcal{B}}"] & \mathbb{F}^n
\end{tikzcd}$$

Careful, even though $K_{\mathcal{B}}$ has to be invertible, L and $L_{\mathcal{B}}$ do not need to be.

The mapping $L_{\mathcal{B}}$ maps $\mathbb{F}^n$ to $\mathbb{F}^n$ and so can be viewed as a matrix mapping. We can construct this matrix as before by constructing the individual columns. Denote by $\mathcal{B} := \{\vec{b}_1, \ldots, \vec{b}_n\}$ the given basis of the vector space V. Another diagram showing the effect of the mappings will help us to develop an algorithm.

$$
\begin{tikzcd}
\vec{b}_i \arrow[rrr, mapsto] \arrow[ddd, mapsto] & & & L(\vec{b}_i) \arrow[ddd, mapsto] \\
& V \arrow[r, "L"] \arrow[d, "K_{\mathcal{B}}"'] & V \arrow[d, "K_{\mathcal{B}}"] & \\
& \mathbb{F}^n \arrow[r, "L_{\mathcal{B}}"] & \mathbb{F}^n & \\
\vec{e}_i \arrow[rrr, mapsto] & & & K_{\mathcal{B}}(L(\vec{b}_i))
\end{tikzcd}
$$

Algorithm (matrix representative)

Let V be an n-dimensional vector space over the field $\mathbb{F}$ with basis $\mathcal{B} := \{\vec{b}_1, \ldots, \vec{b}_n\}$. The **matrix representative** $L_{\mathcal{B}}$ of L wrt $\mathcal{B}$ can be calculated columnwise for $i = 1, \ldots, n$ as follows.

1. The only element in the preimage of the i-th standard basis vector wrt $K_{\mathcal{B}}$ is $\vec{b}_i$: $K_{\mathcal{B}}^{-1}(\vec{e}_i) = \vec{b}_i$.
2. Map the i-th basis vector in $\mathcal{B}$ using L: $L(\vec{b}_i)$.
3. Determine the coordinate vector of $L(\vec{b}_i)$ using $K_{\mathcal{B}}$: $K_{\mathcal{B}}(L(\vec{b}_i))$.
4. The result from step 3 is the i-th column of the matrix $L_{\mathcal{B}}$.

Comments

• The coordinate mapping does not need to be explicitly constructed if we are only interested in the matrix representative. Sometimes it is possible to “see” what the coordinate vector is in step 3. In the examination, you will want to write down a suitable equation that explain “what you see”.

• The size of $L_{\mathcal{B}}$ is $n \times n$, whereby n is the dimension of V.

Example (matrix representative)

Given the vector space $\mathbb{R}_{\leq 1}[x]$ and the basis $\mathcal{B}_1 := \{3x + 7, 2x + 5\}$, find the matrix representative $L_{\mathcal{B}_1}$ of the linear mapping

$$L : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x];\ ax + b \mapsto (13a - 5b)x + (30a - 12b)$$

wrt the given basis $\mathcal{B}_1$.

Using the information $K_{\mathcal{B}_1}(ax + b) = \begin{bmatrix} 5a - 2b \\ -7a + 3b \end{bmatrix}$ from page 74, we apply the algorithm:

<u>$i = 1$</u> $L(3x + 7) = (39 - 35)x + (90 - 84) = 4x + 6,\ K_{\mathcal{B}_1}(4x + 6) = \begin{bmatrix} 8 \\ -10 \end{bmatrix}$,

<u>$i = 2$</u> $L(2x + 5) = (26 - 25)x + (60 - 60) = x,\ K_{\mathcal{B}_1}(x) = \begin{bmatrix} 5 \\ -7 \end{bmatrix}$,

$$L_{\mathcal{B}_1} = \begin{bmatrix} 8 & 5 \\ -10 & -7 \end{bmatrix}.$$

Exercise 4 (learning check)
Given the vector space $\mathbb{R}_{\leq 1}[x]$ and the basis $\mathcal{B}_2 := \{x+2, x+3\}$ of $\mathbb{R}_{\leq 1}[x]$, find the matrix representative $L_{\mathcal{B}_2}$ of the linear mapping

$$L : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x];\ ax+b \mapsto (13a-5b)x + (30a-12b)$$

wrt the basis $\mathcal{B}_2$.

What is the connection between $L_{\mathcal{B}_1}$ and $L_{\mathcal{B}_2} = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$? Let's extend the commutative diagram.

$$\begin{tikzcd}
 & \mathbb{R}_{\leq 1}[x] \arrow[r, "L"] \arrow[d, "K_{\mathcal{B}_1}"] \arrow[ddl, "K_{\mathcal{B}_2}"'] & \mathbb{R}_{\leq 1}[x] \arrow[d, "K_{\mathcal{B}_1}"'] \arrow[ddr, "K_{\mathcal{B}_2}"] & \\
 & \mathbb{R}^2 \arrow[r, "L_{\mathcal{B}_1}"] \arrow[dl, "T"] & \mathbb{R}^2 \arrow[dr, "T"'] & \\
\mathbb{R}^2 \arrow[rrr, "L_{\mathcal{B}_2}"'] & & & \mathbb{R}^2
\end{tikzcd}$$

To calculate $L_{\mathcal{B}_2}$ from $L_{\mathcal{B}_1}$, follow the arrows in the diagram associated with $T \circ L_{\mathcal{B}_1} \circ T^{-1}$ from right to left (begin with T^{-1} followed by $L_{\mathcal{B}_1}$ and finally T). This is the same as if we had directly calculated the effect of $L_{\mathcal{B}_2}$ on the standard basis vectors because, in both cases, we are beginning at the same starting point and ending in the same place.

Example (matrix representative)
Consider the vector space $\mathbb{R}_{\leq 1}[x]$ and the two bases $\mathcal{B}_1 := \{3x+7, 2x+5\}$ and $\mathcal{B}_2 := \{x+2, x+3\}$. The matrix representative of the linear mapping $L : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x]$ wrt the basis $\mathcal{B}_1$ is given by $L_{\mathcal{B}_1} := \begin{bmatrix} 8 & 5 \\ -10 & -7 \end{bmatrix}$. Determine the matrix representative $L_{\mathcal{B}_2}$ of the linear mapping L wrt the basis $\mathcal{B}_2$.

Since $T, L_{\mathcal{B}_1}, T^{-1}$ are all matrix mappings and we know the corresponding matrices, the composition $T \circ L_{\mathcal{B}_1} \circ T^{-1}$ can be determined using matrix multiplication:

$$L_{\mathcal{B}_2} = TL_{\mathcal{B}_1}T^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 8 & 5 \\ -10 & -7 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$$

The aim of the following exercises is to show that there are many tasks for which the commutative diagram is useful. In the homework or in the exam, the required matrices or mappings are usually unknown. Here we practice in known situations to help build your knowledge and confidence. From the given information you should find the objects you are trying to calculate.

Exercise 5 (learning check)
Consider the vector space $\mathbb{R}_{\leq 1}[x]$ and the two bases $\mathcal{B}_1 := \{3x+7, 2x+5\}$ and $\mathcal{B}_2 := \{x+2, x+3\}$. The matrix representative of a linear mapping $L : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x]$ wrt the basis $\mathcal{B}_2$ is given by $L_{\mathcal{B}_2} := \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$. Determine the matrix representative $L_{\mathcal{B}_1}$ of the same linear mapping L wrt the basis $\mathcal{B}_1$.

Exercise 6
Consider the vector space $\mathbb{R}_{\leq 1}[x]$ with the given basis $\mathcal{B}_2 := \{x+2, x+3\}$. The matrix representative of a linear mapping $L : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x]$ wrt the basis $\mathcal{B}_2$ is given by $L_{\mathcal{B}_2} := \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$. Using this information, find the linear mapping L and write it as a mapping without using matrices.

7.4 Repeat with vector spaces of matrices

In this section we repeat all notions introduced in this chapter when the underlying vector space is $V = \mathbb{R}^{2,2}$, i.e. the matrices of size 2×2. Before reading further, answer the following questions. After you've worked your way through the section, make sure to check your answers.

• What is the dimension of the vector space $V = \mathbb{R}^{2,2}$?

• The coordinate mapping from V wrt a given basis is a mapping from which vector space to which vector space (i.e. determine the domain and image spaces)?

• What size is the matrix representative of the linear mapping $L : V \to V$ wrt a basis of V?

Example (coordinate vectors for matrices)

Consider the vector space $\mathbb{R}^{2,2}$ with the basis $\mathcal{B}_1 := \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}, \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{-1} \end{bmatrix} \right\}$.

Determine the coordinate vector of $\begin{bmatrix} \mathbf{1} & \mathbf{2} \\ \mathbf{3} & \mathbf{4} \end{bmatrix}$ wrt the basis $\mathcal{B}_1$.

1. linear combination: $\boldsymbol{\alpha_1} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \boldsymbol{\alpha_2} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \boldsymbol{\alpha_3} \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{bmatrix} + \boldsymbol{\alpha_4} \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{-1} \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{2} \\ \mathbf{3} & \mathbf{4} \end{bmatrix}$

   or $\begin{bmatrix} \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 & \alpha_2 \\ \alpha_3 & \alpha_3 - \alpha_4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

2. Compare components to get SLE: $\left\{ \begin{array}{ccccccccc} \alpha_1 & + & \alpha_2 & + & \alpha_3 & + & \alpha_4 & = & 1 \\ & & \alpha_2 & & & & & = & 2 \\ & & & & \alpha_3 & & & = & 3 \\ & & & & \alpha_3 & - & \alpha_4 & = & 4. \end{array} \right.$

3. Bring the AM into rref (write down your steps in the examination). $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 & 4 \end{array} \right] \longrightarrow \ldots \longrightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right]$.

   The coordinate vector of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ wrt $\mathcal{B}_1$ is $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{\mathcal{B}_1} = \begin{bmatrix} \mathbf{-3} \\ \mathbf{2} \\ \mathbf{3} \\ \mathbf{-1} \end{bmatrix}$ $(\in \mathbb{R}^4 = \mathbb{R}^{\dim \mathbb{R}^{2,2}})$.

(4.) Check solution: $\mathbf{-3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \mathbf{2} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \mathbf{3} \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{bmatrix} \mathbf{-1} \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{-1} \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{2} \\ \mathbf{3} & \mathbf{4} \end{bmatrix}$. ✓

Example (coordinate mapping with matrices)

Consider the vector space $\mathbb{R}^{2,2}$ with the basis $\mathcal{B}_1 := \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$. Determine the coordinate mapping of $\mathbb{R}^{2,2}$ wrt the basis $\mathcal{B}_1$.

1. linear combination: $\alpha_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \alpha_4 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

2. SLE: $\left\{ \begin{array}{ccccccccc} \alpha_1 & + & \alpha_2 & + & \alpha_3 & + & \alpha_4 & = & a \\ & & \alpha_2 & & & & & = & b \\ & & & & \alpha_3 & & & = & c \\ & & & & \alpha_3 & - & \alpha_4 & = & d \end{array} \right.$

3. AM $\to$ rref:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 0 & c \\ 0 & 0 & 1 & -1 & d \end{array}\right] \longrightarrow \ldots \longrightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & a-b-2c+d \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 0 & c \\ 0 & 0 & 0 & 1 & c-d \end{array}\right]$$

coordinate vector of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ wrt $\mathcal{B}_1$: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}_{\mathcal{B}_1} = \begin{bmatrix} a-b-2c+d \\ b \\ c \\ c-d \end{bmatrix}$

The coordinate mapping from $\mathbb{R}^{2,2}$ wrt the basis $\mathcal{B}_1$ is given by

$$K_{\mathcal{B}_1}: \quad \mathbb{R}^{2,2} \to \mathbb{R}^4, \qquad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a-b-2c+d \\ b \\ c \\ c-d \end{bmatrix}.$$

Exercise 7 (learning check)

Consider the vector space $\mathbb{R}^{2,2}$ with the basis $\mathcal{B}_2 := \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \right\}$. Determine the coordinate mapping of $\mathbb{R}^{2,2}$ wrt the basis $\mathcal{B}_2$.

Example (inverse of a coordinate mapping)

Find $K_{\mathcal{B}_1}^{-1}$, the inverse of the coordinate mapping $K_{\mathcal{B}_1}$ of $\mathbb{R}^{2,2}$ wrt the basis $\mathcal{B}_1 := \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$.

$$K_{\mathcal{B}_1}^{-1}: \mathbb{R}^4 \to \mathbb{R}^{2,2}; \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \mapsto a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + d \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} a+b+c+d & b \\ c & c-d \end{bmatrix}$$

(See page 75 for an explanation.)

Exercise 8 (learning check)

Find $K_{\mathcal{B}_2}^{-1}$, the inverse of the coordinate mapping $K_{\mathcal{B}_2}$ of $\mathbb{R}^{2,2}$ wrt the basis $\mathcal{B}_2 := \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \right\}$.

Example (transition matrix)

Calculate the transition matrix T under a change of basis from $\mathcal{B}_1$ of $\mathbb{R}^{2,2}$ to the basis $\mathcal{B}_2$ of $\mathbb{R}^{2,2}$ with

$$\mathcal{B}_1 := \left\{ B_1 := \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B_2 := \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}, B_3 := \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}, B_4 := \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{-1} \end{bmatrix} \right\}$$

$$\mathcal{B}_2 := \left\{ C_1 := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, C_2 := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C_3 := \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, C_4 := \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \right\}.$$

We calculate the columns of T as in the alternative solution method on page 76. Denote the i-th column of T by T_i. T_i contains the coordinate vector of B_i wrt the basis $\mathcal{B}_2$. (This works because e.g. the 1st column of T

is $T\vec{e}_1 = (K_{\mathcal{B}_2} \circ K_{\mathcal{B}_1}^{-1})(\vec{e}_1) = K_{\mathcal{B}_2}(K_{\mathcal{B}_1}^{-1}(\vec{e}_1)) = K_{\mathcal{B}_2}(B_1))$.
You have hopefully already determined $K_{\mathcal{B}_2}$ in the last exercise:

$$K_{\mathcal{B}_2} : \mathbb{R}^{2,2} \to \mathbb{R}^4;\ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a \\ -a+b \\ -2a+c+d \\ a-d \end{bmatrix}.$$

Thus the first column of T is

$$T_1 = K_{\mathcal{B}_2}\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} \mathbf{1} \\ \mathbf{-1} \\ \mathbf{-2} \\ \mathbf{1} \end{bmatrix}.$$

The remaining columns can be obtained by plugging in the other basis elements in $K_{\mathcal{B}_2}$:

$$T_2 = K_{\mathcal{B}_2}(\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}) = \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{-2} \\ \mathbf{1} \end{bmatrix},\ T_3 = K_{\mathcal{B}_2}(\begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}) = \begin{bmatrix} \mathbf{1} \\ \mathbf{-1} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix},\ T_4 = K_{\mathcal{B}_2}(\begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{-1} \end{bmatrix}) = \begin{bmatrix} \mathbf{1} \\ \mathbf{-1} \\ \mathbf{-3} \\ \mathbf{2} \end{bmatrix}.$$

The transition matrix is therefore $T = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{0} & \mathbf{-1} & \mathbf{-1} \\ \mathbf{-2} & \mathbf{-2} & \mathbf{0} & \mathbf{-3} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{2} \end{bmatrix}$.

Exercise 9 (learning check)
Determine the transition matrix S under a change of basis from $\mathcal{B}_2$ to $\mathcal{B}_1$ with

$$\mathcal{B}_1 := \left\{ B_1 := \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B_2 := \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B_3 := \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B_4 := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$$

$$\mathcal{B}_2 := \left\{ C_1 := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, C_2 := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C_3 := \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, C_4 := \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \right\}.$$

What other options are there to solve the task? For example, $K_{\mathcal{B}_1} \circ K_{\mathcal{B}_2}^{-1}$ can be explicitly calculated or the inverse of S can be determined.

Example (matrix representative)
The linear mapping $L : \mathbb{R}^{2,2} \to \mathbb{R}^{2,2};\ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} 2b & 2b \\ -11a+2b+3c+8d & 2b \end{bmatrix}$ is given.
Determine the matrix representative of L wrt $\mathcal{B}_1$ resp. $\mathcal{B}_2$ whereby

$$\mathcal{B}_1 := \left\{ B_1 := \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B_2 := \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B_3 := \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B_4 := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\},$$

$$\mathcal{B}_2 := \left\{ C_1 := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, C_2 := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C_3 := \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, C_4 := \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \right\}.$$

Following the algorithm for calculating the matrix representative of L wrt the basis $\mathcal{B}_1$, the details are as follows.

1) Compute the images of the basis vectors:

$$
\begin{aligned}
L(B_1) &= L\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ -11 & 0 \end{bmatrix}, \quad & L(B_2) &= L\left(\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 2 & 2 \\ -9 & 2 \end{bmatrix}, \\
L(B_3) &= L\left(\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad & L(B_4) &= L\left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ -19 & 0 \end{bmatrix}.
\end{aligned}
$$

2) Find the coordinate vectors of the images of the basis vectors in $\mathcal{B}_1$ using the coordinate mapping

$$
K_{\mathcal{B}_1} : \mathbb{R}^{2,2} \to \mathbb{R}^4; \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a-b-2c+d \\ b \\ c \\ c-d \end{bmatrix}.
$$

$$
K_{\mathcal{B}_1}(L(B_1)) = \begin{bmatrix} 22 \\ 0 \\ -11 \\ -11 \end{bmatrix}, \quad K_{\mathcal{B}_1}(L(B_2)) = \begin{bmatrix} 20 \\ 2 \\ -9 \\ -11 \end{bmatrix}, \quad K_{\mathcal{B}_1}(L(B_3)) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad K_{\mathcal{B}_1}(L(B_4)) = \begin{bmatrix} 38 \\ 0 \\ -19 \\ -19 \end{bmatrix}.
$$

3) Write down the matrix representative:

$$
L_{\mathcal{B}_1} = \begin{bmatrix} 22 & 20 & 0 & 38 \\ 0 & 2 & 0 & 0 \\ -11 & -9 & 0 & -19 \\ -11 & -11 & 0 & -19 \end{bmatrix}.
$$

Do the same thing for $L_{\mathcal{B}_2}$. Compare your answer with the one given in the next step.
The connections between the different mappings are shown in the diagram.

$$
\begin{tikzcd}
 & \mathbb{R}^{2,2} \arrow[r, "L"] \arrow[d, "K_{\mathcal{B}_1}"] \arrow[ddl, "K_{\mathcal{B}_2}"'] & \mathbb{R}^{2,2} \arrow[d, "K_{\mathcal{B}_1}"'] \arrow[ddr, "K_{\mathcal{B}_2}"] & \\
 & \mathbb{R}^4 \arrow[r, "L_{\mathcal{B}_1}"] \arrow[dl, "T"] & \mathbb{R}^4 \arrow[dr, "T"'] & \\
\mathbb{R}^4 \arrow[rrr, "L_{\mathcal{B}_2}"'] & & & \mathbb{R}^4
\end{tikzcd}
$$

Draw both ways taken in the formula below in the diagram.

$$
\begin{aligned}
L_{\mathcal{B}_2} = T\,L_{\mathcal{B}_1}\,T^{-1} &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & -1 & -1 \\ -2 & -2 & 0 & -3 \\ 1 & 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 22 & 20 & 0 & 38 \\ 0 & 2 & 0 & 0 \\ -11 & -9 & 0 & -19 \\ -11 & -11 & 0 & -19 \end{bmatrix} \begin{bmatrix} -1 & -1 & -2 & -3 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & -1 & -1 \\ -2 & -2 & 0 & -3 \\ 1 & 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} -2 & -2 & -6 & 10 \\ 2 & 2 & 0 & 0 \\ 2 & 2 & 3 & -5 \\ 0 & 0 & 3 & -5 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}
\end{aligned}
$$

7.5 Suggested solutions to the exercises in Chapter 7

Exercise 1

The coordinate mapping of $\mathbb{R}_{\leq 1}[x]$ wrt the basis $\mathcal{B}_2$ maps an arbitrary polynomial $p \in \mathbb{R}_{\leq 1}[x], p(x) = ax + b$ onto its coordinate vector wrt the basis $\mathcal{B}_2 := \{x+2, x+3\}$, i.e.

$$\begin{array}{rccc} K_{\mathcal{B}_2}: & \mathbb{R}_{\leq 1}[x] & \to & \mathbb{R}^2 \\ & ax+b & \mapsto & \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \end{array} \qquad \text{for } \alpha_1(x+2) + \alpha_2(x+3) = ax + b.$$

Comparing the coefficients leads to the following SLE:

$$\begin{array}{rcccccc} x^1: & \alpha_1 & + & \alpha_2 & = & a \\ x^0 = 1: & 2\alpha_1 & + & 3\alpha_2 & = & b \end{array}.$$

Set up the AM and bring it into rref.

$$\left[\begin{array}{cc|c} 1 & 1 & a \\ 2 & 3 & b \end{array}\right] \overset{\text{II}-2\text{I}}{\to} \left[\begin{array}{cc|c} 1 & 1 & a \\ 0 & 1 & -2a+b \end{array}\right] \overset{\text{I}-\text{II}}{\to} \left[\begin{array}{cc|c} 1 & 0 & 3a-b \\ 0 & 1 & -2a+b \end{array}\right].$$

The desired coordinate mapping is $K_{\mathcal{B}_2}: \mathbb{R}_{\leq 1}[x] \to \mathbb{R}^2;\ ax + b \mapsto \begin{bmatrix} 3a-b \\ -2a+b \end{bmatrix}$.

Plugging the basis polynomials into the coordinate mapping yields the standard basis vectors in $\mathbb{R}^2$ so that the answer is correct.

Caution! The coordinate mapping is, in general, not a matrix. This problem shows that a matrix would not make sense because a matrix cannot be multiplied with a polynomial.

Exercise 2 $K_{\mathcal{B}_2}^{-1}: \mathbb{R}^2 \to \mathbb{R}_{\leq 1}[x];\ \begin{bmatrix} c \\ d \end{bmatrix} \mapsto c(x+2) + d(x+3) = (c+d)x + (2c+3d)$

Exercise 3 Applying the method leads to $S = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$. ST is the identity matrix because S and T are inverses of each other.

Exercise 4 $K_{\mathcal{B}_2}$ is not explicitly given in this exercise because the coordinate vectors of the images are easy to determine:

$\underline{i=1}$ $L(x+2) = 3x + 6 = 3(x+2) + 0(x+3),\ K_{\mathcal{B}_2}(3x+6) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$,

$\underline{i=2}$ $L(x+3) = -2x - 6 = 0(x+2) - 2(x+3),\ K_{\mathcal{B}_2}(-2x-6) = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$,

$L_{\mathcal{B}_2} = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$.

Exercise 5 Look in the diagram to see how to calculate the matrix representative $L_{\mathcal{B}_1} := \begin{bmatrix} 8 & 5 \\ -10 & -7 \end{bmatrix}$ using only $L_{\mathcal{B}_2}, T, T^{-1}$.

Exercise 6 Tips: First determine $L(ax+b)$. For that, look at the extended commutative diagram (see page 79) and think about where to start and where you want to go. Now choose a suitable route for which you have the information so that you can compute the mapping explicitly.
Of course your answer should $L: \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x];\ ax + b \mapsto (13a - 5b)x + (30a - 12b)$.

Exercise 7 Follow the steps in the example “coordinate mapping with matrices in Section 7.4:

$$\begin{array}{cccc} K_{\mathcal{B}_2}: & \mathbb{R}^{2,2} & \to & \mathbb{R}^4 \\ & \begin{bmatrix} a & b \\ c & d \end{bmatrix} & \mapsto & \begin{bmatrix} a \\ -a+b \\ -2a+c+d \\ a-d \end{bmatrix}. \end{array}$$

Exercise 8 $K_{\mathcal{B}_2}^{-1}: \mathbb{R}^4 \to \mathbb{R}^{2,2}; \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \mapsto a \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} a & a+b \\ a+c+d & a-d \end{bmatrix}$

Exercise 9 Follow the steps in the example “transition matrix” in Section 7.4:

$$T = \begin{bmatrix} -1 & -1 & -2 & -3 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

8 Euclidean and unitary vector spaces

Up until now, we have not mentioned how to do geometry in a vector space. In this chapter, we will define a norm mapping, which helps us to measure lengths, and a scalar product, which allows us to measure angles.

In Chapter 3 we defined the notion of an orthogonal matrix as one in which the transpose of the matrix is the inverse of the matrix. In this chapter we will learn how to construct such matrices using the Gram-Schmidt algorithm.

8.1 Norm

Definition (norm)
Let V be a vector space over $\mathbb{F}$. A mapping

$$\|\cdot\| : V \to \mathbb{R}^+ := \{x \in \mathbb{R} \,|\, x \geq 0\}; \vec{v} \mapsto \|\vec{v}\|$$

is called a **norm** if the following properties hold for all $\vec{u}, \vec{v} \in V$ and for all $\alpha \in \mathbb{K}$.

1. $\|\vec{v}\| \geq 0$ with $\|\vec{v}\| = 0 \Leftrightarrow \vec{v} = \vec{0}$,
2. $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$,
3. $\|\alpha\vec{v}\| = |\alpha|\,\|\vec{v}\|$.

Remarks

• The first property ensures that the "length" of a vector is nonnegative. Only the zero vector has length 0.

• Even if V is a vector space over $\mathbb{C}$, the image of the norm mapping is a nonnegative *real* number.

• The second property is called the triangle inequality.

• In the third property, $|\alpha|$ is just the absolute value of α. Recall that if
$\alpha := a + bi \in \mathbb{C}$, then

$$|\alpha| := \sqrt{(a+bi)(a-bi)} = \sqrt{a^2+b^2}$$

is its absolute value. It is the distance of the complex number from the origin.

Here are three examples of real norms.

Example (norm mappings for $\mathbb{R}^n$)

1. $\vec{x} \in \mathbb{R}^n$, **standard norm** $\quad \|\vec{x}\| := \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$
2. $\vec{x} \in \mathbb{R}^n$, **maximum norm** $\quad \|\vec{x}\|_{\text{max}} = \max\{|x_1|, |x_2|, \ldots, |x_n|\}$
3. $\vec{x} \in \mathbb{R}^n$, L^1 **norm** $\quad \|\vec{x}\|_{L^1} := |x_1| + |x_2| + \cdots + |x_n|$

a) Find the norm of the vector $\begin{bmatrix} -4 \\ -7 \\ 4 \end{bmatrix}$ with respect to
1. the standard norm, 2. the maximum norm, 3. the L^1 norm.

b) Make a sketch of the unit disc including the boundary[a] in $\mathbb{R}^2$ wrt
1. the standard norm, 2. the maximum norm, 3. the L^1 norm.

[a] This is the set of all vectors in $\mathbb{R}^2$ of norm less than or equal to one. $\{\vec{x} \in \mathbb{R}^2 \mid \|\vec{x}\| \leq 1\}$

a)

1. $\left\| \begin{bmatrix} -4 \\ -7 \\ 4 \end{bmatrix} \right\| = \sqrt{(-4)^2 + (-7)^2 + 4^2} = \sqrt{81} = 9 \qquad$ 2. $\left\| \begin{bmatrix} -4 \\ -7 \\ 4 \end{bmatrix} \right\|_{\text{max}} = \max\{|-4|, |-7|, |4|\} = 7$

$$3. \left\|\begin{bmatrix}-4\\-7\\4\end{bmatrix}\right\|_{L^1} = |-4| + |-7| + |4| = 15$$

b) The norm (or length) of a vector $\vec{x} = \begin{bmatrix}x\\y\end{bmatrix}$ is the distance between the origin and the point (x, y). The vectors with norm 1 define the boundary of the unit disc. The vectors that are closer to the origin build up the interior of the disc.

1. The standard norm corresponds to our usual notion of length. In $\mathbb{R}^2$ the boundary of the unit disc is just a circle of radius one centered at the origin because $\sqrt{1} = \sqrt{x^2 + y^2}$ implies $x^2 + y^2 = 1$. The unit disc is that circle plus its interior.

2. A vector in $\mathbb{R}^2$ is in the unit disc with respect to the maximum norm if the absolute value of every component is less than or equal to 1. The boundary is made up of those points in the unit disc that have at least one component of 1 or -1.

3. To determine the boundary of the unit disc wrt the L^1 norm, consider the first quadrant. Since $|x| + |y| = x + y = 1$ has to be satisfied, we know that $y = 1 - x$. This gives us the line segment from the point $(0, 1)$ to the point $(1, 0)$. A similar observation gives us a diamond centered at the origin with tips at $(0, \pm 1)$ and $(\pm 1, 0)$. The unit disc is thus the diamond with its interior.

Exercise 1 (learning check)

Determine the norm of $\begin{bmatrix}3\\4\end{bmatrix}$ wrt the a) standard norm b) maximum norm c) L^1 norm.

Example (norm in $\mathbb{C}^{2,2}$)

Determine the norm of $\begin{bmatrix}4-3i & 8+6i\\2 & -7i\end{bmatrix} \in \mathbb{C}^{2,2}$ wrt the given norm.

a) $V := \mathbb{C}^{2,2}$, $A := (a_{ik}) \in \mathbb{C}^{2,2}$: $\;||A||_1 := \sum_{i,k} |a_{ik}|$

b) $V := \mathbb{C}^{2,2}$, $A := (a_{ik}) \in \mathbb{C}^{2,2}$: $\;||A||_\infty := \max|a_{ik}|$

a)

$$\left\|\begin{bmatrix}4-3i & 8+6i\\2 & -7i\end{bmatrix}\right\|_1 = |4-3i| + |8+6i| + |2| + |-7i|$$

$$= \sqrt{4^2+3^2} + \sqrt{8^2+6^2} + 2 + \sqrt{7^2} = \sqrt{25} + \sqrt{100} + 2 + 7 = 24$$

b)

$$\left\|\begin{bmatrix}4-3i & 8+6i\\2 & -7i\end{bmatrix}\right\|_\infty = \max\{|4-3i|, |8+6i|, |2|, |-7i|\}$$
$$= \max\{5, 10, 2, 7\} = 10$$

Exercise 2

Let V be a nontrivial real vector space and $|| \cdot || : V \to \mathbb{R}^+$ a norm on V. Check whether $|| \cdot ||$ is a linear mapping.

8.2 Scalar product

The scalar product is the key to measuring the angle between two vectors.

Definition (euclidean scalar product)

Let V be a vector space over $\mathbb{R}$. The mapping $\langle\cdot\,,\cdot\rangle : V \times V \to \mathbb{R}$ is called a **euclidean scalar product** if the following properties are satisfied for all $\vec{u}, \vec{v}, \vec{w} \in V$, $\alpha \in \mathbb{R}$.

(1). $\langle \vec{u}+\vec{v}, \vec{w}\rangle = \langle \vec{u}, \vec{w}\rangle + \langle \vec{v}, \vec{w}\rangle$,

(2). $\langle \alpha\vec{u}, \vec{v}\rangle = \alpha\langle \vec{u}, \vec{v}\rangle$,

(3). $\langle \vec{v}, \vec{v}\rangle \geq 0$ with $\langle \vec{v}, \vec{v}\rangle = 0 \Leftrightarrow \vec{v} = \vec{0}$ (positive definite),

(4). $\langle \vec{u}, \vec{v}\rangle = \langle \vec{v}, \vec{u}\rangle$ (symmetric).

A vector space over $\mathbb{R}$ equipped with a scalar product is called a **euclidean vector space**.

Example (standard scalar product in $\mathbb{R}^n$)

The **standard scalar product** is defined by

$$\langle \vec{u}, \vec{v}\rangle := u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

for all $\vec{u}, \vec{v} \in \mathbb{R}^n$. It is often referred to as the **dot product**.

Determine the scalar products $\left\langle \begin{bmatrix} -3 \\ 4 \end{bmatrix}, \begin{bmatrix} -5 \\ -6 \end{bmatrix} \right\rangle$, $\left\langle \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ \frac{-4}{5} \end{bmatrix} \right\rangle$, $\left\langle \begin{bmatrix} \frac{3}{5} \\ \frac{-4}{5} \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ \frac{-4}{5} \end{bmatrix} \right\rangle$.

$$\left\langle \begin{bmatrix} -3 \\ 4 \end{bmatrix}, \begin{bmatrix} -5 \\ -6 \end{bmatrix} \right\rangle = (-3)(-5) + (4)(-6) = -9$$

$$\left\langle \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ \frac{-4}{5} \end{bmatrix} \right\rangle = \frac{12}{25} + \frac{-12}{25} = 0 \qquad \left\langle \begin{bmatrix} \frac{3}{5} \\ \frac{-4}{5} \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ \frac{-4}{5} \end{bmatrix} \right\rangle = \frac{9}{25} + \frac{16}{25} = 1$$

Comment: There are many different scalar products that can be defined on $\mathbb{R}^n$. For example, when $k > 0$ the mapping $\langle\ ,\ \rangle_k : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}; (\vec{u}, \vec{v}) \mapsto ku_1v_1 + ku_2v_2 + \cdots + ku_nv_n$ defines a scalar product.

Example (scalar product in $\mathbb{R}_{\leq n}[x]$)

Given the scalar product

$$\langle p, q\rangle := \int_0^1 p(x)q(x)dx, \quad p, q \in \mathbb{R}_{\leq n}[x],$$

evaluate $\langle 3x^2, 2-4x\rangle$ and $\langle 3x^2, 3x^2\rangle$.

$$\langle 3x^2, 2-4x\rangle = \int_0^1 3x^2(2-4x)dx = \int_0^1 (6x^2 - 12x^3)dx = [2x^3 - 3x^4]_0^1 = 2 - 3 = -1$$

$$\langle 3x^2, 3x^2\rangle = \int_0^1 9x^4 dx = \left[\frac{9}{5}x^5\right]_0^1 = \frac{9}{5}$$

Exercise 3 (learning check)

Given the scalar product

$$\langle p_2x^2 + p_1x + p_0, q_2x^2 + q_1x + q_0\rangle := p_2q_2 + 3p_1q_1 + p_0q_0$$

in $\mathbb{R}_{\leq 2}[x]$, find $\langle 4x^2 + 2x + 1, -5x + 6\rangle$.

Exercise 4
Let $\langle\ ,\ \rangle : V \times V \to \mathbb{R}$ be a euclidean scalar product. For an arbitrary $\vec{v}_0 \in V$ define

$$L_{\vec{v}_0} : V \to \mathbb{R}, \quad \vec{v} \mapsto \langle \vec{v}, \vec{v}_0 \rangle. \tag{7}$$

Is $L_{\vec{v}_0}$ a linear mapping?

Definition (unitary scalar product)

Let V be a vector space over $\mathbb{C}$. The mapping $\langle \cdot\,,\,\cdot \rangle : V \times V \to \mathbb{C}$ is called a unitary scalar product if the following properties hold for all $\vec{u}, \vec{v}, \vec{w} \in V$, $\alpha \in \mathbb{C}$.

(1). $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$,
(2). $\langle \alpha\vec{u}, \vec{v} \rangle = \alpha \langle \vec{u}, \vec{v} \rangle$,
(3). $\langle \vec{v}, \vec{v} \rangle \geq 0$ with $\langle \vec{v}, \vec{v} \rangle = 0 \Leftrightarrow \vec{v} = \vec{0}$ (positive definite),
(4). $\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$.

A vector space V over $\mathbb{C}$ with a scalar product is called a **unitary vector space**.

8.3 Connection between the norm and a scalar product

Definition (associated norm)

Let V be a euclidean or unitary vector space. Using the scalar product, define (the **associated norm**) as

$$||\vec{v}|| := \sqrt{\langle \vec{v}, \vec{v} \rangle}.$$

Remarks

• If a scalar product is explicitly defined, then the associated norm is implicitly defined.

• The standard norm is the associated norm of the standard scalar product.

Definition (angle)

The angle (in radians) between the vectors $\vec{u}, \vec{v} \in V \setminus \{\vec{0}\}$, denoted by $\angle(\vec{u}, \vec{v})$, can be found using the formula

$$\cos \angle(\vec{u}, \vec{v}) := \frac{\langle \vec{u}, \vec{v} \rangle}{||\vec{u}||\, ||\vec{v}||}.$$

Remarks

• It can be shown that the formula is correct in $\mathbb{R}^2$ and in $\mathbb{R}^3$ using geometric considerations when the standard scalar product is used.

• Two nonzero vectors $\vec{u}, \vec{v} \in \mathbb{R}^2$ equipped with the standard scalar product are perpendicular to each other when the angle between them is $90°$ (or $\frac{\pi}{2}$ in radians) so that $\cos \angle(\vec{u}, \vec{v}) = 0$. In more general settings, we speak of vectors being **orthogonal** when the scalar product of the vectors is zero.

Exercise 5

Let $\vec{v}_0$ be a vector in $\mathbb{R}^2$. Determine whether the set $M_{\vec{v}_0} := \{\vec{v} \in \mathbb{R}^2 \mid \langle \vec{v}, \vec{v}_0 \rangle = 0\}$ is a subspace of $\mathbb{R}^2$.

Remark: The set $M_{\vec{v}_0}$ consists of all vectors in $\mathbb{R}^2$ that are orthogonal to the vector $\vec{v}_0$. This is the kernel of the linear mapping $L_{\vec{v}_0}$ (see the mapping (7) on page 89).

8.4 Orthonormal basis, Gram-Schmidt algorithm

Definition (orthonormal basis)

Let $\mathcal{B} := \{\vec{b}_1, \ldots, \vec{b}_n\}$ be a basis of a finite-dimensional euclidean vector space V. $\mathcal{B}$ is an **orthonormal** basis (abbr. ONB) when

$$\langle \vec{b}_i, \vec{b}_j \rangle = \delta_{i,j} := \begin{cases} \mathbf{0} & \text{für} \quad i \neq j \\ \mathbf{1} & \text{für} \quad i = j \end{cases}$$

holds for all $i, j \in \{1, \ldots, n\}$.

Remarks

• When $i \neq j$, the scalar product $\langle \vec{b}_i, \vec{b}_j \rangle$ is zero so that $\vec{b}_i, \vec{b}_j$ are orthogonal. Furthermore, every basis element has length 1 because $\sqrt{\langle \vec{b}_i, \vec{b}_i \rangle} = \sqrt{1} = 1$. We say that the vectors are **normalized**.

• The symbol $\delta_{i,j}$ is called the Kronecker delta symbol.

Example (orthonormal basis - ONB)

Verify that $\mathcal{B} := \left\{ \vec{b}_1 := \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}, \vec{b}_2 := \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \right\}$ is an ONB wrt the standard scalar product in $\mathbb{R}^2$.

We first argue that $\mathcal{B}$ is indeed a basis of $\mathbb{R}^2$.
Each basis of $\mathbb{R}^2$ consists of 2 linearly independent vectors. $\mathcal{B}$ has exactly 2 elements and they are linearly independent (with two vectors it is sufficient to note that the first vector is not a multiple of the second vector, which is the case here). Hence $\mathcal{B}$ is a basis of $\mathbb{R}^2$.

Next, we check that $\boldsymbol{\vec{b}_1, \vec{b}_2}$ are **orthogonal** vectors:

$$\langle \boldsymbol{\vec{b}_1}, \boldsymbol{\vec{b}_2} \rangle = \langle \vec{b}_2, \vec{b}_1 \rangle = \left\langle \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix} \right\rangle = \frac{12}{25} - \frac{12}{25} = \mathbf{0}\,.$$

The last task is to check that each vector $\boldsymbol{\vec{b}_1, \vec{b}_2}$ in the basis has length 1:

$$\langle \vec{b}_1, \vec{b}_1 \rangle = \left\langle \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix} \right\rangle = \frac{9}{25} + \frac{16}{25} = 1 \;\Rightarrow\; \|\boldsymbol{\vec{b}_1}\| = \sqrt{\langle \boldsymbol{\vec{b}_1}, \boldsymbol{\vec{b}_1} \rangle} = \mathbf{1},$$

$$\langle \vec{b}_2, \vec{b}_2 \rangle = \left\langle \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}, \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \right\rangle = \frac{9}{25} + \frac{16}{25} = 1 \;\Rightarrow\; \|\boldsymbol{\vec{b}_2}\| = \sqrt{\langle \boldsymbol{\vec{b}_2}, \boldsymbol{\vec{b}_2} \rangle} = \mathbf{1}.$$

$\mathcal{B}$ satisfies the definition of an ONB.

The next examples show two applications of ONBs. The first application is the connection between ONBs and orthogonal matrices. The key lies in the connection between the standard scalar product and matrix multiplication:

$$\langle \vec{u}, \vec{v} \rangle = \vec{u}^T \vec{v}.$$

We calculate the standard scalar product $\langle \vec{u}, \vec{v} \rangle$ and then the matrix multiplication $\vec{u}^T \vec{v}$ for $\vec{u}, \vec{v} \in \mathbb{R}^2$ to make this understandable:

$$\langle \vec{u}, \vec{v} \rangle = \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle = u_1 v_1 + u_2 v_2,$$

$$\vec{u}^T \vec{y} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = u_1 v_1 + u_2 v_2.$$

Example (ONB, orthogonal matrices)

Consider the ONB $\mathcal{B} := \left\{ \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}, \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \right\}$ of $\mathbb{R}^2$ equipped with the standard scalar product. Show that the matrix $A := \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$ is orthogonal.

A matrix A is orthogonal when $AA^T = I = A^T A$. $\mathcal{B} := \{\vec{b}_1, \vec{b}_2\}$ is an ONB means that $\langle \vec{b}_1, \vec{b}_2 \rangle = \vec{b}_1^T \vec{b}_2 = \delta_{i,j}$. Because the columns of A build an orthonormal basis of $\mathbb{R}^2$ and the columns of A are represented as rows in A^T in the corresponding order, we obtain $A^T A = I_2$:

$$A^T A = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

Since $A^T = A^{-1}$, the equation $AA^T = I_2$ holds.

Remark: If $\mathcal{B} := \{\vec{q}_1, \vec{q}_2, \ldots, \vec{q}_n\}$ is an ONB of the vector space $\mathbb{R}^n$ equipped with the standard scalar product, then the matrix Q, whose columns are the basis vectors, is an orthogonal matrix ($QQ^T = I_n$). The matrices Q and Q^T are called **orthogonal matrices** (and *not* orthonormal matrices).

As a second application we consider coordinate vectors wrt an ONB. These are especially easy to determine. Let $\mathcal{B} := \{\vec{b}_1, \vec{b}_2, \ldots, \vec{b}_n\}$ be an ONB of the euclidean vector space V. A vector $\vec{v} \in V$ can be uniquely represented as a linear combination of the basis vectors:

$$\vec{v} = \alpha_1 \vec{b}_1 + \boldsymbol{\alpha_2} \boldsymbol{\vec{b}_2} + \cdots + \boldsymbol{\alpha_n} \boldsymbol{\vec{b}_n}.$$

The scalar product of $\vec{v}$ with the first basis vector $\vec{b}_1$ yields the first coordinate α_1:

$$\begin{aligned}
\langle \vec{v}, \vec{b}_1 \rangle &= \langle \alpha_1 \vec{b}_1 + \alpha_2 \vec{b}_2 + \cdots + \alpha_n \vec{b}_n, \vec{b}_1 \rangle \\
&= \langle (\alpha_1 \vec{b}_1) + (\alpha_2 \vec{b}_2 + \cdots + \alpha_n \vec{b}_n), \vec{b}_1 \rangle && \text{Group appropriately.} \\
&= \langle \alpha_1 \vec{b}_1, \vec{b}_1 \rangle + \langle \alpha_2 \vec{b}_2 + \cdots + \alpha_n \vec{b}_n, \vec{b}_1 \rangle && \text{Apply property (1) of a scalar product (see page 88) until there are no more summands within } \langle\, , \rangle. \\
&\;\;\vdots \\
&= \langle \alpha_1 \vec{b}_1, \vec{b}_1 \rangle + \langle \alpha_2 \vec{b}_2, \vec{b}_1 \rangle + \cdots + \langle \alpha_n \vec{b}_n, \vec{b}_1 \rangle \\
&= \alpha_1 \underbrace{\langle \vec{b}_1, \vec{b}_1 \rangle}_{=1} + \alpha_2 \underbrace{\langle \vec{b}_1, \vec{b}_2 \rangle}_{=0} + \cdots + \alpha_n \underbrace{\langle \vec{b}_1, \vec{b}_n \rangle}_{=0} && \text{Apply property (2) of a scalar product. properties of an ONB} \\
&= \alpha_1
\end{aligned}$$

Analogously, $\langle \vec{v}, \vec{b}_i \rangle = \alpha_i$, $i = 1, \ldots, n$.

Example (ONB and coordinates)

Let $\mathcal{B} := \left\{ \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}, \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \right\}$ be an ONB of $\mathbb{R}^2$ equipped with the standard scalar product. Determine the coordinate vector of $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$ wrt $\mathcal{B}$.

$$\alpha_1 = \left\langle \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \end{bmatrix} \right\rangle = \frac{-22}{5}, \quad \alpha_2 = \left\langle \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \end{bmatrix} \right\rangle = \frac{29}{5} \quad \Rightarrow \quad \begin{bmatrix} 2 \\ 7 \end{bmatrix}_{\mathcal{B}} = \begin{bmatrix} \frac{-22}{5} \\ \frac{29}{5} \end{bmatrix}$$

Because of the connection between the standard scalar product and matrix multiplication, we could have alternatively calculated as follows.

$$B^T \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} \frac{-22}{5} \\ \frac{29}{5} \end{bmatrix}$$

Exercise 6 (learning check and review)

Consider the following orthonormal basis (ONB) of $\mathbb{R}^4$ equipped with the standard scalar product.

$$\mathcal{B} := \left\{ \vec{q}_1 := \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \vec{q}_2 := \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \vec{q}_3 := \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}, \vec{q}_4 := \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \right\}.$$

Let Q be the matrix $Q := [\vec{q}_1\ \vec{q}_2\ \vec{q}_3\ \vec{q}_4]$.

a) Write down the matrix Q and determine its inverse Q^{-1}.

b) Determine the coordinate mapping from $\mathbb{R}^4$ wrt the basis $\mathcal{B}$.

c) Find $K_{\mathcal{B}}^{-1}\left(\begin{bmatrix} 2 \\ 4 \\ -6 \\ -2 \end{bmatrix}\right)$.

Algorithm (Gram-Schmidt algorithm – ONB)

Given the basis $\{\vec{b}_1, \ldots, \vec{b}_n\}$ of a *euclidean* vector space V, produce an ONB $\{\vec{q}_1, \ldots, \vec{q}_n\}$ of V.

---

1. Normalize the vector $\vec{b}_1$: $\quad \vec{q}_1 := \dfrac{\vec{b}_1}{||\vec{b}_1||}$.

2. Compute a perpendicular to the line determined by $\vec{q}_1$ (and $\vec{0}$) using $\vec{b}_2$:
$\vec{l}_2 := \vec{b}_2 - \langle \vec{b}_2, \vec{q}_1 \rangle \cdot \vec{q}_1$.

3. Normalize $\vec{l}_2$: $\quad \vec{q}_2 := \dfrac{\vec{l}_2}{||\vec{l}_2||}$.

The following two steps are to be repeated for $k = 3, \ldots, n$.

4. Compute a perpendicular to the plane spanned by $\vec{q}_1, \ldots, \vec{q}_{k-1}$ using $\vec{b}_k$:

$$\vec{l}_k := \vec{b}_k - \langle \vec{b}_k, \vec{q}_1 \rangle \cdot \vec{q}_1 - \cdots - \langle \vec{b}_k, \vec{q}_{k-1} \rangle \cdot \vec{q}_{k-1}.$$

5. Normalize $\vec{l}_k$: $\quad \vec{q}_k := \dfrac{\vec{l}_k}{||\vec{l}_k||}$.

What is this algorithm doing? It constructs vectors that are orthogonal to each other and it normalizes them. In the first step, we just normalize the vector to get the first vector $\vec{q}_1$ in an ONB. $\langle \vec{q}_1, \vec{q}_1 \rangle = 1$.

In the second step, we need to find a vector orthogonal to $\vec{q}_1$ using $\vec{b}_2$.
Using the properties of the scalar product, we check that the formula for $\vec{l}_2$ is indeed orthogonal to $\vec{q}_1$.

$$
\begin{aligned}
\langle \vec{l}_2, \vec{q}_1 \rangle &= \langle \vec{b}_2 - \langle \vec{b}_2, \vec{q}_1 \rangle \cdot \vec{q}_1, \vec{q}_1 \rangle \\
&= \langle \vec{b}_2, \vec{q}_1 \rangle - \langle \langle \vec{b}_2, \vec{q}_1 \rangle \cdot \vec{q}_1, \vec{q}_1 \rangle \\
&= \langle \vec{b}_2, \vec{q}_1 \rangle - \langle \vec{b}_2, \vec{q}_1 \rangle \langle \vec{q}_1, \vec{q}_1 \rangle \\
&= \langle \vec{b}_2, \vec{q}_1 \rangle - \langle \vec{b}_2, \vec{q}_1 \rangle \cdot 1 \\
&= 0.
\end{aligned}
$$

Going from the 3rd to the 4th line, we see why it is necessary to normalize the vectors as we go along. If $\langle \vec{q}_1, \vec{q}_1 \rangle \neq 1$, then $\vec{l}_2$ would not be orthogonal to $\vec{q}_1$.

The vector $\vec{l}_2$ is then normalized and added to the new basis.

The rest of the vectors in the new basis are constructed in the same manner. We have to ensure that at each step, the vector we calculate is orthogonal to all previously constructed vectors in the new basis.

Exercise 7 (learning check)

Show that the k-th vector $\vec{l}_k$ is orthogonal to the orthonormal vectors $\vec{q}_1, \dots, \vec{q}_{k-1}$.

Example (Gram-Schmidt algorithm)

Let $\mathbb{R}^3$ be equipped with the standard scalar product. Apply the Gram-Schmidt algorithm to the basis $\mathcal{B} := \left\{ \vec{b}_1 := \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}, \vec{b}_2 := \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}, \vec{b}_3 := \begin{bmatrix} 4 \\ -1 \\ -3 \end{bmatrix} \right\}$ of $\mathbb{R}^3$ to get an ONB.

$$
\vec{q}_1 = \frac{\vec{b}_1}{||\vec{b}_1||}, \; ||\vec{b}_1|| = \sqrt{\langle \vec{b}_1, \vec{b}_1 \rangle} = \sqrt{9+16} = 5 \quad \Rightarrow \quad \vec{q}_1 = \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix}
$$

$$
\vec{l}_2 = \vec{b}_2 - \langle \vec{b}_2, \vec{q}_1 \rangle \vec{q}_1 = \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix} - \left\langle \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix} \right\rangle \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix} - \frac{31}{5} \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix} = \begin{bmatrix} \frac{-68}{25} \\ 0 \\ \frac{51}{25} \end{bmatrix}
$$

$$
||\vec{l}_2|| = \sqrt{\tfrac{68^2}{25^2} + \tfrac{51^2}{25^2}} = \tfrac{17}{5} \quad \Rightarrow \quad \vec{q}_2 = \tfrac{5}{17} \begin{bmatrix} \frac{-68}{25} \\ 0 \\ \frac{51}{25} \end{bmatrix} = \begin{bmatrix} \frac{-4}{5} \\ 0 \\ \frac{3}{5} \end{bmatrix}
$$

$$
\begin{aligned}
\vec{l}_3 &= \vec{b}_3 - \langle \vec{b}_3, \vec{q}_1 \rangle \vec{q}_1 - \langle \vec{b}_3, \vec{q}_2 \rangle \vec{q}_2 \\
&= \begin{bmatrix} 4 \\ -1 \\ -3 \end{bmatrix} - \langle \begin{bmatrix} 4 \\ -1 \\ -3 \end{bmatrix}, \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix} \rangle \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix} - \langle \begin{bmatrix} 4 \\ -1 \\ -3 \end{bmatrix}, \begin{bmatrix} \frac{-4}{5} \\ 0 \\ \frac{3}{5} \end{bmatrix} \rangle \begin{bmatrix} \frac{-4}{5} \\ 0 \\ \frac{3}{5} \end{bmatrix} \\
&= \begin{bmatrix} 4 \\ -1 \\ -3 \end{bmatrix} - 0 \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix} + 5 \begin{bmatrix} \frac{-4}{5} \\ 0 \\ \frac{3}{5} \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \vec{q}_3
\end{aligned}
$$

since $\vec{l}_3$ is already normalized. Thus $\mathcal{B}_{\text{ONB}} = \left\{ \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix}, \begin{bmatrix} \frac{-4}{5} \\ 0 \\ \frac{3}{5} \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \right\}$ is an orthonormalbasis of $\mathbb{R}^3$.

Exercise 8

The vector space $\mathbb{R}^{2,2}$ is given with the scalar product

$$\left\langle \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}, \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \right\rangle := a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2.$$

Show that the matrices $B_1 := \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B_2 := \frac{1}{\sqrt{6}}\begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}, B_3 := \frac{1}{2\sqrt{3}}\begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix}$ are orthonormal.

Example (Gram-Schmidt with matrices)

The vector space $\mathbb{R}^{2,2}$ is equipped with the scalar product

$$\left\langle \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}, \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \right\rangle := a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2.$$

Apply the Gram-Schmidt algorithm to the basis

$$\mathcal{B} := \left\{ B_1 := \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B_2 := \frac{1}{\sqrt{6}}\begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}, B_3 := \frac{1}{2\sqrt{3}}\begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix}, B_4 := \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

of $\mathbb{R}^{2,2}$ to construct an orthonormal basis of $\mathbb{R}^{2,2}$.

The Gram-Schmidt algorithm does not have any effect on the first three matrices since these are already orthonormal (see the previous exercise). The first three vectors in the ONB are $Q_1 = B_1, Q_2 = B_2, Q_3 = B_3$.

We compute L_4 as follows. $L_4 := B_4 - \langle B_4, B_1\rangle B_1 - \langle B_4, B_2\rangle B_2 - \langle B_4, B_3\rangle B_3$:

$$L_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} - \left\langle \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} - \left\langle \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \frac{1}{\sqrt{6}}\begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} \right\rangle \frac{1}{\sqrt{6}}\begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}$$

$$- \left\langle \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \frac{1}{2\sqrt{3}}\begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix} \right\rangle \frac{1}{2\sqrt{3}}\begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} - 0\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} + \frac{1}{12}\begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Normalize L_4 : $\quad Q_4 := \dfrac{L_4}{||L_4||}$

$$\left\langle \frac{1}{4}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \frac{1}{4}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \right\rangle = \frac{4}{16} = \frac{1}{4} \quad \Rightarrow \quad ||L_4|| = \frac{1}{2} \quad \Rightarrow \quad Q_4 = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

An orthonormal basis of $\mathbb{R}^{2,2}$ wrt the given scalar product is

$$\left\{ \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \frac{1}{\sqrt{6}}\begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}, \frac{1}{2\sqrt{3}}\begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix}, \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \right\}.$$

8.5 Suggested solutions to the exercises in Chapter 8

Exercise 1 a) $\left\| \begin{bmatrix} -3 \\ -4 \end{bmatrix} \right\| = \sqrt{(-3)^2 + (-4)^2} = \sqrt{25} = 5$

b) $\left\| \begin{bmatrix} -3 \\ -4 \end{bmatrix} \right\|_{\max} = \max\{|-3|, |-4|\} = 4$ c) $\left\| \begin{bmatrix} -3 \\ -4 \end{bmatrix} \right\|_{L} = |-3| + |-4| = 7$

Exercise 2 A linear mapping $L : V \to W$ is compatible with addition and multiplication with scalar:

- $L(\vec{v}_1 + \vec{v}_2) = L(\vec{v}_1) + L(\vec{v}_2)$
- $L(\alpha\vec{v}) = \alpha L(\vec{v})$

for all $\alpha \in \mathbb{R}$ and $\vec{v}, \vec{v}_1, \vec{v}_2 \in V$.

[Idea: Because of the first property of a norm mapping, we know that $\|\vec{v}\|$ cannot be negative. This indicates that the second criterion in the definition of a linear mapping probably will not hold. To find a counterexample, we have to choose $\vec{v}$ carefully since $\|\vec{0}\| = 0 = -\|\vec{0}\|$.]

The vector space V is assumed to be nontrivial, so there is a vector in $\vec{v} \in V$ that is not the zero vector.
The first property of a norm mapping says that $\|\vec{v}\|$ and $\| - \vec{v}\|$ are positive numbers. This means that $\|(-1) \cdot \vec{v}\| \neq (-1) \cdot \|\vec{v}\|$. Therefore a norm mapping is **not** a linear mapping when V is a nontrivial vector space.

Exercise 3 $\langle 4x^2 + 2x + 2, -5x + 12\rangle = 4 \cdot 0 + 3 \cdot 2 \cdot (-5) + 2 \cdot 12 = -6$

Exercise 4 Let $\vec{u}, \vec{v} \in V$ and $\alpha \in \mathbb{R}$. According to the definition of a linear mapping, we need to check if

$$L_{\vec{b}}(\vec{u} + \vec{v}) = L_{\vec{b}}(\vec{u}) + L_{\vec{b}}(\vec{v}) \text{ and } L_{\vec{b}}(\alpha\vec{v}) = \alpha L_{\vec{b}}(\vec{v})$$

hold. Applying the properties of a scalar product we obtain

$$L_{\vec{b}}(\vec{u} + \vec{v}) = \langle \vec{u} + \vec{v}, \vec{v}_0\rangle = \langle \vec{u}, \vec{v}_0\rangle + \langle \vec{v}, \vec{v}_0\rangle = L_{\vec{v}_0}(\vec{u}) + L_{\vec{v}_0}(\vec{v}),$$

$$L_{\vec{v}_0}(\alpha\vec{v}) = \langle \alpha\vec{v}, \vec{v}_0\rangle = \alpha\langle \vec{v}, \vec{v}_0\rangle = \alpha L_{\vec{v}_0}(\vec{v}).$$

$L_{\vec{v}_0}$ is thus a linear mapping. (Caution! The scalar product $\langle\ ,\ \rangle$ is **not** a linear mapping! The linearity works only when one argument (here $\vec{v}_0$) is fixed, as is the case with $L_{\vec{v}_0}$.

Exercise 5 Set $M := M_{\vec{v}_0}$. For M to be a subspace of $\mathbb{R}^2$, the set must be nonempty. Furthermore, M has to be closed wrt the addition and multiplication with scalars. We check these three conditions.

- M is nonempty: $\quad \langle \vec{0}, \vec{v}_0\rangle = 0 \quad \Rightarrow \vec{0} \in M$.
- M is closed wrt addition: Let $\vec{u}, \vec{v} \in M$, i.e. $\langle \vec{u}, \vec{v}_0\rangle = 0 = \langle \vec{v}, \vec{v}_0\rangle$. Because of the first property of a scalar product:
$$\langle \vec{u} + \vec{v}, \vec{v}_0\rangle = \langle \vec{u}, \vec{v}_0\rangle + \langle \vec{v}, \vec{v}_0\rangle = 0 + 0 = 0 \quad \Rightarrow \vec{u} + \vec{v} \in M.$$
- We show that M is closed wrt multiplication with scalars. Let $\vec{v} \in M, \alpha \in \mathbb{R}$. Due to the second property of a scalar product:
$$\langle \alpha\vec{v}, \vec{v}_0\rangle = \alpha\langle \vec{v}, \vec{v}_0\rangle = \alpha \cdot 0 = 0 \quad \Rightarrow \alpha\vec{v} \in M.$$

M satisfies the definition of a subspace of $\mathbb{R}^2$.

Exercise 6 a) Since we are dealing with an ONB wrt the standard scalar product, the matrix

$$Q = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

is orthogonal; i.e. $Q^TQ = I$ and therefore

$$Q^T = Q^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

(Do not try to do this using Gaussian elimination in the test. You will lose way too much valuable time!)

b) The coordinate mapping of $\mathbb{R}^4$ wrt the basis $\mathcal{B}$ is $K_{\mathcal{B}} : \mathbb{R}^4 \to \mathbb{R}^4; \vec{v} \mapsto \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$ with $\vec{v} = \alpha_1\vec{q}_1 + \alpha_2\vec{q}_2 + \alpha_3\vec{q}_3 + \alpha_4\vec{q}_4$.

Since $\mathcal{B}$ is an ONB,

$$\alpha_i = \langle \vec{q}_i, \vec{v} \rangle = \vec{q}_i^{\,T}\vec{v},\ i = 1, 2, 3, 4$$

holds so that $\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = Q^T\vec{v}$. The coordinate mapping of $\mathbb{R}^4$ wrt $\mathcal{B}$ is therefore:

$$K_{\mathcal{B}} : \mathbb{R}^4 \to \mathbb{R}^4; \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \mapsto Q^T\vec{v} = \begin{bmatrix} \frac{1}{2}v_1 + \frac{1}{2}v_2 + \frac{1}{2}v_3 + \frac{1}{2}v_4 \\ \frac{1}{2}v_1 - \frac{1}{2}v_2 - \frac{1}{2}v_3 + \frac{1}{2}v_4 \\ \frac{1}{2}v_1 + \frac{1}{2}v_2 - \frac{1}{2}v_3 - \frac{1}{2}v_4 \\ \frac{1}{2}v_1 - \frac{1}{2}v_2 + \frac{1}{2}v_3 - \frac{1}{2}v_4 \end{bmatrix}.$$

c) $K_{\mathcal{B}}^{-1}\left(\begin{bmatrix} 2 \\ 4 \\ -6 \\ -2 \end{bmatrix}\right) = 2\vec{q}_1 + 4\vec{q}_2 - 6\vec{q}_3 - 2\vec{q}_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ -2 \\ -2 \\ 2 \end{bmatrix} + \begin{bmatrix} -3 \\ -3 \\ 3 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ 1 \\ 7 \end{bmatrix}$

Exercise 7 The perpendicular $\vec{l}_k$ is orthogonal to $\vec{q}_i$, $i = 1, \dots, k-1$ if $\langle \vec{l}_k, \vec{q}_i \rangle = 0$. We verify this.

$$\begin{aligned} \langle \vec{l}_k, \vec{q}_i \rangle &= \langle \vec{b}_k - \langle \vec{b}_k, \vec{q}_1 \rangle \vec{q}_1 - \langle \vec{b}_k, \vec{q}_2 \rangle \vec{q}_2 - \cdots - \langle \vec{b}_k, \vec{q}_i \rangle \vec{q}_i - \cdots - \langle \vec{b}_k, \vec{q}_{k-1} \rangle \vec{q}_{k-1}\ ,\ \vec{q}_i \rangle \\ &= \langle \vec{b}_k, \vec{q}_i \rangle - \langle \langle \vec{b}_k, \vec{q}_1 \rangle \vec{q}_1, \vec{q}_i \rangle - \langle \langle \vec{b}_k, \vec{q}_2 \rangle \vec{q}_2, \vec{q}_i \rangle - \cdots - \langle \langle \vec{b}_k, \vec{q}_i \rangle \vec{q}_i, \vec{q}_i \rangle - \cdots - \langle \langle \vec{b}_k, \vec{q}_{k-1} \rangle \vec{q}_{k-1}, \vec{q}_i \rangle \\ &= \langle \vec{b}_k, \vec{q}_i \rangle - \langle \vec{b}_k, \vec{q}_1 \rangle \langle \vec{q}_1, \vec{q}_i \rangle - \langle \vec{b}_k, \vec{q}_2 \rangle \langle \vec{q}_2, \vec{q}_i \rangle - \cdots - \langle \vec{b}_k, \vec{q}_i \rangle \langle \vec{q}_i, \vec{q}_i \rangle - \cdots - \langle \vec{b}_k, \vec{q}_{k-1} \rangle \langle \vec{q}_{k-1}, \vec{q}_i \rangle \end{aligned}$$

It follows from construction that $\langle \vec{q}_i, \vec{q}_j \rangle = \delta_{i,j}$ for $1 \le i, j \le k-1$, i.e. :

$$\langle \vec{l}_k, \vec{q}_i \rangle = \langle \vec{b}_k, \vec{q}_i \rangle - \langle \vec{b}_k, \vec{q}_i \rangle = 0.$$

Exercise 8 The matrices $B_i, B_j, 1 \le i, j \le 3, i \ne j$ are orthogonal to each other if $\langle B_i, B_j \rangle = 0$. Due to the symmetry of the scalar product, it suffices to show this for all matrices B_i, B_j with $i < j$.

B_1, B_2 are orthogonal: $\left\langle \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \frac{1}{\sqrt{6}} \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} \right\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{6}}(1 \cdot (-1) + 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 2) = 0.$

B_1, B_3 are orthogonal: $\left\langle \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \frac{1}{2\sqrt{3}} \begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix} \right\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{2\sqrt{3}}(1 \cdot (-1) + 1 \cdot 1 + 0 \cdot 3 + 0 \cdot (-1)) = 0.$

B_2, B_3 are orthogonal: $\frac{1}{\sqrt{6}} \left\langle \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}, \frac{1}{2\sqrt{3}} \begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix} \right\rangle = \frac{1}{\sqrt{6}} \cdot \frac{1}{2\sqrt{3}}((-1) \cdot (-1) + 1 \cdot 1 + 0 \cdot 3 + 2 \cdot (-1)) = 0.$

The matrix $B_i, 1 \le i \le 3$ is normalized if $\|B_i\| = \sqrt{\langle B_i, B_i \rangle} = 1$ or, in this case, $\langle B_i, B_i \rangle = 1$.

B_1 is normalized: $\left\langle \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle = \left(\frac{1}{\sqrt{2}}\right)^2 (1^2 + 1^2) = \frac{1}{2}(2) = 1.$

B_2 is normalized: $\left\langle \frac{1}{\sqrt{6}} \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}, \frac{1}{\sqrt{6}} \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} \right\rangle = \frac{1}{6}((-1)^2 + 1^2 + 2^2) = 1.$

B_3 is normalized: $\left\langle \frac{1}{2\sqrt{3}} \begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix}, \frac{1}{2\sqrt{3}} \begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix} \right\rangle = \frac{1}{12}((-1)^2 + 1^2 + 3^2 + (-1)^2) = 1.$

9 Orthogonal and unitary mappings

9.1 Basics

Definition (orthogonal transformation)

Let V be a euclidean vector space and $Q : V \to V$ an invertible linear mapping. Then Q is an orthogonal transformation (or mapping) if for all $\vec{u}, \vec{v} \in V$

$$\langle Q\vec{u}, Q\vec{v}\rangle = \langle \vec{u}, \vec{v}\rangle.$$

Remarks

• Since the scalar product of vector is invariant under an orthogonal mapping, the norm of the vectors and the angle between them also remain the same. We say that orthogonal mappings are length and angle preserving.

• An orthogonal transformation maps an ONB onto an ONB because of the length and angle preserving properties.

Example (rotation of $\mathbb{R}^2$)

Let $\mathbb{R}^2$ be equipped with the standard scalar product. Determine the effect of the linear mapping

$$R_\varphi := \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$$

on the standard basis vectors.

$$R_\varphi \vec{e}_1 = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\varphi \\ \sin\varphi \end{bmatrix}$$

$$R_\varphi \vec{e}_2 = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\varphi \\ \cos\varphi \end{bmatrix}$$

Draw yourself a picture of the results to the side (or wait for class).

Remarks

• R_φ is an orthogonal matrix:

$$R_\varphi^T R_\varphi = \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \cos^2\varphi + \sin^2\varphi & 0 \\ 0 & \cos^2\varphi + \sin^2\varphi \end{bmatrix} = I_2.$$

R_φ rotates all vectors in $\mathbb{R}^2$ by the angle φ in the counterclockwise direction and is thus called a **rotation matrix**.

• Rotations in $\mathbb{R}^2$ are length, angle, and orientation preserving. In this context, orientation preserving means that if the vector $\vec{u}$ lies closer to $\vec{v}$ in the counterclockwise direction than in the clockwise direction, then $R_\varphi(\vec{u})$ is also closer to $R_\varphi(\vec{v})$ in the counterclockwise than in the clockwise direction.

• Note that $R(\varphi_1) \cdot R(\varphi_2) = R_{\varphi_1+\varphi_2}$ so that $R_{\varphi_1} \cdot R_{-\varphi_1} = R_0 = I_2$.

Exercise 1 (reflections im $\mathbb{R}^2$)

Let $\mathbb{R}^2$ be equipped with the standard scalar product. A matrix $A \in \mathbb{R}^{2,2}$ is called a **reflection** if the associated matrix mapping $A : \mathbb{R}^2 \to \mathbb{R}^2; \vec{x} \mapsto A\vec{x}$ produces the mirror image of each vector $\mathbb{R}^2$ about a fixed line through the origin (axis of reflection). About which axis do the following matrix mappings reflect? $X := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $Y := \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ $S_{1,1} := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Hint: Apply the mapping to an arbitrary vector $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$.

Remarks

• Reflections in $\mathbb{R}^2$ are length and angle preserving.

• If A is a reflection, is it true that $A^2 = I_2$.

• When $\mathbb{R}^n$ is equipped with the standard scalar product, orthogonal mappings can be identified with orthogonal matrices. (Recall: A square matrix $Q \in \mathbb{R}^{n,n}$ is called orthogonal if $Q^TQ = I_n$.) The columns of an orthogonal matrix form an ONB wrt the standard scalar product.

• The product of orthogonal matrices $P, Q \in \mathbb{R}^{n,n}$ is again orthogonal ($PP^T = I_n, QQ^T = I_n \Rightarrow PQ(PQ)^T = PQQ^TP^T = PI_nP^T = PP^T = I_n$).

Example (rotations in the (i,j)-coordinate plane)

$$R_{2,3}(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix}$$

is a rotation about the first axis (i.e. a rotation in the $(2,3)$-coordinate plane) because the first column is the first standard basis vector and the first row is its transpose. Because the mapping leaves the first coordinate of a vector invariant, the entire x-axis is invariant. The other entries form a rotation matrix in $\mathbb{R}^{2,2}$.

$$R_{1,3}(\varphi) = \begin{bmatrix} \cos\varphi & 0 & -\sin\varphi \\ 0 & 1 & 0 \\ \sin\varphi & 0 & \cos\varphi \end{bmatrix}$$

is a rotation about the y-axis.

• These rotations are orthogonal. For example

$$R_{2,3}(\varphi)^T \cdot R_{2,3}(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & \sin\varphi \\ 0 & -\sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos^2\varphi + \sin^2\varphi & 0 \\ 0 & 0 & \cos^2\varphi + \sin^2\varphi \end{bmatrix} = I_3.$$

What does $R_{1,2}(\varphi)$, a rotation of angle φ about the z-axis, look like?

Exercise 2 (learning check)

Verify using the matrices $R_{1,2}\left(\frac{\pi}{3}\right)$ and $R_{1,3}\left(\frac{\pi}{6}\right)$ that the multiplication of rotation matrices in $\mathbb{R}^{3,3}$ is not commutative, i.e.

$$\boldsymbol{R_{1,2}\left(\frac{\pi}{3}\right) \cdot R_{1,3}\left(\frac{\pi}{6}\right) \neq R_{1,3}\left(\frac{\pi}{6}\right) \cdot R_{1,2}\left(\frac{\pi}{3}\right)}.$$

In complex vector spaces, mappings that do not change the scalar product of vectors are called unitary transformations.

Definition (unitary transformation)

Let V be a unitary vector space and $U : V \to V$ an invertible linear mapping. Then U is a unitary transformation if and only if for all $\vec{v}, \vec{w} \in V$

$$\langle U\vec{v}, U\vec{w}\rangle = \langle \vec{v}, \vec{w}\rangle.$$

Remarks

• Unitary transformations map ONBs onto ONBs.

• Analogous to orthogonal matrices, we defined unitary matrices in Chapter 3: $U \in \mathbb{C}^{n,n}$ is called unitary if $U^*U = I_n$ holds. These matrices are the associated matrices for unitary mappings from $\mathbb{C}^n$ to $\mathbb{C}^n$ wrt the complex standard scalar product: $\left\langle \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \right\rangle := u_1\overline{v_1} + \cdots + u_n\overline{v_n}.$

9.2 The QR decomposition

Definition (QR decomposition)

A QR decomposition of a matrix $A \in \mathbb{R}^{m,m}$ is a factorization of the form $A = QR$ with $Q \in \mathbb{R}^{m,m}$ an orthogonal matrix and $R \in \mathbb{R}^{m,m}$ an upper triangular matrix.

Motivation:

If $A\vec{x} = \vec{b}$ is an SLE and $A = QR$ with Q, R as described above, we can multiply both sides of the SLE with $(QR)^{-1} = R^{-1}Q^{-1}$ to solve for $\vec{x} = R^{-1}Q^{-1}\vec{b}$:

$$\begin{aligned} QR\vec{x} &= \vec{b} \\ (QR)^{-1}QR\vec{x} &= (QR)^{-1}\vec{b} \\ R^{-1}Q^{-1}QR\vec{x} &= R^{-1}Q^{-1}\vec{b} \\ \vec{x} &= R^{-1}Q^{-1}\vec{b}. \end{aligned}$$

Because of the form of the matrices Q, R , they are easy to invert: $Q^{-1} = Q^T$ and R is already in ref, so inverting it only requires relativey few operations.

Example (QR decomposition)

Determine a QR decomposition of the invertible matrix $A := \begin{bmatrix} 3 & 1 & 4 \\ 0 & 0 & -1 \\ 4 & 7 & -3 \end{bmatrix}$.

Since A is invertible, the columns of A $\vec{a}_1 := \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$, $\vec{a}_2 := \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}$, $\vec{a}_3 := \begin{bmatrix} 4 \\ -1 \\ -3 \end{bmatrix}$ form a basis $\mathcal{B}$ of $\mathbb{R}^3$.

Idea: Apply Gram-Schmidt to the basis $\mathcal{B}$ to obtain an ONB $\mathcal{Q} := \{\vec{q}_1, \vec{q}_2, \vec{q}_3\}$. The matrix $Q := [\vec{q}_1\ \vec{q}_2\ \vec{q}_3]$ is orthogonal wrt the standard scalar product. The matrix $R := [[\vec{a}_1]_{\mathcal{Q}}\ [\vec{a}_2]_{\mathcal{Q}}\ [\vec{a}_3]_{\mathcal{Q}}]$ of the coordinate vectors of $\vec{a}_i$ wrt to the ONB $\mathcal{Q}$ is by construction an upper triangular matrix.

We have already applied the Gram-Schmidt algorithm to $\mathcal{B}$ in Chapter 8:

$$\mathcal{B}_{\text{ONB}} = \left\{ \vec{q}_1 = \begin{bmatrix} \frac{3}{5} \\ 0 \\ \frac{4}{5} \end{bmatrix}, \vec{q}_2 = \begin{bmatrix} \frac{-4}{5} \\ 0 \\ \frac{3}{5} \end{bmatrix}, \vec{q}_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \right\}.$$

Now we need to find R. We can multiply the equation $A = QR$ on both sides with $Q^{-1} = Q^T$, i.e.

$$Q^T A = Q^T QR = R,$$

so that

$$\begin{aligned} Q^T A &= \begin{bmatrix} \frac{3}{5} & 0 & \frac{4}{5} \\ -\frac{4}{5} & 0 & \frac{3}{5} \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 1 & 4 \\ 0 & 0 & -1 \\ 4 & 7 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 5 & \frac{31}{5} & 0 \\ 0 & \frac{17}{5} & -5 \\ 0 & 0 & 1 \end{bmatrix} \\ &= R. \end{aligned}$$

Verify that the columns of R are just the coordinate vectors of $\vec{a_i}$ wrt the ONB $\mathcal{Q} = \{\vec{q}_1, \vec{q}_2, \vec{q}_3\}$. The matrix R defined in this way is always upper triangular. (Why? Look again at the Gram-Schmidt algorithm if necessary.)

Exercise 3 (learning check)

a) Show that

$$\mathcal{B} := \left\{ \vec{b}_1 := \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \vec{b}_2 := \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{b}_3 := \begin{bmatrix} 2 \\ -1 \\ 4 \\ -3 \end{bmatrix}, \vec{b}_4 := \begin{bmatrix} 0 \\ 2 \\ 0 \\ -1 \end{bmatrix} \right\}$$

is a basis of $\mathbb{R}^4$.

b) Apply the Gram-Schmidt algorithm to the basis $\mathcal{B}$ to obtain an orthonormal basis (ONB) $\{\vec{q}_1,\ \vec{q}_2,\ \vec{q}_3,\ \vec{q}_4\}$ wrt the standard scalar product.

c) Show that the matrix $Q = [\vec{q}_1\ \vec{q}_2\ \vec{q}_3\ \vec{q}_4]$ defines an orthogonal matrix mapping.

d) Find a QR decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 1 & -1 & 2 \\ 3 & 1 & 4 & 0 \\ 4 & 1 & -3 & -1 \end{bmatrix}$.

Example (linear regression)
Find the line $p(t) = a_1 t + a_2$ best describing the points (measured values)

t_i	**1**	**2**	**3**	**4**
y_i	**5**	**−3**	**−2**	**6**

in the sense that the sum of the squared errors

$$F(\vec{m}) = \sum_{i=1}^{4}((m_1 t_i + m_2) - y_i)^2, \ \vec{m} = \begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$$

is minimized.
a) Write F in the form $F(\vec{m}) = \|A\vec{m} - \vec{y}\|^2$ with the standard norm. Explicitly state A and $\vec{y}$.
b) Use the basis

$$\mathcal{B} := \left\{ \vec{b}_1 := \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \vec{b}_2 := \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \vec{b}_3 := \begin{bmatrix} 2 \\ -1 \\ 4 \\ -3 \end{bmatrix}, \vec{b}_4 := \begin{bmatrix} 0 \\ 2 \\ 0 \\ -1 \end{bmatrix} \right\}$$

of $\mathbb{R}^4$ to produce a QR decomposition of A.
c) Calculate the solution to the linear regression problem and the sum of the squared errors.

a) The matrix A is defined by $A\vec{m} = \begin{bmatrix} m_1 t_1 + m_2 \\ m_1 t_2 + m_2 \\ m_1 t_3 + m_2 \\ m_1 t_4 + m_2 \end{bmatrix} = \begin{bmatrix} t_1 & 1 \\ t_2 & 1 \\ t_3 & 1 \\ t_4 & 1 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$. Plugging in the measured values, we get $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}$. Set $\vec{y} := \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ -2 \\ 6 \end{bmatrix}$. This yields

$$F(\vec{m}) = \left\| \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} - \begin{bmatrix} 5 \\ -3 \\ -2 \\ 6 \end{bmatrix} \right\|^2.$$

b) Since the standard norm is given, the associated scalar product is the standard scalar product in $\mathbb{R}^4$. The Gram-Schmidt algorithm has already applied to the basis B to get the ONB with the associated orthogonal matrix

$$Q := \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{6}} & 0 & -\frac{3}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{4}{\sqrt{30}} \\ \frac{3}{\sqrt{30}} & 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{30}} \\ \frac{4}{\sqrt{30}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{30}} \end{bmatrix}.$$

Remark: Since A is not quadratic but Q has to be (Q is an orthogonal matrix), the matrix R in the QR decomposition will not be quadratic. The sizes of Q and R are therefore 4×4 resp. 4×2.

So $Q := [\vec{q}_1 \ \vec{q}_2 \ \vec{q}_3 \ \vec{q}_4]$ and $R = Q^T[\vec{b}_1 \ \vec{b}_2 \ \vec{b}_3 \ \vec{b}_4]$ are to be a QR decomposition of A. To get R , we calculate $Q^T A$:

$$R = Q^T A = \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{3}{\sqrt{30}} & \frac{4}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ -\frac{3}{\sqrt{30}} & \frac{4}{\sqrt{30}} & \frac{1}{\sqrt{30}} & -\frac{2}{\sqrt{30}} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} \frac{30}{\sqrt{30}} & \frac{10}{\sqrt{30}} \\ 0 & \frac{2}{\sqrt{6}} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

A QR decomposition of A is thus

$$A = \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{6}} & 0 & -\frac{3}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{4}{\sqrt{30}} \\ \frac{3}{\sqrt{30}} & 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{30}} \\ \frac{4}{\sqrt{30}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{30}} \end{bmatrix} \begin{bmatrix} \frac{30}{\sqrt{30}} & \frac{10}{\sqrt{30}} \\ 0 & \frac{2}{\sqrt{6}} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

c) We are looking for a vector $\vec{m}$ with $F(\vec{m})$ minimal; i.e.

$$F(\vec{m}) = \|A\vec{m} - \vec{y}\|^2 \overset{\text{orth. transform.}}{=} \|Q^T(A\vec{m} - \vec{y})\|^2 = \|Q^T A\vec{m} - Q^T\vec{y}\|^2 = \|R\vec{m} - Q^T\vec{y}\|^2$$

needs to be minimized. Now the following is minimized

$$\begin{aligned} \|R\vec{m} - Q^T\vec{y}\|^2 &= \left\| \begin{bmatrix} \frac{30}{\sqrt{30}} & \frac{10}{\sqrt{30}} \\ 0 & \frac{2}{\sqrt{6}} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} - \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{3}{\sqrt{30}} & \frac{4}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ -\frac{3}{\sqrt{30}} & \frac{4}{\sqrt{30}} & \frac{1}{\sqrt{30}} & -\frac{2}{\sqrt{30}} \end{bmatrix} \begin{bmatrix} 5 \\ -3 \\ -2 \\ 6 \end{bmatrix} \right\|^2 \\ &= \left\| \begin{bmatrix} \frac{30}{\sqrt{30}} m_1 + \frac{10}{\sqrt{30}} m_2 \\ \frac{2}{\sqrt{6}} m_2 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{17}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} \\ -\frac{7}{\sqrt{6}} \\ -\frac{41}{\sqrt{30}} \end{bmatrix} \right\|^2 \end{aligned}$$

when

$$\begin{bmatrix} \frac{30}{\sqrt{30}} m_1 + \frac{10}{\sqrt{30}} m_2 \\ \frac{2}{\sqrt{6}} m_2 \end{bmatrix} = \begin{bmatrix} \frac{17}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}.$$

Thus we form the AM and bring it in rref.

$$\left[\begin{array}{cc|c} \frac{30}{\sqrt{30}} & \frac{10}{\sqrt{30}} & \frac{17}{\sqrt{30}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{array}\right] \xrightarrow{\frac{\sqrt{30}}{30}\text{I},\ \frac{\sqrt{6}}{2}\text{II}} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & \frac{17}{30} \\ 0 & 1 & \frac{1}{2} \end{array}\right] \xrightarrow{\text{I-}\frac{1}{3}\text{II}} \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{5} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

This leads to the line $y = \frac{2}{5}t + \frac{1}{2}$, which is the solution of the linear regression problem. The sum of the squared errors is just

$$F(\vec{m}) = F\left(\begin{bmatrix} \frac{2}{5} \\ \frac{1}{2} \end{bmatrix}\right) = \sum_{i=1}^{4} \left(\left(\frac{2}{5}t_i + \frac{1}{2}\right) - y_i\right)^2 = 64.2 = \left\| \begin{bmatrix} -\frac{7}{\sqrt{6}} \\ -\frac{41}{\sqrt{30}} \end{bmatrix} \right\|^2.$$

Remark: In reality, there will be hundreds of measured values that need to be worked into the problem, which can then be solved using software such as MATLAB.

The next example shows how to generalize the method to obtain a model using a higher degree polynomial (here with degree 2).

Example (linear regression with a quadratic polynomial

Determine the quadratic polynomial $p(t) = m_0 + m_1 t + m_2 t^2$ that best describes the four measured values

t_i	-1	0	1	2
y_i	2	1	2	3

in the sense that the sum of the squared errors

$$F(m_0, m_1, m_2) = \sum_{i=1}^{4} (p(t_i) - y_i)^2$$

is minimized.

a) The sum of the squared errors F can be written as $F(\vec{m}) = \|A\vec{m} - \vec{y}\|^2$ if we assume that the standard norm is being used. Find the matrix A and the vectors $\vec{m}, \vec{y}$.

b) An approximation (rounded to one place after the decimal) to the QR decomposition is:

$$Q = \frac{1}{10}\begin{bmatrix} -5 & 7 & 5 & 2 \\ -5 & 2 & -5 & -7 \\ -5 & -2 & -5 & 7 \\ -5 & -7 & 5 & -2 \end{bmatrix}, \; R = \begin{bmatrix} -2 & -1 & -3 \\ 0 & -2 & -2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Calculate the solution to the regression problem and the sum of the squared errors.

a) $F(\vec{m}) = \|A\vec{m} - \vec{y}\|^2$ mit

$$A := \begin{bmatrix} 1 & t_1 & t_1^2 \\ 1 & t_2 & t_2^2 \\ 1 & t_3 & t_3^2 \\ 1 & t_4 & t_4^2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}, \; \vec{m} := \begin{bmatrix} m_0 \\ m_1 \\ m_2 \end{bmatrix}, \; \vec{y} := \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 3 \end{bmatrix}.$$

b)

$$\begin{aligned} R\vec{m} - Q^T\vec{y} &= \begin{bmatrix} -2 & -1 & -3 \\ 0 & -2 & -2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} m_0 \\ m_1 \\ m_2 \end{bmatrix} - \frac{1}{10}\begin{bmatrix} -5 & -5 & -5 & -5 \\ 7 & 2 & -2 & -7 \\ 5 & -5 & -5 & 5 \\ 2 & -7 & 7 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} -2m_0 - m_1 - 3m_2 \\ -2m_1 - 2m_2 \\ 2m_2 \\ 0 \end{bmatrix} - \frac{1}{10}\begin{bmatrix} -40 \\ -9 \\ 10 \\ 5 \end{bmatrix}, \end{aligned}$$

$F(\vec{m})$ is minimal for $\begin{bmatrix} -2m_0 - m_1 - 3m_2 \\ -2m_1 - 2m_2 \\ 2m_2 \end{bmatrix} - \frac{1}{10}\begin{bmatrix} -40 \\ -9 \\ 10 \end{bmatrix} = \vec{0}.$

Solve the SLE

$$\begin{bmatrix} -2 & -1 & -3 \\ 0 & -2 & -2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} m_0 \\ m_1 \\ m_2 \end{bmatrix} = \begin{bmatrix} -4 \\ -0.9 \\ 1 \end{bmatrix}.$$

The coefficients of the polynomial are

$$m_2 = 0.5; \; m_1 = -0.05; \; m_0 = 1.275.$$

Thus

$$F(m_0, m_1, m_2) = 0.5t^2 - 0.05t + 1.275.$$

The sum of the squared errors is $\|\frac{5}{10}\|^2 = 0.25$.

9.3 Suggested solutions to the exercises in Chapter 9

Exercise 1 $X\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}1&0\\0&-1\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}x\\-y\end{bmatrix}$ (reflection about the x-axis)

$Y\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}-1&0\\0&1\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}-x\\y\end{bmatrix}$ (reflection about the y-axis)

$S_{1,1}\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}0&1\\1&0\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}y\\x\end{bmatrix}$ (reflection about the line through $\begin{bmatrix}1\\1\end{bmatrix}$ and the zero vector)

Tip: Make a sketch to better understand the answer.

Exercise 2 As per definition:

$$\boldsymbol{R_{1,2}\left(\frac{\pi}{3}\right)}=\begin{bmatrix}\cos\left(\frac{\pi}{3}\right)&-\sin\left(\frac{\pi}{3}\right)&0\\\sin\left(\frac{\pi}{3}\right)&\cos\left(\frac{\pi}{3}\right)&0\\0&0&1\end{bmatrix}=\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}&0\\\frac{\sqrt{3}}{2}&\frac{1}{2}&0\\0&0&1\end{bmatrix},$$

$$\boldsymbol{R_{1,3}\left(\frac{\pi}{6}\right)}=\begin{bmatrix}\cos\left(\frac{\pi}{6}\right)&0&-\sin\left(\frac{\pi}{6}\right)\\0&1&0\\\sin\left(\frac{\pi}{6}\right)&0&\cos\left(\frac{\pi}{6}\right)\end{bmatrix}=\begin{bmatrix}\frac{\sqrt{3}}{2}&0&-\frac{1}{2}\\0&1&0\\\frac{1}{2}&0&\frac{\sqrt{3}}{2}\end{bmatrix}.$$

We need to calculate the products $R_{1,2}\left(\frac{\pi}{3}\right)\cdot R_{1,3}\left(\frac{\pi}{6}\right)$ and $R_{1,3}\left(\frac{\pi}{6}\right)\cdot R_{1,2}\left(\frac{\pi}{3}\right)$.

$$\boldsymbol{R_{1,2}\left(\frac{\pi}{3}\right)\cdot R_{1,3}\left(\frac{\pi}{6}\right)}=\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}&0\\\frac{\sqrt{3}}{2}&\frac{1}{2}&0\\0&0&1\end{bmatrix}\begin{bmatrix}\frac{\sqrt{3}}{2}&0&-\frac{1}{2}\\0&1&0\\\frac{1}{2}&0&\frac{\sqrt{3}}{2}\end{bmatrix}=\begin{bmatrix}\frac{\sqrt{3}}{4}&-\frac{\sqrt{3}}{2}&-\frac{1}{4}\\\frac{3}{4}&\frac{1}{2}&-\frac{\sqrt{3}}{4}\\\frac{1}{2}&0&\frac{\sqrt{3}}{2}\end{bmatrix}=:\boldsymbol{A}$$

$$\neq$$

$$\boldsymbol{R_{1,3}\left(\frac{\pi}{6}\right)\cdot R_{1,2}\left(\frac{\pi}{3}\right)}=\begin{bmatrix}\frac{\sqrt{3}}{2}&0&-\frac{1}{2}\\0&1&0\\\frac{1}{2}&0&\frac{\sqrt{3}}{2}\end{bmatrix}\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}&0\\\frac{\sqrt{3}}{2}&\frac{1}{2}&0\\0&0&1\end{bmatrix}=\begin{bmatrix}\frac{\sqrt{3}}{4}&-\frac{3}{4}&-\frac{1}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}&0\\\frac{1}{4}&-\frac{\sqrt{3}}{4}&\frac{\sqrt{3}}{2}\end{bmatrix}=:\boldsymbol{B}$$

Since the products are not equal ($\boldsymbol{A}\neq\boldsymbol{B}$), the multiplication of rotation matrices in $\mathbb{R}^{3,3}$ is not commutative (even though the rotation matrices in $\mathbb{R}^{2,2}$ are).

Exercise 3 a) Every basis of $\mathbb{R}^4$ consists of four vectors. If the four vectors $\vec{b}_1,\vec{b}_2,\vec{b}_3,\vec{b}_4\in\mathbb{R}^4$ are linearly independent, then they must also form a generating set (otherwise the dimension of $\mathbb{R}^4$ would have to be greater than four). Calculating the rank of the matrix $B:=[\vec{b}_1\ \vec{b}_2\ \vec{b}_3\ \vec{b}_4]$, we find the maximal number of linearly independent rows resp. columns, i.e. the number of **leading coefficient variables** of $[\vec{b}_1\ \vec{b}_2\ \vec{b}_3\ \vec{b}_4]$ in **ref**. The ref is found using the Gauss algorithm:

$$\begin{bmatrix}1&1&2&0\\2&1&-1&2\\3&1&4&0\\4&1&-3&-1\end{bmatrix}\xrightarrow{\text{II-2I, III-3I, IV-4I}}\begin{bmatrix}1&1&2&0\\0&-1&-5&2\\0&-2&-2&0\\0&-3&-11&-1\end{bmatrix}$$

$$\xrightarrow{\text{III-2II, IV-3II}}\begin{bmatrix}1&1&2&0\\0&-1&-5&2\\0&0&8&-4\\0&0&4&-7\end{bmatrix}\xrightarrow{\text{2IV-III}}\begin{bmatrix}1&1&2&0\\0&-1&-5&2\\0&0&8&-4\\0&0&0&-10\end{bmatrix}.$$

There are 4 leading coefficient variables so that $\text{rank}(B)=4$. The set $\{\vec{b}_1,\ \vec{b}_2,\ \vec{b}_3,\ \vec{b}_4\}$ is thus a basis of $\mathbb{R}^4$.

b) Normalize $\vec{b}_1$.

$\vec{q}_1 := \frac{\vec{b}_1}{\|\vec{b}_1\|}$ mit $\|\vec{b}_1\| = \sqrt{\langle \vec{b}_1, \vec{b}_1 \rangle} = \sqrt{1^2+2^2+3^2+4^2} = \sqrt{30} \quad \Rightarrow \quad \boldsymbol{\vec{q}_1 = \frac{1}{\sqrt{30}}\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix}}$

Find the perpendicular $\vec{\ell}_2$ to the line spanned by the vector $\vec{q}_1$.

$$\vec{\ell}_2 := \vec{b}_2 - \langle \vec{b}_2, \vec{q}_1 \rangle \vec{q}_1 = \begin{bmatrix} 1\\1\\1\\1 \end{bmatrix} - \left\langle \begin{bmatrix} 1\\1\\1\\1 \end{bmatrix}, \frac{1}{\sqrt{30}}\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} \right\rangle \frac{1}{\sqrt{30}}\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} = \begin{bmatrix} 1\\1\\1\\1 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix}$$

Normalize $\vec{\ell}_2$.

$\vec{q}_2 = \frac{\vec{\ell}_2}{\|\vec{\ell}_2\|}$ with $\|\vec{\ell}_2\| = \sqrt{\langle \vec{\ell}_2, \vec{\ell}_2 \rangle} = \sqrt{\frac{1}{9}(4+1+1)} = \sqrt{\frac{2}{3}} \Rightarrow \boldsymbol{\vec{q}_2} = \frac{\sqrt{3}}{3\sqrt{2}}\begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix} = \boldsymbol{\frac{1}{\sqrt{6}}\begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix}}$

We continue with the algorithm.

$$\vec{\ell}_3 = \vec{b}_3 - \langle \vec{b}_3, \vec{q}_1 \rangle \vec{q}_1 - \langle \vec{b}_3, \vec{q}_2 \rangle \vec{q}_2$$

$$= \begin{bmatrix} 2\\-1\\4\\-3 \end{bmatrix} - \left\langle \begin{bmatrix} 2\\-1\\4\\-3 \end{bmatrix}, \frac{1}{\sqrt{30}}\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} \right\rangle \frac{1}{\sqrt{30}}\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} - \left\langle \begin{bmatrix} 2\\-1\\4\\-3 \end{bmatrix}, \frac{1}{\sqrt{6}}\begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix} \right\rangle \frac{1}{\sqrt{6}}\begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix}$$

$$= \begin{bmatrix} 2\\-1\\4\\-3 \end{bmatrix} - \frac{1}{30}(0)\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} - \frac{1}{6}(6)\begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix} = \begin{bmatrix} 0\\-2\\4\\-2 \end{bmatrix}$$

$\vec{q}_3 = \frac{\vec{\ell}_3}{\|\vec{\ell}_3\|}$ with $\|\vec{\ell}_3\| = \sqrt{\langle \vec{\ell}_3, \vec{\ell}_3 \rangle} = \sqrt{4+16+4} = \sqrt{24} = 2\sqrt{6} \Rightarrow \boldsymbol{\vec{q}_3 = \frac{1}{\sqrt{6}}\begin{bmatrix} 0\\-1\\2\\-1 \end{bmatrix}}$

$$\vec{\ell}_4 = \vec{b}_4 - \langle \vec{b}_4, \vec{q}_1 \rangle \vec{q}_1 - \langle \vec{b}_4, \vec{q}_2 \rangle \vec{q}_2 - \langle \vec{b}_4, \vec{q}_3 \rangle \vec{q}_3$$

$$= \begin{bmatrix} 0\\2\\0\\-1 \end{bmatrix} - \frac{1}{30}\left\langle \begin{bmatrix} 0\\2\\0\\-1 \end{bmatrix}, \begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} \right\rangle \begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} - \frac{1}{6}\left\langle \begin{bmatrix} 0\\2\\0\\-1 \end{bmatrix}, \begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix} \right\rangle \begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix} - \frac{1}{6}\left\langle \begin{bmatrix} 0\\2\\0\\-1 \end{bmatrix}, \begin{bmatrix} 0\\-1\\2\\-1 \end{bmatrix} \right\rangle \begin{bmatrix} 0\\-1\\2\\-1 \end{bmatrix}$$

$$= \begin{bmatrix} 0\\2\\0\\-1 \end{bmatrix} - \frac{1}{30}(0)\begin{bmatrix} 1\\2\\3\\4 \end{bmatrix} - \frac{1}{6}(3)\begin{bmatrix} 2\\1\\0\\-1 \end{bmatrix} - \frac{1}{6}(-1)\begin{bmatrix} 0\\-1\\2\\-1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -3\\4\\1\\-2 \end{bmatrix}$$

$\vec{q}_4 = \frac{\vec{\ell}_4}{\|\vec{\ell}_4\|}$ with $\|\vec{\ell}_4\| = \sqrt{\langle \vec{\ell}_4, \vec{\ell}_4 \rangle} = \sqrt{\frac{1}{9}(9+16+1+4)} = \sqrt{\frac{30}{9}} = \frac{\sqrt{30}}{3} \Rightarrow \boldsymbol{\vec{q}_4 = \frac{1}{\sqrt{30}}\begin{bmatrix} -3\\4\\1\\-2 \end{bmatrix}}$

We have constructed the ONB

$$\mathcal{B}_{\text{ONB}} := \left\{ \vec{q}_1 = \frac{1}{\sqrt{30}}\begin{bmatrix}1\\2\\3\\4\end{bmatrix}, \vec{q}_2 = \frac{1}{\sqrt{6}}\begin{bmatrix}2\\1\\0\\-1\end{bmatrix}, \vec{q}_3 = \frac{1}{\sqrt{6}}\begin{bmatrix}0\\-1\\2\\-1\end{bmatrix}, \vec{q}_4 = \frac{1}{\sqrt{30}}\begin{bmatrix}-3\\4\\1\\-2\end{bmatrix} \right\}$$

of $\mathbb{R}^4$.

c) A matrix mapping $Q : \mathbb{R}^4 \to \mathbb{R}^4$ is an orthogonal mapping if $\langle Q\vec{u}, Q\vec{v}\rangle = \langle \vec{u}, \vec{v}\rangle$ for all $\vec{u}, \vec{v} \in \mathbb{R}^4$ holds. Using algebraic manipulations on $\langle Q\vec{u}, Q\vec{v}\rangle$ we get

$$\langle Q\vec{v}, Q\vec{w}\rangle \overset{\text{standard scalar product}\downarrow}{=} (Q\vec{v})^T Q\vec{w} \overset{\downarrow\text{properties of the transpose}}{=} \vec{v}^T Q^T Q\vec{w} = \vec{v}^T\vec{w} = \langle \vec{v}, \vec{w}\rangle.$$

The last equality holds if and only if $Q^T Q = I$. Thus Q is an orthogonal matrix mapping wrt to the standard scalar product $\langle\ ,\ \rangle_{\text{Std}}$ exactly when Q is an orthogonal matrix. This can be explicitly verified with a matrix multiplication. Alternatively, we can argue as follows. For all $1 \le i, j \le 4$ we get

$$(\langle \vec{q}_i, \vec{q}_j\rangle_{\text{Std}} = \delta_{ij}) \Leftrightarrow \left(\vec{q}_i^{\,T}\vec{q}_j = \delta_{ij}\right) \Leftrightarrow \left(\vec{q}_i^{\,T}\vec{q}_j = 0 \ \ i \neq j \text{ and } \vec{q}_i^{\,T}\vec{q}_i = 1\right).$$

That means the entries on the diagonal of the matrix $Q^T Q$ are all 1s and all other entries are 0s. This is just the identity matrix so that Q (regarded as a matrix) is orthogonal. Consequently, the mapping Q is orthogonal.

d) The columns of the matrix A form the basis $\mathcal{B}$ from part a). The first step is thus already done. We can take

$$Q := \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{6}} & 0 & -\frac{3}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{4}{\sqrt{30}} \\ \frac{3}{\sqrt{30}} & 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{30}} \\ \frac{4}{\sqrt{30}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{30}} \end{bmatrix}$$

from part c). To find R, consider the equation $A = QR$. Multiply with Q^T on both sides of the equation to get $Q^T A = Q^T QR$. Since Q is an orthogonal matrix, we know that $Q^T A = R$. We calculate R and look to see that it is an upper triangular matrix:

$$R = \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{3}{\sqrt{30}} & \frac{4}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ \frac{-3}{\sqrt{30}} & \frac{4}{\sqrt{30}} & \frac{1}{\sqrt{30}} & -\frac{2}{\sqrt{30}} \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 1 & -1 & 2 \\ 3 & 1 & 4 & 0 \\ 4 & 1 & -3 & -1 \end{bmatrix} = \begin{bmatrix} \sqrt{30} & \frac{10}{\sqrt{30}} & 0 & 0 \\ 0 & \frac{2}{\sqrt{6}} & \frac{6}{\sqrt{6}} & \frac{3}{\sqrt{6}} \\ 0 & 0 & \frac{12}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & 0 & 0 & \frac{10}{\sqrt{30}} \end{bmatrix}.$$

The matrix R is indeed an upper triangular matrix so that we are a bit more confident that are calculations are correct. The QR decomposition is thus

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 1 & -1 & 2 \\ 3 & 1 & 4 & 0 \\ 4 & 1 & -3 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{6}} & 0 & -\frac{3}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{4}{\sqrt{30}} \\ \frac{3}{\sqrt{30}} & 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{30}} \\ \frac{4}{\sqrt{30}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{30}} \end{bmatrix} \begin{bmatrix} \sqrt{30} & \frac{10}{\sqrt{30}} & 0 & 0 \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{3}{\sqrt{6}} \\ 0 & 0 & \frac{12}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & 0 & 0 & \frac{10}{\sqrt{30}} \end{bmatrix}.$$

10 Determinants

10.1 Motivation in $\mathbb{R}^{1,1}$, $\mathbb{R}^{2,2}$ and $\mathbb{R}^{3,3}$

Definition (determinant von $[u] \in \mathbb{R}^{1,1}$)
Let $u \in \mathbb{R}$. The determinant of the matrix $[u] \in \mathbb{R}^{1,1}$ is defined as $\det([u]) := u$.

Remarks

- $|\det([u])| =$ the length of u
- The sign indicates the orientation of u.

Definition (determinant of $A \in \mathbb{R}^{2,2}$)

For $A := \begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix} \in \mathbb{R}^{2,2}$ define $\det(A) = \det\left(\begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix}\right) := \begin{vmatrix} \boldsymbol{u_1} & & \boldsymbol{v_1} \\ & \times & \\ \boldsymbol{u_2} & & \boldsymbol{v_2} \end{vmatrix} := \boldsymbol{u_1 v_2} - \boldsymbol{u_2 v_1}$.

Remarks:

- For $\vec{u}, \vec{v} \in \mathbb{R}^2$, the determinant $|\det[\vec{u}\ \vec{v}]|$ provides the area F of the parallelogram spanned by $\vec{u}$ and $\vec{v}$. Let h be the height of the parallelogram.

$$F = \|\vec{u}\|h \Rightarrow$$

$$\begin{aligned}
F^2 &= \|\vec{u}\|^2\|\vec{v}\|^2\sin^2(\alpha) = \|\vec{u}\|^2\|\vec{v}\|^2(1-\cos^2(\alpha)) \\
&= \|\vec{u}\|^2\|\vec{v}\|^2 - \|\vec{u}\|^2\|\vec{v}\|^2\cos^2(\alpha) \\
&= \|\vec{u}\|^2\|\vec{v}\|^2 - (\langle\vec{u},\vec{v}\rangle)^2 \quad \text{(follows from } \cos(\alpha) := \tfrac{\langle\vec{u},\vec{v}\rangle}{\|\vec{u}\|\,\|\vec{v}\|}\text{)} \\
&= \langle\vec{u},\vec{u}\rangle\langle\vec{v},\vec{v}\rangle - (\langle\vec{u},\vec{v}\rangle)^2 \quad \text{(follows } \|\vec{w}\| := \sqrt{\langle\vec{w},\vec{w}\rangle}\text{)} \\
&= (u_1^2+u_2^2)(v_1^2+v_2^2) - (u_1v_1+u_2v_2)^2 \\
&= \cdots = u_1^2v_2^2 - 2u_1u_2v_1v_2 + u_2^2v_1^2 \\
&= (u_1v_2 - u_2v_1)^2
\end{aligned}$$

$$\Rightarrow F = |u_1v_2 - u_2v_1|$$

- The sign indicates the orientation of the vectors.

Example (determinant of 2×2 matrices)
Calculate the determinants of the following matrices:

$$I := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, P := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, N := \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, O := \begin{bmatrix} 3 & 2 \\ 0 & -8 \end{bmatrix}.$$

Remark: Some rules that hold true for certain $n \times n$ matrices will be illustrated here.

$\det\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = 1\cdot 1 - 0\cdot 0 = 1$	The determinant of the identity matrix is 1.
$\det\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = 0\cdot 0 - 1\cdot 1 = -1$	Exchanging two rows changes the sign of the determinant.
$\det\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix} = 1\cdot 4 - 2\cdot 2 = 0$	The determinant is zero $\Leftrightarrow$ the rows (resp. columns) of the matrix are linearly dependent
$\det\begin{bmatrix}3 & 2\\ 0 & -8\end{bmatrix} = 3\cdot(-8) - 0\cdot 2 = -24$	The determinant of an upper triangular matrix is the product of the elements on the diagonal.

Exercise 1 (learning check)
Compute the determinant of the matrices $A = \begin{bmatrix}3 & -5\\ -4 & 6\end{bmatrix}$, A^T, A^2, A^{-1}.

Definition (vector product; determinant for $A \in \mathbb{R}^{3,3}$)
The determinant of a matrix $A := [\vec{u}\ \ \vec{v}\ \ \vec{w}] \in \mathbb{R}^{3,3}$ can be defined using the standard scalar product $\langle\, ,\, \rangle$ and the **vector product** of $\vec{v}, \vec{w} \in \mathbb{R}^3$

$$v \times w := \begin{bmatrix} v_2w_3 - v_3w_2\\ v_3w_1 - v_1w_3\\ v_1w_2 - v_2w_1\end{bmatrix}.$$

The **determinant** of A is given by

$$\det(A) = \det\left([\vec{u}\ \ \vec{v}\ \ \vec{w}]\right) := \langle u, v \times w\rangle.$$

Remarks

• $|\det[\vec{u}\ \ \vec{v}\ \ \vec{w}]\,|$ is the volume of the parallelepiped spanned by the vectors.

• $\|v \times w\|$ is the area of the parallelogram spanned by $\vec{v}$ and $\vec{w}$ (analogous to the 2-dimensional case). If β is the angle between the vectors $\vec{v}$ and $\vec{w}$, calculate

$$F^2 = \|\vec{v}\|^2\|\vec{w}\|^2\sin^2(\beta) = \|\vec{v}\|^2\|\vec{w}\|^2 - \langle\vec{v},\vec{w}\rangle^2 \text{ and } \|\vec{v}\times\vec{w}\|^2.$$

The results have to be the same after algebraic manipulations.

• The vector product of the vectors $\vec{v}$ and $\vec{w}$ is orthogonal to the plane spanned by the vectors $\vec{v}$ and $\vec{w}$. The volume of the parallelepiped is therefore $V = \|\vec{v}\times\vec{w}\|\,\|\vec{u}\|\cos\gamma$, whereby γ is the angle between the vectors $\vec{v}\times\vec{w}$ and $\vec{u}$.

• The sign indicates the orientation of the vectors.

Per definition and with the notation $|A| := \det(A)$ we have the following results:

$$\begin{aligned}
\det[\vec{u}\ \ \vec{v}\ \ \vec{w}] &= \left\langle \begin{bmatrix}u_1\\u_2\\u_3\end{bmatrix}, \begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix} \times \begin{bmatrix}w_1\\w_2\\w_3\end{bmatrix}\right\rangle\\
&= \left\langle \begin{bmatrix}u_1\\u_2\\u_3\end{bmatrix}, \begin{bmatrix}v_2w_3 - v_3w_2\\ v_3w_1 - v_1w_3\\ v_1w_2 - v_2w_1\end{bmatrix}\right\rangle\\
&= u_1(v_2w_3 - v_3w_2 + u_2(v_3w_1 - v_1w_3) + u_3(v_1w_2 - v_2w_1)\\
&= u_1\begin{vmatrix}v_2 & w_2\\ v_3 & w_3\end{vmatrix} - u_2\begin{vmatrix}v_1 & w_1\\ v_3 & w_3\end{vmatrix} + u_3\begin{vmatrix}v_1 & w_1\\ v_2 & w_2\end{vmatrix}
\end{aligned}$$

In this way we can define the determinant of a 3×3 matrix in terms of determinants involving 2×2 matrices (the so-called minors). This leads to a recursive definition for determinants of $n\times n$ matrices.

$$\begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}_{S_{1,1}(A)} \quad \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}_{S_{2,1}(A)} \quad \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}_{S_{3,1}(A)}$$

This principle will be generalized to calculate the determinant of $n \times n$ matrices for all $n \in \mathbb{N}^{\geq 2}$ (**Laplace expansion**). To formulate the theorem, we need to define the minors of the matrix.

Definition (minor)

Let $n \in \mathbb{N}^{\geq 2}$, $A \in \mathbb{F}^{n,n}$. The ij-th **minor** $S_{i,j}(A)$ is the $(n-1) \times (n-1)$ matrix that results upon deleting the **i-th row** and the **j-th column** from A .

10.2 The determinant in $\mathbb{R}^{n,n}$

Definition/Theorem (Laplace expansion theorem)

Let A be a square matrix in $\mathbb{F}^{n,n}$.

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(S_{i,j}(A)) \text{ (expansion along the } \boldsymbol{i}\textbf{-th row)}$$

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det(S_{i,j}(A)) \text{ (expansion along the } \boldsymbol{j}\textbf{-th column)}$$

Remarks

- It does not matter which row or column is used for the expansion. The result is always the same. This implies that $\det(A) = \det(A^T)$.
- The determinant is a (nonlinear) mapping $\det : \mathbb{F}^{n,n} \to \mathbb{F}$ for $n > 1$.

Pattern for the signs in the Laplace expansion

The sign of the coefficient $a_{i,k}$ is given by:

$$(-1)^{i+k} = \begin{cases} +1, & i+k \text{ even} \\ -1, & i+k \text{ odd.} \end{cases}$$

For $a_{1,1}$ we have $(-1)^{1+1} = 1$. If we move to an adjacent position in the matrix by moving vertically or horizontally, then we increase exactly one index by 1 so that the power is now odd: $(-1)^{2+1} = (-1)^{1+2} = -1$. With each step, we change the sign. The pattern for the signs is illustrated below and looks like a chessboard if we use the color black for + and white for -.:

$$\begin{bmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}.$$

Example

Let $A_1 := \begin{bmatrix} 1 & -2 & -3 \\ -4 & 5 & 6 \\ -7 & 8 & 9 \end{bmatrix} \in \mathbb{R}^{3,3}$, $A_2 := \begin{bmatrix} 1 & 0 & 2 & 3 \\ 4 & 0 & 5 & 6 \\ 7 & 8 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{bmatrix} \in \mathbb{R}^{4,4}$.

a) Find the determinant of the matrix $A_i, i = 1, 2$.

b) Are the columns of $A_i, i = 1, 2$ linearly dependent?

c) Is the matrix $A_i, i = 1, 2$ invertible?

d) Is the dimension of the kernel of the matrix $A_i, i = 1, 2$ zero?

a) For $\det(A_1)$, we expand along the second row (chosen at random) to find the minors.

$$\begin{bmatrix} 1 & -2 & -3 \\ -4 & 5 & 6 \\ -7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & -2 & -3 \\ -4 & 5 & 6 \\ -7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & -2 & -3 \\ -4 & 5 & 6 \\ -7 & 8 & 9 \end{bmatrix}$$

$$S_{2,1}(A_1) \qquad S_{2,2}(A_1) \qquad S_{2,3}(A_1)$$

The coefficients $a_{i,k}$, of the signs are determined as well:

	$S_{2,1}(A_1)$	$S_{2,2}(A_1)$	$S_{2,3}(A_1)$
$a_{i,j}$:	$\mathbf{-4}$	$\mathbf{5}$	$\mathbf{6}$
sign:	$-$	$+$	$-$

$$\begin{aligned} \det\left(\begin{bmatrix} 1 & -2 & -3 \\ -4 & 5 & 6 \\ -7 & 8 & 9 \end{bmatrix}\right) &= -(\mathbf{-4})\begin{vmatrix} -2 & -3 \\ 8 & 9 \end{vmatrix} + \mathbf{5}\begin{vmatrix} 1 & -3 \\ -7 & 9 \end{vmatrix} - (\mathbf{6})\begin{vmatrix} 1 & -2 \\ -7 & 8 \end{vmatrix} \\ &= 4((-2)9 - 8(-3)) + 5(1\cdot 9 - (-7)\cdot(-3)) - 6(1\cdot 8 - (-7)(-2)) \\ &= 4(-18+24) + 5(9-21) - 6(8-14) = 24 - 60 + 36 = 0. \end{aligned}$$

a) The determinant of A_1 is zero.

b) The columns of A_1 are linearly dependent: $-1\begin{bmatrix} 1 \\ -4 \\ -7 \end{bmatrix} - 2\begin{bmatrix} -2 \\ 5 \\ 8 \end{bmatrix} + \begin{bmatrix} -3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

c) The matrix A_1 is not invertible.

d) The dimension of the kernel of the matrix A_1 is not zero. For example, $\begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$ is in $\ker(A_1)$.

$\det(A_2)$: We begin by expanding along the second column (because of the zeros)

$$\det\begin{bmatrix} 1 & 0 & 2 & 3 \\ 4 & 0 & 5 & 6 \\ 7 & 8 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{bmatrix} = -8\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 11 & 12 \end{vmatrix}$$

(Expand along the third row)

$$= -8\left(-11\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} + 12\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}\right) = -8(-11(6-12) + 12(5-8)) = -240$$

a) The determinant of A_2 is nonzero.
b) The columns of A_2 are linearly independent. (The rref of A_2 is the identity matrix.)
c) The matrix A_2 is invertible.
d) The dimension of the kernel of the matrix A_2 is zero.

Exercise 2 (learning check)

Calculate the determinant of $B_1 := \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & -2 \\ 0 & 2 & 5 \end{bmatrix}$ and of $B_2 := \begin{bmatrix} 0 & 0 & 3 \\ 0 & -2 & 0 \\ 5 & 0 & 0 \end{bmatrix}$.

Exercise 3 (determinant; linearly independent; basis)

Check whether $\mathcal{B} := \left\{ \vec{b}_1 := \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \vec{b}_2 := \begin{bmatrix} 0 \\ 0 \\ 3 \\ 4 \end{bmatrix}, \vec{b}_3 := \begin{bmatrix} 0 \\ 5 \\ 6 \\ 0 \end{bmatrix}, \vec{b}_4 := \begin{bmatrix} 7 \\ 0 \\ 0 \\ 8 \end{bmatrix} \right\}$

is a basis of $\mathbb{R}^4$. Use the determinant.

Example (upper triangular matrix)

Find the determinant of the matrix $\begin{bmatrix} 2 & -1 & -3 \\ 0 & -5 & 6 \\ 0 & 0 & 9 \end{bmatrix}$.

$$\det \begin{bmatrix} 2 & -1 & -3 \\ 0 & -5 & 6 \\ 0 & 0 & 9 \end{bmatrix} = 2 \begin{vmatrix} -5 & 6 \\ 0 & 9 \end{vmatrix} = 2(-5 \cdot 9 - 0 \cdot 6) = (2)(-5)(9) = -90$$

The determinant of an *upper triangular matrix* is the product of the elements on the diagonal.

Example (determinant is multiplicative but not additive)

Verify using the matrices $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$ that

$$\det(AB) = \det(A) \cdot \det(B)$$

but that

$$\det(A + B) \neq \det(A) + \det(B).$$

$$\det(A) = \det\left(\begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}\right) = 10 \quad \det(B) = \det\left(\begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}\right) = -3$$

$$AB = \begin{bmatrix} -1 & 5 \\ 7 & -5 \end{bmatrix}: \qquad \det(AB) = \left(\begin{bmatrix} -1 & 5 \\ 7 & -5 \end{bmatrix}\right) = -30 = \det(A)\det(B)$$

$$A + B = \begin{bmatrix} 2 & -1 \\ 4 & 2 \end{bmatrix}: \qquad \det(A + B) = 8 \neq 10 - 3 = \det(A) + \det(B)$$

Remark: The determinant is only applicable for square matrices. If A and B are square but of different sizes, then the product is of course not defined. The formula $\det(AB) = \det(A)\det(B)$ is only valid for matrices A and B with $A, B \in \mathbb{F}^{n,n}$.

Theorem (invertible matrices and the determinant)

If $A \in \mathbb{F}^{n,n}$ is an invertible matrix, then $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

A invertible means that $\det(A) \neq 0$ and $AA^{-1} = I_n$. Since the determinant is multiplicative and $\det(I_n) = 1$, we find that:

$$\det(AA^{-1}) = \det I_n,$$
$$\det(A)\det(A^{-1}) = 1,$$
$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Consequence:

For matrices $A, B \in \mathbb{F}^{n,n}$ with A invertible:

$$\det(B) = \det(B)\det(I) = \det(B)\det(AA^{-1}) = \mathbf{\det(B)\det(A)}\det(A^{-1})$$

$$\stackrel{*}{=} \mathbf{\det(A)\det(B)}\det(A^{-1}) = \det(ABA^{-1}).$$

* We know that matrix multiplication is not commutative (AB is not necessarily BA). But $\det(B)$ and $\det(A)$ are just numbers, so that $\det(B)\det(A) = \det(A)\det(B)$.

In particular, if we have the *matrix representation* $L_{\mathcal{B}_1}$ of a linear mapping $L : V \to V$ wrt the basis $\mathcal{B}_1$ and the transition matrix $T = K_{\mathcal{B}_2}(K_{\mathcal{B}_1})^{-1}$ under a change of basis from $\mathcal{B}_1$ to the basis $\mathcal{B}_2$, we can easily calculate $L_{\mathcal{B}_2}$ as

$$L_{\mathcal{B}_2} = T L_{\mathcal{B}_1} T^{-1}$$

and therfore also

$$\det(L_{\mathcal{B}_2}) = \det(L_{\mathcal{B}_1}).$$

This implies that the determinant is an invariant of a linear mapping. It remains the same regardless of which basis we choose. Consequently, we can define the deteminant of a linear mapping through any matrix representative: $\det(L) := \det(L_{\mathcal{B}_1})$.

Exercise 4 Let $Q := [\vec{q}_1 \ \cdots \ \vec{q}_n] \in \mathbb{R}^{n,n}$ be an orthogonal matrix. Determine the volume of the parallelepiped spanned by $\vec{q}_1, \ldots, \vec{q}_n$.

Exercise 5 Let $A \in \mathbb{R}^{3,3}$ and consider A as a matrix mapping $A : \mathbb{R}^3 \to \mathbb{R}^3, \vec{v} \mapsto A\vec{v}$.
The following information is known:

$$A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \ A \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \ A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}.$$

a) Determine the matrix A.
b) Find $\det(A)$.
c) Is $A : \mathbb{R}^3 \to \mathbb{R}^3, \vec{x} \mapsto A\vec{x}$ an injective / surjective / bijective mapping?
d) Find $\ker(A)$ and $\operatorname{im}(A)$.

Further properties are illustrated using Gaussian elimination.

Example (determinant using Gaussian elimination)
Find $\det\left(\begin{bmatrix} 1 & 4 & 3 \\ 2 & 3 & -6 \\ 4 & 16 & -2 \end{bmatrix}\right)$ using Gaussian elimination.

$$\det\left(\begin{bmatrix} 1 & 4 & 3 \\ 2 & 3 & -6 \\ 4 & 16 & -2 \end{bmatrix}\right) = \begin{vmatrix} 1 & 4 & 3 \\ 2 & 3 & -6 \\ 4 & 16 & -2 \end{vmatrix}$$

One row $\pm$ a multiple of another row does not change the determinant.

$$\overset{\text{III-2II}}{=} \begin{vmatrix} 1 & 4 & 3 \\ 2 & 3 & -6 \\ 0 & 10 & 10 \end{vmatrix}$$

Exchanging two rows changes the sign of the determinant.

$$\overset{\text{III}\leftrightarrow\text{II}}{=} - \begin{vmatrix} 1 & 4 & 3 \\ 0 & 10 & 10 \\ 2 & 3 & -6 \end{vmatrix}$$

multiplication of a row with a nonzero number $\frac{1}{k}, k \neq 0$, changes the determinant by a factor of k.

$$\overset{\frac{1}{10}\text{II}}{=} -10 \begin{vmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 2 & 3 & -6 \end{vmatrix}$$

$$\overset{\text{III-2I}}{=} -10 \begin{vmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & -5 & -12 \end{vmatrix}$$

upper triangular matrix form is found

$$\overset{\text{III+5II}}{=} -10 \begin{vmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -7 \end{vmatrix}$$

$$\det\left(\begin{bmatrix} 1 & 4 & 3 \\ 2 & 3 & -6 \\ 4 & 16 & -2 \end{bmatrix}\right) = (-10)(1)(1)(-7) = 70$$

Exercise 6 (learning check)

Calculate the determinant of the matrix $\begin{bmatrix} 1 & 0 & 0 & 7 \\ 2 & 0 & 5 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & 4 & 0 & 8 \end{bmatrix}$ using Gaussian elimination.

10.3 Suggested solutions to the exercises in Chapter 10

Exercise 1

• $\det(A) = \det\left(\begin{bmatrix} 3 & -5 \\ -4 & 6 \end{bmatrix}\right) = 18 - 20 = -2$

• $\det(A^T) = \det\left(\begin{bmatrix} 3 & -4 \\ -5 & 6 \end{bmatrix}\right) = 18 - 20 = -2$

(The exercise illustrates the fact the $\det(A) = \det(A^T)$. If $\det(A)$ is known, then we do not have to calculate $\det(A^T)$.)

• $\det(A^2) = \det\left(\begin{bmatrix} 3 & -5 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} 3 & -5 \\ -4 & 6 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 29 & -45 \\ -36 & 56 \end{bmatrix}\right) = 1624 - 1620 = 4$

In this exercise, we note that it is often easier to use the fact that $\det(AB) = \det(A)\det(B)$ than it is to first multiply A and B and then find the determinant, especially if A or B has lots of zero entries, but AB does not.

$$\det(A^2) = \det(A)\det(A) = (-2)(-2) = 4.$$

• $A^{-1} = \begin{bmatrix} -3 & -\frac{5}{2} \\ -2 & -\frac{3}{2} \end{bmatrix}$, $\quad \det(A^{-1}) = \frac{9}{2} - 5 = -\frac{1}{2}$.

In this case, it is a lot easier to just apply the properties of the determinant since we already know the determinant of A: $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{-2}$.

Exercise 2 $\det(B_1) = \det\left(\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & -2 \\ 0 & 2 & 5 \end{bmatrix}\right) = \cdots = 15, \quad \det(B_2) = \det\left(\begin{bmatrix} 0 & 0 & 3 \\ 0 & -2 & 0 \\ 5 & 0 & 0 \end{bmatrix}\right) = \cdots = 30.$

Exercise 3 Every basis of the four dimensional vector space $\mathbb{R}^4$ consists of four elements. The set $\mathcal{B}$ contains four elements, so it is enough to show that the vectors are linearly independent (alternatively a generating set). The vectors $\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4$ are linearly independent if and only if $\det[\vec{b}_1\ \vec{b}_2\ \vec{b}_3\ \vec{b}_4] \neq 0$. Let us calculate the determinant using the Laplace expansion theorem.

$$\det\left(\begin{bmatrix} 1 & 0 & 0 & 7 \\ 2 & 0 & 5 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & 4 & 0 & 8 \end{bmatrix}\right) \overset{\text{Exp. 1st row}}{=} 1 \cdot \begin{vmatrix} 0 & 5 & 0 \\ 3 & 6 & 0 \\ 4 & 0 & 8 \end{vmatrix} + (-7)\begin{vmatrix} 2 & 0 & 5 \\ 0 & 3 & 6 \\ 0 & 4 & 0 \end{vmatrix}$$

$$\overset{\text{Exp. 1st row resp. 1st column}}{=} 1 \cdot (-5)\begin{vmatrix} 3 & 0 \\ 4 & 8 \end{vmatrix} + (-7) \cdot 2\begin{vmatrix} 3 & 6 \\ 4 & 0 \end{vmatrix}$$

$$= -5 \cdot 24 + (-14) \cdot (-24) = \underset{\neq 0}{24} \cdot \underbrace{(-5 + 14)}_{\neq 0} \neq 0$$

The vectors $\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4$ are therefore linearly independent. It follows that $\mathcal{B} = \{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4\}$ is a basis of $\mathbb{R}^4$.

Exercise 4 The volume of the parallelepiped spanned by $\vec{q}_1, \ldots, \vec{q}_n$ is given by $|\det(Q)|$. From the definition of an orthogonal matrix and the properties of the determinant, we obtain the following information:

$\det(Q) = \det(Q^T)$
Q is orthogonal $\Rightarrow Q^TQ = I \Rightarrow Q^T = Q^{-1}$ ($\Rightarrow Q$ invertible)
Q invertible $\Rightarrow \det(Q) = \frac{1}{\det(Q^{-1})}$

$$|\det(Q^T)| = |\det(Q)| = \frac{1}{|\det(Q^{-1})|} = \frac{1}{|\det(Q^T)|}$$

It follows that $|\det(Q^T)|^2 = 1 \Rightarrow |\det(Q)|^2 = 1 \Rightarrow |\det(Q)| = 1$. Thus the volume is 1.

Exercise 5 a) Since $A\vec{e}_i = i$-th column of A, we have $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 2 & -2 & 0 \end{bmatrix}$.

b) Since the three vectors are linearly dependent $\left(\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\right)$, we know that $\det(A) = 0$.

Alternatives:

1) $\det(A) = \det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 2 & -2 & 0 \end{bmatrix} \overset{\text{II-2I, III-2I}}{=} \det \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & -6 & -6 \end{bmatrix} \overset{\text{III-2II}}{=} \det \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} = 1\cdot(-3)\cdot 0 = 0$

2) $\det(A) \overset{\text{Entw. nach der 3. Zeile}}{=} 2\begin{vmatrix} 2 & 3 \\ 1 & 3 \end{vmatrix} - (-2)\begin{vmatrix} 1 & 3 \\ 2 & 3 \end{vmatrix} = 2(6-3) + 2(3-6) = 6 - 6 = 0$

c) Since $\det(A) = 0$, it follows that the rows of A are linearly dependent. We are dealing with a square matrix, which means that the rref $\tilde{A}$ of the matrix A has to contain a row of zeros. This means that the number of leading coefficient variables (lcv's) of the matrix $\tilde{A}$ has to be less than 3.

The dimension of the kernel of A is (3$-$ number of lcv's of $\tilde{A}$), and therefore it follows that dim(ker$(A)) > 0$. Thus A is not an injective mapping (which would be the case if the dimension of the kernel of A were 0), so it cannot be bijective.

The dimension of the image of A is the number of lcv's in $\tilde{A}$, which is less than 3. This means that A is not surjective because dim(im$(A)) \neq 3 =$ dimension of the image space $\mathbb{R}^3$.

Remark: There are lots of ways to solve this problem!

d) To find ker(A) and im(A), bring A in rref:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 2 & -2 & 0 \end{bmatrix} \overset{\text{II-2I, III-2I}}{\to} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & -6 & -6 \end{bmatrix} \overset{\text{III-2II}}{\to} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\overset{-\frac{1}{3}\text{II}}{\to} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \overset{\text{I-2II}}{\to} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} =: \tilde{A}.$$

ker(A): solve for the lcv's x_1, x_2:

$$x_1 = -x_3, \quad x_2 = -x_3.$$

Set $x_3 = 1$. Thus $\ker(A) = \operatorname{span}\left\{\begin{bmatrix}-1\\-1\\1\end{bmatrix}\right\}$.

im(A): The lcv's of $\tilde{A}$ are x_1, x_2, so that the first two columns of A form a basis of the image of A:

$$\operatorname{im}(A) = \operatorname{span}\left\{\begin{bmatrix}1\\2\\2\end{bmatrix}, \begin{bmatrix}2\\1\\-2\end{bmatrix}\right\}.$$

Exercise 6

$$\det\left(\begin{bmatrix}1&0&0&7\\2&0&5&0\\0&3&6&0\\0&4&0&8\end{bmatrix}\right) \overset{\text{II-2I}}{=} \begin{vmatrix}1&0&0&7\\0&0&5&-14\\0&3&6&0\\0&4&0&8\end{vmatrix}$$

A row $\pm$ a multiple of another row does not change the determinant.

$$\overset{\text{II}\leftrightarrow\text{III}}{=} -\begin{vmatrix}1&0&0&7\\0&3&6&0\\0&0&5&-14\\0&4&0&8\end{vmatrix}$$

Exchanging two rows changes the sign of the determinant.

$$\overset{\frac{1}{3}\text{II}}{=} -(\mathbf{3})\begin{vmatrix}1&0&0&7\\0&1&2&0\\0&0&5&-14\\0&4&0&8\end{vmatrix}$$

Multiplication of a row with $\frac{1}{\boldsymbol{k}}$, $k \neq 0$, changes the determinant by a factor $\boldsymbol{k}$.

$$\overset{\text{IV-4II}}{=} -3\begin{vmatrix}1&0&0&7\\0&1&2&0\\0&0&5&-14\\0&0&-8&8\end{vmatrix}$$

$$\overset{-\frac{1}{8}\text{IV}}{=} -3\cdot(-8)\begin{vmatrix}1&0&0&7\\0&1&2&0\\0&0&5&-14\\0&0&1&-1\end{vmatrix}$$

$$\overset{\text{III}\leftrightarrow\text{IV}}{=} 3\cdot(-8)\begin{vmatrix}1&0&0&7\\0&1&2&0\\0&0&1&-1\\0&0&5&-14\end{vmatrix}$$

$$\overset{\text{IV-5III}}{=} \mathbf{3\cdot(-8)}\begin{vmatrix}\mathbf{1}&0&0&7\\0&\mathbf{1}&2&0\\0&0&\mathbf{1}&-1\\0&0&0&\mathbf{-9}\end{vmatrix}$$

upper triangular form

$$= \mathbf{-24}\cdot\mathbf{1}\cdot\mathbf{1}\cdot\mathbf{1}\cdot(\mathbf{-9}) = 216$$

The determinant of an upper triangular matrix is the product of the elements on the diagonal.

11 Eigenvalues, eigenvectors, characteristic polynomial

11.1 Eigenvalues, eigenvectors

Definition (eigenvalue; eigenvector)

Let V be a vector space over the field $\mathbb{F}$ and $L : V \to V$ a linear mapping. The scalar $\boldsymbol{\lambda} \in \mathbb{F}$ is called an **eigenvalue** (abbr.: **EVAL**) of L, if there is a nontrivial vector $\boldsymbol{\vec{v}} \in V \setminus \{\vec{0}\}$ (i.e. $\boldsymbol{\vec{v}} \in V$, $\boldsymbol{\vec{v}} \neq \vec{0}$) such that

$$L(\boldsymbol{\vec{v}}) = \boldsymbol{\lambda}\boldsymbol{\vec{v}}.$$

The nontrivial vector $\boldsymbol{\vec{v}}$ is called an **eigenvector** (abbr.: **EVEC**) for the eigenvalue λ. In the other direction, we say that λ is the eigenvalue of the eigenvector $\vec{v}$ von L.

Comments

• The abbreviations EVAL and EVEC can be used in the examination.
• The linear mapping takes $\vec{v}$ to $\lambda\vec{v}$ means that the linear mapping scales the vector $\vec{v}$. The image $L(\vec{v})$ lies on the same line as $\vec{v}$ (i.e., which is the line through the point with coordinates in $\vec{v}$ and the origin).

For example, let V be the vector space $\mathbb{R}^2$ and $L : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear mapping with

$$L(\vec{e}_1) = 3\vec{e}_1 = \begin{bmatrix} \mathbf{3} \\ \mathbf{0} \end{bmatrix}$$

and

$$L\left(\begin{bmatrix} -\mathbf{1} \\ \mathbf{1} \end{bmatrix}\right) = -\frac{\mathbf{1}}{\mathbf{2}}\begin{bmatrix} -\mathbf{1} \\ \mathbf{1} \end{bmatrix} = \begin{bmatrix} \frac{\mathbf{1}}{\mathbf{2}} \\ -\frac{\mathbf{1}}{\mathbf{2}} \end{bmatrix}$$

Then 3 is the EVAL for the EVEC $\vec{e}_1$ and $-\frac{1}{2}$ is the EVAL for the EVEC $\begin{bmatrix} -\mathbf{1} \\ \mathbf{1} \end{bmatrix}$.

L stretches the vector $\vec{e}_1$ by a factor of 3. L shrinks the vector $\begin{bmatrix} -\mathbf{1} \\ \mathbf{1} \end{bmatrix}$ by a factor of $\frac{1}{2} = \left|-\frac{1}{2}\right|$ and because of the negative sign of the eigenvalue, it switches the orientation of the vector on the line through $\begin{bmatrix} -\mathbf{1} \\ \mathbf{1} \end{bmatrix}$ and $\vec{0}$.

Consequences:

Let $L : V \to V$ be a linear mapping with V a vector space over $\mathbb{F}$. The following statements are valid.

1. The zero vector is not an eigenvector!

The zero vector solves the equation for any $\lambda \in \mathbb{F}$ since $L(\vec{0}) = \vec{0} = \lambda\vec{0}$. There is nothing special about λ in this case. This is why we exclude $\vec{0}$ as an eigenvector.

2. L is **injective** if and only if 0 is **not** an eigenvalue of L.

If L is injective, then the kernel of L can only contain the zero vector. The equation $L(\vec{v}) = \vec{0} = 0\vec{v}$ is therefore only satisfied when $\vec{v} = \vec{0}$ so that 0 is not an eigenvalue of L. In the other direction, if 0 is not an eigenvalue of L, then there are no nontrivial vectors $\vec{v} \neq \vec{0}$ with $L(\vec{v}) = 0\vec{v} = \vec{0}$. The kernel of L consists only of the zero vector so that L is an injective mapping.

3. If L is **not injective**, then 0 is an eigenvalue of L. The **eigenvectors** of the **eigenvalue 0** are just the elements in $\mathbf{ker}\, \boldsymbol{L} \setminus \{\vec{\mathbf{0}}\}$: $\quad \vec{v} \in \ker(L), \vec{v} \neq \vec{0} \Rightarrow L(\boldsymbol{\vec{v}}) = \vec{0} = \mathbf{0}\boldsymbol{\vec{v}}$.

4. If $L : V \to V; \vec{v} \mapsto \vec{0}$ is the **zero mapping**, then every element $\boldsymbol{\vec{v}} \in \boldsymbol{V} \setminus \{\vec{\mathbf{0}}\}$ is an **EVEC** of L for the **EVAL** $\boldsymbol{\lambda} = \mathbf{0}$.

5. If $\boldsymbol{\vec{v}}$ is an **EVEC** of L for the **EVAL** $\boldsymbol{\lambda} \in \mathbb{F}$, then all elements of the form $\boldsymbol{\alpha\vec{v}}$ $(\alpha \in \mathbb{F} \setminus \{0\})$ are also

EVECs of L for the same **EVAL** $\boldsymbol{\lambda}$:

$\vec{v} \neq \vec{0}$ **EVEC** for EVAL $\boldsymbol{\lambda}$	$\Rightarrow L(\vec{v}) = \lambda\vec{v}$	definition EVAL, EVEC
	$\Rightarrow \alpha L(\vec{v}) = \alpha\lambda\vec{v}$	multiplication with α
	$\Rightarrow L(\alpha\vec{v}) = \lambda(\alpha\vec{v})$	linearity of L

$$\Rightarrow \text{span}\{\vec{v}\} \setminus \{\vec{0}\} \text{ are eigenvectors of } L \text{ for the eigenvalue } \lambda.$$

6. If $\vec{u}, \vec{v}$ are linearly independent **EVECs** for the same **EVAL** $\boldsymbol{\lambda}$, i.e.

$$L(\vec{u}) = \lambda\vec{u}, \ \ L(\vec{v}) = \lambda\vec{v},$$

then every linear combination $\alpha\vec{u} + \beta\vec{v}$ with $\alpha, \beta \in \mathbb{F}$ (not both zero) is an **EVEC** for the **EVAL** $\boldsymbol{\lambda}$:

$$\begin{aligned} L(\alpha\vec{u} + \beta\vec{v}) &= L(\alpha\vec{u}) + L(\beta\vec{v}) && \text{linearity} \\ &= \alpha L(\vec{u}) + \beta L(\vec{v}) && \text{linearity} \\ &= \alpha(\lambda\vec{u}) + \beta(\lambda\vec{v}) && \text{apply mapping} \\ &= \lambda(\alpha\vec{u}) + \lambda(\beta\vec{v}) && \text{commutativity of scalars} \\ &= \lambda(\alpha\vec{u} + \beta\vec{v}). && \text{factoring} \end{aligned}$$

To illustrate this better, consider a linear mapping $L : \mathbb{R}^3 \to \mathbb{R}^3$ with exactly two linearly independent eigenvectors $\vec{u}, \vec{v}$ for the eigenvalue -2. All possible linear combinations of $\vec{u}, \vec{v}$ (span$\{\vec{u}, \vec{v}\}$) define a plane through $\vec{u}, \vec{v}$ and $\vec{0}$, which is a subspace of $\mathbb{R}^3$. All vectors in this plane (except for the zero vector) are eigenvectors for the same eigenvalue. Make a sketch of this situation using the standard basis vectors e_2 and e_3 as two linearly independent eigenvectors.

Example (eigenvalues and eigenvectors of a matrix mapping)

Let $V = \mathbb{C}^2$ and $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$. Verify that $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ are eigenvectors of the linear mapping $A : V \to V; \vec{v} \mapsto A\vec{v}$.

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i^2 \\ i \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ i \end{bmatrix}, \quad \begin{bmatrix} 1 \\ i \end{bmatrix} \text{ is EVEC of } A \text{ for EVAL } \lambda_1 = 1$$

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} i^2 \\ i \end{bmatrix} = \begin{bmatrix} -1 \\ i \end{bmatrix} = -1 \cdot \begin{bmatrix} 1 \\ -i \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -i \end{bmatrix} \text{ is EVEC of } A \text{ for EVAL } \lambda_2 = -1$$

Example (eigenvalues and eigenvectors of a mapping with polynomials)

Consider the vector space $V = \mathbb{R}_{\leq 3}[x]$ and the linear mapping $L : V \to V; \ p \mapsto x \cdot p'$. Show that $x^3, x^2, x, 1$ are eigenvectors of L.

$$\begin{aligned} L(x^3) &= x \cdot 3x^2 = 3 \cdot x^3 && \text{EVAL}: \lambda_3 = 3 && \text{EVEC}: x^3 \\ L(x^2) &= x \cdot 2x = 2 \cdot x^2 && \text{EVAL}: \lambda_2 = 2 && \text{EVEC}: x^2 \\ L(x) &= x \cdot 1 = 1 \cdot x && \text{EVAL}: \lambda_1 = 1 && \text{EVEC}: x \\ L(1) &= x \cdot 0 = 0 \cdot 1 && \text{EVAL}: \lambda_0 = 0 && \text{EVEC}: 1 \end{aligned}$$

Exercise 1 (learning check)

Detemine the eigenvalue for each of the eigenvectors $\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}2\\3\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$ of the matrix mapping $A : \mathbb{R}^3 \to \mathbb{R}^3$ with $A := \begin{bmatrix}-16 & 7 & 11\\-6 & 2 & 6\\-16 & 7 & 11\end{bmatrix}$.

Exercise 2

Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear mapping with an eigenvector $\vec{b}_1 := \begin{bmatrix}-1\\2\end{bmatrix}$ for the eigenvalue $\lambda_1 := 3$ and another eigenvector $\vec{b}_2 := \begin{bmatrix}3\\-1\end{bmatrix}$ for the eigenvalue $\lambda_2 := -2$.

(a) Determine $L\left(\vec{b}_1\right)$ and $L\left(\vec{b}_2\right)$.

(b) Find the matrix representative of L wrt the basis $\mathcal{B} := \left\{\vec{b}_1, \vec{b}_2\right\}$ of $\mathbb{R}^2$.

(c) Find the matrix representative $L_{\mathcal{B}_{\text{std}}}$ for the standard basis $\mathcal{B}_{\text{std}}$ of $\mathbb{R}^2$.

(d) Determine $L\left(\begin{bmatrix}a\\b\end{bmatrix}\right)$.

Definition (eigenspace; geometric multiplicity of an eigenvalue)

The **eigenspace** (abbr.: ES) of the eigenvalue λ of the linear mapping $L : V \to V$ is the set of all solutions to the equation

$$V_\lambda = \{\text{eigenvectors for } \lambda\} \cup \{\vec{0}\}.$$

The **geometric multiplicity** (abbr: geoM) of the eigenvalue λ is given by $\dim V_\lambda$ = maximal number of *linearly independent* EVEC for the EVAL λ.

Comments:

• In particular, $\vec{0}$ is an element of every eigenspace since the eigenvalue equation is satisfied using the zero vector.

• The geometric multiplicity is defined for each individual eigenvalue.

Theorem (eigenvalues and linear independence of eigenvectors)

Let $L : V \to V$ be a linear mapping with eigenvector $\vec{v}_1$ for the eigenvalue λ_1 and EVEC $\vec{v}_2$ for EVAL λ_2.

The following statement holds. $\lambda_1 \neq \lambda_2 \Rightarrow \vec{v}_1, \vec{v}_2$ linearly independent

The statement (using the same assumptions) is equivalent to the following statement, which is easier to prove: $\vec{v}_1, \vec{v}_2$ linearly dependent $\Rightarrow \lambda_1 = \lambda_2$

Assume $\vec{v}_1, \vec{v}_2$ are linearly dependent, i.e. there exist $\alpha, \beta \in \mathbb{F}$ (not both zero) with

$$\alpha\vec{v}_1 + \beta\vec{v}_2 = \vec{0} \Leftrightarrow \beta\vec{v}_2 = -\alpha\vec{v}_1.$$

Since the eigenvectors $\vec{v}_1, \vec{v}_2$ are not the zero vector, α and β both have to be nonzero.

On the one hand, we have that $\vec{v}_1(\neq \vec{0})$ is an EVEC for the EVAL λ_1, i.e. $-\alpha\vec{v}_1$ is also an EVEC for the EVAL λ_1 (see point 5 on page 117) so that the eigenvalue equation

$$L(-\alpha\vec{v}_1) = \lambda_1(-\alpha\vec{v}_1)$$

is satisfied. Since $\boldsymbol{\beta}\boldsymbol{\vec{v}_2} = -\alpha\vec{v}_1$, we can set $\boldsymbol{\beta}\boldsymbol{\vec{v}_2}$ for $-\alpha\vec{v}_1$ in the equation:

$$L(\boldsymbol{\beta}\boldsymbol{\vec{v}_2}) = \lambda_1(\boldsymbol{\beta}\boldsymbol{\vec{v}_2}).$$

On the other hand, it follows from $\boldsymbol{\vec{v}_2}(\neq \vec{0})$ is an EVEC for the EVAL $\boldsymbol{\lambda_2}$ that $\boldsymbol{\beta}\boldsymbol{\vec{v}_2}$ is an EVEC for the EVAL $\boldsymbol{\lambda_2}$:

$$L(\boldsymbol{\beta}\boldsymbol{\vec{v}_2}) = \boldsymbol{\lambda_2}(\boldsymbol{\beta}\boldsymbol{\vec{v}_2}).$$

Therefore

$$\lambda_1(\boldsymbol{\beta}\boldsymbol{\vec{v}_2}) = \boldsymbol{\lambda_2}(\boldsymbol{\beta}\boldsymbol{\vec{v}_2}).$$

Since $\beta \neq 0, \vec{v}_2 \neq \vec{0}$ hold, it must be the case that $\lambda_1 = \boldsymbol{\lambda_2}$.

Consequences

- Let $\vec{v}_1, \ldots, \vec{v}_k$ be eigenvectors of L for the eigenvalues $\lambda_1, \ldots, \lambda_k$.

  $\lambda_1, \ldots, \lambda_k$ pairwise different eigenvalues (i.e. all individual eigenvalues are different)

  $\Rightarrow \vec{v}_1, \ldots, \vec{v}_k$ linearly independent.

- If V is finite dimensional and $L : V \to V$ is a linear mapping, then L has at most $\dim V$ eigenvalues.

Example (eigenspace; geometric multiplicity)

The matrix $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (considered as a matrix mapping $A : \mathbb{R}^3 \to \mathbb{R}^3; \vec{v} \mapsto A\vec{v}$) has two known eigenvectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ for the eigenvalue λ_1 and one known eigenvector $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ for the eigenvalue λ_2. What are the values for λ_1 and λ_2? What are the eigenspaces associated with λ_1 and λ_2? State the geometric multipicities of λ_1 and λ_2.

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \lambda_1 = 1$$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \lambda_2 = 0$$

The eigenspace for λ_1 contains $\operatorname{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. Since $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are two vectors that are not multiples of each other, they are linearly independent so that the dimension of the eigenspace is at least 2. It follows that $(\operatorname{geoM} \lambda_1) \geq 2$.

The eigenspace for λ_2 contains $\operatorname{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$. There is at least one linearly independent vector in $V_{\lambda_2 = 0}$ so that the dimension of the eigenspace is at least one. Therefore $(\operatorname{geoM} \lambda_2) \geq 1$.

Eigenvectors for different eigenvalues are known to be linearly independent. We have already found three linearly independent eigenvectors in $\mathbb{R}^3$. There cannot be any further linearly independent vectors because otherwise the dimension of $\mathbb{R}^3$ would have to be greater than 3, which is absurd. The eigenspaces are thus

$$V_{\lambda_1=1} = \text{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}, \quad V_{\lambda_2=0} = \text{span}\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix}\right\}$$

and $(\text{geoM}\,\lambda_1) = 2$, $(\text{geoM}\,\lambda_2) = 1$.

11.2 Characteristic polynomial

Let V be a finite-dimensional vector space over the field $\mathbb{F}$. For a given linear mapping $L : V \to V$, we want to find all $\boldsymbol{\lambda} \in \mathbb{F}$ so that there exists a vector $\vec{\boldsymbol{v}} \in V \setminus \{\vec{0}\}$ satisfying the eigenvalue equation: $L(\vec{\boldsymbol{v}}) = \boldsymbol{\lambda}\vec{\boldsymbol{v}}$.

To determine how to solve this problem, consider first the case of a matrix mapping. Let $A \in \mathbb{F}^{n,n}$ and z a variable with values in $\mathbb{F}$. The eigenvalue equation takes the form

$$A\vec{v} = z\vec{v} \text{ and } \vec{v} \neq \vec{0} \Leftrightarrow A\vec{v} - z\vec{v} = \vec{0} \text{ and } \vec{v} \neq \vec{0}.$$

Note that the vector $\vec{v}$ cannot be factored out from $A\vec{v} - z\vec{v}$ because that would lead to $(A - z)\vec{v}$ with A a matrix and z a number. That does not make sense. Multiplication of $\vec{v}$ with the diagonal matrix zI_n from the left (i.e. $(zI_n)\vec{v}$) has the same effect as the scalar multiplication of $\vec{v}$ with z, namely $z\vec{v} = \begin{bmatrix}zv_1\\ \vdots \\ zv_n\end{bmatrix}$, so that for $\vec{v} \neq \vec{0}$

$$A\vec{v} - z\vec{v} = \vec{0} \Leftrightarrow A\vec{v} - (zI_n)\vec{v} = \vec{0} \Leftrightarrow (A - zI_n)\vec{v} = \vec{0}.$$

We are searching for a $z \in \mathbb{F}$, so that $\ker(A - zI_n) \neq \{\vec{0}\}$. This holds if and only if $\det(A - zI_n) = 0$.

Definition (characteristic polynomial of a matrix mapping; algebraic multiplicity of an eigenvalue)
$p_A(z) = \det(A - zI_n)$ is the **characteristic polynomial** of A. (abbr: char poly)
The zeros $\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_k$, $(\lambda_i \neq \lambda_j, i \neq j)$ of the characteristic polynomial $p_A(z)$ are the eigenvalues of A; i.e.

$$p_A(z) = (\boldsymbol{\lambda}_1 - z)^{\boldsymbol{r}_1} \cdots (\boldsymbol{\lambda}_k - z)^{\boldsymbol{r}_k}$$

The **algebraic multiplicity** of $\boldsymbol{\lambda}_i$ is $\boldsymbol{r}_i$ (abbr: $(\text{algM}\,\boldsymbol{\lambda}_i) = \boldsymbol{r}_i$), i.e. how often the EVAL λ_i is represented as a zero in the characteristic polynomial.

Example (char poly; EVAL; algM)

Let A be the matrix mapping defined by $A : \mathbb{C}^3 \to \mathbb{C}^3;\ \vec{v} \mapsto \begin{bmatrix}3 & 1 & 0\\0 & 3 & 0\\0 & 0 & 0\end{bmatrix}\vec{v}$. Determine the eigenvalues of A and their algebraic multiplicities.

$$p_A(z) = \det(A - zI_3) = \det\left(\begin{bmatrix}3 & 1 & 0\\0 & 3 & 0\\0 & 0 & 0\end{bmatrix} - \begin{bmatrix}z & 0 & 0\\0 & z & 0\\0 & 0 & z\end{bmatrix}\right)$$

$$= \det\left(\begin{bmatrix}3-z & 1 & 0\\0 & 3-z & 0\\0 & 0 & -z\end{bmatrix}\right) = (-z)(3-z)^2$$

$$p_A(z) = 0 \Leftrightarrow (z = \mathbf{0} \text{ or } z = \mathbf{3})$$

The char poly is $p_A(z) = (-z)^{\mathbf{1}}(3-z)^{\mathbf{2}}$. The eigenvalues of A are: $\boldsymbol{\lambda_1 = 0}$ with algM $\mathbf{1}$, $\boldsymbol{\lambda_{2/3} = 3}$ with algM $\mathbf{2}$.

Tip: The determinant of an upper triangular matrix is the product of the entries on the diagonal. If A is an upper triangular matrix, then so is $A - zI_n$. The eigenvalues are theus the entries on the diagonal of A. (Why?)

Example (EVEC; geoM)

Let A be the matrix mapping $A : \mathbb{C}^3 \to \mathbb{C}^3;\ \vec{v} \mapsto \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \vec{v}$. Determine the eigenvectors of A and the geometric multiplicity of the eigenvalues 0 and 3.

The eigenvalues are the numbers λ_i that make $\det(A - \lambda_i I_n) = 0$. The determinant of $A - \lambda_i I_n$ is 0 if $\ker(A - \lambda_i I_n)$ is nontrivial. To calculate the eigenvectors, we need to find the nonzero elements in the kernel of $A - \lambda_i I$, i.e. the solution set to the homogeneous SLE $(A - \lambda_i I_n)\vec{v} = 0$ without the zero vector.

For each EVAL, we calculate the kernel of the matrix $\begin{bmatrix} 3-\lambda_i & 1 & 0 \\ 0 & 3-\lambda_i & 0 \\ 0 & 0 & -\lambda_i \end{bmatrix}$.

$\lambda_1 = 0$:
$$\begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \overset{\text{rref}}{\to} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We can follow the algorithm for determining a basis of the kernel of a matrix (see page 41): $x_1 = 0, x_2 = 0$ and $x_3 = 1$, since x_3 is a free variable (nlcv): $V_{\lambda_1} = \left\{ \alpha \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \;\middle|\; \alpha \in \mathbb{C} \right\} = \text{span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$.

Thus the set of EVECs (for the EVAL λ_1) is $V_{\lambda_1} \setminus \{\vec{0}\}$, so (geoM λ_1)$= 1$.
Note: (algM λ_1) $=$ (geoM λ_1)

$\lambda_{2/3} = 3$: From $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{bmatrix} \overset{\text{rref}}{\to} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ it follows that $V_{\lambda_{2/3}} = \left\{ \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \;\middle|\; \beta \in \mathbb{C} \right\} = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$.

The set of EVECs (for the EVAL $\lambda_{2/3}$) is $V_{\lambda_{2/3}} \setminus \{\vec{0}\}$, so (geoM λ_2)$= 1$.
Note: (algM λ_2) $\neq$ (geoM λ_2)

Tip: The λ_i were chosen so that the kernel of $A - \lambda_i I$ contains more elements than the zero vector. Since $A - \lambda_i I$ is a square matrix, it means that the rref of $A - \lambda_i I$ has to contain a row of zeros, leading to at least one nlcv. If you only get the zero vector in $\ker(A - \lambda_i I)$ (only leading coefficient variables in (r)ref), then you have made an error somewhere in your computations.

Remarks

• In this example the algebraic multiplicity of the eigenvalue $\lambda_{2/3}$ is different from its geometric multiplicity. For λ_1 the algebraic and the geometric multiplicities agree. To determine if the algM is equal to the geoM for each eigenvalue, you have to check every single one. It is not enough to check one individual value, as is demonstrated in the example.

• In general, the geometric multiplicity of an eigenvalue has to be at least 1, otherwise we would not have found that eigenvalue with the eigenvalue equation. The geometric multiplicity of an EVAL is bounded above by its algebraic multiplicity (of the same EVAL):

$$1 \leq \text{ geoM } \lambda_i \leq \text{ algM } \lambda_i \tag{8}$$

If the algebraic multiplicity of an eigenvalue is 1, then the geometric multiplicity of that eigenvalue is also 1.

Here is a summary for determining eigenvalues and eigenvectors.

Algorithm (eigenvalues; eigenspaces; eigenvectors of a matrix mapping)

Given the vector space $\mathbb{F}^n$ and the matrix mapping $A : \mathbb{F}^n \to \mathbb{F}^n; \vec{v} \mapsto A\vec{v}$, the eigenvalues and associated eigenvectors can be calculated using the following steps.

1. Find the characteristic polynomial $p_A(z) = \det(A - zI)$.
2. Set the characteristic polynomial to 0 and solve. The eigenvalues $\lambda_1, \ldots, \lambda_n$ are just the roots (or zeros).
3. Each eigenspace V_{λ_i} is found by solving $\ker(A - \lambda_i I)$ $(i = 1, \ldots, n)$. The eigenvectors are the vectors in $V_{\lambda_i} \setminus \{\vec{0}\}$.

Example (eigenvalues; eigenspaces)

Find the eigenvalues and eigenspaces of the linear mapping

$$A : \mathbb{C}^3 \to \mathbb{C}^3;\ \vec{v} \mapsto \begin{bmatrix} 1 & 0 & -4 \\ 0 & 5 & 4 \\ -4 & 4 & 3 \end{bmatrix} \vec{v}.$$

Step 1 (char poly)

$$\begin{aligned} p_A(z) &= \det \begin{bmatrix} 1-z & 0 & -4 \\ 0 & 5-z & 4 \\ -4 & 4 & 3-z \end{bmatrix} \\ &= (1-z) \begin{vmatrix} 5-z & 4 \\ 4 & 3-z \end{vmatrix} + (-4) \begin{vmatrix} 0 & 5-z \\ -4 & 4 \end{vmatrix} \\ &= (1-z)((5-z)(3-z) - 16) + (-4)(0 - (-4)(5-z)) \\ &= -z^3 + 9z^2 + 9z - 81 \end{aligned}$$

Step 2 (EVAL = roots of the char poly)

Set $p_A(z) = 0$ and solve for z:

$$p_A(z) = -z^3 + 9z^2 + 9z - 81 = 0$$

$$(z - 3)^1 (z - (-3))^1 (z - 9)^1 = 0$$

$$\text{EVAL} : \boldsymbol{\lambda_1 = 3},\ \boldsymbol{\lambda_2 = -3},\ \boldsymbol{\lambda_3 = 9}$$

Note: The algM **of each** EVAL is 1 (red tinted numbers in the exponents), i.e. the geoM **of each** EVAL is 1 (compare the inequality (8) on page 122).

Step 3 (eigenspace for λ_i: V_{λ_i})

$$\lambda_1 = 3 : \begin{bmatrix} 1-3 & 0 & -4 \\ 0 & 5-3 & 4 \\ -4 & 4 & 3-3 \end{bmatrix} \overset{\text{rref}}{\to} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \Rightarrow \quad V_{\lambda_1} = \left\{ \alpha \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \,\middle|\, \alpha \in \mathbb{C} \right\}$$

$$\lambda_2 = -3 : \begin{bmatrix} 1+3 & 0 & -4 \\ 0 & 5+3 & 4 \\ -4 & 4 & 3+3 \end{bmatrix} \overset{\text{ref}}{\to} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \Rightarrow \quad V_{\lambda_2} = \left\{ \beta \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} \,\middle|\, \beta \in \mathbb{C} \right\}$$

$$\lambda_3 = 9 : \begin{bmatrix} 1-9 & 0 & -4 \\ 0 & 5-9 & 4 \\ -4 & 4 & 3-9 \end{bmatrix} \overset{\text{ref}}{\to} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \Rightarrow \quad V_{\lambda_3} = \left\{ \gamma \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \,\middle|\, \gamma \in \mathbb{C} \right\}$$

Exercise 3 (learning check)

Determine the eigenvalues and eigenspaces of the matrix $B := \begin{bmatrix} -1 & 12 & -28 \\ 0 & 9 & -18 \\ 0 & 3 & -6 \end{bmatrix}$.

Example (complex eigenvalues)

Find the eigenvalues and the associated eigenspaces for the matrix $A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$.

Find the characteristic polynomial and eigenvalues on your own for practice.
Your result should be $p_A(z) = z^2 - 4z + 5$ with eigenvalues $\lambda_{1/2} = 2 \pm i$.

$$\ker(A - \lambda_1 I_2) = \ker \begin{bmatrix} 1-(2+i) & -2 \\ 1 & 3-(2+i) \end{bmatrix} = \ker \begin{bmatrix} -1-i & -2 \\ 1 & 1-i \end{bmatrix}$$

Apply Gaussian elimination:

$$\begin{bmatrix} -1-i & -2 \\ 1 & 1-i \end{bmatrix} \xrightarrow{\mathrm{I} \leftrightarrow \mathrm{II}} \begin{bmatrix} 1 & 1-i \\ -1-i & -2 \end{bmatrix} \xrightarrow{\mathrm{II}+(1+i)I} \begin{bmatrix} 1 & 1-i \\ 0 & 0 \end{bmatrix}.$$

(lcv: x_1, nlcv: x_2)

$$\begin{aligned} x_1 &= (-1+i)x_2 \\ x_2 &= r \in \mathbb{C} \end{aligned} \qquad \text{leads to } V_{\lambda_1} = \left\{ r \begin{bmatrix} -1+i \\ 1 \end{bmatrix} \middle| \; r \in \mathbb{C} \right\}$$

Perform a similar calculation for V_{λ_2}.

Result: $V_{\lambda_2} = \left\{ s \begin{bmatrix} -1-i \\ 1 \end{bmatrix} \middle| \; s \in \mathbb{C} \right\}$

Example (complex eigenvalues; eigenspaces; rotations in $\mathbb{R}^2$)

Let $R_\varphi = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$ be a rotation matrix. Find the eigenvalues and the associated eigenspaces of R_φ for $\varphi = \frac{\pi}{4}$.

The **eigenvalues (EVAL)** of the matrix $R_{\frac{\pi}{4}} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$ are the **roots of the characteristic polynomial** $p_{R_{\frac{\pi}{4}}}(z) = \det(R_{\frac{\pi}{4}} - zI_2)$. The characteristic polynomial is:

$$\begin{aligned} p_{R_{\frac{\pi}{4}}}(z) = \det\left(\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} - z \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) &= \begin{vmatrix} \frac{\sqrt{2}}{2} - z & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} - z \end{vmatrix} \\ &= \left(\frac{\sqrt{2}}{2} - z \right)^2 + \left(\frac{\sqrt{2}}{2} \right)^2 = z^2 - \sqrt{2}z + 1. \end{aligned}$$

The zeros of $p_{R_{\frac{\pi}{4}}}(z)$ are $\lambda_1 = \frac{\sqrt{2}+i\sqrt{2}}{2}, \quad \lambda_2 = \frac{\sqrt{2}-i\sqrt{2}}{2}$.

Remarks

• A rotation by $\frac{\pi}{4}$ has no real eigenvalues or eigenvectors since no vector gets sent to a vector on the same line except the zero vector. (Make a sketch to illustrate this fact.) There are however **complex** eigenvalues. If we consider $R_{\frac{\pi}{4}}$ as a **complex** mapping $R_{\frac{\pi}{4}} : \mathbb{C}^2 \to \mathbb{C}^2$ (which will be our convention unless explicitly stated otherwise), then $R_{\frac{\pi}{4}}$ can no longer be interpreted as a rotation of the plane.

• The characteristic polynomial has real coefficients. The fundamental theorem of algebra says that the characteristic polynomial can be factored into linear factors over the complex numbers. The complex roots come in complex conjugate pairs. Thus if $a+bi, b \neq 0$ is a root of a char poly p with real coefficients, then $a-bi$ is also a root of p. (*Caution!* Polynomials with complex coefficients can also be factorized into linear factors. The roots no longer need to come in conjugate pairs. For example, the polynomial $p(z) = z - i$ has only the one root $z_1 = i$.)

To determine the **eigenspaces** $V_{\lambda_i}, i = 1, 2$, calculate $\mathbf{ker}(R_{\frac{\pi}{4}} - \lambda_i I_2)$.

$$\boxed{\lambda_1} \quad \ker(R_{\frac{\pi}{4}} - \lambda_1 I_2) = \text{Kern}\begin{bmatrix} \frac{\sqrt{2}}{2} - \frac{\sqrt{2}+i\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} - \frac{\sqrt{2}+i\sqrt{2}}{2} \end{bmatrix} = \ker\begin{bmatrix} -\frac{i\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{i\sqrt{2}}{2} \end{bmatrix}$$

Apply Gaussian elimination:

$$\begin{bmatrix} -\frac{i\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{i\sqrt{2}}{2} \end{bmatrix} \xrightarrow{\frac{2i}{\sqrt{2}}\mathrm{I},\, \frac{2}{\sqrt{2}}\mathrm{II}} \begin{bmatrix} -i^2 & -i \\ 1 & -i \end{bmatrix} = \begin{bmatrix} 1 & -i \\ 1 & -i \end{bmatrix} \xrightarrow{\mathrm{II}-\mathrm{I}} \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}.$$

(lcv: x_1, nlcv: x_2)

$$x_1 = ix_2$$
$$x_2 = r \in \mathbb{C}$$

$V_{\lambda_1} = \left\{ r \begin{bmatrix} i \\ 1 \end{bmatrix} \middle| \; r \in \mathbb{C} \right\}$ is the eigenspace of EVAL $\lambda_1 = \frac{\sqrt{2}+i\sqrt{2}}{2}$.

$$\boxed{\lambda_2} \quad \ker(R_{\frac{\pi}{4}} - \lambda_2 I_2) = \text{Kern}\begin{bmatrix} \frac{\sqrt{2}}{2} - \frac{\sqrt{2}-i\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} - \frac{\sqrt{2}-i\sqrt{2}}{2} \end{bmatrix} = \ker\begin{bmatrix} \frac{i\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{i\sqrt{2}}{2} \end{bmatrix}$$

Apply Gaussian elimination:

$$\begin{bmatrix} \frac{i\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{i\sqrt{2}}{2} \end{bmatrix} \xrightarrow{-\frac{2i}{\sqrt{2}}\mathrm{I},\, \frac{2}{\sqrt{2}}\mathrm{II}} \begin{bmatrix} -i^2 & i \\ 1 & i \end{bmatrix} = \begin{bmatrix} 1 & i \\ 1 & i \end{bmatrix} \xrightarrow{\mathrm{II}-\mathrm{I}} \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}.$$

(lcv: x_1, nlcv: x_2)

$$x_1 = -ix_2$$
$$x_2 = s \in \mathbb{C}$$

$V_{\lambda_2} = \left\{ s \begin{bmatrix} -i \\ 1 \end{bmatrix} \middle| \; s \in \mathbb{C} \right\}$ is the eigenspace of EVAL $\lambda_2 = \frac{\sqrt{2}-i\sqrt{2}}{2}$.

Exercise 4 (learning check)

Consider the matrix $R_\varphi = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \in \mathbb{C}^{2,2}$. Find the EVALs and the associated ESs of R_φ for a general $\varphi \in]0, \pi[$.

11.3 Connection to matrix representatives

In this section we consider finite-dimensional vector spaces V over the field $\mathbb{F}$ and linear mappings $L : V \to V$.

Let $A, S \in \mathbb{C}^{n,n}$ be matrices with S invertible.

- $p_A(z) = p_{A^T}(z)$, whereby A^T is the transpose of A (compare with the Laplace expansion theorem).

- $p_A(z) = p_{SAS^{-1}}(z)$.

If S is interpreted as a transition matrix under a change of basis we get the following theorem.

Theorem (characteristic polynomial and matrix representatives)

Let V be a finite-dimensional vector space with two given bases $\mathcal{B}_1$ and $\mathcal{B}_2$. If $L : V \to V$ is a linear mapping with representation matrices $L_{\mathcal{B}_1}$ and $L_{\mathcal{B}_2}$ wrt the bases $\mathcal{B}_1, \mathcal{B}_2$, then it follows that $p_{L_{\mathcal{B}_1}}(z) = p_{L_{\mathcal{B}_2}}(z)$ since $L_{\mathcal{B}_2} = SL_{\mathcal{B}_1}S^{-1}$.

The theorem tells us that the characteristic polynomial is invariant under a change of basis, i.e. the characteristic polynomial of $L : V \to V$ is independent of the choice of basis. Thus we can also define the characteristic polynomial for mappings that are not given as a matrix mapping. This is done by using any matrix representative of the mapping.

Definition (characteristic polynomial of a general linear mapping)

Let V be a finite-dimensional vector space with $\dim(V) = n$. The **characteristic polynomial** of a linear mapping $L : V \to V$ wrt a basis $\mathcal{B}$ of V is the characteristic polynomial of $L_{\mathcal{B}}$, the matrix representative of L wrt the basis $\mathcal{B}$, i.e. $p_L(z) := \det(L_{\mathcal{B}} - zI_n)$.

Example (characteristic polynomial; eigenvalues)

Determine the characteristic polynomial and the eigenvalues of the linear mapping

$$L : \begin{array}{ccc} \mathbb{R}_{\leq 3}[x] & \to & \mathbb{R}_{\leq 3}[x] \\ p & \mapsto & x \cdot p' \end{array}.$$

Use the basis $\mathcal{B} := \{x^3, x^2, x, 1\}$.

The matrix representative is $L_{\mathcal{B}} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. (Why?) The characteristic polynomial is thus

$$\begin{aligned} p_{L_{\mathcal{B}}}(z) = \det(L_{\mathcal{B}} - zI_4) &= \det \begin{bmatrix} 3-z & 0 & 0 & 0 \\ 0 & 2-z & 0 & 0 \\ 0 & 0 & 1-z & 0 \\ 0 & 0 & 0 & -z \end{bmatrix} \\ &= (3-z)(2-z)(1-z)(-z). \end{aligned}$$

The EVAL are the roots of $p_{L_{\mathcal{B}}}(z)$: $3, 2, 1, 0$ (just like they were given in the example on page 118). Find the eigenvectors of $L_{\mathcal{B}}$. How can you turn these into the EVECs of L?

Example (eigenvalues and eigenvectors of a general mapping)

Determine the eigenvalues and an associated eigenvector for the linear mapping $L : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}_{\leq 1}[x]$; $ax + b \mapsto (-12a + 15b)x + (-10a + 13b)$.

First choose an easy basis $\mathcal{B}$ of $\mathbb{R}_{\leq 1}[x]$ to get a matrix representative $L_{\mathcal{B}}$. According to the theorem on page 126, the characteristic polynomial of $L_{\mathcal{B}}$ is independent of the choice of basis so that the eigenvalues of the matrix mapping $L_{\mathcal{B}}$ (i.e. the roots of the chacteristic polynomial of the matrix $L_{\mathcal{B}}$) are also the eigenvalues of L.

<u>matrix representative wrt the basis:</u>

Choose $\mathcal{B} := \{x, 1\}$. The matrix representative of the mapping L wrt the basis $\mathcal{B}$ is given by $L_{\mathcal{B}} :=$

$[L(x)_{\mathcal{B}} \quad L(1)_{\mathcal{B}}]$. So the columns are the coordinate vectors of the images of the basis vectors. For the calculation it is helpful to know the coordinate mapping $K_{\mathcal{B}}$ of $\mathbb{R}_{\leq 1}[x]$ wrt the basis $\mathcal{B}$ as well as the inverse $K_{\mathcal{B}}^{-1}$:

$$K_{\mathcal{B}} : \mathbb{R}_{\leq 1}[x] \to \mathbb{R}^2;\ ax+b \mapsto \begin{bmatrix} a \\ b \end{bmatrix} \qquad K_{\mathcal{B}}^{-1} : \mathbb{R}^2 \to \mathbb{R}_{\leq 1}[x];\ \begin{bmatrix} a \\ b \end{bmatrix} \mapsto ax+b. \qquad \text{(Why?)}$$

$L(x) = -12x - 10$

$K_{\mathcal{B}}(-12x-10) = \begin{bmatrix} -12 \\ -10 \end{bmatrix} \leftarrow$ 1st column of $L_{\mathcal{B}}$

$L(1) = 15x + 13$

$K_{\mathcal{B}}(15x+13) = \begin{bmatrix} 15 \\ 13 \end{bmatrix} \leftarrow$ 2nd column of $L_{\mathcal{B}}$

Therefore $L_{\mathcal{B}} = \begin{bmatrix} -12 & 15 \\ -10 & 13 \end{bmatrix}$, so that the characteristic polynomial is

$$\begin{aligned} p_L(z) = p_{L_{\mathcal{B}}} = \det(L_{\mathcal{B}} - zI_2) &= \det\left(\begin{bmatrix} -12-z & 15 \\ -10 & 13-z \end{bmatrix}\right) \\ &= (-12-z)(13-z) + 150 = z^2 - z - 6 = (z+2)(z-3). \end{aligned}$$

The roots of $p_L(z)$ are $\lambda_1 := -2$ and $\lambda_2 := 3$. These are the EVALs of the matrix mapping $L_{\mathcal{B}}$ and the linear mapping L.

<u>eigenvectors of the linear matrix mapping $L_{\mathcal{B}}$</u>

Show that $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is an EVEC for the EVAL $\lambda_1 = -2$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an EVEC for the EVAL $\lambda_2 = 3$ of $L_{\mathcal{B}}$.

<u>eigenvectors of the linear mapping L</u>

An EVEC for the EVAL $\lambda_1 = -2$ of L is given by $K_{\mathcal{B}}^{-1}\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = 3x + 2$.

An EVEC for the EVAL $\lambda_2 = 3$ of L is given by $K_{\mathcal{B}}^{-1}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = x + 1$.

11.4 Suggested solutions to the exercises in Chapter 11

Exercise 1 $A \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \\ -5 \end{bmatrix} = -5 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow -5$ is the EVAL of the EVEC $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ of A.

$A \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \Rightarrow 0$ is the EVAL of the EVEC $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ of A.

$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow 2$ is the EVAL of the EVEC $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ of A.

Exercise 2 (a) $\vec{b}_i$ is an eigenvector for the eigenvalue λ_i, $i = 1, 2 \quad \Rightarrow \quad L(\vec{b}_i) = \lambda_i \vec{b}_i$:

$$L\left(\begin{bmatrix} \mathbf{-1} \\ \mathbf{2} \end{bmatrix}\right) = \mathbf{3}\begin{bmatrix} \mathbf{-1} \\ \mathbf{2} \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}, \qquad L\left(\begin{bmatrix} \mathbf{3} \\ \mathbf{-1} \end{bmatrix}\right) = \mathbf{-2}\begin{bmatrix} \mathbf{3} \\ \mathbf{-1} \end{bmatrix} = \begin{bmatrix} -6 \\ 2 \end{bmatrix}.$$

(b) The ith column of $L_{\mathcal{B}}$ is the coordinate vector of $L(\vec{b}_i)$ wrt the basis $\mathcal{B}$.

$$L\left(\begin{bmatrix}-1\\2\end{bmatrix}\right)=\begin{bmatrix}-3\\6\end{bmatrix}=\mathbf{3}\begin{bmatrix}-1\\2\end{bmatrix}+\mathbf{0}\begin{bmatrix}3\\-1\end{bmatrix}\Rightarrow \text{ 1. Spalte von } L_{\mathcal{B}} \text{ ist } \begin{bmatrix}\mathbf{3}\\\mathbf{0}\end{bmatrix}.$$

$$L\left(\begin{bmatrix}3\\-1\end{bmatrix}\right)=\begin{bmatrix}-6\\2\end{bmatrix}=\mathbf{0}\begin{bmatrix}-1\\2\end{bmatrix}+(\mathbf{-2})\begin{bmatrix}3\\-1\end{bmatrix}\Rightarrow \text{ 2. Spalte von } L_{\mathcal{B}} \text{ ist } \begin{bmatrix}\mathbf{0}\\\mathbf{-2}\end{bmatrix}.$$

The matrix representative of L wrt $\mathcal{B}$ is $\begin{bmatrix}\mathbf{3}&\mathbf{0}\\\mathbf{0}&\mathbf{-2}\end{bmatrix}$.

(c) The ith column of $L_{\mathcal{B}_{\text{std}}}$ is the coordinate vector of $L(\vec{e}_i)$ wrt the basis $\mathcal{B}_{\text{std}}$. This is just $L(\vec{e}_i)$.

(Alternative 1) Write $\vec{e}_i$ as a linear combination of the vectors $\vec{b}_1, \vec{b}_2$ and use the linearity of L.

$$\alpha_1\begin{bmatrix}\mathbf{-1}\\\mathbf{2}\end{bmatrix}+\alpha_2\begin{bmatrix}\mathbf{3}\\\mathbf{-1}\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}$$

leads to the SLE

$$\begin{bmatrix}-1&3\\2&-1\end{bmatrix}\begin{bmatrix}\alpha_1\\\alpha_2\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}$$

with the AM $\left[\begin{array}{rr|rr}-1&3&1&0\\2&-1&0&1\end{array}\right]$. Bring the AM into rref:

$$\xrightarrow{\text{II}+2\text{I}}\left[\begin{array}{rr|rr}-1&3&1&0\\0&5&2&1\end{array}\right]\xrightarrow{-\text{I},\frac{1}{5}\text{II}}\left[\begin{array}{rr|rr}1&-3&-1&0\\0&1&\frac{2}{5}&\frac{1}{5}\end{array}\right]\xrightarrow{\text{I}+3\text{II}}\left[\begin{array}{rr|rr}1&0&\frac{1}{5}&\frac{3}{5}\\0&1&\frac{2}{5}&\frac{1}{5}\end{array}\right].$$

$$\begin{bmatrix}1\\0\end{bmatrix}=\frac{\mathbf{1}}{\mathbf{5}}\begin{bmatrix}\mathbf{-1}\\\mathbf{2}\end{bmatrix}+\frac{\mathbf{2}}{\mathbf{5}}\begin{bmatrix}\mathbf{3}\\\mathbf{-1}\end{bmatrix},\qquad\begin{bmatrix}0\\1\end{bmatrix}=\frac{\mathbf{3}}{\mathbf{5}}\begin{bmatrix}\mathbf{-1}\\\mathbf{2}\end{bmatrix}+\frac{\mathbf{1}}{\mathbf{5}}\begin{bmatrix}\mathbf{3}\\\mathbf{-1}\end{bmatrix}$$

$$L\left(\begin{bmatrix}1\\0\end{bmatrix}\right)=L\left(\frac{1}{5}\begin{bmatrix}-1\\2\end{bmatrix}+\frac{2}{5}\begin{bmatrix}3\\-1\end{bmatrix}\right)=\frac{1}{5}L\left(\begin{bmatrix}-1\\2\end{bmatrix}\right)+\frac{2}{5}L\left(\begin{bmatrix}3\\-1\end{bmatrix}\right)$$

$$=\frac{1}{5}\begin{bmatrix}-3\\6\end{bmatrix}+\frac{2}{5}\begin{bmatrix}-6\\2\end{bmatrix}=\begin{bmatrix}-\frac{3}{5}\\\frac{6}{5}\end{bmatrix}+\begin{bmatrix}-\frac{12}{5}\\\frac{4}{5}\end{bmatrix}=\begin{bmatrix}\mathbf{-3}\\\mathbf{2}\end{bmatrix}$$

Verify with a similar approach that $L\left(\begin{bmatrix}0\\1\end{bmatrix}\right)=\begin{bmatrix}-3\\4\end{bmatrix}$ so that $L_{\mathcal{B}_{\text{std}}}=\begin{bmatrix}\mathbf{-3}&-3\\\mathbf{2}&4\end{bmatrix}$.

(Alternative 2) Use $L_{\mathcal{B}_{\text{std}}}=K_{\mathcal{B}}^{-1}L_{\mathcal{B}}K_{\mathcal{B}}$ to change the basis.

Remark: $K_{\mathcal{B}}$ and $K_{\mathcal{B}}^{-1}$ are mappings between $\mathbb{R}^2$ and $\mathbb{R}^2$ so that each can be regarded as a matrix mapping.

$$K_{\mathcal{B}}:\mathbb{R}^2\to\mathbb{R}^2;\quad\begin{bmatrix}u\\v\end{bmatrix}\mapsto\begin{bmatrix}\alpha_1\\\alpha_2\end{bmatrix},\text{ whereby } \alpha_1\vec{\boldsymbol{b}}_{\mathbf{1}}+\alpha_2\vec{\boldsymbol{b}}_{\mathbf{2}}=\begin{bmatrix}u\\v\end{bmatrix}.$$

Write down the SLE:

$$\alpha_1\begin{bmatrix}\mathbf{-1}\\\mathbf{2}\end{bmatrix}+\alpha_2\begin{bmatrix}\mathbf{3}\\\mathbf{-1}\end{bmatrix}=\begin{bmatrix}u\\v\end{bmatrix}\quad\Rightarrow\quad\begin{array}{rcrcl}-\alpha_1&+&3\alpha_2&=&u\\2\alpha_1&-&\alpha_2&=&v.\end{array}$$

Bring the AM into rref:

$$\left[\begin{array}{rr|r} -1 & 3 & u \\ 2 & -1 & v \end{array}\right] \xrightarrow{\text{II}+2\text{I}} \left[\begin{array}{rr|c} -1 & 3 & u \\ 0 & 5 & 2u+v \end{array}\right] \xrightarrow{-\text{I},\,\frac{1}{5}\text{II}} \left[\begin{array}{rr|c} 1 & -3 & -u \\ 0 & 1 & \frac{2}{5}u+\frac{1}{5}v \end{array}\right] \xrightarrow{\text{I}+3\text{II}} \left[\begin{array}{rr|c} 1 & 0 & \frac{1}{5}u+\frac{3}{5}v \\ 0 & 1 & \frac{2}{5}u+\frac{1}{5}v \end{array}\right].$$

$$K_{\mathcal{B}} : \mathbb{R}^2 \to \mathbb{R}^2;\ \begin{bmatrix} u \\ v \end{bmatrix} \mapsto \begin{bmatrix} \frac{1}{5}u+\frac{3}{5}v \\ \frac{2}{5}u+\frac{1}{5}v \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{3}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}.$$

Calculate $K_{\mathcal{B}}^{-1}$:

$$K_{\mathcal{B}}^{-1}\left(\begin{bmatrix} u \\ v \end{bmatrix}\right) = u\vec{b}_1 + v\vec{b}_2 = u\begin{bmatrix} -1 \\ 2 \end{bmatrix} + v\begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}.$$

(Or just invert the matrix $\begin{bmatrix} \frac{1}{5} & \frac{3}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}$.)

Determine $L_{\mathcal{B}_{\text{std}}}$:

$$L_{\mathcal{B}_{\text{Std}}} = K_{\mathcal{B}}^{-1} L_{\mathcal{B}} K_{\mathcal{B}} = \begin{bmatrix} -1 & 3 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} \frac{1}{5} & \frac{3}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \mathbf{-3} & \mathbf{-3} \\ \mathbf{2} & \mathbf{4} \end{bmatrix}.$$

(What does the commutative diagram look like? Look at Chapter 7 if you cannot produce the diagram on your own.)

(d) $L\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} \mathbf{-3} & \mathbf{-3} \\ \mathbf{2} & \mathbf{4} \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -3a-3b \\ 2a+4b \end{bmatrix}$

Exercise 3 Results to verify your calculations:

eigenvalues: $0, -1, 3$

eigenspaces: $V_{\lambda_1=0} = \text{span}\left\{\begin{bmatrix} -4 \\ 2 \\ 1 \end{bmatrix}\right\}$, $V_{\lambda_2=-1} = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right\}$, $V_{\lambda_3=3} = \text{span}\left\{\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}\right\}$

Exercise 4 Since φ lies between 0 and π, we know that $0 < \sin\varphi < 1$.

Remark: R_φ rotates the vectors in the plane by a positive angle less than π so that we are expecting only complex eigenvalues.

The characteristic polynomial $p_{R_\varphi}(z)$ is given by

$$\det(R_\varphi - zI_2) = \begin{vmatrix} \cos\varphi - z & -\sin\varphi \\ \sin\varphi & \cos\varphi - z \end{vmatrix} = (\cos\varphi - z)^2 + \sin^2\varphi$$
$$= z^2 - (2\cos\varphi)z + \cos^2\varphi + \sin^2\varphi = z^2 - (2\cos\varphi)z + 1.$$

The eigenvalues of R_φ are the zeros (or roots) of the characteristic polynomial:

$$\lambda_{1/2} = \cos\varphi \pm \sqrt{\cos^2\varphi - 1} = \cos\varphi \pm \sqrt{-\sin^2\varphi},$$
$$\overset{\sin\varphi>0}{\Rightarrow} \qquad \lambda_1 = \cos\varphi + i\,\sin\varphi, \quad \lambda_2 = \cos\varphi - i\,\sin\varphi.$$

The eigenspace for the eigenvalue λ_1 is $\ker(R_\varphi - \lambda_1 I_2)$:

$$\ker\begin{bmatrix} \cos\varphi - (\cos\varphi + i\,\sin\varphi) & -\sin\varphi \\ \sin\varphi & \cos\varphi - (\cos\varphi + i\,\sin\varphi) \end{bmatrix} = \ker\begin{bmatrix} -i\,\sin\varphi & -\sin\varphi \\ \sin\varphi & -i\,\sin\varphi \end{bmatrix}.$$

We apply Gaussian elimination using the information $\sin\varphi \neq 0$:

$$\begin{bmatrix} -i\,\sin\varphi & -\sin\varphi \\ \sin\varphi & -i\,\sin\varphi \end{bmatrix} \xrightarrow{\frac{i}{\sin\varphi}\mathrm{I},\frac{1}{\sin\varphi}\mathrm{II}} \begin{bmatrix} 1 & -i \\ 1 & -i \end{bmatrix} \xrightarrow{\mathrm{II}-\mathrm{I}} \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$$

$$\text{with}\ \begin{array}{ll} \text{LV:} & x_1 = ix_2, \\ \text{FV:} & x_2 = r \in \mathbb{C}. \end{array}$$

$V_{\lambda_1} = \left\{ r \begin{bmatrix} i \\ 1 \end{bmatrix} \mid r \in \mathbb{C} \right\}$ is the eigenspace for the eigenvalue $\lambda_1 = \cos\varphi + i\,\sin\varphi$.

Analogously, the eigenspace for the eigenvalue $\lambda_2 = \cos\varphi - i\,\sin\varphi$ is given by $V_{\lambda_2} = \left\{ s \begin{bmatrix} -i \\ 1 \end{bmatrix} \mid s \in \mathbb{C} \right\}$.

12 Diagonalizability of matrices

12.1 Motivation and definition

Computing with diagonal matrices is easy. Compare:

$$\begin{bmatrix} 3 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}^8 = \begin{bmatrix} 3^8 & 0 \\ 0 & \left(\frac{1}{2}\right)^8 \end{bmatrix} = \begin{bmatrix} 6561 & 0 \\ 0 & \frac{1}{256} \end{bmatrix},$$

$$\begin{aligned} \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}^8 &= \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}^4 \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}^2 \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}^2 \\ = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}^4 \begin{bmatrix} -1 & -10 \\ 5 & 14 \end{bmatrix} \begin{bmatrix} -1 & -10 \\ 5 & 14 \end{bmatrix} &= \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}^4 \begin{bmatrix} -49 & -130 \\ 65 & 146 \end{bmatrix} \\ = \begin{bmatrix} -49 & -130 \\ 65 & 146 \end{bmatrix} \begin{bmatrix} -49 & -130 \\ 65 & 146 \end{bmatrix} &= \begin{bmatrix} -6049 & -12610 \\ 6305 & 12866 \end{bmatrix}. \end{aligned}$$

Goal: Let V be a vector space over the field $\mathbb{F}$. For a given linear mapping $L : V \to V$ choose (if possible) a basis $\mathcal{B}$ of V so that the matrix representation of the mapping L wrt $\mathcal{B}$ is a diagonal matrix.

Question: Which basis is "correct"?

Answer: If the vector space V has an **eigenbasis** (basis of eigenvectors) $\mathcal{B} := \{\vec{b}_1, \ldots, \vec{b}_n\}$ wrt the linear mapping L, which means that $L(\vec{b}_i) = \lambda_i \vec{b}_i, i = 1, \ldots, n$, than A is **diagonalizable** (and vice versa) ! The **algM** and the **geoM** of each eigenvalue have to agree if there is to be a basis of eigenvectors. (If there is an eigenbasis of L then there are infinitely many since our base field is infinite. So there is then no one basis that is correct. Any basis of EVECs will do.)

Why is the statement true? The columns of the matrix $D := L_{\mathcal{B}}$ are $D\vec{e}_i$. On the other hand, the i-th column of D is

$$\begin{aligned} K_{\mathcal{B}}(L(K_{\mathcal{B}}^{-1}(\vec{e}_i))) &= K_{\mathcal{B}}(L(\vec{b}_i)) \\ &= K_{\mathcal{B}}(\lambda_i \vec{b}_i) \\ &= \lambda_i \vec{e}_i, \end{aligned}$$

as we see in the commutative diagram (compare with Chapter 7):

$$\begin{tikzcd} V \arrow[r, "L"] \arrow[d, "K_{\mathcal{B}}"] & V \arrow[d, "K_{\mathcal{B}}"'] \\ \mathbb{F}^n \arrow[r, "L_{\mathcal{B}}=D"'] & \mathbb{F}^n \end{tikzcd}$$

The matrix D is then the diagonal matrix with the EVALs on the diagonal:

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

put in the corresponding order to match with the EVECs.

In particular, we can read off the simple formula $A = SDS^{-1}$ from the extended commutative diagram, whereby A is the matrix associated to L and D is the matrix representative of L wrt the basis $\mathcal{B}$:

$$\begin{tikzcd} & V \arrow[r, "L"] \arrow[d, "K_{\mathcal{B}}"] \arrow[ddl, "K_{\mathcal{B}_{\text{std}}}=I_{\mathbb{F}^n}"'] & V \arrow[d, "K_{\mathcal{B}}"'] \arrow[ddr, "K_{\mathcal{B}_{\text{std}}}=I_{\mathbb{F}^n}"] & \\ & \mathbb{F}^n \arrow[r, "L_{\mathcal{B}}=D"'] \arrow[dl, "T"] & \mathbb{F}^n \arrow[dr, "T"'] & \\ \mathbb{F}^n \arrow[rrr, "L_{\mathcal{B}_{\text{std}}}=A"'] & & & \mathbb{F}^n \end{tikzcd}$$

The matrix $I_{\mathbb{F}^n}$ is the identity mapping $I_{\mathbb{F}^n} : \mathbb{F}^n \to \mathbb{F}^n; \vec{v} \to \vec{v}$. The transition matrix T under a change of basis from $\mathcal{B}$ to $\mathcal{B}_{\text{std}}$ contains the eigenvectors of the linear mapping A since the i-th column of S is

$$T\vec{e}_i = K_{\mathcal{B}_{\text{std}}}(K_{\mathcal{B}}^{-1}(\vec{e}_i)) = K_{\mathcal{B}_{\text{std}}}(\vec{b}_i) = \vec{b}_i.$$

Definition (diagonalizability)

Let V be a vector space over the field $\mathbb{F}$ and $L : V \to V$ be a linear mapping. L is called diagonalizable if V has an eigenbasis wrt L.

Remark: The algebraic multiplicity and the geometric multiplicity of each EVAL have to agree.

Example (diagonalization)

Is the matrix mapping $A := \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$ diagonalizable (i.e. does A possess two linearly independent eigenvectors)? If so, which basis $\mathcal{B}$ can be chosen so that the matrix representative $D := A_{\mathcal{B}}$ of A wrt $\mathcal{B}$ is diagonal?

Step 1: Compute EVALs and EVECs.

char poly: $p_A(z) = \det(A - zI) = \begin{vmatrix} 1-z & -2 \\ 1 & 4-z \end{vmatrix} = (1-z)(4-z) + 2 = z^2 - 5z + 6 = (z - \mathbf{2})(z - \mathbf{3})$

EVAL: $p_A(z) = (z-2)(z-3) = 0 \Rightarrow \boldsymbol{\lambda_1 := 2}, \boldsymbol{\lambda_2 := 3}$

The matrix A has two pairwise different EVALs
$\Rightarrow$ (algM von λ_i) = (geoM von λ_i) = 1
$\Rightarrow$ there exists a basis of $\mathbb{C}^2$ consisting of linearly independent EVECs of A
$\Rightarrow$ *A is diagonalizable*!

eigenvectors:

- for $\boldsymbol{\lambda_1 = 2}$

$$\ker \begin{bmatrix} 1-2 & -2 \\ 1 & 4-2 \end{bmatrix} = \ker \begin{bmatrix} -1 & -2 \\ 1 & 2 \end{bmatrix} = \ker \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

The EVECs for $\boldsymbol{\lambda_1 = 2}$ are span $\left\{ \begin{bmatrix} \mathbf{-2} \\ \mathbf{1} \end{bmatrix} \right\} \setminus \{\vec{0}\}$.

- for $\boldsymbol{\lambda_2 = 3}$

$$\ker \begin{bmatrix} 1-3 & -2 \\ 1 & 4-3 \end{bmatrix} = \ker \begin{bmatrix} -2 & -2 \\ 1 & 1 \end{bmatrix} = \ker \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

The EVECs for $\boldsymbol{\lambda_2 = 3}$ are span$\left\{ \begin{bmatrix} \mathbf{1} \\ \mathbf{-1} \end{bmatrix} \right\} \setminus \{\vec{0}\}$.

Step 2: Define a basis.

$\begin{bmatrix} \mathbf{-2} \\ \mathbf{1} \end{bmatrix}, \begin{bmatrix} \mathbf{1} \\ \mathbf{-1} \end{bmatrix}$ are linearly independent EVECs (because they are EVECs for different EVALs). Every basis of $\mathbb{C}^2$ contains two vectors ($\dim \mathbb{C}^2 = 2$) so that $\mathcal{B} := \left\{ \boldsymbol{\vec{b}_1} := \begin{bmatrix} \mathbf{-2} \\ \mathbf{1} \end{bmatrix}, \vec{b}_2 := \begin{bmatrix} \mathbf{1} \\ \mathbf{-1} \end{bmatrix} \right\}$ is a basis of $\mathbb{C}^2$ consisting of EVALs (of A).

Step 3: Find the matrix representative of $A_{\mathcal{B}}$.

The coordinate vectors of the image of the basis vectors are the columns of $A_{\mathcal{B}}$:

$$\begin{aligned} A\begin{bmatrix} -2 \\ 1 \end{bmatrix} &= \mathbf{2}\begin{bmatrix} -2 \\ 1 \end{bmatrix} = \mathbf{2}\boldsymbol{\vec{b}_1} + 0\vec{b}_2 \Rightarrow [A\boldsymbol{\vec{b}_1}]_{\mathcal{B}} = \begin{bmatrix} \mathbf{2} \\ 0 \end{bmatrix} \\ A\begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \mathbf{3}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0\boldsymbol{\vec{b}_1} + \mathbf{3}\vec{b}_2 \Rightarrow [A\vec{b}_2]_{\mathcal{B}} = \begin{bmatrix} 0 \\ \mathbf{3} \end{bmatrix} \end{aligned} \quad \Rightarrow A_{\mathcal{B}} = \begin{bmatrix} \mathbf{2} & 0 \\ 0 & \mathbf{3} \end{bmatrix}.$$

General situation: Let $A \in \mathbb{C}^{n,n}$ be the matrix representative of a linear mapping wrt a basis, whereby A possesses n linearly independent EVECs $\vec{b}_1, \ldots, \vec{\boldsymbol{b}}_n$ for the (not necessarily pairwise different) EVALs $\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_n$. Define the basis $\mathcal{B} := \left\{\vec{\boldsymbol{b}}_1, \ldots, \vec{b}_n\right\}$ and the matrix $S := \left[\vec{\boldsymbol{b}}_1 \cdots \vec{b}_n\right]$. Then S is invertible (why?) and with $D = \operatorname{diag}\{\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_n\}$: $D = S^{-1}AS$ and $A = SDS^{-1}$.

Example (*non*-diagonalizable matrix)

Show that the matrix $C := \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is not diagonalizable.

char poly: $p_C(z) = (z-3)^2$

EVAL: $\lambda_{1/2} = 3$ (algM of $\lambda_{1/2}$ is 2)

ES: $V_{\lambda_{1/2}} = \ker \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \operatorname{span}\left\{\vec{b}_1 := \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right\}$ (geoM of $\lambda_{1/2}$ is 1)

Note: It is not possible to find a linearly independent vector $\vec{b}_2$ to $\vec{b}_1$ so that the coordinate vector of $\vec{b}_2$ wrt $\{\vec{b}_1, \vec{b}_2\}$ has the form $[C\vec{b}_2]_{\{\vec{b}_1,\vec{b}_2\}} = \begin{bmatrix} 0 \\ \alpha \end{bmatrix}$. If it were possible, it would have been found while computing the kernel of $C - 3I$.

12.2 Diagonalization, eigenvectors, and eigenvalues

Theorem (diagonalizability and eigenvectors)

$A \in \mathbb{C}^{n,n}$ **diagonalizable**

$\Leftrightarrow$ the matrix mapping $A : \mathbb{C}^n \to \mathbb{C}^n$ has a **basis of eigenvectors**

$\Leftrightarrow$ for each EVAL λ of the mapping $A : \mathbb{C}^n \to \mathbb{C}^n$ we have **(algM λ) = (geoM λ)**

$\Leftrightarrow$ there exists an **invertible matrix** $S \in \mathbb{C}^{n,n}$ (the columns of the square matrix S are linearly independent eigenvectors) with $D := S^{-1}AS$ a diagonal matrix.

12.3 Algorithm to compute the diagonalization

Algorithm (diagonalization of matrices)

Input: $A \in \mathbb{F}^{n,n}$.

Output: diagonal matrix D (if it exists) and a diagonalizing matrix $S \in \mathbb{C}^{n,n}$

Step 1: Compute the eigenvalues $\lambda_1, \ldots, \lambda_k$ and associated eigenvectors (resp. associated eigenspaces) of A.

Step 2: If algM $=$ geoM for all λ_i, choose a basis $\mathcal{B}$ of $\mathbb{C}^n$ made up of eigenvectors of A.

(If not, A is not diagonalizable – break!)

Step 3: $D = S^{-1}AS$, whereby the EVALs are on the diagonal of D and the EVECs in $\mathcal{B}$ form the columns of S in a correct order.

Remark: The matrix S^{-1} does not need to be explicitly computed unless otherwise stated.

Example (diagonalizability)

Decide if the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \in \mathbb{C}^{3,3}$ is diagonalizable. Determine if applicable (invertible) matrices S, S^{-1} and a diagonal matrix D, so that $D = S^{-1}AS$ holds.

Step 1. Compute EVALs (= roots of $p_A(z)$).

$$p_A(z) = \det \begin{bmatrix} 1-z & 0 & 0 \\ 0 & -z & 1 \\ 0 & 1 & -z \end{bmatrix} = (1-z)(z^2-1) = -(z-\mathbf{1})^{\mathbf{2}}(z-(\mathbf{-1}))^{\mathbf{1}}$$

EVALs: $\boldsymbol{\lambda_{1/2} := 1}$, $\boldsymbol{\lambda_3 := -1}$.

Calculate ESs:

$$V_{\lambda_{1/2}} = \ker \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = \ker \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \\ \mathbf{1} \end{bmatrix} \right\}$$

$$V_{\lambda_3} = \ker \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \ker \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \operatorname{span}\left\{ \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \\ \mathbf{-1} \end{bmatrix} \right\}$$

Step 2: Check the algMs and geoMs.

$\boldsymbol{\lambda_{1/2}}$ has **algM 2** (double root of the char poly) and geoM 2 (due to the construction of the basis vectors in the algorithm that finds a basis for the kernel, the vectors that span the ES are always linearly independent, so $\dim V_{\lambda_{1/2}} = 2$).

$\boldsymbol{\lambda_3}$ has **algM 1** $\Rightarrow$ $\boldsymbol{\lambda_3}$ has geoM 1.

The algM and geoM of each EVAL of A agree. It follows that A is diagonalizable.

Choose $\mathcal{B} := \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \\ \mathbf{1} \end{bmatrix}, \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \\ \mathbf{-1} \end{bmatrix} \right\}$.

Step 3: With $S := \begin{bmatrix} 1 & \mathbf{0} & \mathbf{0} \\ 0 & \mathbf{1} & \mathbf{1} \\ 0 & \mathbf{1} & \mathbf{-1} \end{bmatrix}$ we form $D = \begin{bmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{-1} \end{bmatrix}$. S can be inverted as follows. S^{-1} : $[S|I] \overset{\text{rref}}{\to} [I|S^{-1}]$. This results in $S^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$.

Remark: If we had chosen $\tilde{\mathcal{B}} := \left\{ \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \\ \mathbf{-1} \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{0} \\ \mathbf{1} \\ \mathbf{1} \end{bmatrix} \right\}$, then we would have gotten $\tilde{S} = \begin{bmatrix} \mathbf{0} & 1 & \mathbf{0} \\ \mathbf{1} & 0 & \mathbf{1} \\ \mathbf{-1} & 0 & \mathbf{1} \end{bmatrix}$ and $\tilde{D} = \begin{bmatrix} \mathbf{-1} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(Step 4). Check answer.

$$S^{-1}AS = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = D$$

Remark: $A = SDS^{-1} \Rightarrow p_A(z) = p_D(z)$

Exercise 1 (learning check)

Decide whether $A := \begin{bmatrix} -16 & -14 & 0 \\ 21 & 19 & 0 \\ -21 & -14 & 5 \end{bmatrix}$ is a diagonalizable matrix. If possible, find matrices S, S^{-1} and D diagonal so that $A = SDS^{-1}$ holds.

Exercise 2 (learning check)

Which of the following matrices are diagonalizable? Determine, if possible, a diagonal matrix D and invertible matrices , S, S^{-1} with $D = S^{-1}AS$.

$$A := \begin{bmatrix} 2 & 3 & 1 & 1 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}, B := \begin{bmatrix} 3 & 3 & 1 & 1 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}, C := \begin{bmatrix} 3 & 3 & 0 & 1 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Exercise 3 (learning check)

Consider the rotation matrices $R_\varphi = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$.

(a) Ist R_φ diagonalizable for $R_\varphi \in \mathbb{R}^{2,2}$ when $\varphi = \frac{\pi}{4}$? Determine, if possible, a diagonalization of $R_{\frac{\pi}{4}}$ (i.e. find a diagonal matrix D and an invertible matrix S so that $A = SDS^{-1}$ holds).

(b) Determine a diagonalization of $R_\varphi : \mathbb{C}^2 \to \mathbb{C}^2$ for $\varphi \in]0, \pi[$.

12.4 Computing with diagonalizable matrices

Example (computing with diagonalizable matrices)

Calculate A^8 for $A := \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$ using the diagonalization of A:

$$A = \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix}$$

$$\begin{aligned} A^8 = (SDS^{-1})(SDS^{-1})\cdots(SDS^{-1}) &= SD^8S^{-1} \\ &= \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}^8 \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2^8 & 0 \\ 0 & 3^8 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 256 & 0 \\ 0 & 6561 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -256 & -256 \\ -6561 & -13122 \end{bmatrix} \\ &= \begin{bmatrix} -6049 & -12610 \\ 6305 & 12866 \end{bmatrix} \end{aligned}$$

Analogous to the formula $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ from Analysis we can define the notion of an exponential function for matrices.

Definition (exponential function of a matrix A)

The exponential function of a quadratic matrix $A \in \mathbb{F}^{n,n}$ is defined by $e^A := \sum_{k=0}^{\infty} \frac{A^k}{k!}$.

Remarks • The sequence of partial sums $\left(\sum_{k=0}^{N} \frac{A^k}{k!} \right)_{N=0}^{\infty}$ converges for every matrix $A \in \mathbb{F}^{n,n}$.

• If A is diagonalizable with $D = S^{-1}AS = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$, then e^A is given by

$$e^A = S e^{D} S^{-1} = S \begin{bmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{bmatrix} S^{-1}.$$

Proof:

$$\begin{aligned} e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = \sum_{k=0}^{\infty} \frac{(SDS^{-1})^k}{k!} = \sum_{k=0}^{\infty} \frac{SD^kS^{-1}}{k!} &= S \left(\sum_{k=0}^{\infty} \frac{D^k}{k!} \right) S^{-1} \\ &= S \left(\begin{bmatrix} \sum_{k=0}^{\infty} \frac{\lambda_1^k}{k!} & & 0 \\ & \ddots & \\ 0 & & \sum_{k=0}^{\infty} \frac{\lambda_n^k}{k!} \end{bmatrix} \right) S^{-1} \\ &= S \begin{bmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{bmatrix} S^{-1} = S e^D S^{-1} \end{aligned}$$

Example (exponential function)

Find the exponential function e^A for the matrix $A = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$.

We have already given a diagonalization for A in the last example.

$$\begin{aligned} e^A = Se^DS^{-1} &= \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} e^{\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}} \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e^2 & 0 \\ 0 & e^3 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -e^2 & -e^2 \\ -e^3 & -2e^3 \end{bmatrix} \\ &= \begin{bmatrix} 2e^2 - e^3 & 2e^2 - 2e^3 \\ -e^2 + e^3 & -e^2 + 2e^3 \end{bmatrix} \end{aligned}$$

12.5 Suggested solutions to the exercises in Chapter 12

Exercise 1

Step 1.

$$\det(A-\lambda I) = \begin{vmatrix} -16-\lambda & -14 & 0 \\ 21 & 19-\lambda & 0 \\ -21 & -14 & 5-\lambda \end{vmatrix} = (5-\lambda)[(-16-\lambda)(19-\lambda) - 21(-14)]$$
$$= (5-\lambda)[\lambda^2 - 3\lambda - 10] = -(\lambda-5)^2(\lambda+2) \stackrel{!}{=} 0$$

$\lambda_{1/2} = 5$ with algM 2, $\lambda_3 = -2$ with geoM 1

ESs

$V_{\lambda_{1/2}=5}$

$$\ker(A-5I) = \ker \begin{bmatrix} -21 & -14 & 0 \\ 21 & 14 & 0 \\ -21 & -14 & 0 \end{bmatrix} = \ker \begin{bmatrix} -21 & -14 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$
$$= \ker \begin{bmatrix} 3 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \text{span}\left\{ \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$V_{\lambda_3=-2}$

$$\ker(A-(-2)I) = \ker \begin{bmatrix} -14 & -14 & 0 \\ 21 & 21 & 0 \\ -21 & -14 & 7 \end{bmatrix} = \ker \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ -3 & -2 & 1 \end{bmatrix}$$
$$= \ker \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \ker \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Step 2.

$\lambda_{1/2}$: geoM $= \dim V_{\lambda_{1/2}} = 2 =$ algM
λ_3 : geoM $= \dim V_{\lambda_3} = 1 =$ algM

$\Rightarrow A$ is diagonalizable.

$$\mathcal{B} := \left\{ \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Step 3.

$$S = \begin{bmatrix} 2 & 0 & 1 \\ -3 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$(S^{-1} : [S|I] \stackrel{\text{NZSF}}{\longrightarrow} [I|S^{-1}])\ S^{-1} = \begin{bmatrix} -1 & -1 & 0 \\ -3 & -2 & 1 \\ 3 & 2 & 0 \end{bmatrix}$$

Check answer

Is $S^{-1}AS = D$?

$$S^{-1}AS = \begin{bmatrix} -1 & -1 & 0 \\ -3 & -2 & 1 \\ 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} -16 & -14 & 0 \\ 21 & 19 & 0 \\ -21 & -14 & 5 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ -3 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} = \cdots = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -2 \end{bmatrix} = D$$

Exercise 2

The EVALs of a matrix are the roots of the characteristic polynomial, and these are the elements on the diagonal when we have an upper triangular matrix, as is the case for each subproblem here. We also know that the inequality $1 \leq \text{geoM } \lambda_i \leq \text{algM } \lambda_i$ holds.

$\boxed{A}$ The EVALs of A are $\lambda_1 := 2, \lambda_2 := -1, \lambda_3 := 3, \lambda_4 := 4$. The algM of each EVAL is 1 so that $1 \leq \text{geoM } \lambda_i \leq \text{algM } \lambda_i = 1$, i.e. the geoM of each EVAL is also 1. It follows that A is diagonalizable.

To find a diagonalizing matrix S, we need to find an EVEC for each EVAL.

$\boxed{\lambda_1}$ A nontrivial element in $\ker(A-2I) = \ker \begin{bmatrix} 0 & 3 & 1 & 1 \\ 0 & -3 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ is clearly $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$. $(\text{geoM } \lambda_1) = 1 = \dim V_{\lambda_1} \Rightarrow$ all other EVECs for λ_1 lie in span$\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}\right\}$.

$\boxed{\lambda_2}$ A nontrivial element of $\ker(A+I) = \ker \begin{bmatrix} 3 & 3 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$ is $\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ (bring the matrix in rref $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and determine the kernel for practice.)

$\boxed{\lambda_3}$ A nontrivial element of $\ker(A-3I) = \ker \begin{bmatrix} -1 & 3 & 1 & 1 \\ 0 & -4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is $\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ (provide details).

$\boxed{\lambda_4}$ A nontrivial element of $\ker(A-4I) = \ker \begin{bmatrix} -2 & 3 & 1 & 1 \\ 0 & -5 & 0 & 2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is $\begin{bmatrix} 11 \\ 4 \\ 0 \\ 10 \end{bmatrix}$ (provide details).

$A = SDS^{-1}$ mit $S = \begin{bmatrix} 1 & -1 & 1 & 11 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$, $S^{-1} = \begin{bmatrix} 1 & 1 & -1 & \frac{-3}{2} \\ 0 & 1 & 0 & \frac{-2}{5} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{10} \end{bmatrix}$ (provide details)

$\boxed{B, C}$

EVAL λ_i	algM λ_i	geoM λ_i
$\lambda_{1/2} := 3$	2	1 or 2
$\lambda_3 := -1$	1	1
$\lambda_4 := 4$	1	1

$\boxed{B}$ The geoM of $\lambda_{1/2}$ is the dimension of the eigenspace $V_{\lambda_{1/2}} = \ker(B - \lambda_{1/2}I)$.

$$\ker(B-\lambda_{1/2}I) = \ker\left(\begin{bmatrix} 3 & 3 & 1 & 1 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}\right)$$

$$= \ker \begin{bmatrix} 0 & 3 & 1 & 1 \\ 0 & -4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \overset{\mathrm{II}+\frac{4}{3}\,\mathrm{I},\mathrm{III}\leftrightarrow\mathrm{IV}}{=} \ker \begin{bmatrix} 0 & 3 & 1 & 1 \\ 0 & 0 & \frac{4}{3} & \frac{10}{3} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

An ref for $B - \lambda_{1/2}I$ has only one leading variable (x_1). Therefore the dimension of the kernel of $B - \lambda_{1/2}I$ is only 1. This is the dimension of the associated ES:

$$(\text{geoM } \lambda_{1/2}) = \dim V_{\lambda_{1/2}} = 1 \neq 2 = (\text{algM } \lambda_{1/2}).$$

There is no basis of EVECs for $\mathbb{R}^4$ so that the matrix B is not diagonalizable.

$\boxed{C}$ The geoM of $\lambda_{1/2}$ is the dimension of the eigenspace $V_{\lambda_{1/2}} = \ker(C - \lambda_{1/2}I)$.

$$\ker(C-\lambda_{1/2}I) = \ker\left(\begin{bmatrix} 3 & 3 & 0 & 1 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}\right)$$

$$\begin{bmatrix} 0 & 3 & 0 & 1 \\ 0 & -4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \text{compute this!}$$

$x_2 = 0, x_4 = 0, x_1, x_3$ arbitrary $\Rightarrow V_{\lambda_{1/2}} = \operatorname{span}\left\{\begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\end{bmatrix}\right\}$

so that the geoM ($= 2$) of $\lambda_{1/2}$ agrees with the algM of $\lambda_{1/2}$. The other EVALs have algM 1, forcing the geoM to also be one. Thus, C is diagonalizable. To find a diagonalizing matrix S, we still need a nontrivial eigenvector for each of the eigenvalues λ_3 and λ_4:

$\boxed{\lambda_3}$

$$\ker(C-\lambda_3 I) = \ker\left(\begin{bmatrix} 3 & 3 & 0 & 1 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = \ker\begin{bmatrix} 4 & 3 & 0 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Show that $\begin{bmatrix}3\\-4\\0\\0\end{bmatrix}$ is an EVEC for the EVAL $\lambda_3 = -1$.

$\boxed{\lambda_4}$

$$\ker(C-\lambda_4 I) = \ker\left(\begin{bmatrix} 3 & 3 & 0 & 1 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}\right) = \ker\begin{bmatrix} -1 & 3 & 0 & 1 \\ 0 & -5 & 0 & 2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\overset{-\mathrm{I},-\frac{1}{5}\mathrm{II},-\mathrm{III}}{=} \ker\begin{bmatrix} 1 & -3 & 0 & -1 \\ 0 & 1 & 0 & -\frac{2}{5} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \overset{\mathrm{I}+3\mathrm{II}}{=} \ker\begin{bmatrix} 1 & 0 & 0 & -\frac{11}{5} \\ 0 & 1 & 0 & -\frac{2}{5} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = \frac{11}{5}x_4, x_2 = \frac{2}{5}x_4, x_3 = 0, x_4$ arbitrary. Set $x_4 = 5$ to get the EVEC $\begin{bmatrix}11\\2\\0\\5\end{bmatrix}$ for the EVAL $\lambda_4 = 4$.

$A = SDS^{-1}$ mit $S = \begin{bmatrix} 1 & 0 & 3 & 11 \\ 0 & 0 & -4 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$, $D = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$, $S^{-1} = \begin{bmatrix} 1 & \frac{3}{4} & 0 & \frac{-5}{2} \\ 0 & 0 & 1 & 0 \\ 0 & \frac{-1}{4} & 0 & \frac{1}{10} \\ 0 & 0 & 0 & \frac{1}{5} \end{bmatrix}$. (Show yourself that S^{-1} is the inverse of S.)

Exercise 3 (a) The (EVAL) of the matrix $R_{\frac{\pi}{4}} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$ are the zeros of the characteristic polynomial $p_{R_{\frac{\pi}{4}}}(z) = \det(R_{\frac{\pi}{4}} - zI_2) = z^2 - \sqrt{2}z + 1$, which we computed in an example in Section 11.2:

$\boldsymbol{\lambda_1 = \frac{\sqrt{2}+i\sqrt{2}}{2}}, \quad \boldsymbol{\lambda_2 = \frac{\sqrt{2}-i\sqrt{2}}{2}}.$

These are not **real** EVALs, so there can be no real EVECs. The matrix $R_{\frac{\pi}{4}} : \mathbb{R}^2 \to \mathbb{R}^2$ is not diagonalizable as a real mapping.

If we consider $R_{\frac{\pi}{4}}$ as a **complex** mapping $R_{\frac{\pi}{4}} : \mathbb{C}^2 \to \mathbb{C}^2$, then $R_{\frac{\pi}{4}}$ is diagonalizable because the two eigenvalues $\lambda_1 = \frac{\sqrt{2}+i\sqrt{2}}{2}$, $\lambda_2 = \frac{\sqrt{2}-i\sqrt{2}}{2}$ are distinct.

To find S, determine the ESs $V_{\lambda_i}, i = 1, 2,$. We have already done this in Chapter 11:

$$V_{\boldsymbol{\lambda_1}} = \left\{ r \begin{bmatrix} \boldsymbol{i} \\ \mathbf{1} \end{bmatrix} \mid r \in \mathbb{C} \right\} \text{ is the ES for the EVAL } \boldsymbol{\lambda_1 = \frac{\sqrt{2}+i\sqrt{2}}{2}},$$

$$V_{\boldsymbol{\lambda_2}} = \left\{ s \begin{bmatrix} -i \\ 1 \end{bmatrix} \mid s \in \mathbb{C} \right\} \text{ is the ES for the EVAL } \boldsymbol{\lambda_2 = \frac{\sqrt{2}-i\sqrt{2}}{2}}.$$

Choosing as basis of eigenvectors $\mathcal{B} = \left\{ \begin{bmatrix} \boldsymbol{i} \\ \mathbf{1} \end{bmatrix}, \begin{bmatrix} -i \\ 1 \end{bmatrix} \right\}$, we get

$$S = \begin{bmatrix} \boldsymbol{i} & -i \\ \mathbf{1} & 1 \end{bmatrix}$$

as the diagonalizing matrix. The associated diagonal matrix has the eigenvalues on the diagonal:

$$D = \begin{bmatrix} \boldsymbol{\frac{\sqrt{2}+i\sqrt{2}}{2}} & 0 \\ 0 & \boldsymbol{\frac{\sqrt{2}-i\sqrt{2}}{2}} \end{bmatrix}.$$

(b) The characteristic polynomial, the eigenvalues and associated eigenvectors were already found in the first example in Chapter 11:

$$p_{R_\varphi}(z) = z^2 - (2\cos\varphi)z + 1.$$

$$\text{EVAL:} \quad \boldsymbol{\lambda_1 = \cos\varphi + i\,\sin\varphi}, \quad \boldsymbol{\lambda_2 = \cos\varphi - i\,\sin\varphi}$$

$$\text{ER:} \quad V_{\boldsymbol{\lambda_1}} = \left\{ r \begin{bmatrix} \boldsymbol{i} \\ \mathbf{1} \end{bmatrix} \mid r \in \mathbb{C} \right\} \quad V_{\boldsymbol{\lambda_2}} = \left\{ s \begin{bmatrix} -i \\ 1 \end{bmatrix} \mid s \in \mathbb{C} \right\}$$

A diagonalization of R_φ (considered as a complex matrix) is therefore

$$S = \begin{bmatrix} \boldsymbol{i} & -i \\ \mathbf{1} & 1 \end{bmatrix}, \quad D = \begin{bmatrix} \boldsymbol{\cos\varphi + i\sin\varphi} & 0 \\ 0 & \boldsymbol{\cos\varphi - i\sin\varphi} \end{bmatrix}.$$

Index